

Complex Analysis

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Contents

Preface	7
1 Algebraic properties of complex numbers	9
1.1 Exercises	18
2 Topological properties of $\mathbb{C}$	20
2.1 Exercises	28
3 Differentiation	30
3.1 Exercises	41
4 Path integrals	44
4.1 Exercises	49
5 Infinite series and products	50
5.1 Exercises	64
6 Sequences of functions	66
7 Power series	74
7.1 Exercises	81
8 Local Cauchy theorems	86
8.1 Exercises	93
9 Power series representations	94
9.1 Exercises	98
10 Applications of Cauchy's theory	99
10.1 Exercises	105
11 Zeros of holomorphic functions	108
12 The Maximum Modulus Theorem	114
12.1 Exercises	117
13 The Open Mapping Theorem	119
13.1 Exercises	124

14 Elementary functions	125
14.1 Exercises	132
15 The Riemann sphere	138
15.1 Exercises	144
16 Isolated singularities	146
16.1 Exercises	159
17 Möbius transformations	161
17.1 Exercises	174
18 Schwarz's Lemma	178
18.1 Exercises	186
19 Winding numbers	189
19.1 Exercises	198
20 Global Cauchy theorems	203
20.1 Exercises	209
21 Laurent series	210
21.1 Exercises	218
22 The Residue Theorem	223
22.1 Exercises	232
23 Normal families	236
23.1 Exercises	241
24 The Riemann Mapping Theorem	243
24.1 Exercises	252
25 The Cauchy transform	260
26 Runge's Approximation Theorem	269
27 Characterizations of simply connected regions	277
28 The Gamma Function	281

<i>CONTENTS</i>	5
29 Solutions and hints for selected exercises	284
References	289

Preface

These notes cover the material of a course on complex analysis that I taught repeatedly at UCLA. In many respects I closely follow Rudin's book on "Real and Complex Analysis".

The notes give a fairly accurate account of the material covered in class. They are rather terse as oral discussions that gave further explanations or put results into perspective are mostly omitted. In addition, all pictures and diagrams are currently missing as it is much easier to produce them on the blackboard than to put them into print.

1 Algebraic properties of complex numbers

1.1. Intuitive idea. Complex numbers are expressions of the form $a + bi$ with real numbers a and b . Here i is the *imaginary unit*. One computes (i.e., adds and multiplies) with complex numbers as usual, but sets $i^2 = i \cdot i := -1$. For example,

$$(3 + 5i)(2 + i) = 6 + 10i + 3i + 5i^2 = 6 + 13i - 5 = 1 + 13i.$$

1.2. Definition of the complex numbers. For a rigorous definition we let the set of *complex numbers* be

$$\mathbb{C} := \{(a, b) : a, b \in \mathbb{R}\}$$

with the correspondence $(a, b) \cong a + bi$ in mind. *Addition* and *multiplication* are defined accordingly:

$$\begin{aligned}(a, b) + (c, d) &:= (a + c, b + d), \\ (a, b) \cdot (c, d) &:= (ac - bd, ad + bc).\end{aligned}$$

One often omits the multiplication sign and writes $zw := z \cdot w$ for $z, w \in \mathbb{C}$. One also uses the convention that multiplication binds stronger than addition. So $u + vw = u + (v \cdot w)$ for $u, v, w \in \mathbb{C}$, etc.

That one computes with complex numbers “as usual” is mathematically expressed by the following fact.

Theorem 1.3. $(\mathbb{C}, +, \cdot)$ is a field, that is:

1. $(\mathbb{C}, +)$ is an abelian group, which means that
 - 1.1 the addition $+$ is associative,
 - 1.2 there exists a neutral element $0 := (0, 0) \in \mathbb{C}$ with respect to addition,
 - 1.3 every element $(a, b) \in \mathbb{C}$ has an (additive) inverse $(-a, -b) \in \mathbb{C}$,
 - 1.4 the addition $+$ is commutative.
2. $(\mathbb{C}, +, \cdot)$ is a commutative ring, which means that
 - 2.1 $(\mathbb{C}, +)$ is an abelian group,

2.2 the multiplication $\cdot$ is associative and commutative,

2.3 the distributive law holds.

3. $(\mathbb{C}^*, \cdot)$ is a group, where $\mathbb{C}^* := \mathbb{C} \setminus \{0\}$.

The proof is straightforward, but tedious, and so we skip it.

Remark 1.4. If we write (for the moment), $\tilde{a} := (a, 0) \in \mathbb{C}$ for $a \in \mathbb{R}$, then

$$\begin{aligned}\tilde{a} + \tilde{b} &= (a, 0) + (b, 0) \\ &= (a + b, 0) = \widetilde{a + b},\end{aligned}$$

and

$$\begin{aligned}\tilde{a} \cdot \tilde{b} &= (a, 0) \cdot (b, 0) \\ &= (ab, 0) = \widetilde{ab}.\end{aligned}$$

This means that with expressions $\tilde{a}$, $\tilde{b}$, etc., one can compute in exactly the same way as with the underlying real numbers a , b , etc. More precisely, the map

$$\varphi: \mathbb{R} \rightarrow \mathbb{C}, a \in \mathbb{R} \mapsto \varphi(a) := \tilde{a},$$

is a *field isomorphism* of $\mathbb{R}$ onto its image in $\mathbb{C}$.

Accordingly, one “identifies” the image of $\mathbb{R}$ under φ with $\mathbb{R}$, writes a instead of $\tilde{a}$, and considers $\mathbb{R}$ as subset of $\mathbb{C}$.

Now let $i := (0, 1)$. Then for $a, b \in \mathbb{R}$ we have

$$\begin{aligned}(a, b) &= (a, 0) + (0, b) \\ &= (a, 0) + (b, 0) \cdot (0, 1) \\ &= \tilde{a} + \tilde{b}i = a + bi.\end{aligned}$$

So every complex number $z \in \mathbb{C}$ can uniquely be written as $z = a + bi$, where $a, b \in \mathbb{R}$. Note also that we have

$$\begin{aligned}i^2 := i \cdot i &= (0, 1) \cdot (0, 1) \\ &= (-1, 0) \\ &= \widetilde{-1} = -1.\end{aligned}$$

Using these conventions we reconcile the precise definition of complex numbers and their operations with the intuitive notion that we took as our starting point.

Definition 1.5. Let $z = a + bi \in \mathbb{C}$, where $a, b \in \mathbb{R}$. We define

- (i) $\operatorname{Re}(z) := a$ (the *real part* of z),
- (ii) $\operatorname{Im}(z) := b$ (the *imaginary part* of z),
- (iii) $\bar{z} := a - bi$ (the *complex conjugate* of z),
- (iv) $|z| := \sqrt{a^2 + b^2}$ (the *absolute value* of z).

1.6. Geometric interpretations. These concepts and also addition of complex numbers have obvious geometric interpretations if one identifies $z = a + bi$ with the point (a, b) in the plane $\mathbb{R}^2$.

1.7. Subtraction and division. As in every field one can define a notion of *subtraction* and *division* of complex numbers. Namely, if $z \in \mathbb{C}$, one denotes the additive inverse of z by $-z$ and defines

$$w - z := w + (-z)$$

for $z, w \in \mathbb{C}$. If $z = a + bi$, $w = c + di$, then $-z = (-a) + (-b)i$, and so

$$w - z = (c - a) + (d - b)i.$$

If $z \neq 0$, then we denote by z^{-1} the (*multiplicative*) *inverse* of z . For $z = a + bi \neq 0$ we have

$$z^{-1} = \frac{a}{a^2 + b^2} - \frac{b}{a^2 + b^2}i.$$

For $w \in \mathbb{C}$ one defines

$$w/z = \frac{w}{z} := w \cdot z^{-1}.$$

One can compute with fractions of complex numbers as usual (as in any field). Using the fact that $z\bar{z} = |z|^2$, one can simplify fractions of complex numbers by multiplying in numerator and denominator by the complex conjugate of the denominator. For example,

$$\begin{aligned} \frac{2+i}{3+5i} &= \frac{(2+i)(3-5i)}{(3+5i)(3-5i)} \\ &= \frac{(6+5) + (-10+3)i}{3^2 + 5^2} \\ &= \frac{11-7i}{34} = \frac{11}{34} - \frac{7}{34}i. \end{aligned}$$

Theorem 1.8. *Let $z, w \in \mathbb{C}$. Then the following statements are true:*

- (i) $\operatorname{Re}(z) = \frac{1}{2}(z + \bar{z})$,
- (ii) $\operatorname{Im}(z) = \frac{1}{2i}(z - \bar{z})$,
- (iii) $\operatorname{Re}(z + w) = \operatorname{Re}(z) + \operatorname{Re}(w)$,
- (iv) $\operatorname{Im}(z + w) = \operatorname{Im}(z) + \operatorname{Im}(w)$,
- (v) $z \in \mathbb{R}$ iff $\operatorname{Im}(z) = 0$ iff $z = \bar{z}$,
- (vi) $\bar{\bar{z}} = z$,
- (vii) $\overline{z + w} = \bar{z} + \bar{w}$, $\overline{z - w} = \bar{z} - \bar{w}$,
- (viii) $\overline{z \cdot w} = \bar{z} \cdot \bar{w}$,
- (ix) $\overline{\left(\frac{z}{w}\right)} = \frac{\bar{z}}{\bar{w}}$,
- (x) $z \cdot \bar{z} = |z|^2$,
- (xi) $|z| = 0$ iff $z = 0$,
- (xii) $|z \cdot w| = |z| \cdot |w|$,
- (xiii) $\left|\frac{z}{w}\right| = \frac{|z|}{|w|}$,
- (xiv) $|z + w| \leq |z| + |w|$ (triangle inequality).

Proof. The proofs of these facts are straightforward, often tedious, and we omit the details; as an example, we will prove (xii).

If $z = a + bi$ and $w = c + di$, where $a, b, c, d \in \mathbb{R}$, then

$$z \cdot w = (ac - bd) + (ad + bc)i,$$

and so

$$\begin{aligned} |z \cdot w|^2 &= (ac - bd)^2 + (ad + bc)^2 \\ &= a^2c^2 + b^2d^2 + a^2d^2 + b^2c^2 \\ &= (a^2 + b^2)(c^2 + d^2) \\ &= |z|^2 \cdot |w|^2. \end{aligned}$$

Hence

$$|z \cdot w| = |z| \cdot |w|.$$

(For this conclusion it is important that the terms on both sides are non-negative). $\square$

Definition 1.9 (The exponential function for complex arguments). For $z = x + iy$, where $x, y \in \mathbb{R}$, we define

$$e^z = \exp(z) := e^x(\cos y + i \sin y).$$

Note that for $z \in \mathbb{R}$, when $y = \operatorname{Im}(z) = 0$, this agrees with the usual exponential function. So the exponential function for complex arguments is an extension of the exponential function for real arguments. Choosing this particular extension seems arbitrary and unmotivated at this point. It will become more natural after we have introduced power series, because we will see that the complex exponential function can be represented by the usual power series for the real exponential function. A deeper reason for choosing this particular extension is that the complex exponential function is *holomorphic* and by the *uniqueness theorem* every function on $\mathbb{R}$ has at most one holomorphic extension to $\mathbb{C}$. All this will be discussed later in the course.

Theorem 1.10. *The exponential function $\exp: \mathbb{C} \rightarrow \mathbb{C}$ has the following properties:*

- (i) $e^{it} = \cos t + i \sin t$ for $t \in \mathbb{R}$ (Euler–Moivre formula),
- (ii) $e^{z+w} = e^z \cdot e^w$ for $z, w \in \mathbb{C}$ (functional equation of $\exp$),
- (iii) $e^{z+2\pi i} = e^z$ for $z \in \mathbb{C}$ ($\exp$ is $2\pi i$ -periodic),
- (iv) $e^z = 1$ iff $z = 2\pi i k$ with $k \in \mathbb{Z}$,
- (v) $e^w = e^z$ iff $w = z + 2\pi i k$ with $k \in \mathbb{Z}$.

Proof. (i) Obvious from the definition.

(ii) If $z = x + iy$ and $w = u + iv$ with $x, y, u, v \in \mathbb{R}$, then

$$z + w = (x + u) + i(y + v).$$

Note that

$$\begin{aligned}\cos(y+v) &= \cos y \cos v - \sin y \sin v, \\ \sin(y+v) &= \sin y \cos v + \cos y \sin v.\end{aligned}$$

Hence

$$\begin{aligned}e^z \cdot e^w &= e^x(\cos y + i \sin y)e^u(\cos v + i \sin v), \\ &= e^{x+u}((\cos y \cos v - \sin y \sin v) + i(\sin y \cos v + \cos y \sin v)) \\ &= e^{x+u}(\cos(y+v) + i \sin(y+v)) \\ &= e^{(x+u)+i(y+v)} = e^{z+w}.\end{aligned}$$

(iv) Let $z = x + iy$, $x, y \in \mathbb{R}$. Then

$$\begin{aligned}e^z = 1 &\Leftrightarrow e^x(\cos y + i \sin y) = 1 \\ &\Leftrightarrow e^x \cos y = 1 \text{ and } e^x \sin y = 0, \\ &\Leftrightarrow e^x \cos y = 1 \text{ and } \sin y = 0, \\ &\Leftrightarrow e^x \cos y = 1 \text{ and } y = n\pi \text{ for } n \in \mathbb{Z}, \\ &\Leftrightarrow e^x(-1)^n = 1 \text{ and } y = n\pi \text{ for } n \in \mathbb{Z}, \\ &\Leftrightarrow e^x = 1 \text{ and } y = n\pi \text{ for some } n \in \mathbb{Z} \text{ even}, \\ &\Leftrightarrow x = 0 \text{ and } y = n\pi \text{ for some } n \in \mathbb{Z} \text{ even}, \\ &\Leftrightarrow z = x + iy = 2\pi ik \text{ for } k \in \mathbb{Z}.\end{aligned}$$

(iii) Using (ii) and (iv) we have

$$e^{z+2\pi i} = e^z \cdot e^{2\pi i} = e^z \cdot 1 = e^z$$

for $z \in \mathbb{C}$.

(v) Note that $e^z \neq 0$ for $z \in \mathbb{C}$, because $e^z \cdot e^{-z} = e^0 = 1 \neq 0$. So for $z, w \in \mathbb{C}$ we have by (iv),

$$\begin{aligned}e^w = e^z &\Leftrightarrow e^{w-z} = e^w \cdot e^{-z} = e^z \cdot e^{-z} = e^0 = 1, \\ &\Leftrightarrow w - z = 2\pi ik \text{ for some } k \in \mathbb{Z}, \\ &\Leftrightarrow w = z + 2\pi ik \text{ for some } k \in \mathbb{Z}. \quad \square\end{aligned}$$

1.11. Mapping properties of $\exp$. The exponential function maps lines parallel to the real axis to rays starting at 0; lines parallel to the imaginary axis are mapped to circles centered at 0.

The exponential function maps the strip

$$S = \{x + iy : x \in \mathbb{R}, 0 < y < 2\pi\}$$

bijectively onto $\mathbb{C} \setminus [0, \infty)$.

1.12. Polar coordinates. If $z = x + iy$, $x, y \in \mathbb{R}$, then by using polar coordinates we can write

$$\begin{aligned} x &= r \cos \varphi, \\ y &= r \sin \varphi, \end{aligned}$$

where $r \geq 0$ and $\varphi \in \mathbb{R}$. Hence every complex number can be written as

$$z = x + iy = r(\cos \varphi + i \sin \varphi) = re^{i\varphi},$$

where $r \geq 0$ and $\varphi \in \mathbb{R}$. Note that $r = |z|$ is the absolute value of z . The angle φ is called the *argument* of z , written $\varphi = \arg(z)$. It is only determined up to integer multiples of 2π . If $\operatorname{Re}(z) \neq 0$, then

$$\tan \varphi = \frac{\operatorname{Im}(z)}{\operatorname{Re}(z)}.$$

This formular allows the computation of φ for given z .

1.13. Geometric interpretation of multiplication and division of complex numbers. Let $z = re^{i\alpha}$ and $w = se^{i\beta}$, where $r, s \geq 0$ and $\alpha, \beta \in \mathbb{R}$. Then

$$z \cdot w = re^{i\alpha}se^{i\beta} = rse^{i(\alpha+\beta)},$$

and

$$\frac{z}{w} = \frac{re^{i\alpha}}{se^{i\beta}} = \frac{r}{s}e^{i\alpha}e^{-i\beta} = \frac{r}{s}e^{i(\alpha-\beta)}, \quad w \neq 0.$$

So complex numbers are multiplied by multiplying their absolute values and adding their arguments. One divides them by dividing their absolute values and subtracting their arguments.

These facts and their geometric interpretations will be used throughout; for example, multiplication by $i = e^{i\pi/2}$, i.e., the map $z \in \mathbb{C} \mapsto iz$ correspond to counterclockwise rotation in the plane by 90° .

1.14. Computation of n -th roots. For $z \in \mathbb{C}$ and $n \in \mathbb{N}$ we set

$$z^n := \underbrace{z \cdots z}_{n \text{ factors}}.$$

We use the convention $z^0 := 1$ for $z \in \mathbb{C}$, and set $z^n := (z^{-1})^{|n|}$ for $n \in \mathbb{Z}$, $n < 0$, $z \neq 0$. One then has the usual computational rules

$$\begin{aligned} z^n z^k &= z^{n+k}, \\ (z^n)^k &= z^{nk}, \\ (zw)^n &= z^n w^n \end{aligned}$$

for $z, w \in \mathbb{C}$, $z, w \neq 0$, $n, k \in \mathbb{Z}$.

Now let $n \in \mathbb{N}$, $a \in \mathbb{C}$, $a \neq 0$. Every solution z of the equation $z^n = a$ (for given n and a) is called an n -th root of a . As we will see momentarily, every $a \neq 0$ has precisely n distinct n -th roots; hence for complex numbers a we will usually not use the ambiguous notation $\sqrt[n]{a}$, but we will use it for positive real numbers a (where it denotes the unique positive real number with $(\sqrt[n]{a})^n = a$).

We write $a = re^{i\varphi}$ with $r > 0$, $\varphi \in \mathbb{R}$, and use the *ansatz* $z = \rho e^{i\alpha}$, where $\rho > 0$ and $\alpha \in \mathbb{R}$. Then

$$z^n = \underbrace{z \cdots z}_{n \text{ factors}} = \rho^n e^{in\alpha} = re^{i\varphi}.$$

Hence $\rho^n = r$ and $n\alpha = \varphi + 2\pi k$ for $k \in \mathbb{Z}$. This implies $\rho = \sqrt[n]{r}$ and

$$\alpha = \frac{\varphi}{n} + \frac{2\pi}{n}k, \quad k \in \mathbb{Z}.$$

We conclude that a complex number $a = re^{i\varphi}$, $a \neq 0$, has n distinct n -th roots

$$z_k = \sqrt[n]{r} e^{i\alpha_k},$$

where

$$\alpha_k = \frac{\varphi}{n} + \frac{2\pi}{n}k \quad \text{with} \quad k \in \{0, \dots, n-1\}.$$

1.15. Examples. (a) Third roots of $a = -8$: We have $a = -8 = 8e^{i\pi}$; so

$$z_k = \sqrt[3]{8} e^{i\alpha_k} = 2e^{i\alpha_k},$$

$$\alpha_k = \frac{\pi}{3} + \frac{2\pi}{3}k \quad \text{with} \quad k \in \{0, 1, 2\}.$$

Hence

$$\alpha_0 = \frac{\pi}{3}, \quad \alpha_1 = \pi, \quad \alpha_2 = \frac{5\pi}{3},$$

and

$$\begin{aligned} z_0 &= 2e^{i\pi/3} = 2(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}) = 2(\frac{1}{2} + i\frac{\sqrt{3}}{2}) = 1 + i\sqrt{3}, \\ z_1 &= 2e^{i\pi} = -2, \\ z_2 &= 2e^{i5\pi/3} = 2(\cos \frac{5\pi}{3} + i \sin \frac{5\pi}{3}) = 2(-\frac{1}{2} - i\frac{\sqrt{3}}{2}) = -1 - i\sqrt{3}. \end{aligned}$$

(b) Computation of square roots by a different method: To solve the equation

$$z^2 = -3 + 4i,$$

for example, we use the *ansatz* $z = a + bi$ with $a, b \in \mathbb{R}$ and solve for a and b :

$$z^2 = (a + bi)^2 = a^2 - b^2 + 2abi = -3 + 4i.$$

Hence

$$a^2 - b^2 = -3 \quad \text{and} \quad 2ab = 4. \tag{1.1}$$

Squaring both equations and adding leads to

$$(a^2 - b^2)^2 + 4a^2b^2 = a^4 + 2a^2b^2 + b^4 = (a^2 + b^2)^2 = (-3)^2 + 4^2 = 25.$$

Thus,

$$a^2 + b^2 = 5.$$

Combining this with (1.1) gives

$$2a^2 = 2 \quad \Rightarrow \quad a^2 = 1 \quad \Rightarrow \quad a_1 = 1, a_2 = -1$$

and

$$b_1 = 2, b_2 = -2.$$

So we get the solutions

$$z_1 = 1 + 2i \quad \text{and} \quad z_2 = -1 - 2i.$$

Note that $z_2 = -z_1$ as it should be.

(c) Solutions of quadratic equations can be computed as usual by completing the square, etc. A more general fact is true: Every polynomial equation

$$z^n + a_{n-1}z^{n-1} + \cdots + a_1z + a_0 = 0,$$

where $a_0, \dots, a_{n-1} \in \mathbb{C}$ has a solution $z \in \mathbb{C}$. This *Fundamental Theorem of Algebra* will be proved later in this course.

1.1 Exercises

Problem 1.1. Find all solutions z of the following equations:

- (a) $z^2 = -15 - 8i$,
- (b) $(3 + 5i)z^2 + (3 + 5i)z + (8 + 2i) = 0$.

Problem 1.2. Sketch the following sets:

- (a) $\left\{ z \in \mathbb{C} : \frac{|z - 3i|}{|z + 3i|} \leq 2 \right\}$,
- (b) $\left\{ z \in \mathbb{C} : \operatorname{Im} \left(\frac{z - 1}{1 + i} \right) \geq 0 \right\}$,
- (c) $\{ z \in \mathbb{C} : |z - 1|/|z + 2i| < \sqrt{2} \}$.

Problem 1.3. Find $f(M)$ for the following maps $f: \mathbb{C} \rightarrow \mathbb{C}$ and the following sets $M \subseteq \mathbb{C}$:

- (a) $f(z) = e^z$ and $M = \{ z \in \mathbb{C} : 2 \leq \operatorname{Re}(z) \leq 3 \text{ and } |\operatorname{Im}(z)| \leq \pi/2 \}$,
- (b) $f(z) = z^2$ and $M = \{ z \in \mathbb{C} : \operatorname{Im}(z) > 0 \text{ and } 1 \leq |z| \leq 2 \}$.

Problem 1.4. (a) Show that $(\cos x + i \sin x)^n = \cos(nx) + i \sin(nx)$ for all $n \in \mathbb{N}$, $x \in \mathbb{R}$.

- (b) Express $\cos(5x)$ in terms of $\cos x$, and $\sin(5x)$ in terms of $\sin x$.
- (c) Express $\cos\left(\frac{2\pi}{5}\right)$ and $\sin\left(\frac{2\pi}{5}\right)$ in terms of roots of real numbers.

Problem 1.5. Suppose that $z, w \in \mathbb{C}$ and $|z|, |w| < 1$. Show that then $\left| \frac{z - w}{1 - \bar{z}w} \right| < 1$.

Problem 1.6. Let $P(z) = a_0 + a_1z + \cdots + a_{n-1}z^{n-1} + z^n$ be a polynomial of degree $n \in \mathbb{N}$ with real coefficients $a_0, \dots, a_{n-1} \in [0, 1]$.

Show that if $P(z_0) = 0$ for some $z_0 \in \mathbb{C}$, then $\operatorname{Re}(z_0) < 0$ or $|z_0| < (1 + \sqrt{5})/2$.

Problem 1.7. (a) Let $P(z) = a_0 + a_1z + \cdots + a_{n-1}z^{n-1} + z^n$ be a polynomial of degree $n \in \mathbb{N}$ with complex coefficients $a_0, \dots, a_{n-1}$. Suppose that $P(z) = \prod_{j=1}^n (z - z_j)$. Find an expression for $\sum_{j=1}^n z_j^2$ in terms of the coefficients of P .

(b) Let $n \in \mathbb{N}$ and $\zeta_k = \exp\left(\frac{2\pi i k}{n}\right)$ for $k = 1, \dots, n-1$. Find a simple expression for

$$\sum_{k=1}^{n-1} \frac{1}{(\zeta_k - 1)^2}$$

in terms of n .

Problem 1.8. Let $n \in \mathbb{N}$. Find all (complex) roots of the polynomial

$$P(z) = (z + 1)^n - z^n$$

and express them explicitly using trigonometric functions (applied to real arguments).

Problem 1.9. (a) Show that a map $L: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is linear (as a map on the vector space $\mathbb{R}^2$ over $\mathbb{R}$) if and only if there are constants $\alpha, \beta \in \mathbb{C}$ such that L can be written in the form

$$L(z) = \alpha z + \beta \bar{z}, \quad z \in \mathbb{C}, \quad (1.2)$$

under the identification of points $w \in \mathbb{C}$ with $(\operatorname{Re}(w), \operatorname{Im}(w)) \in \mathbb{R}^2$.

Hint: The map L can be represented by a matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with respect to the standard basis in $\mathbb{R}^2$. Express α and β in terms of the matrix coefficients a, b, c, d and vice versa.

(b) When is L a $\mathbb{C}$ -linear map, that is, linear as a map on the vector space $\mathbb{C}$ over $\mathbb{C}$? Find necessary and sufficient conditions in terms of α, β and in terms of a, b, c, d .

(c) Express $\det(L)$ and the operator norm

$$\|L\| := \sup\{|L(z)| : z \in \mathbb{C}, |z| \leq 1\}$$

in terms of α and β .

2 Topological properties of $\mathbb{C}$

In this section we summarize some standard facts from point-set topology. We will mostly omit proofs or only give a brief outline.

Definition 2.1. A *metric space* (X, d) is a set X together with a function $d: X \times X \rightarrow [0, \infty)$ such that for all $x, y, z \in X$ the following properties are true:

- (i) $d(x, y) = 0$ iff $x = y$,
- (ii) $d(x, y) = d(y, x)$ (symmetry),
- (iii) $d(x, z) \leq d(x, y) + d(y, z)$ (triangle inequality).

If these axioms hold, then d is called a *metric* on X .

In a metric space (X, d) we define for $a \in X$ and $r > 0$,

$$B(a, r) := \{x \in X : d(x, a) < r\},$$
$$\overline{B}(a, r) := \{x \in X : d(x, a) \leq r\},$$

and call this the *open ball* and the *closed ball* of radius r centered at a , respectively.

Often one also calls $B(a, r)$ the *(open) r -neighborhood* of a .

Example 2.2. If we define $d(z, w) := |z - w|$ for $z, w \in \mathbb{C}$, then d is a metric on $\mathbb{C}$, called the *Euclidean metric*. From now on we consider $\mathbb{C}$ as a metric space equipped with the Euclidean metric.

Definition 2.3. Let (X, d) be a metric space, and $M \subseteq X$. A point $x \in X$ is called

- (i) an *interior point* of M if there exists $\epsilon > 0$ such that $B(x, \epsilon) \subseteq M$,
- (ii) an *exterior point* of M if there exists $\epsilon > 0$ such that $B(x, \epsilon) \subseteq X \setminus M$.
- (iii) a *boundary point* of M if $M \cap B(x, \epsilon) \neq \emptyset$ and $(X \setminus M) \cap B(x, \epsilon) \neq \emptyset$ for all $\epsilon > 0$.

The set of interior points of M is denoted by $\text{int}(M)$, and the set of boundary points by ∂M .

Remark 2.4. Every point in X is an interior point, an exterior point, or a boundary point of M , and these cases are mutually exclusive. An interior point of M always belongs to M , while an exterior point always lies in the complement of M in X . A boundary point of M may or may not belong to M .

Definition 2.5. Let (X, d) be a metric space, and $M \subseteq X$. Then M is called *open* if it has only interior points, i.e., for all $x \in M$ there exists $\epsilon > 0$ such that $B(x, \epsilon) \subseteq M$.

The set M is called *closed* if it contains all of its boundary points.

Example 2.6. $\mathbb{D} := \{z \in \mathbb{C} : |z| < 1\}$ is an open set in $\mathbb{C}$, called the *open unit disk*. The set $\overline{\mathbb{D}} := \{z \in \mathbb{C} : |z| \leq 1\}$ is a closed set in $\mathbb{C}$, called the *closed unit disk*. The set $M = \mathbb{D} \cup \{1\}$ is neither open nor closed.

Definition 2.7. Let (X, d) be a metric space, and $M \subseteq X$. Then $\overline{M} := M \cup \partial M$ is called *the closure* of M .

One can show that $\overline{M}$ is the smallest closed set in X containing M .

Our notation for the closed unit disk $\overline{\mathbb{D}}$ was motivated by the fact that this set is the closure of $\mathbb{D}$.

Theorem 2.8. Let (X, d) be a metric space. Then the following statements are true:

- (i) a set in X is $\begin{Bmatrix} \text{open} \\ \text{closed} \end{Bmatrix}$ if its complement is $\begin{Bmatrix} \text{closed} \\ \text{open} \end{Bmatrix}$,
- (ii) $\begin{Bmatrix} \text{a union} \\ \text{an intersection} \end{Bmatrix}$ of a family of $\begin{Bmatrix} \text{open} \\ \text{closed} \end{Bmatrix}$ sets in X is $\begin{Bmatrix} \text{open} \\ \text{closed} \end{Bmatrix}$,
- (iii) $\begin{Bmatrix} \text{an intersection} \\ \text{a union} \end{Bmatrix}$ of a finite family of $\begin{Bmatrix} \text{open} \\ \text{closed} \end{Bmatrix}$ sets in X is $\begin{Bmatrix} \text{open} \\ \text{closed} \end{Bmatrix}$.

Remark 2.9. Suppose X is a set together with a family $\mathcal{O}$ of its subsets called *open sets*. If $\emptyset, X \in \mathcal{O}$ and if the properties (ii) and (iii) in the previous theorem are satisfied, then $(X, \mathcal{O})$ is called a *topological space* and the system $\mathcal{O}$ a *topology* on X .

By what we have seen, every metric d on a set X determines a natural system $\mathcal{O}$ of open sets that form a topology on X . One calls this the topology *induced* by d .

Definition 2.10. Let (X, d) be a metric space, and $\{x_n\}$ be a sequence of points in X .

- (i) The sequence $\{x_n\}$ is called *convergent* if there exists a point $x \in X$ (the *limit* of $\{x_n\}$) such that for all $\epsilon > 0$ there exists $N \in \mathbb{N}$ such that for all $n \in \mathbb{N}$ with $n \geq N$ we have $d(x, x_n) < \epsilon$.

One can show that if the limit x exists, then it is unique, and one writes $x = \lim_{n \rightarrow \infty} x_n$ or simply $x_n \rightarrow x$.

- (ii) The sequence $\{x_n\}$ is called a *Cauchy sequence* if for all $\epsilon > 0$ there exists $N \in \mathbb{N}$ such that for all $n, k \in \mathbb{N}$ with $n, k \geq N$ we have $d(x_n, x_k) < \epsilon$.

- (iii) A point $x \in X$ is called a *sublimit* of $\{x_n\}$ if there exists a subsequence $\{x_{n_k}\}$ of $\{x_n\}$ such that $\lim_{k \rightarrow \infty} x_{n_k} = x$. In this case, we say that $\{x_n\}$ *subconverges* to x .

Proposition 2.11. A sequence $\{z_n\}$ in $\mathbb{C}$ converges if and only if the sequences $\{\operatorname{Re}(z_n)\}$ and $\{\operatorname{Im}(z_n)\}$ of real numbers converge.

In case of convergence we have

$$\lim_{n \rightarrow \infty} z_n = \lim_{n \rightarrow \infty} \operatorname{Re}(z_n) + i \lim_{n \rightarrow \infty} \operatorname{Im}(z_n).$$

Proof. $\Rightarrow$: Suppose $\{z_n\}$ converges, and let $z := \lim_{n \rightarrow \infty} z_n$. Note that $|\operatorname{Re}(w)| \leq |w|$ for $w \in \mathbb{C}$. Hence

$$|\operatorname{Re}(z_n) - \operatorname{Re}(z)| \leq |z_n - z|$$

for $n \in \mathbb{N}$. This implies that $\lim_{n \rightarrow \infty} \operatorname{Re}(z_n) = \operatorname{Re}(z)$. Similarly, $\lim_{n \rightarrow \infty} \operatorname{Im}(z_n) = \operatorname{Im}(z)$.

$\Leftarrow$: Suppose $\{\operatorname{Re}(z_n)\}$ and $\{\operatorname{Im}(z_n)\}$ converge. Let $a := \lim_{n \rightarrow \infty} \operatorname{Re}(z_n)$, $b := \lim_{n \rightarrow \infty} \operatorname{Im}(z_n)$, and $z := a + bi$. Note that for all $w \in \mathbb{C}$ we have

$$|w| = \sqrt{\operatorname{Re}(w)^2 + \operatorname{Im}(w)^2} \leq |\operatorname{Re}(w)| + |\operatorname{Im}(w)|.$$

Hence

$$|z_n - z| \leq |\operatorname{Re}(z_n) - a| + |\operatorname{Im}(z_n) - b|.$$

It follows that $\lim_{n \rightarrow \infty} z_n = z$. □

Remark 2.12. The usual computational rules for limits are true for sequences in $\mathbb{C}$. If $\{z_n\}$ and $\{w_n\}$ are sequences in $\mathbb{C}$ with $z_n \rightarrow z$ and $w_n \rightarrow w$, then

- (i) $z_n + w_n \rightarrow z + w$,
- (ii) $z_n w_n \rightarrow zw$,
- (iii) $\operatorname{Re}(z_n) \rightarrow \operatorname{Re}(z)$, $\operatorname{Im}(z_n) \rightarrow \operatorname{Im}(z)$, $\bar{z}_n \rightarrow \bar{z}$, $|z_n| \rightarrow |z|$,
- (iv) $\frac{w_n}{z_n} \rightarrow \frac{w}{z}$, if, in addition, $z \neq 0$ and $z_n \neq 0$ for $n \in \mathbb{N}$.

Definition 2.13. A metric space (X, d) is called *complete* if every Cauchy sequence in X converges.

Theorem 2.14. *The space $\mathbb{C}$ (equipped with the Euclidean metric) is complete.*

Proof. This follows from the completeness of $\mathbb{R}$ and Proposition 2.11. $\square$

Proposition 2.15. *A subset A of a metric space (X, d) is closed if and only if every convergent sequence in A has its limit also in A .*

Definition 2.16. A subset K of a metric space (X, d) is called *compact* if every open cover of K has a finite subcover; i.e., whenever $\{U_i : i \in I\}$ is a family of open sets in X with $K \subseteq \bigcup_{i \in I} U_i$, then there exist $i_1, \dots, i_n \in I$ such that $K \subseteq U_{i_1} \cup \dots \cup U_{i_n}$.

Theorem 2.17. *Let (X, d) be a metric space. Then $K \subseteq X$ is compact if and only if every sequence $\{x_n\}$ in K has a sublimit in K , i.e., there exists a subsequence $\{x_{n_k}\}$ of $\{x_n\}$ that converges and $x = \lim_{n \rightarrow \infty} x_n \in K$.*

The last condition is called *sequential compactness*. So a set in a metric space is compact if and only if it is sequentially compact.

Proposition 2.18. (a) *Every compact metric space is complete.*

(b) *Every compact subset of a metric space is closed.*

Theorem 2.19 (Heine-Borel). *A subset $K \subseteq \mathbb{C}$ is compact if and only if it is closed and bounded.*

Remark 2.20. A subset M of a metric space is called *bounded* if its *diameter* defined as

$$\text{diam}(M) := \sup\{d(x, y) : x, y \in M\}$$

is finite. This is equivalent to the requirement that there exists $a \in X$ and $r > 0$ such that $M \subseteq B(a, r)$.

Definition 2.21. Let (X, d) and (Y, ρ) be metric spaces, and $f: X \rightarrow Y$ be a map.

- (i) We say that f *approaches the limit* $y \in Y$ as x approaches $a \in X$, written as $\lim_{x \rightarrow a} f(x) = y$, if for all $\epsilon > 0$ there exists $\delta > 0$ such that for all $x \in X$ we have that

$$0 < d(x, a) < \delta \quad \text{implies} \quad \rho(f(x), y) < \epsilon.$$

- (ii) The map f is called *continuous at* $a \in X$ if $\lim_{x \rightarrow a} f(x) = f(a)$.
- (iii) The map f is called *continuous (on X)* if it is continuous at all points $a \in X$.

Note that in (i) it does not matter what happens for $x = a$.

Proposition 2.22. Let (X, d) and (Y, ρ) be metric spaces, and $f: X \rightarrow Y$ be a map. Then the following conditions are equivalent:

- (i) f is continuous,
- (ii) for every convergent sequence $\{x_n\}$ in X , the sequence $\{f(x_n)\}$ also converges and we have

$$\lim_{n \rightarrow \infty} f(x_n) = f\left(\lim_{n \rightarrow \infty} x_n\right),$$

- (iii) for all $x \in X$ and all $\epsilon > 0$ there exists $\delta > 0$ such that

$$f(B(x, \delta)) \subseteq B(f(x), \epsilon),$$

- (iv) preimages of open sets are open, i.e., $f^{-1}(V)$ is open in X whenever $V \subseteq Y$ is open in Y .
- (v) preimages of closed sets are closed.

Remark 2.23. We will be mostly interested in limits of functions $f: M \rightarrow \mathbb{C}$, where $M \subseteq \mathbb{R}$ or $M \subseteq \mathbb{C}$. In this case, one has the usual computational rules for function limits. For example, if $f: M \rightarrow \mathbb{C}$ and $g: M \rightarrow \mathbb{C}$ are functions, and $a \in M$, then

$$\lim_{x \rightarrow a} f(x)g(x) = \left(\lim_{x \rightarrow a} f(x)\right)\left(\lim_{x \rightarrow a} g(x)\right)$$

if the limits on the right hand side exist, etc. Based on this one can prove that if $f: M \rightarrow \mathbb{C}$ and $g: M \rightarrow \mathbb{C}$ are continuous, then $f + g$ is continuous, fg is continuous, etc.

Theorem 2.24. *Let (X, d) and (Y, ρ) be metric spaces, and $f: X \rightarrow Y$ be a continuous map. If $K \subseteq X$ is compact, then $f(K) \subseteq Y$ is compact (continuous images of compact sets are compact).*

Proof. Suppose that $K \subseteq X$ is compact, and let $\{y_n\}$ be an arbitrary sequence in $f(K)$. Then for all $n \in \mathbb{N}$ there exists $x_n \in K$ such that $y_n = f(x_n)$. Then $\{x_n\}$ is a sequence in K . Since K is compact, by Theorem 2.17 the sequence $\{x_n\}$ has a convergent subsequence $\{x_{n_k}\}$ with a limit in K , say $x_{n_k} \rightarrow x \in K$. By continuity of f it follows that

$$y_{n_k} = f(x_{n_k}) \rightarrow f(x) \in f(K).$$

So $\{y_n\}$ has a sublimit in $f(K)$. Hence $f(K)$ is compact by Theorem 2.17. $\square$

Remark 2.25. There are other important statements involving compactness and continuity: a real-valued function $f: X \rightarrow \mathbb{R}$ on a compact metric space (X, d) attains maximum and minimum. A continuous map $f: X \rightarrow Y$ between a compact metric space (X, d) and a metric space (Y, ρ) is *uniformly continuous*, i.e., for all $\epsilon > 0$ that exists $\delta > 0$ such that for all $x, y \in X$ we have that

$$d(x, y) < \delta \quad \text{implies} \quad \rho(f(x), f(y)) < \epsilon.$$

Definition 2.26. A subset M of a metric space (X, d) is called *connected* if the following condition is true: if $U, V \subseteq X$ are open sets with $M \subseteq U \cup V$ and $U \cap V = \emptyset$, then $M \subseteq U$ or $M \subseteq V$.

Remark 2.27. (a) For $M = X$ this statement can be reformulated in equivalent form in the following way: a metric space X is connected if every decomposition $X = U \cup V$, $U \cap V = \emptyset$, into open subsets $U, V \subseteq X$ is trivial,

i.e., $U = M$ or $V = M$. Equivalently, X is connected if $\emptyset$ and X are the only subsets of X that are both open and closed.

(b) For subsets M of a metric space (X, d) one can characterize connectedness in a similar way if one uses relatively closed and open sets. By definition a set $A \subseteq M$ is called *relatively closed* or *relatively open* in M if there exists a closed or an open set $B \subseteq X$, respectively, such that $A = B \cap M$.

The *restriction* $d_M := d|_{M \times M}$ of d to M is a metric on M . It is not hard to see that $A \subseteq M$ is relatively closed or relatively open in M if A is closed or open in the metric space (M, d_M) , respectively.

Using this terminology one can show that a subset M of a metric space is connected if and only if $\emptyset$ and M are the only subsets of M that are both relatively closed and relatively open in M (exercise!).

Proposition 2.28. *A non-empty subset $M \subseteq \mathbb{R}$ is connected if and only if M is an interval (possibly degenerate).*

Here we call an interval *degenerate* if it consists of only one point. All intervals in $\mathbb{R}$ are of the form $[a, b]$, $[a, b)$, $(a, b]$, (a, b) , $(-\infty, a]$, $(-\infty, a)$, $[a, +\infty)$, $(a, +\infty)$, or $(-\infty, +\infty) = \mathbb{R}$, where $a, b \in \mathbb{R}$, $a \leq b$.

Proposition 2.29. *Let (X, d) and (Y, ρ) be metric spaces, and $f: X \rightarrow Y$ be a continuous map. If $M \subseteq X$ is connected, then $f(M) \subseteq Y$ is connected (continuous images of connected sets are connected).*

Proof. Under the given assumptions, assume that $M \subseteq X$ is connected. To show that $f(M)$ is connected, let $U, V \subseteq Y$ be arbitrary open sets with $f(M) \subseteq U \cup V$ and $U \cap V = \emptyset$. Since f is continuous, the sets $U' := f^{-1}(U)$ and $V' := f^{-1}(V)$ are open. Moreover,

$$U' \cup V' = f^{-1}(U) \cup f^{-1}(V) = f^{-1}(U \cup V) \supseteq f^{-1}(f(M)) \supseteq M,$$

and

$$U' \cap V' = f^{-1}(U) \cap f^{-1}(V) = f^{-1}(U \cap V) = f^{-1}(\emptyset) = \emptyset.$$

Since M is connected, we have $M \subseteq U'$ or $M \subseteq V'$, say $M \subseteq U'$. Then

$$f(M) \subseteq f(U') = f(f^{-1}(U)) \subseteq U.$$

Hence $f(M)$ is connected. □

If $z, w \in \mathbb{C}$ we denote by

$$[z, w] := \{(1 - t)z + tw : t \in [0, 1]\}$$

the closed *line segment* with endpoints z and w . A *polygonal path* P in $\mathbb{C}$ is a set of the form

$$P = [z_0, z_1] \cup \cdots \cup [z_{n-1}, z_n],$$

where $n \in \mathbb{N}$, $z_1, \dots, z_n \in \mathbb{C}$. Such a path *joins* two points z and w if $z_0 = z$ and $w = z_n$.

Definition 2.30. A non-empty set $\Omega \subseteq \mathbb{C}$ is called a *region* if it is open and connected.

Theorem 2.31. A non-empty open set $\Omega \subseteq \mathbb{C}$ is a region if and only if any two points $a, b \in \Omega$ can be joined by a polygonal path in Ω .

Proof. $\Rightarrow$: Assume that Ω is a region. Fix a point $a \in \Omega$, and consider the points set M of points $b \in \Omega$ that can be joined to a by a polygonal path in Ω . It suffices to show that $M = \Omega$.

Note that $a \in M$, and so $M \neq \emptyset$. In order to conclude that $M = \Omega$, it is enough to show that M is open and relatively closed in Ω .

1. M is open.

If $y \in M \subseteq \Omega$, then, since Ω is open, there exists $\epsilon > 0$ such that $B(y, \epsilon) \subseteq \Omega$. If $z \in B(y, \epsilon)$ we can find a polygonal path in Ω joining a and y and the line segment $[y, z] \subseteq B(y, \epsilon) \subseteq \Omega$. Hence $B(y, \epsilon) \subseteq M$, and so M is open.

2. M is relatively closed.

Let $\{x_n\}$ be an arbitrary sequence in M that converges in the “ambient” space Ω . So $x_n \rightarrow x \in \Omega$. We have to show that $x \in M$.

Since Ω is open, there exists $\epsilon > 0$ such that $B(x, \epsilon) \subseteq \Omega$. Since $x_n \rightarrow x$, there exists $k \in \mathbb{N}$ such that $x_k \in B(x, \epsilon)$. Now $x_k \in M$ and $[x_k, x] \subseteq B(x, \epsilon)$. So we can find a polygonal path in Ω joining a and x by first joining a and x_k by a polygonal path in Ω followed by the line segment $[x_k, x]$. So $x \in M$, and M is relatively closed.

$\Leftarrow$: Suppose any two points in Ω can be joined by a polygonal path in Ω . Fix $a \in \Omega$. Then for every point $b \in \Omega$ there exists a polygonal path $P_b \subseteq \Omega$ joining a and b . Then $\Omega = \bigcup_{b \in \Omega} P_b$ and $a \in \bigcap_{b \in \Omega} P_b$. Moreover, each polygonal path is the image of an interval in $\mathbb{R}$ and hence connected.

We conclude that Ω is connected, because a union of a family of connected sets with non-empty intersection is connected (exercise!). $\square$

2.1 Exercises

Problem 2.1. Let (X, d) be a metric space. Show that if $A, B \subseteq X$, then $\overline{A \cup B} = \overline{A} \cup \overline{B}$.

Problem 2.2. Let (X, d) be a metric space and $M \subseteq X$. Recall that a set $A \subseteq M$ is called *relatively closed (in M)* if there exists a closed set $A' \subseteq X$ such that $A = A' \cap M$.

- (a) Show that $A \subseteq M$ is relatively closed if and only if $A = \overline{A} \cap M$.
- (b) Show that $A \subseteq M$ is relatively closed if and only if the following condition is true: whenever $\{x_n\}$ is a convergent sequence in A and $x_n \rightarrow x \in M$, then $x \in A$.

Problem 2.3. (a) Show that if $A \subseteq \mathbb{C}$ is a closed set contained in an open disk $D = B(z_0, R)$, where $z_0 \in \mathbb{C}$ and $R > 0$, then there exists a number $r < R$ such that $|z - z_0| \leq r$ for all $z \in A$.

- (b) For non-empty sets A and B in $\mathbb{C}$ we define their distance as

$$\text{dist}(A, B) := \inf\{|z - w| : z \in A \text{ and } w \in B\}.$$

Show that if A is compact and B is closed, then there exist points $u \in A$ and $v \in B$ such that $\text{dist}(A, B) = |u - v|$.

Problem 2.4. (a) Show that the function $z \mapsto \frac{z^2+3}{z+1}$ is continuous on the set $\mathbb{C} \setminus \{-1\}$ by using the ϵ - δ -definition of continuity.

- (b) Suppose $\{z_n\}$ and $\{w_n\}$ are convergent sequences in $\mathbb{C}$, say $z_n \rightarrow z$ and $w_n \rightarrow w$. Assume $w_n \neq 0$ for $n \in \mathbb{N}$ and $w \neq 0$. Show that $z_n/w_n \rightarrow z/w$ by using the ϵ -definition for convergence.

Problem 2.5. Suppose $f: \overline{\mathbb{D}} \rightarrow \mathbb{C}$ is a continuous function with $\exp(f(z)) = 1$ for all $z \in \overline{\mathbb{D}}$. Show that then f is constant.

Problem 2.6. Let $U \subseteq \mathbb{C}$ be an open set with $0 \in U$. Show that then it is impossible to define a continuous square root function on U ; more precisely, show that there exists no continuous function $S: U \rightarrow \mathbb{C}$ such that $S(z)^2 = z$ for each $z \in U$. Hint: Consider a suitable parametrization of a small circle centered at 0.

Problem 2.7. Let (X, d) be a metric space. Recall that $M \subseteq X$ is called *connected* if the following condition is true: if $U, V \subseteq X$ are open, $M \subseteq U \cup V$, and $U \cap V = \emptyset$, then $M \subseteq U$ or $M \subseteq V$.

Show that $M \subseteq X$ is connected if and only if $\emptyset$ and M are the only subsets of M that are both relatively open and relatively closed.

Note that this statement would be very easy to prove if we replaced the condition $U \cap V = \emptyset$ with $U \cap V \cap M = \emptyset$ in the definition of connectedness.

Problem 2.8. Let (X, d) be a metric space, and $A \subseteq X$ be a connected set. Show that every set $B \subseteq X$ with $A \subseteq B \subseteq \overline{A}$ is also connected.

Problem 2.9. Let (X, d) be a metric space, I be an index set, and $M_i \subseteq X$ be connected for $i \in I$. Show that if $\bigcap_{i \in I} M_i \neq \emptyset$, then $\bigcup_{i \in I} M_i$ is connected.

Problem 2.10. Let M be a non-empty subset of $\mathbb{C}$. A (*connected*) *component* C of M is a maximal (with respect to inclusion) connected subset of M . Show that:

- (a) Each component C of M is relatively closed in M .
- (b) If C and C' are distinct components of M , then $C \cap C' = \emptyset$.
- (c) Every connected subset A of M lies in a unique component C of M .
- (d) M is the (disjoint) union of its components.
- (e) There exists an equivalence relation $\sim$ on M whose equivalence classes are the components of M . Give a concise description of $\sim$ that does not use the concept of a component!

3 Differentiation

Remark 3.1. An obvious extension of differential calculus of real-valued functions of one real variable is to complex-valued functions of one real variable; so if $f: I \rightarrow \mathbb{C}$ is a complex-valued function on an interval $I \subseteq \mathbb{R}$, we call f *differentiable* at $x_0 \in I$, if the limit

$$\lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0}$$

exists. This limit is denoted by $f'(x_0)$ and called the *derivative of f at x_0* .

It (essentially) follows from Proposition 2.11 that f is differentiable at x_0 if and only if the functions $\operatorname{Re}(f)$ and $\operatorname{Im}(f)$ are differentiable at x_0 . In this case

$$f'(x_0) = (\operatorname{Re}(f))'(x_0) + i(\operatorname{Im}(f))'(x_0).$$

For example, let $f: \mathbb{R} \rightarrow \mathbb{C}$, $f(x) := e^{ix}$ for $x \in \mathbb{R}$. Then $f(x) = \cos x + i \sin x$, and so

$$\begin{aligned} f'(x) &= -\sin x + i \cos x \\ &= i(\cos x + i \sin x) = ie^{ix}. \end{aligned}$$

One is tempted to use the chain rule here:

$$\frac{d}{dx} e^{ix} = \left. \frac{de^z}{dz} \right|_{z=ix} \cdot \frac{d(ix)}{dx} = e^z|_{z=ix} \cdot i = ie^{ix}.$$

At the moment this is not justified, because we have not yet defined derivatives of complex-valued functions of a *complex variable*.

Definition 3.2. Let $U \subseteq \mathbb{C}$ be open and $f: U \rightarrow \mathbb{C}$ be a function.

(i) f is called *differentiable at $z_0 \in U$* if

$$\lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0}$$

exists. This limit is called the *derivative of f at z_0* and denoted by $f'(z_0)$, $\frac{df}{dz}(z_0)$, etc.

(ii) f is called *holomorphic* (on U) if it is differentiable at every point $z_0 \in U$.

(iii) We denote by $H(U)$ the set of all holomorphic functions on U .

Theorem 3.3. *Let $U, V \subseteq \mathbb{C}$ be open sets.*

(a) *If $f \in H(U)$, then f is continuous on U .*

(b) *Let $a, b \in \mathbb{C}$, and $f, g \in H(U)$. Then*

(i) *$af + bg \in H(U)$, and $(af + bg)' = af' + bg'$,*

(ii) *$fg \in H(U)$ and $(fg)' = f'g + fg'$ (product rule),*

(iii) *$\frac{f}{g} \in H(U)$ and $\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$ if we assume in addition that $g(z) \neq 0$ for $z \in U$ (quotient rule).*

(c) *Let $f \in H(U)$, $g \in H(V)$, and $f(U) \subseteq V$. Then $g \circ f \in H(U)$ and $(g \circ f)' = (g' \circ f) \cdot f'$ (chain rule).*

Proof. The standard proofs from real analysis transfer to this setting. For illustration we will prove (a) and (c).

(a) Let $z_0 \in U$ be arbitrary. Then

$$\begin{aligned} \lim_{z \rightarrow z_0} (f(z) - f(z_0)) &= \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0} (z - z_0) \\ &= \left(\lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0} \right) \cdot \left(\lim_{z \rightarrow z_0} (z - z_0) \right) \\ &= f'(z_0) \cdot 0 = 0. \end{aligned}$$

Hence

$$\lim_{z \rightarrow z_0} f(z) = f(z_0),$$

and so f is continuous at z_0 .

(c) A function $h: W \rightarrow \mathbb{C}$ on an open set $W \subseteq \mathbb{C}$ is differentiable at $z_0 \in W$ if and only if there exists a constant $c \in \mathbb{C}$, and a function $s: W \rightarrow \mathbb{C}$ with $\lim_{z \rightarrow z_0} s(z) = 0$ such that

$$h(z) - h(z_0) = (z - z_0)(c + s(z))$$

for all $z \in W$. If this is true, then $c = h'(z_0)$.

Now suppose f and g are as in the statement, and let $z_0 \in U$ be arbitrary. Set $w_0 = f(z_0)$. Then there exist functions $s: U \rightarrow \mathbb{C}$ and $t: V \rightarrow \mathbb{C}$ with $\lim_{z \rightarrow z_0} s(z) = 0$ and $\lim_{w \rightarrow w_0} t(w) = 0$ such that

$$\begin{aligned} f(z) - f(z_0) &= (z - z_0)(f'(z_0) + s(z)), \quad z \in U, \\ g(w) - g(w_0) &= (w - w_0)(g'(w_0) + t(w)), \quad w \in V. \end{aligned}$$

Hence for all $z \in U$ we have

$$\begin{aligned} \underbrace{g(f(z))}_w - \underbrace{g(f(z_0))}_{w_0} &= (f(z) - f(z_0))(g'(w_0) + t(f(z))), \\ &= (z - z_0)(f'(z_0) + s(z))(g'(w_0) + t(f(z))) \\ &= (z - z_0)(g'(w_0)f'(z_0) + \underbrace{g'(w_0)s(z) + f'(z_0)t(f(z)) + s(z)t(f(z))}_{r(z)}). \end{aligned}$$

Note that $f(z) \rightarrow f(z_0) = w_0$ as $z \rightarrow z_0$ by continuity of f , and so

$$\begin{aligned} \lim_{z \rightarrow z_0} r(z) &= \lim_{z \rightarrow z_0} (g'(w_0)s(z) + f'(z_0)t(f(z)) + s(z)t(f(z))) \\ &= g'(w_0) \lim_{z \rightarrow z_0} s(z) + f'(z_0) \lim_{w \rightarrow w_0} t(w) + \left(\lim_{z \rightarrow z_0} s(z) \right) \left(\lim_{w \rightarrow w_0} t(w) \right) \\ &= g'(w_0) \cdot 0 + f'(z_0) \cdot 0 + 0 \cdot 0 = 0. \end{aligned}$$

Hence $g \circ f$ is differentiable at z_0 and

$$(g \circ f)'(z_0) = g'(w_0)f'(z_0) = g'(f(z_0))f'(z_0).$$

The claim follows. □

Remark 3.4. Similar statements hold for complex-valued functions on intervals. One can also prove a chain rule for a function given by a (post-) composition of a complex-valued function on interval by a holomorphic function.

Theorem 3.5. *Let $U \subseteq \mathbb{C}$ be a region, and $f \in H(U)$. If $f' = 0$, then f is a constant function.*

Proof. Fix $z_0 \in U$. We will show that $f(z) = f(z_0)$ for all $z \in U$.

Let $z \in U$ be arbitrary. By Theorem 2.31 there exists a polygonal path

$$P = [z_0, z_1] \cup \cdots \cup [z_{n-1}, z_n] \subseteq U,$$

where $z_n = z$. So it is enough to show that f is constant on each segment $[u, v] \subseteq U$.

To see this, let $u, v \in U$ be arbitrary, and consider $h: [0, 1] \rightarrow \mathbb{C}$, $h(t) := f(u + (v - u)t)$ for $t \in [0, 1]$. Then by the chain rule,

$$h'(t) = f'(u + (v - u)t)(v - u) = 0$$

for all $t \in [0, 1]$. Hence h is constant on $[0, 1]$, and so f is constant on $[u, v]$. $\square$

Remark 3.6. By using the previous statements, one can produce a certain, although limited, supply of holomorphic functions; namely, it follows from the definitions that a constant function $z \mapsto c \in \mathbb{C}$, and the function $z \mapsto z$ are holomorphic in $\mathbb{C}$, and we have

$$\frac{dc}{dz} = 0 \quad \text{and} \quad \frac{dz}{dz} = 1.$$

By induction it follows from the product rule that $z \mapsto z^n$ is holomorphic in $\mathbb{C}$ for all $n \in \mathbb{N}$, and

$$\frac{dz^n}{dz} = nz^{n-1}.$$

This implies that each polynomial P , i.e., each function $P: \mathbb{C} \rightarrow \mathbb{C}$ of the form

$$z \mapsto P(z) := a_0 + a_1z + \cdots + a_nz^n,$$

where $n \in \mathbb{N}_0$, $a_0, \dots, a_n \in \mathbb{C}$, is holomorphic on $\mathbb{C}$. Finally, every rational function

$$R(z) = \frac{P(z)}{Q(z)},$$

where P and $Q \neq 0$ are polynomials, is holomorphic on the complement $\mathbb{C} \setminus Q^{-1}(0)$ of the zero-set of Q .

Remark 3.7. For the rest of this section we identify $\mathbb{C}$ with $\mathbb{R}^2$ by the correspondence $z = x + iy \in \mathbb{C} \cong (x, y) \in \mathbb{R}^2$. We write a complex-valued function $f: U \rightarrow \mathbb{C}$ on $U \subseteq \mathbb{C}$ in the form $f = u + iv$, where $u = \operatorname{Re}(f)$ and $v = \operatorname{Im}(f)$. Then u and v are considered as functions of $(x, y) \in U \subseteq \mathbb{R}^2$, and we have

$$f(x + iy) = u(x, y) + iv(x, y),$$

where $x + iy \cong (x, y) \in U$.

We will denote by $u_x, u_y, \dots$ partial derivatives; so $u_x = \frac{\partial u}{\partial x}$, etc.

Theorem 3.8. *Let $U \subseteq \mathbb{C}$ be open, $f: U \rightarrow \mathbb{C}$, $f = u + iv$.*

Then f is holomorphic on U if and only if u and v are C^1 -smooth on U and the so-called Cauchy–Riemann equations

$$u_x = v_y, \quad u_y = -v_x$$

are valid on U .

In this case,

$$\begin{aligned} f'(x + iy) &= u_x(x, y) + iv_x(x, y) \\ &= \frac{1}{i}(u_y(x, y) + iv_y(x, y)), \end{aligned}$$

whenever $x + iy \in U$.

For the proof of the implication “ $\Rightarrow$ ” we will use the following

Fact: *If $U \subseteq \mathbb{C}$ is open, and $f \in H(U)$, then $f' \in H(U)$.*

This will be proved later in this course (independently of the previous theorem of course); see Corollary 9.2.

Proof. “ $\Rightarrow$ ” Suppose f is holomorphic on U . Then for each $z = x + iy \in U$ we have,

$$f'(z) = \lim_{w \rightarrow 0} \frac{f(z + w) - f(z)}{w}.$$

Setting $w = h$ or $w = ik$, where $h, k \in \mathbb{R}$, and $h \rightarrow 0$ or $k \rightarrow 0$, we obtain

$$\begin{aligned} f'(x + iy) &= \lim_{h \rightarrow 0} \frac{f(z + h) - f(z)}{h} \\ &= \lim_{h \rightarrow 0} \frac{u(x + h, y) - u(x, y)}{h} + i \lim_{h \rightarrow 0} \frac{v(x + h, y) - v(x, y)}{h} \\ &= u_x(x, y) + iv_x(x, y), \end{aligned}$$

and

$$\begin{aligned} f'(x + iy) &= \lim_{k \rightarrow 0} \frac{f(z + ik) - f(z)}{ik} \\ &= \frac{1}{i} \lim_{k \rightarrow 0} \frac{u(x, y + k) - u(x, y)}{k} + \frac{1}{i} \cdot i \lim_{k \rightarrow 0} \frac{v(x, y + k) - v(x, y)}{k} \\ &= \frac{1}{i}(u_y(x, y) + iv_y(x, y)). \end{aligned}$$

In particular, the partial derivatives of u and v exist, and

$$u_x + iv_x = \frac{1}{i}(u_y + iv_y) = v_y - iu_y$$

on U ; so $u_x = v_y$ and $v_x = -u_y$.

Moreover, we have that

$$u_x = \operatorname{Re}(f'), \quad v_x = \operatorname{Im}(f'), \quad u_y = -\operatorname{Im}(f'), \quad v_y = \operatorname{Re}(f').$$

Since by the fact mentioned above, f' is holomorphic and hence continuous on U , the partial derivatives of u and v are also continuous on U . It follows that u and v are C^1 -smooth.

“ $\Leftarrow$ ” Suppose that u and v are C^1 -smooth and that the Cauchy–Riemann equations hold. Since the functions u and v are C^1 -smooth, they are differentiable on U ; this means that for each point $(x, y) \in U$ and $(h, k) \in \mathbb{R}^2$ with $|(h, k)| := \sqrt{h^2 + k^2}$ small we have

$$u(x + h, y + k) = u(x, y) + u_x(x, y)h + u_y(x, y)k + r(h, k),$$

$$v(x + h, y + k) = v(x, y) + v_x(x, y)h + v_y(x, y)k + s(h, k),$$

where

$$\frac{r(h, k)}{|(h, k)|} \rightarrow 0 \quad \text{and} \quad \frac{s(h, k)}{|(h, k)|} \rightarrow 0$$

as $|(h, k)| \rightarrow 0$.

If we set $z = x + iy$, $w = h + ik$, and $c = u_x(x, y) + iv_x(x, y)$, then for small $|w| \neq 0$, we have by the Cauchy–Riemann equations,

$$\begin{aligned} & \left| \frac{f(z + w) - f(z)}{w} - c \right| = \left| \frac{f(z + w) - f(z) - (h + ik)c}{w} \right| \\ &= \frac{1}{|w|} |f(z + w) - f(z) - hu_x(x, y) - hv_x(x, y) - ku_y(x, y) - kv_y(x, y)| \\ &\leq \frac{1}{|w|} |u(x + h, y + k) - u(x, y) - hu_x(x, y) - ku_y(x, y)| + \\ &\quad \frac{1}{|w|} |v(x + h, y + k) - v(x, y) - hv_x(x, y) - kv_y(x, y)| \\ &\leq \frac{|r(h, k)| + |s(h, k)|}{|(h, k)|} \rightarrow 0 \quad \text{as} \quad w = h + ik \rightarrow 0. \end{aligned}$$

It follows that f is differentiable at z . Since $z \in U$ was arbitrary, f is holomorphic on U . $\square$

Example 3.9. By using the previous theorem, one can see that the exponential function is holomorphic on $\mathbb{C}$; indeed,

$$e^z = e^{x+iy} = e^x \cos y + ie^x \sin y,$$

and so

$$u(x, y) = e^x \cos y \quad \text{and} \quad v(x, y) = e^x \sin y.$$

Hence u and v are C^1 -smooth in $\mathbb{R}^2$. Moreover, we have

$$u_x = e^x \cos y = v_y \quad \text{and} \quad u_y = -e^x \sin y = -v_x,$$

and so the Cauchy–Riemann equations hold.

For the derivative of the exponential function we obtain

$$\frac{d}{dz}e^z = u_x + iv_x = e^x \cos y + ie^x \sin y = e^z.$$

Remark 3.10. If $f = u + iv$ is holomorphic, then u and v are C^1 -smooth and the first partial derivatives of u and v are again real and imaginary parts of a holomorphic function ($u_x = \operatorname{Re}(f')$, etc.). It follows that the first partial derivatives of u and v are C^1 -smooth, and so u and v are C^2 -smooth. If we repeat this argument, we conclude that u and v are C^∞ -smooth.

Definition 3.11. Let $U \subseteq \mathbb{R}^2$ be open, and $h: U \rightarrow \mathbb{R}$ be C^2 -smooth. The *Laplacian* Δh of h is defined as

$$\Delta h = h_{xx} + h_{yy} = \frac{\partial^2 h}{\partial x^2} + \frac{\partial^2 h}{\partial y^2}.$$

The function h is called *harmonic* on U if its Laplacian vanishes on U , i.e., if $\Delta h = 0$ on U .

Theorem 3.12. Let $U \subseteq \mathbb{C}$ be open, $f \in H(U)$, $f = u + iv$. Then u and v are harmonic on U .

Proof. The functions u and v are C^∞ -smooth by Remark 3.10. Moreover, $u_x = v_y$ and $u_y = -v_x$ by the Cauchy–Riemann equations. Hence

$$\Delta u = u_{xx} + u_{yy} = v_{yx} - v_{xy} = 0,$$

$$\Delta v = v_{xx} + v_{yy} = -u_{yx} + u_{xy} = 0.$$

□

Remark 3.13. If $f = u + iv$ is a complex-valued function of $(x, y) \in U \subseteq \mathbb{R}^2$, one defines its partial derivatives as

$$\frac{\partial f}{\partial x} := \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}, \quad \text{and} \quad \frac{\partial f}{\partial y} := \frac{\partial u}{\partial y} + i \frac{\partial v}{\partial y}.$$

Usually, it is better to work with the so-called z - and $\bar{z}$ -derivatives or *Wirtinger derivatives* defined as

$$\frac{\partial f}{\partial z} := \frac{1}{2} \left(\frac{\partial f}{\partial x} - i \frac{\partial f}{\partial y} \right), \quad \text{and} \quad \frac{\partial f}{\partial \bar{z}} := \frac{1}{2} \left(\frac{\partial f}{\partial x} + i \frac{\partial f}{\partial y} \right).$$

So symbolically,

$$\frac{\partial}{\partial z} = \frac{1}{2} \left(\frac{\partial}{\partial x} - i \frac{\partial}{\partial y} \right), \quad \text{and} \quad \frac{\partial}{\partial \bar{z}} = \frac{1}{2} \left(\frac{\partial}{\partial x} + i \frac{\partial}{\partial y} \right).$$

Note that

$$\begin{aligned} \frac{\partial z}{\partial z} &= \frac{1}{2} \left(\frac{\partial(x + iy)}{\partial x} - i \frac{\partial(x + iy)}{\partial y} \right) = \frac{1}{2}(1 + 1) = 1, \\ \frac{\partial \bar{z}}{\partial z} &= \frac{1}{2} \left(\frac{\partial(x - iy)}{\partial x} - i \frac{\partial(x - iy)}{\partial y} \right) = \frac{1}{2}(1 - 1) = 0. \end{aligned}$$

Similarly,

$$\frac{\partial \bar{z}}{\partial z} = 0 \quad \text{and} \quad \frac{\partial \bar{z}}{\partial \bar{z}} = 1.$$

So with respect to the operator $\frac{\partial}{\partial z}$ the function $z \mapsto z$ behaves as expected, and $z \mapsto \bar{z}$ like a constant, and we have a similar behavior for the operator $\frac{\partial}{\partial \bar{z}}$.

Moreover, the usual computational rules hold for these operators (product rule, quotient rule, etc.). This allows computations such as

$$\frac{\partial |z|^2}{\partial z} = \frac{\partial(z\bar{z})}{\partial z} = \frac{\partial z}{\partial z} \bar{z} + \frac{\partial \bar{z}}{\partial z} z = \bar{z}.$$

We also have a version of the chain rule: suppose that $h = h(w, \bar{w})$ and $w = w(z, \bar{z})$. Then

$$\frac{\partial h}{\partial z} = \frac{\partial h}{\partial w} \cdot \frac{\partial w}{\partial z} + \frac{\partial h}{\partial \bar{w}} \cdot \frac{\partial \bar{w}}{\partial z}, \quad \text{and} \quad \frac{\partial h}{\partial \bar{z}} = \frac{\partial h}{\partial w} \cdot \frac{\partial w}{\partial \bar{z}} + \frac{\partial h}{\partial \bar{w}} \cdot \frac{\partial \bar{w}}{\partial \bar{z}}.$$

Note also

$$\overline{\left(\frac{\partial w}{\partial z} \right)} = \frac{\partial \bar{w}}{\partial \bar{z}}, \quad \text{and} \quad \overline{\left(\frac{\partial w}{\partial \bar{z}} \right)} = \frac{\partial \bar{w}}{\partial z}.$$

Theorem 3.14. Let $U \subseteq \mathbb{C}$ be open, $f: U \rightarrow \mathbb{C}$, $f = u + iv$. Then f is holomorphic on U if and only if f is C^1 -smooth (i.e., u and v are C^1 -smooth) and

$$\frac{\partial f}{\partial \bar{z}} = 0$$

on U (complex version of the Cauchy–Riemann equations).

In this case, $f' = \frac{\partial f}{\partial z}$.

Proof. We have

$$\begin{aligned} \frac{\partial f}{\partial \bar{z}} &= \frac{1}{2}(f_x + if_y) = \frac{1}{2}(u_x + iv_x) + \frac{i}{2}(u_y + iv_y) \\ &= \frac{1}{2}(u_x - v_y) + \frac{i}{2}(v_x + u_y). \end{aligned}$$

So the condition $\frac{\partial f}{\partial \bar{z}} = 0$ is equivalent to the Cauchy–Riemann equations. The first part of the statement now follows from Theorem 3.8.

In case of holomorphicity we have

$$f' = u_x + iv_x = \frac{1}{i}(u_y + iv_y).$$

Hence

$$f' = \frac{1}{2}(u_x + iv_x) - \frac{i}{2}(u_y + iv_y) = \frac{\partial u}{\partial z} + i\frac{\partial v}{\partial z} = \frac{\partial f}{\partial z}. \quad \square$$

Example 3.15. Let $f(z) = z|z|^2$ for $z \in \mathbb{C}$. Is f anywhere holomorphic in $\mathbb{C}$? Obviously, $f \in C^\infty(\mathbb{C})$, and we have

$$\frac{\partial f}{\partial \bar{z}} = \frac{\partial(z^2 \bar{z})}{\partial \bar{z}} = z^2.$$

So the Cauchy–Riemann equations are true only for $z = 0$. Hence there is no open set $U \subseteq \mathbb{C}$ where $f|_U$ is holomorphic.

Definition 3.16. Let $U \subseteq \mathbb{C}$ be open, $u: U \rightarrow \mathbb{R}$ be C^2 -smooth and harmonic. A harmonic function $v: U \rightarrow \mathbb{R}$ is called a *harmonic conjugate* of u in U if $f = u + iv$ is holomorphic.

Remark 3.17. (a) It easily follows from the Cauchy–Riemann equations that if a harmonic conjugate exists on a region U , then it is unique up to an additive constant (exercise!).

(b) In general a harmonic conjugate need not exist. If U satisfies a special geometric condition, then *every* harmonic function u on U has a harmonic conjugate. The requirement is that U should have “no holes” and is *simply connected*. This concept will be later discussed in detail in this course.

Example 3.18. (a) Let $u(x, y) = x^2 - y^2$ for $(x, y) \in \mathbb{R}^2$. Then $u \in C^\infty(\mathbb{R}^2)$, $u_x = 2x$, $u_y = -2y$, and $u_{xx} = 2$, $u_{yy} = -2$. Hence $\Delta u = 0$, and so u is harmonic.

Does u have a harmonic conjugate, and how to find it? We use the Cauchy–Riemann equation that a harmonic conjugate v of u must satisfy:

$$v_x = -u_y = 2y, \quad v_y = u_x = 2x.$$

By the first equation we have

$$v(x, y) = \int 2y \, dx + C(y) = 2xy + C(y).$$

Using this, we get

$$v_y = 2x + C'(y) = u_x = 2x.$$

Hence $C'(y) = 0$, and so $C(y) \equiv \text{const.} =: c \in \mathbb{R}$. So $v(x, y) = 2xy + c$. If we set

$$f(\underbrace{x + iy}_z) := u(x, y) + iv(x, y) = x^2 - y^2 + 2ixy + c = z^2 + c,$$

then f is indeed holomorphic.

(b) Let $u(x, y) = \frac{1}{2} \log(x^2 + y^2)$ for $(x, y) \in U := \mathbb{R}^2 \setminus \{0\}$. Then $u \in C^\infty(\mathbb{R}^2 \setminus \{0\})$,

$$u_x = \frac{x}{x^2 + y^2}, \quad u_y = \frac{y}{x^2 + y^2}, \quad u_{xx} = \frac{-x^2 + y^2}{(x^2 + y^2)^2}, \quad u_{yy} = \frac{x^2 - y^2}{(x^2 + y^2)^2}.$$

Hence $\Delta u = 0$, and so u is harmonic on U .

Does u have a harmonic conjugate v ? If so, then

$$v_y = u_x = \frac{x}{x^2 + y^2},$$

and so at least locally, where $x \neq 0$,

$$v = \int \frac{x}{x^2 + y^2} dy + C(x) = \arctan(y/x) + C(x).$$

Then

$$v_x = -\frac{y}{x^2 + y^2} + C'(x) = -u_y = -\frac{y}{x^2 + y^2},$$

so $C'(x) \equiv \text{const.} =: c$, and

$$v(x, y) = \arctan(y/x) + c$$

locally near points (x, y) where $x \neq 0$.

If we introduce polar coordinates, and set $z = x + iy = re^{i\varphi}$, then locally near points (x, y) where $x \neq 0$,

$$v(x, y) = \arctan(y/x) + c = \varphi + c',$$

where c' is locally constant.

If u had a harmonic conjugate, then by continuity necessarily

$$v(x, y) = \varphi + c',$$

for *all* $(x, y) \in U$; but there is *no* well-defined function $(x, y) \mapsto \varphi$ on *all* of U (it exists locally near each point, or on the *slit plane* $\mathbb{C} \setminus (-\infty, 0]$, for example).

The conclusion is that u has no harmonic conjugate *on* U ; on a more intuitive level, u has a locally harmonic conjugate (unique up to constant), but it changes globally by the period 2π if we run around 0 counter-clockwise once.

This is an instance of a more general fact: on arbitrary regions U every harmonic function has a local harmonic conjugate v , but v changes by certain additive *periods* if we run around the “holes” of U .

3.1 Exercises

Problem 3.1. Let $D = \bar{B}(a, r)$ be a closed Euclidean disk in $\mathbb{C}$ of radius $r > 0$ centered at $a \in \mathbb{C}$.

- (a) Suppose $z_1, \dots, z_n$ are points in D , and $\lambda_1, \dots, \lambda_n$ are real numbers in the interval $[0, 1]$ with $\lambda_1 + \dots + \lambda_n = 1$. Show that

$$\lambda_1 z_1 + \dots + \lambda_n z_n \in D.$$

- (b) Suppose $P(z) = \prod_{k=1}^n (z - z_k)$ is a complex polynomial whose zeros $z_1, \dots, z_n$ lie in D . Show that the zeros of P' also lie in D .

Problem 3.2. (a) Let $\mathbb{S}^1 := \{w \in \mathbb{C} : |w| = 1\}$. Show that the map $f: \mathbb{R} \rightarrow \mathbb{S}^1$, $t \mapsto f(t) := e^{it}$, is a *covering map* of $\mathbb{R}$ onto $\mathbb{S}^1$, i.e., for each point $w_0 \in \mathbb{S}^1$ there exists an open neighborhood $U \subseteq \mathbb{S}^1$ of w_0 such that $f^{-1}(U)$ can be written as a countable union

$$f^{-1}(U) = \bigcup_{k \in \mathbb{Z}} U_k$$

of pairwise disjoint open sets $U_k \subseteq \mathbb{R}$ such that $f|_{U_k}$ is a homeomorphism of U_k onto U for each $k \in \mathbb{Z}$.

- (b) Show that the exponential function $\exp: \mathbb{C} \rightarrow \mathbb{C}^* = \mathbb{C} \setminus \{0\}$ is a covering map.

Problem 3.3. Let $P(z) = a_0 + a_1 z + \dots$ and $Q(z) = b_0 + b_1 z + \dots$ be polynomials with complex coefficients. In Problem 1.7 it was implicitly used that if P and Q represent the same functions of z , that is, $P(z) = Q(z)$ for all z , then one can *compare coefficients* and conclude that $a_0 = b_0$, $a_1 = b_1$, etc. Provide a proof of this fact.

Problem 3.4. Let $U \subseteq \mathbb{C}$ be an open set and $z_0 \in U$. Suppose that $f: U \rightarrow \mathbb{C}$ is differentiable at z_0 and $f(z_0) \neq 0$. Prove directly from the definitions that there exists a neighborhood V of z_0 such that $g(z) = 1/f(z)$ is defined for $z \in V$, that g is differentiable at z_0 , and that

$$g'(z_0) = -\frac{f'(z_0)}{f(z_0)^2}.$$

Hint: Of course, you are not supposed to apply the quotient rule here, but establish this special case of the quotient rule directly.

Problem 3.5. Let $U \subseteq \mathbb{C}$ be a region and $u: U \rightarrow \mathbb{R}$ be a harmonic function. Suppose v and $\tilde{v}$ are harmonic conjugates of u on U . Show that then $v - \tilde{v}$ is a constant function on U .

Problem 3.6. Let $U \subseteq \mathbb{C}$ be a region and $f \in H(U)$. By using the Cauchy–Riemann equations show that:

- (a) if $\operatorname{Re}(f)$ or $\operatorname{Im}(f)$ is a constant function on U , then f is a constant function.
- (b) if $|f|$ is a constant function on U , then f is a constant function.

Problem 3.7. Let $U, V \subseteq \mathbb{C}$ be open sets, $h: V \rightarrow \mathbb{R}$ be a harmonic function, and $f: U \rightarrow V$ be a holomorphic function. Show that then the function $g := h \circ f: U \rightarrow \mathbb{R}$ is also harmonic.

Hint: You may disregard all smoothness issues. To show that $\Delta g = 0$, use Wirtinger derivatives throughout.

Problem 3.8. Let $U \subseteq \mathbb{C}$ be an open set. A C^2 -smooth function $s: U \rightarrow \mathbb{R}$ is called *subharmonic* if $\Delta s \geq 0$ on U (note: there are other equivalent definitions of subharmonic functions valid under weaker smoothness assumptions).

Show that if $f: U \rightarrow \mathbb{C}$ is a holomorphic function on U , then the function $s: U \rightarrow \mathbb{R}$ defined as

$$s(z) = \log(1 + |f(z)|^2), \quad z \in U,$$

is subharmonic on U .

Problem 3.9. Let u be a C^2 -smooth function defined on some open subset of $\mathbb{R}^2$. Then we can consider $u = u(x, y)$ as a function of two real variables x and y . If we introduce polar coordinates in the x - y -plane by $x = r \cos(\varphi)$ and $y = r \sin(\varphi)$, then u can also be considered as a function $(r, \varphi) \mapsto u(r \cos(\varphi), r \sin(\varphi))$ of the variables r and φ . In the following, we denote differentiation with respect to any of these variables x, y, r, φ by a subscript.

- (a) Show that

$$u_r = \cos(\varphi)u_x + \sin(\varphi)u_y \quad \text{and} \quad u_\varphi = -r \sin(\varphi)u_x + r \cos(\varphi)u_y.$$

- (b) Find similar expressions for the second derivatives u_{rr} and $u_{\varphi\varphi}$.
- (c) By using the expressions in (a) and (b), show that the Laplacian

$$\Delta u = u_{xx} + u_{yy}$$

can be expressed as

$$\Delta u = u_{rr} + \frac{1}{r}u_r + \frac{1}{r^2}u_{\varphi\varphi}. \quad (3.1)$$

4 Path integrals

Definition 4.1. A *path* is a continuous map $\gamma: [a, b] \rightarrow \mathbb{C}$ defined on a compact interval $[a, b] \subseteq \mathbb{R}$. The *image* or *trace* of the path is the set $\gamma([a, b])$ denoted by γ^* . The *endpoints* of γ are $\gamma(a)$ and $\gamma(b)$. The path is a *loop* if $\gamma(a) = \gamma(b)$. If $\gamma^* \subseteq U \subseteq \mathbb{C}$, then we say γ is a path *in* U .

The path γ is called *piecewise smooth* if there exists a partition $t_0 = a < t_1 < \dots < t_n = b$ of $[a, b]$ such that γ is differentiable on each interval $[t_{k-1}, t_k]$, $k \in \{1, \dots, n\}$, with continuous derivative γ' .

Definition 4.2. Let $h: [a, b] \rightarrow \mathbb{C}$ be a function. We call h (*Riemann*) *integrable* if $\operatorname{Re}(h)$ and $\operatorname{Im}(h)$ are (Riemann) integrable on $[a, b]$. If h is integrable, we define

$$\int_a^b h(t) dt := \int_a^b (\operatorname{Re} h)(t) dt + i \int_a^b (\operatorname{Im} h)(t) dt.$$

Proposition 4.3. Let $[a, b] \subseteq \mathbb{R}$ be compact interval, $f, g, h: [a, b] \rightarrow \mathbb{C}$ be integrable functions, and $\alpha, \beta \in \mathbb{C}$. Then the following statements are true:

$$(a) \quad \int_a^b (\alpha f + \beta g)(t) dt = \alpha \int_a^b f(t) dt + \beta \int_a^b g(t) dt,$$

$$(b) \quad \left| \int_a^b h(t) dt \right| \leq \int_a^b |h(t)| dt,$$

(c) if h has a primitive H , i.e., H is differentiable on $[a, b]$ and $H' = h$, then

$$\int_a^b h(t) dt = H(b) - H(a).$$

Proof. The statements (a) and (c) follow from the corresponding statements for integrals of real-valued functions if we split the relevant expressions up into real and imaginary parts.

(b) Assume $h = u + iv$, where $u = \operatorname{Re}(h)$ and $v = \operatorname{Im}(h)$. Define $\alpha := \int_a^b u(t) dt$ and $\beta := \int_a^b v(t) dt$. Then by the Cauchy–Schwarz inequality we

have

$$\begin{aligned} \left| \int_a^b h(t) dt \right|^2 &= \alpha^2 + \beta^2 = \int_a^b (\alpha u(t) + \beta v(t)) dt \\ &\leq (\alpha^2 + \beta^2)^{1/2} \int_a^b (u(t)^2 + v(t)^2)^{1/2} dt \\ &= \left| \int_a^b h(t) dt \right| \cdot \int_a^b |h(t)| dt. \end{aligned}$$

Inequality (b) follows. $\square$

Definition 4.4 (Path integrals). Let $\gamma: [a, b] \rightarrow \mathbb{C}$ be a piecewise smooth path, and $f: \gamma^* \rightarrow \mathbb{C}$ be a continuous function. Then we define the (*path*) *integral of f over γ* (denoted by $\int_\gamma f(z) dz$, $\int_\gamma f$, etc.), as

$$\int_\gamma f(z) dz := \int_a^b f(\gamma(t)) \gamma'(t) dt.$$

Example 4.5. Let $a \in \mathbb{C}$, $r > 0$, and

$$\gamma(t) = a + re^{it}, \quad t \in [0, 2\pi].$$

This is a positively-oriented parametrization of the circle of radius r centered at a . Let $f(z) = (z - a)^n$, $n \in \mathbb{Z}$. Then $\gamma'(t) = ire^{it}$, and so

$$\begin{aligned} \int_\gamma (z - a)^n dz &= \int_0^{2\pi} (re^{it})^n ire^{it} dt = ir^{n+1} \int_0^{2\pi} e^{i(n+1)t} dt \\ &= \begin{cases} i \int_0^{2\pi} dt = 2\pi i, & n = -1, \\ \frac{ir^{n+1}}{i(n+1)} e^{i(n+1)t} \Big|_0^{2\pi} = 0, & n \neq -1. \end{cases} \end{aligned}$$

If we run through the circle with different speed (e.g., $\gamma(t) = a + re^{2\pi it}$, $t \in [0, 1]$), or with different orientation ($\gamma(t) = a + re^{-it}$, $t \in [0, 2\pi]$), then the integrals are unchanged or change sign, respectively.

4.6. Reparametrization of paths. Let $\alpha: [a, b] \rightarrow \mathbb{C}$ and $\beta: [c, d] \rightarrow \mathbb{C}$ be piecewise smooth paths. Then β is called a *reparametrization* of α if

there exists a bijection $\varphi: [c, d] \rightarrow [a, b]$ such that φ and φ^{-1} are C^1 -smooth, $\varphi(c) = a$, $\varphi(d) = b$, and $\beta = \alpha \circ \varphi$.

Integrals do not change under reparametrizations: if β is a reparametrization of α and $f: \alpha^* = \beta^* \rightarrow \mathbb{C}$ is continuous, then

$$\int_{\beta} f = \int_{\alpha} f.$$

Indeed,

$$\begin{aligned} \int_{\beta} f &= \int_{\beta} f(w) dw = \int_c^d f(\beta(s)) \beta'(s) ds \\ &= \int_c^d f(\alpha(\varphi(s))) \alpha'(\varphi(s)) \varphi'(s) ds \quad (t = \varphi(s)) \\ &= \int_a^b f(\alpha(t)) \alpha'(t) dt = \int_{\alpha} f(z) dz = \int_{\alpha} f. \end{aligned}$$

The paths α and β are called *equivalent* if β is a reparametrization of α (this is indeed an equivalence relation for paths!).

The *oriented interval* $[a, b]$ for $a, b \in \mathbb{C}$ is the path $\gamma: [0, 1] \rightarrow \mathbb{C}$, $\gamma(t) := (1-t)a + tb$, $t \in [0, 1]$ (or any equivalent path). In this case, one writes $\int_{[a,b]} f(z) dz$ instead of $\int_{\gamma} f(z) dz$.

4.7. Computational rules for path integrals. (a) Let $\gamma: [a, b] \rightarrow \mathbb{C}$ be a piecewise smooth path, $f, g: \gamma^* \rightarrow \mathbb{C}$ be continuous, and $c, d \in \mathbb{C}$. Then

$$\int_{\gamma} (cf(z) + dg(z)) dz = c \int_{\gamma} f(z) dz + d \int_{\gamma} g(z) dz.$$

(b) Suppose γ_1 and γ_2 are piecewise smooth paths such that the endpoint of γ_1 is equal to the initial point of γ_2 . Then we may without loss of generality assume that $\gamma_1: [0, 1] \rightarrow \mathbb{C}$, $\gamma_2: [1, 2] \rightarrow \mathbb{C}$ and $\gamma_1(1) = \gamma_2(1)$. Then the *concatenation* of γ_1 and γ_2 is the path $\gamma: [0, 2] \rightarrow \mathbb{C}$ defined as

$$\gamma(t) = \begin{cases} \gamma_1(t), & t \in [0, 1], \\ \gamma_2(t), & t \in [1, 2]. \end{cases}$$

If $f: \gamma_1^* \cup \gamma_2^* \rightarrow \mathbb{C}$ is continuous, then

$$\int_{\gamma} f(z) dz = \int_{\gamma_1} f(z) dz + \int_{\gamma_2} f(z) dz.$$

(c) Let $U \subseteq \mathbb{C}$ be open, $\gamma: [a, b] \rightarrow U$ be a piecewise smooth path, and $f: U \rightarrow \mathbb{C}$ be a continuous function.

If f has a primitive F , i.e., $F \in H(U)$ and $F' = f$, then

$$\int_{\gamma} f(z) dz = F(\gamma(b)) - F(\gamma(a)).$$

In particular, $\int_{\gamma} f(z) dz = 0$ if γ is a loop. Indeed,

$$\frac{d}{dt} F(\gamma(t)) = F'(\gamma(t))\gamma'(t) = f(\gamma(t))\gamma'(t),$$

and so

$$\int_{\gamma} f(z) dz = \int_a^b f(\gamma(t))\gamma'(t) dt = F(\gamma(t)) \Big|_a^b = F(\gamma(b)) - F(\gamma(a)).$$

4.8. Lengths of paths. Let $\gamma: [a, b] \rightarrow X$ be a path in a metric space (X, d) . One defines its *length* $\ell(\gamma) \in [0, \infty]$ as

$$\ell(\gamma) := \sup_{a=t_0 < t_1 < \dots < t_n=b} \sum_{k=1}^n d(\gamma(t_{k-1}), \gamma(t_k)),$$

where the supremum is taken over all partitions of $[a, b]$. One says that γ is *rectifiable* if $\ell(\gamma) < \infty$.

If $\gamma: [a, b] \rightarrow \mathbb{C}$ is piecewise smooth path, one can show that γ is rectifiable and

$$\ell(\gamma) = \int_a^b |\gamma'(t)| dt. \quad (4.1)$$

Since we are only interested in piecewise smooth paths in $\mathbb{C}$, we can take (4.1) as the definition of the length of such a path.

This notion of length has the expected properties (such as invariance under reparametrizations, or additivity under concatenation of paths).

Lemma 4.9. *Let $\gamma: [a, b] \rightarrow \mathbb{C}$ be a piecewise smooth path, $f: \gamma^* \rightarrow \mathbb{C}$ be a continuous function, and $M \geq 0$ be such that $|f(z)| \leq M$ for $z \in \gamma^*$. Then*

$$\left| \int_{\gamma} f(z) dz \right| \leq M \ell(\gamma).$$

Proof.

$$\begin{aligned}\left| \int_{\gamma} f(z) \, dz \right| &= \left| \int_a^b f(\gamma(t)) \gamma'(t) \, dt \right| \\ &\leq \int_a^b |f(\gamma(t))| \cdot |\gamma'(t)| \, dt \quad (\text{by Prop. 4.3 (b)}) \\ &\leq M \int_a^b |\gamma'(t)| \, dt = M\ell(\gamma). \quad \square\end{aligned}$$

4.1 Exercises

Problem 4.1. Let $\Omega \subseteq \mathbb{C}$ be a convex set and $f \in H(\Omega)$. Show that if $\operatorname{Re}(f') > 0$ on Ω , then f is injective.

Problem 4.2. The purpose of this problem is to establish a simple special case of Fubini's Theorem in measure theory.

(a) Let $[a, b], [c, d] \subseteq \mathbb{R}$ be compact intervals, and $F: [a, b] \times [c, d] \rightarrow \mathbb{C}$ be a continuous function. Show that the functions

$$x \mapsto \int_c^d F(x, y) dy \quad \text{and} \quad y \mapsto \int_a^b F(x, y) dx$$

are continuous on $[a, b]$ and $[c, d]$, respectively.

(b) Show that

$$\int_a^b \left(\int_c^d F(x, y) dy \right) dx = \int_c^d \left(\int_a^b F(x, y) dx \right) dy. \quad (4.2)$$

Hint: Complete the following outline: the integrals in (4.2) exist by part (a). To prove that they are equal, first note that this is true for a constant function F . Derive the general case by subdividing R into small rectangles, and approximating F in suitable way..

Problem 4.3. Let $\alpha, \beta: [0, 1] \rightarrow \mathbb{C}$ be piecewise smooth paths, and $G: \alpha^* \times \beta^* \rightarrow \mathbb{C}$ be continuous. Then we have

$$\int_{\alpha} \left(\int_{\beta} G(z, w) dw \right) dz = \int_{\beta} \left(\int_{\alpha} G(z, w) dz \right) dw.$$

Hint: Use Problem 4.2 to show that these integrals exist and are equal.

5 Infinite series and products

We first discuss infinite series and start with the basic definition.

Definition 5.1. With a given sequence $\{a_n\}$ in $\mathbb{C}$ (indexed by $n \in \mathbb{N}$) we associate another sequence $\{s_n\}$ defined by

$$s_n = a_1 + \cdots + a_n$$

for $n \in \mathbb{N}$. The sequence $\{s_n\}$ is symbolically represented by the symbol $\sum_{n=1}^{\infty} a_n$, called an (*infinite*) *series*. The terms s_n are called the *partial sums*

of this infinite series. We say that $\sum_{n=1}^{\infty} a_n$ *converges* or *diverges* depending on whether the limit $\lim_{n \rightarrow \infty} s_n$ exists or not. In case of convergence, we also denote by $\sum_{n=1}^{\infty} a_n$ the limit $\lim_{n \rightarrow \infty} s_n$.

Instead of $\mathbb{N}$ one often uses other index sets A such as $A = \mathbb{N}_0$, $A = \{N, N+1, \dots\}$ with $N \in \mathbb{N}_0$ or other countably infinite set (see remark below) to label the terms in an infinite series.

Proposition 5.2. Let $a_n \in \mathbb{C}$ for $n \in \mathbb{N}$.

- (a) (Cauchy criterion) *The infinite series $\sum_{n=1}^{\infty} a_n$ converges if and only if for all $\epsilon > 0$ there exists $N \in \mathbb{N}$ such that for all $m, n \in \mathbb{N}$ with $m \geq n$ we have*

$$\left| \sum_{k=n}^m a_k \right| < \epsilon.$$

- (b) *If $\sum_{n=1}^{\infty} a_n$ converges, then $\lim_{n \rightarrow \infty} a_n = 0$.*

Proof. (a) Let $s_n = \sum_{k=1}^n a_k$ for $n \in \mathbb{N}$. Then $\sum_{n=1}^{\infty} a_n$ converges if and only if the sequence $\{s_n\}$ converges. By completeness of $\mathbb{C}$ this is true if and only if

$\{s_n\}$ is a Cauchy sequence. This in turn is equivalent with the condition in the statement, because

$$\left| \sum_{k=n}^m a_k \right| = |s_m - s_{n-1}|$$

for $n, m \in \mathbb{N}$ with $m \geq n \geq 2$.

(b) This immediately follows from (a) if we choose $n = m$. $\square$

Definition 5.3. Let $a_n \in \mathbb{C}$ for $n \in \mathbb{N}$. Then the infinite series $\sum_{n=1}^{\infty} a_n$ is said to *converge absolutely* if $\sum_{n=1}^{\infty} |a_n|$ converges.

Note that partial sums $s_n = |a_1| + \cdots + |a_n|$ of the series $\sum_{n=1}^{\infty} |a_n|$ form an increasing sequence of non-negative numbers. So the sequence $\{s_n\}$ will always have a limit in $[0, \infty]$. The series $\sum_{n=1}^{\infty} a_n$ converges absolutely if this limit is finite which we can indicate by writing $\sum_{n=1}^{\infty} |a_n| < \infty$.

Theorem 5.4. Let $a_n \in \mathbb{C}$ for $n \in \mathbb{N}$.

- (a) If $\sum_{n=1}^{\infty} a_n$ converges absolutely, then it also converges.
- (b) If $\sum_{n=1}^{\infty} a_n$ converges absolutely, then for all $\epsilon > 0$ there exists $N \in \mathbb{N}$ such that $\sum_{n=N}^{\infty} |a_n| < \epsilon$.
- (c) Suppose $\sum_{n=1}^{\infty} a_n$ converges absolutely and let $\varphi: \mathbb{N} \rightarrow \mathbb{N}$ be an arbitrary

bijection. Then $\sum_{n=1}^{\infty} a_{\varphi(n)}$ also converges absolutely and we have

$$\sum_{n=1}^{\infty} a_{\varphi(n)} = \sum_{n=1}^{\infty} a_n.$$

The last statement says that in an absolutely convergent infinite series we can reorder the terms in any way without affecting the convergence behavior and the limit. For an infinite series $\sum_{n=1}^{\infty} a_n$ to converge absolutely, the condition that $a_n \rightarrow 0$ as $n \rightarrow \infty$ is necessary, but not sufficient. For convergence the terms a_n have to tend to 0 “fast enough” as $n \rightarrow \infty$.

Proof. (a) This immediately follows from the Cauchy criterion and the inequality

$$\left| \sum_{k=n}^m a_k \right| \leq \sum_{k=n}^m |a_k|.$$

(b) Note that the partial sums of the infinite series $\sum_{n=N}^{\infty} |a_n|$ form an increasing sequence of non-negative real numbers. Hence this sequence always has a limit in $[0, \infty]$ which is meaning of $\sum_{n=N}^{\infty} |a_n|$.

Now suppose $\sum_{n=1}^{\infty} a_n$ converges absolutely and let $\epsilon > 0$ be arbitrary. Then by the Cauchy criterion there exists $N \in \mathbb{N}$ such that

$$\sum_{n=N}^m |a_n| < \epsilon/2,$$

whenever $m \geq N$. Letting $m \rightarrow \infty$, we see that

$$\sum_{n=N}^{\infty} |a_n| \leq \epsilon/2 < \epsilon.$$

(c) Let $L = \sum_{n=1}^{\infty} a_n \in \mathbb{C}$, and define $s_n = \sum_{k=1}^n a_k$ and $t_n = \sum_{k=1}^n a_{\varphi(k)}$ for $n \in \mathbb{N}$.

Now let $\epsilon > 0$ be arbitrary. Then by (b) we can find $N \in \mathbb{N}$ such that

$$\sum_{n=N}^{\infty} |a_n| < \epsilon.$$

We can then choose $M \in \mathbb{N}$ such that

$$\{1, \dots, N-1\} \subseteq \{\varphi(1), \dots, \varphi(M-1)\}.$$

Then for $m, m' \in \mathbb{N}$ with $m' \geq m \geq M$ we have

$$\{\varphi(m), \dots, \varphi(m')\} \cap \{1, \dots, N-1\} = \emptyset$$

and so

$$\{\varphi(m), \dots, \varphi(m')\} \subseteq \{N, N+1, \dots\}.$$

This implies that

$$\sum_{n=m}^{m'} |a_{\varphi(n)}| \leq \sum_{n=N}^{\infty} |a_k| < \epsilon,$$

and also that for $n \geq M$ we have

$$|t_n - s_{N-1}| \leq \sum_{n=N}^{\infty} |a_k| < \epsilon,$$

which in turn implies

$$|t_n - L| \leq |t_n - s_{N-1}| + |s_{N-1} - L| \leq 2 \sum_{n=N}^{\infty} |a_k| < 2\epsilon.$$

Hence $\sum_{n=1}^{\infty} a_{\varphi(n)}$ converges absolutely by the Cauchy criterion and we also see that $\lim_{n \rightarrow \infty} t_n = L$. □

There are several tests for the absolute convergence of an infinite series.

Theorem 5.5 (Convergence tests). *Let $a_n \in \mathbb{C}$ for $n \in \mathbb{N}$.*

(a) *Let $b_n \geq 0$ for $n \in \mathbb{N}$. If $|a_n| \leq b_n$ for $n \in \mathbb{N}$ and $\sum_{n=1}^{\infty} b_n$ converges, then*

$\sum_{n=1}^{\infty} a_n$ converges absolutely (comparison test).

- (b) Suppose that $a_n \neq 0$ for $n \in \mathbb{N}$ and $\limsup_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| < 1$. Then $\sum_{n=1}^{\infty} a_n$ converges absolutely (ratio test).
- (c) Suppose that $\limsup_{n \rightarrow \infty} \sqrt[n]{|a_n|} < 1$. Then $\sum_{n=1}^{\infty} a_n$ converges absolutely (root test).

Proof. The proofs are similar to the proofs for infinite series of real numbers. We will prove (a) and (c).

(a) Let $s_n = \sum_{k=1}^n a_k$ and $t_n = \sum_{k=1}^n b_k$ for $n \in \mathbb{N}$. By completeness of $\mathbb{C}$ it is enough to show that $\{s_n\}$ is a Cauchy sequence.

To see this let $\epsilon > 0$ be arbitrary. Since $\{t_n\}$ converges, this sequence is a Cauchy sequence. Hence there exists $N \in \mathbb{N}$ such that $|t_n - t_m| < \epsilon$ whenever $m, n \geq N$.

Suppose $n, m \geq N$. Without loss of generality we may also assume that $n \geq m$. Then

$$\begin{aligned} |s_n - s_m| &= \left| \sum_{k=1}^n a_k - \sum_{k=1}^m a_k \right| \\ &= \left| \sum_{k=m+1}^n a_k \right| \leq \sum_{k=m+1}^n |a_k| \\ &\leq \sum_{k=m+1}^n b_k = t_n - t_m < \epsilon. \end{aligned}$$

So $\{s_n\}$ is indeed a Cauchy sequence.

(c) The number $s^* = \limsup_{n \rightarrow \infty} c_n$ for a bounded sequence of real numbers $\{c_n\}$ is characterized by the following properties:

- (i) If $s_1 < s^*$, then there are infinitely many $n \in \mathbb{N}$ with $s_1 \leq c_n$.
- (ii) If $s^* < s_2$, then there are only finitely many $n \in \mathbb{N}$ with $s_2 \leq c_n$.

Now let $q := \limsup_{n \rightarrow \infty} \sqrt[n]{|a_n|} < 1$. Pick s such that $q < s < 1$. Then by (ii) we have

$$s \leq \sqrt[n]{|a_n|}$$

for only finitely many $n \in \mathbb{N}$. Hence there exists $N \in \mathbb{N}$ such that

$$\sqrt[n]{|a_n|} \leq s \quad \text{for all } n \in \mathbb{N}, n \geq N.$$

This implies that

$$|a_n| \leq s^n \quad \text{for all } n \in \mathbb{N}, n \geq N.$$

Since $\sum_{n=N}^{\infty} s^n$ converges (geometric series!), the series $\sum_{n=N}^{\infty} a_n$ also converges

by the comparison test. Hence $\sum_{n=1}^{\infty} a_n$ converges. $\square$

Example 5.6. (a) For $z \in \mathbb{C}$ the series $\sum_{n=0}^{\infty} z^n$ converges precisely if $|z| < 1$, and we have

$$\sum_{n=0}^{\infty} z^n = \frac{1}{1-z}, \quad |z| < 1.$$

This easily follows from the fact that

$$\sum_{k=0}^n z^k = \frac{1 - z^{n+1}}{1 - z}$$

for $z \neq 1$.

(b) The series $\sum_{n=0}^{\infty} \frac{z^n}{n!}$ converges for all $z \in \mathbb{C}$ by the ratio test. Indeed,

convergence is clear for $z = 0$. For $z \neq 0$ put $a_n = \frac{z^n}{n!}$ for $n \in \mathbb{N}_0$. Then

$$\begin{aligned} \limsup_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| &= \limsup_{n \rightarrow \infty} \frac{|z|^{n+1}}{(n+1)!} \cdot \frac{n!}{|z|^n} \\ &= \lim_{n \rightarrow \infty} \frac{|z|}{n+1} = 0 < 1. \end{aligned}$$

Remark 5.7. Often one wants to use some countably infinite index set A to label the terms of an infinite series. So suppose A is such a set and we have numbers $a_\alpha \in \mathbb{C}$ for $\alpha \in A$. Then we say that $\sum_{\alpha \in A} a_\alpha$ *converges absolutely* if

the infinite series $\sum_{n=1}^{\infty} a_{\varphi(n)}$ converges absolutely for some bijection $\varphi: \mathbb{N} \rightarrow A$.

Theorem 5.4 implies that in this case the convergence behavior of $\sum_{n=1}^{\infty} a_{\varphi(n)}$ and its limit is independent of the choice of the bijection $\varphi: \mathbb{N} \rightarrow A$. So if in this case we set

$$\sum_{\alpha \in A} a_{\alpha} := \sum_{n=1}^{\infty} a_{\varphi(n)},$$

then this is well-defined.

It is easy to see that $\sum_{\alpha \in A} a_{\alpha}$ converges absolutely if and only if there exists a number $M \geq 0$ such that

$$\sum_{\alpha \in F} |a_{\alpha}| \leq M \quad (5.1)$$

for all finite subsets $F \subseteq A$ (exercise!). Moreover, the limit $L := \sum_{\alpha \in A} a_{\alpha}$ is characterized by the following property (exercise!): for all $\epsilon > 0$ there exists a finite set $F_0 \subseteq A$ such that

$$\left| \sum_{\alpha \in F} a_{\alpha} - L \right| \leq \epsilon \quad (5.2)$$

for all finite sets $F \subseteq A$ with $F_0 \subseteq F$.

The following statements show that with absolutely convergent series one can compute in the same way as with finite sums.

Proposition 5.8. *Let A be countably infinite index set, let $a_{\alpha} \in \mathbb{C}$ for $\alpha \in A$, and suppose the series $\sum_{\alpha \in A} a_{\alpha}$ converges absolutely.*

(a) *Let $b_{\alpha} \in \mathbb{C}$ for $\alpha \in A$, and suppose that the series $\sum_{\alpha \in A} b_{\alpha}$ also converges*

absolutely. Then $\sum_{\alpha \in A} (a_{\alpha} + b_{\alpha})$ converges absolutely, and we have

$$\sum_{\alpha \in A} (a_{\alpha} + b_{\alpha}) = \sum_{\alpha \in A} a_{\alpha} + \sum_{\alpha \in A} b_{\alpha}.$$

- (b) Suppose the sets A_n , $n \in \mathbb{N}$, form a countable partition of A , that is, these sets are pairwise disjoint and their union is equal to A . Then the series $\sum_{\alpha \in A} a_\alpha$ converges absolutely for each $n \in \mathbb{N}$. Moreover, the

series $\sum_{n \in \mathbb{N}} \left(\sum_{\alpha \in A_n} a_\alpha \right)$ also converges absolutely and we have

$$\sum_{\alpha \in A} a_\alpha = \sum_{n \in \mathbb{N}} \left(\sum_{\alpha \in A_n} a_\alpha \right).$$

- (c) Let B be countably infinite index set, $b_\beta \in \mathbb{C}$ for $\beta \in B$, and suppose that the series $\sum_{\beta \in B} b_\beta$ converges absolutely. Then the series $\sum_{(\alpha, \beta) \in A \times B} a_\alpha b_\beta$ also converges absolutely and we have

$$\sum_{(\alpha, \beta) \in A \times B} a_\alpha b_\beta = \left(\sum_{\alpha \in A} a_\alpha \right) \cdot \left(\sum_{\beta \in B} b_\beta \right).$$

Statement (ii) says that we can group the terms in $\sum_{\alpha \in A} a_\alpha$ by using the sets A_n and compute the “intermediate” terms $c_n = \sum_{\alpha \in A_n} a_\alpha$ for $n \in \mathbb{N}$. Then the series of these intermediate terms c_n gives the value of the whole series.

The proof is left as an exercise!

Proof. We will only prove (ii) and leave the proofs of (i) and (iii) as an exercise.

Since $\sum_{\alpha \in A} a_\alpha$ converges absolutely, there exists $M \geq 0$ such that inequality (5.1) is true for all finite sets $F \subseteq A$. Then it is also true if $n \in \mathbb{N}$ and F is a finite subset of $A_n \subseteq A$. Hence $\sum_{\alpha \in A_n} a_\alpha$ converges absolutely for each $n \in \mathbb{N}$.

Let $L := \sum_{\alpha \in A} a_\alpha$, and define $K_n := \sum_{\alpha \in A_n} |a_\alpha|$ and $L_n := \sum_{\alpha \in A_n} a_\alpha$ for $n \in \mathbb{N}$.

Now let $\epsilon > 0$ be arbitrary. Then it follows from the characterizations of limits as in (5.2) that for each $n \in \mathbb{N}$ there exists a finite set $F_n \subseteq A_n$ that

$$\left| \sum_{\alpha \in F_n} a_\alpha - L_n \right| \leq 2^{-n} \epsilon \quad \text{and} \quad \left| \sum_{\alpha \in F_n} |a_\alpha| - K_n \right| \leq 2^{-n} \epsilon, \quad (5.3)$$

whenever F is a finite set with $F_n \subseteq F \subseteq A_n$.

□

We now turn to infinite products.

Definition 5.9. With a given sequence $\{a_n\}$ in $\mathbb{C}$ (indexed by $n \in \mathbb{N}$, say) we associate another sequence $\{p_n\}$ defined as

$$p_n = a_1 \cdot a_2 \cdots a_n = \prod_{k=1}^n a_k$$

for $n \in \mathbb{N}$. The sequence $\{p_n\}$ is symbolically represented by the symbol $\prod_{n=1}^{\infty} a_n$, called an *infinite product*. The terms p_n are called the *partial products*

of this infinite product. We say that $\prod_{n=1}^{\infty} a_n$ *converges (properly)* if there exists $n_0 \in \mathbb{N}$ such that the limit $\lim_{n \rightarrow \infty} \prod_{k=n_0}^n a_k$ exists and is non-zero. In this case,

the limit $\lim_{n \rightarrow \infty} p_n = \prod_{k=1}^n a_k$ also exists and is denoted by $\prod_{n=1}^{\infty} a_n$.

Lemma 5.10. Let $a_n \in \mathbb{C}$ for $n \in \mathbb{N}$.

(a) If the infinite product $\prod_{n=1}^{\infty} a_n$ converges, then $a_n = 0$ for at most finitely

many $n \in \mathbb{N}$ and $\prod_{n=1}^{\infty} a_n = 0$ if and only if there exists some $n \in \mathbb{N}$ such that $a_n = 0$.

(b) (Cauchy criterion) The infinite product $\prod_{n=1}^{\infty} a_n$ converges if and only if for all $\epsilon > 0$ there exists $N \in \mathbb{N}$ such that for all $m, n \in \mathbb{N}$ with $m \geq n \geq N$ we have

$$\left| \prod_{k=n}^m a_k - 1 \right| < \epsilon.$$

(c) If $\prod_{n=1}^{\infty} a_n$ converges, then $\lim_{n \rightarrow \infty} a_n = 1$.

Proof. (a) This is clear from the definition of (proper) convergence of an infinite product.

(b) “ $\Rightarrow$ ” If $\prod_{n=1}^{\infty} a_n$ converges, then by (a) there are only finitely many $n \in \mathbb{N}$ with $a_n = 0$. As we are only interested in the “tail” of $L := \prod_{n=1}^{\infty} a_n$, we may without loss of generality assume that $a_n \neq 0$ for all $n \in \mathbb{N}$. Then

$$p_n := \prod_{k=1}^n a_k \rightarrow L \neq 0$$

as $n \rightarrow \infty$. So $|p_{n-1}| \geq |L|/2$ for large n and for these n and $m \in \mathbb{N}$ with $m \geq n$ we have

$$\begin{aligned} \left| \prod_{k=n}^m a_k - 1 \right| &= \left| \frac{p_m}{p_{n-1}} - 1 \right| = \frac{1}{|p_{n-1}|} |p_m - p_{n-1}| \\ &\leq \frac{2}{|L|} |p_m - p_{n-1}|. \end{aligned}$$

The desired statement now easily follows from the Cauchy criterion for the sequence $\{p_k\}$.

“ $\Leftarrow$ ” Suppose the Cauchy criterion is satisfied. Again it is enough to consider a tail of the infinite product and so we may assume that $p_n := \prod_{k=1}^n a_k$ satisfies $|p_n - 1| \leq 1/2$ for $n \in \mathbb{N}$. This implies that

$$1/2 \leq |p_n| \leq 2 \text{ for } n \in \mathbb{N}. \quad (5.4)$$

Then for all $n, m \in \mathbb{N}$ with $m \geq n \geq 2$ we have

$$|p_m - p_{n-1}| = |p_{n-1}| \left| \frac{p_m}{p_{n-1}} - 1 \right| \leq 2 \left| \frac{p_m}{p_{n-1}} - 1 \right|.$$

Since $\left| \frac{p_m}{p_{n-1}} - 1 \right|$ is arbitrarily small for sufficiently large n and m by our assumption, this shows that $\{p_n\}$ is a Cauchy sequence and hence converges.

Moreover, it follows from (5.4) that $\prod_{n=1}^{\infty} a_n = \lim_{n \rightarrow \infty} p_n \neq 0$.

(c) This easily follows from (b) if we choose $m = n$. □

Often one writes an infinite product in the form $\prod_{n=1}^{\infty} (1 + b_n)$ with $b_n \in \mathbb{C}$ for $n \in \mathbb{N}$. Then it follows from part (c) of the previous lemma that $b_n \rightarrow 0$ as $n \rightarrow \infty$ is a necessary condition for the convergence of the infinite product. In general, this condition is not sufficient, but for convergence the terms b_n have to tends to 0 “fast enough” as $n \rightarrow \infty$.

Definition 5.11. An infinite product of the form $\prod_{n=1}^{\infty} (1 + b_n)$ with $b_n \in \mathbb{C}$ for $n \in \mathbb{N}$ is said to *converge absolutely* if $\prod_{n=1}^{\infty} (1 + |b_n|)$ converges.

Examples 5.12. (a) The infinite product $\prod_{n=1}^{\infty} (1 + 1/n)$ does not converge. Indeed, we have

$$\begin{aligned} p_n &= \prod_{k=1}^n (1 + 1/k) = \prod_{k=1}^n \frac{k+1}{k} = \underbrace{\frac{2}{1} \cdot \frac{3}{2} \cdots \frac{n+1}{n}}_{\text{“telescoping product”}} \\ &= n + 1 \rightarrow \infty \text{ as } n \rightarrow \infty. \end{aligned}$$

(b) The infinite product $\prod_{n=1}^{\infty} \left(1 + \frac{(-1)^{n-1}}{n}\right)$ converges. Indeed, we have

$$1 + \frac{(-1)^{n-1}}{n} = \frac{n + (-1)^{n-1}}{n} = \begin{cases} (n+1)/n, & n \text{ odd,} \\ (n-1)/n, & n \text{ even.} \end{cases}$$

It follows that

$$\begin{aligned} p_n &= \prod_{k=1}^n \left(1 + \frac{(-1)^{k-1}}{k}\right) = \frac{2}{1} \cdot \frac{1}{2} \cdot \frac{4}{3} \cdot \frac{3}{4} \cdots \frac{n \pm 1}{n} \\ &= \begin{cases} (n+1)/n, & n \text{ odd,} \\ 1, & n \text{ even.} \end{cases} \end{aligned}$$

This shows that $p_n \rightarrow 1 \neq 0$ as $n \rightarrow \infty$, and so the infinite product converges; it follows from (a) that it does not converge absolutely.

Proposition 5.13. Let $b_n \in \mathbb{C}$ for $n \in \mathbb{N}$.

- (a) If $\prod_{n=1}^{\infty} (1 + b_n)$ converges absolutely, then it also converges.
- (b) The infinite product $\prod_{n=1}^{\infty} (1 + b_n)$ converges absolutely if and only if the series $\sum_{n=1}^{\infty} |b_n|$ converges.
- (c) Suppose $\prod_{n=1}^{\infty} (1 + b_n)$ converges absolutely and $\varphi: \mathbb{N} \rightarrow \mathbb{N}$ is an arbitrary bijection. Then $\prod_{n=1}^{\infty} (1 + b_{\varphi(n)})$ also converges absolutely and we have
- $$\prod_{n=1}^{\infty} (1 + b_{\varphi(n)}) = \prod_{n=1}^{\infty} (1 + b_n).$$

The last statement says that in an absolutely convergent infinite product we can reorder the factors in any way without affecting the convergence behavior and the limit.

Proof. (a) Define $p_{n,m} := \prod_{k=n}^m (1 + b_k)$ and $q_{n,m} := \prod_{k=n}^m (1 + |b_k|)$ for $n, m \in \mathbb{N}$, $m \geq n$. Then

$$\begin{aligned} |p_{n,m} - 1| &= |(1 + p_n) \cdots (1 + p_m) - 1| \\ &= \left| \sum_{k=n}^m b_k + \sum_{n \leq k_1 < k_2 \leq m} b_{k_1} b_{k_2} + \cdots \right| \\ &\leq \sum_{k=n}^m |b_k| + \sum_{n \leq k_1 < k_2 \leq m} |b_{k_1}| |b_{k_2}| + \cdots \\ &= (1 + |p_n|) \cdots (1 + |p_m|) - 1 = |q_{n,m} - 1|. \end{aligned}$$

This shows that

$$|p_{n,m} - 1| \leq |q_{n,m} - 1|$$

for all $n, m \in \mathbb{N}$, $m \geq n$. The statement now immediately follows from the Cauchy criterion for infinite products (Lemma 5.10 (b)).

(b) Define $s_{n,m} := \sum_{k=n}^m |b_k|$ for $n, m \in \mathbb{N}$, $m \geq n$. Since $1 + x \leq e^x$ for $x \geq 0$ (why?), we then have

$$\begin{aligned} s_{n,m} &= \sum_{k=n}^m |b_k| \leq (1 + |p_n|) \cdots (1 + |p_m|) - 1 \\ &= q_{n,m} - 1 \leq \exp(|b_n|) \cdots \exp(|b_m|) - 1 = \exp(s_{n,m}) - 1. \end{aligned}$$

Note that $e^x - 1 \rightarrow 0$ as $x \rightarrow 0^+$. Hence the previous inequality implies that $|q_{n,m} - 1|$ is small if and only if $s_{n,m}$ is small. The statement now immediately follows from Cauchy criteria for infinite series and infinite products.

(c) Under the given assumptions, the absolute convergence of $\prod_{n=1}^{\infty} (1 + b_{\varphi(n)})$

follows from (b) and the fact that $\sum_{n=1}^{\infty} |b_{\varphi(n)}| = \sum_{n=1}^{\infty} |b_n| < \infty$.

Now define $L := \prod_{n=1}^{\infty} (1 + b_n)$, $p_n := \prod_{k=1}^n (1 + b_k)$ and $r_n := \prod_{k=1}^n (1 + b_{\varphi(k)})$ for $n \in \mathbb{N}$.

Let $\epsilon \in (0, 1)$ be arbitrary. Then we can find $N \in \mathbb{N}$ such that $|p_N - L| < \epsilon$ and $\sum_{k=N+1}^{\infty} |b_k| < \epsilon$. Moreover, we can then choose $M \in \mathbb{N}$ such that

$$\{1, \dots, N\} \subseteq \{\varphi(1), \dots, \varphi(M)\}.$$

Then for $m \in \mathbb{N}$ with $m \geq M$ we have

$$\{\varphi(1), \dots, \varphi(m)\} = \{1, \dots, N\} \cup E,$$

where $E \subseteq \mathbb{N} \setminus \{1, \dots, N\}$. Hence for $m \geq M$ we have

$$\begin{aligned}
 |r_m - L| &\leq |r_m - p_N| + |p_N - L| \\
 &\leq |p_N| \cdot \left| \prod_{k \in E} (1 + b_k) - 1 \right| + \epsilon \\
 &\leq (|L| + 1) \left(\prod_{k \in E} (1 + |b_k|) - 1 \right) + \epsilon \\
 &\leq (|L| + 1) \left(\exp \left(\sum_{k=N+1}^{\infty} |b_k| \right) - 1 \right) + \epsilon \\
 &\leq (|L| + 1)(\exp(\epsilon) - 1) + \epsilon.
 \end{aligned}$$

Now the last quantity can be made arbitrarily small if $\epsilon > 0$ is small enough. It follows that $r_m \rightarrow L$ as $m \rightarrow \infty$ and the statement follows. $\square$

Remark 5.14. Similarly as for infinite series, one often wants to use some infinite countable index set A to label the elements of an infinite product. So suppose A is such a set and we have numbers $b_\alpha \in \mathbb{C}$ for $\alpha \in A$. Then we say that $\prod_{\alpha \in A} (1 + b_\alpha)$ *converges absolutely* if the infinite product

$\prod_{n=1}^{\infty} (1 + b_{\varphi(n)})$ converges absolutely for some bijection $\varphi: \mathbb{N} \rightarrow A$. Proposition 5.13 implies that in this case the convergence behavior of $\prod_{n=1}^{\infty} (1 + b_{\varphi(n)})$

and its limit is independent of the choice of the bijection $\varphi: \mathbb{N} \rightarrow A$. So if in this case we set

$$\prod_{\alpha \in A} (1 + b_\alpha) := \prod_{n=1}^{\infty} (1 + b_{\varphi(n)}),$$

then this is well-defined.

5.1 Exercises

Problem 5.1. Show that the infinite series

$$\frac{1}{z} + \sum_{n \in \mathbb{N}} \frac{2z}{z^2 - n^2}$$

converges absolutely for each $z \in \mathbb{C} \setminus \mathbb{Z}$.

Problem 5.2. Let Γ be a *rank-2 lattice* in $\mathbb{C}$, that is, a set of the form

$$\Gamma = \{n\omega_1 + k\omega_2 : k, n \in \mathbb{Z}\},$$

where $\omega_1, \omega_2 \in \mathbb{C} \setminus \{0\}$ are linearly independent over $\mathbb{R}$.

Show that

$$\sum_{\gamma \in \Gamma \setminus \{0\}} \frac{1}{\gamma^n}$$

converges absolutely for $n \in \mathbb{N}$, $n \geq 3$.

Problem 5.3. Show that the infinite product $\prod_{n=1}^{\infty} \left(1 + \frac{1}{n}\right) e^{-1/n}$ converges.

Problem 5.4. Let $b_n \geq 0$ for $n \in \mathbb{N}$. Show that then the infinite product $\prod_{n=1}^{\infty} (1 - b_n)$ converges if and only if $\sum_{n=1}^{\infty} b_n$ converges.

Problem 5.5. Let $n \in \mathbb{N}$ and $z_k \in \mathbb{C}$ for $k = 1, \dots, n$. Show that then there exists a set $J \subseteq \{1, \dots, n\}$ such that

$$\left| \sum_{j \in J} z_j \right| \geq \frac{1}{\pi} \sum_{k=1}^n |z_k|.$$

Hint: For $t \in [0, 2\pi]$ define $J(t) := \{j \in \{1, \dots, n\} : \operatorname{Re}(z_j e^{-it}) \geq 0\}$. Find a lower bound for the average of $\left| \sum_{j \in J(t)} z_j \right|$.

Problem 5.6. Let $a_n \in \mathbb{C}$ for $n \in \mathbb{N}$.

(a) Show that the series $\sum_{n=1}^{\infty} a_n$ converges absolutely if and only if the following condition is true: there exists a constant $L \in \mathbb{C}$ such that for all

$\epsilon > 0$ there exists a finite set $F_0 \subseteq \mathbb{N}$ such that for all finite sets $F \subseteq \mathbb{N}$ with $F_0 \subseteq F$ we have

$$\left| \sum_{k \in F} a_k - L \right| < \epsilon.$$

(b) Show that in this case $L = \sum_{n=1}^{\infty} a_n$.

6 Sequences of functions

We now consider sequences and infinite series of functions. The following concept is of crucial importance.

Definition 6.1 (Uniform convergence). Suppose $F: U \rightarrow \mathbb{C}$, and $F_n: U \rightarrow \mathbb{C}$ for $n \in \mathbb{N}$ are functions defined on a set $U \subseteq \mathbb{C}$. We say that the function sequence $\{F_n\}$ *converges uniformly to F on U* , written

$$F_n \rightarrow F \text{ uniformly on } U,$$

if for all $\epsilon > 0$ there exists $N \in \mathbb{N}$ such that for all $n \in \mathbb{N}$ with $n \geq N$ and for all $z \in U$,

$$|F_n(z) - F(z)| < \epsilon.$$

This implies *pointwise convergence*, namely $\lim_{n \rightarrow \infty} F_n(z) = F(z)$ for all $z \in U$, or equivalently: for all $z \in U$ and for all $\epsilon > 0$ there exists $N \in \mathbb{N}$ such that for all $n \in \mathbb{N}$ with $n \geq N$,

$$|F_n(z) - F(z)| < \epsilon.$$

We say that a series $\sum_{n=1}^{\infty} f_n$ of functions $f_n: U \rightarrow \mathbb{C}$ *converges uniformly*

on U if the sequence of partial sum $\{F_n\}$, where $F_n = \sum_{k=1}^n f_k$ for $n \in \mathbb{N}$, converges uniformly on U to some limit function $F: U \rightarrow \mathbb{C}$.

Theorem 6.2 (Weierstrass M -test). Suppose that $f_n: U \rightarrow \mathbb{C}$ for $n \in \mathbb{N}$ are functions defined on $U \subseteq \mathbb{C}$, and that there exist numbers $M_n \geq 0$ such that

$$|f_n(z)| \leq M_n \quad \text{for all } z \in U \text{ and all } n \in \mathbb{N}.$$

If the series $\sum_{n=1}^{\infty} M_n$ converges, then the series $\sum_{n=1}^{\infty} f_n$ converges uniformly on U .

The symbol “ M ” in “Weierstrass M -test” stands for “majorization”.

Proof. By the comparison test, the series $\sum_{n=1}^{\infty} f_n(z)$ converges for all $z \in U$.

Define $F(z) := \sum_{n=1}^{\infty} f_n(z)$ for $z \in U$, and $L := \sum_{n=1}^{\infty} M_n$.

We claim that $\sum_{n=1}^{\infty} f_n$ converges to F uniformly on U . To see this, let $\epsilon > 0$ be arbitrary. Note that

$$\lim_{n \rightarrow \infty} \sum_{k=n+1}^{\infty} M_k = \lim_{n \rightarrow \infty} \left(L - \sum_{k=1}^n M_k \right) = 0$$

(“tails” of a convergent series tend to 0).

Hence there exists $N \in \mathbb{N}$ such that $\sum_{k=N+1}^{\infty} M_k \leq \epsilon$. Now if $z \in U$ and $n \in \mathbb{N}$ with $n \geq N$ are arbitrary,

$$\begin{aligned} \left| F(z) - \sum_{k=1}^n f_k(z) \right| &= \left| \sum_{k=1}^{\infty} f_k(z) - \sum_{k=1}^n f_k(z) \right| \\ &= \left| \sum_{k=n+1}^{\infty} f_k(z) \right| \leq \sum_{k=n+1}^{\infty} |f_k(z)| \\ &\leq \sum_{k=n+1}^{\infty} M_k \leq \epsilon. \end{aligned}$$

The claim follows. $\square$

In complex analysis another type of convergence is very important.

Definition 6.3 (Locally uniform convergence). Suppose that $F: U \rightarrow \mathbb{C}$, and $F_n: U \rightarrow \mathbb{C}$ for $n \in \mathbb{N}$ are functions defined on an open set $U \subseteq \mathbb{C}$. We say that the function sequence $\{F_n\}$ *converges locally uniformly to F on U* , written

$$F_n \rightarrow F \text{ locally uniformly on } U,$$

if for all $p \in U$ there exists a neighborhood $V \subseteq U$ of p such that we have uniform convergence $F_n \rightarrow F$ on V .

We say that a series $\sum_{n=1}^{\infty} f_n$ of functions $f_n: U \rightarrow \mathbb{C}$ *converges locally uniformly on U* if the sequence of partial sum $\{F_n\}$, where $F_n = \sum_{k=1}^n f_k$ for $n \in \mathbb{N}$, converges locally uniformly on U to some limit function $F: U \rightarrow \mathbb{C}$.

Proposition 6.4. *Suppose that $F: U \rightarrow \mathbb{C}$, and $F_n: U \rightarrow \mathbb{C}$ for $n \in \mathbb{N}$ are functions defined on an open set $U \subseteq \mathbb{C}$. Then $F_n \rightarrow F$ locally uniformly on U if and only if $F_n \rightarrow F$ uniformly on K for each compact set $K \subseteq U$.*

Proof. Suppose $F_n \rightarrow F$ locally uniformly on U . Let K be an arbitrary compact subset of U . Then for each $p \in K$ there exists an open neighborhood $U_p \subseteq U$ of p such that $F_n \rightarrow F$ uniformly on U_p . These sets U_p , $p \in K$, form an open cover of K . So by compactness of K there exist finitely many points, say $p_1, \dots, p_m \in K$, such that $K \subseteq U_{p_1} \cup \dots \cup U_{p_m}$. Then it is easy to see (exercise!) that $F_n \rightarrow F$ uniformly on $K \subseteq U_{p_1} \cup \dots \cup U_{p_m}$.

Conversely, suppose that $F_n \rightarrow F$ uniformly on each compact set $K \subseteq U$. Now if $p \in U$ is arbitrary, then, since U is open, there exists a compact neighborhood K of p with $K \subseteq U$. By our assumption, then $F_n \rightarrow F$ uniformly on K . Hence every point has a neighborhood on which the convergence $F_n \rightarrow F$ is uniform, and so $F_n \rightarrow F$ locally uniformly on U . $\square$

Theorem 6.5. *Let $F: U \rightarrow \mathbb{C}$, and $F_n: U \rightarrow \mathbb{C}$ for $n \in \mathbb{N}$ be functions defined on an open set $U \subseteq \mathbb{C}$. Suppose that $F_n \rightarrow F$ locally uniformly on U .*

If each function F_n , $n \in \mathbb{N}$, is continuous on U , then the limit function F is also continuous on U .

Moreover, if $\gamma: [a, b] \rightarrow U$ is a piecewise smooth path, then

$$\lim_{n \rightarrow \infty} \int_{\gamma} F_n(z) dz = \int_{\gamma} F(z) dz.$$

This first statement says that locally uniform limits of continuous functions on an open subset of $\mathbb{C}$ are continuous.

Since $F(z) = \lim_{n \rightarrow \infty} F_n(z)$ for $z \in U \supset \gamma^*$, the second statement can also be written as

$$\lim_{n \rightarrow \infty} \int_{\gamma} F_n(z) dz = \int_{\gamma} \left(\lim_{n \rightarrow \infty} F_n(z) \right) dz.$$

So in this situation one can interchange the function limit and path integration.

Proof. It is well-known that uniform limits of continuous functions are continuous. So our assumptions imply that each point $p \in U$ has an open neighborhood V on which the restriction $F|V$ is continuous. Hence F is continuous at p . Therefore, F is continuous at each point $p \in U$ and hence continuous on U .

To prove the second statement first note that γ^* is a compact subset of U (why?). So our assumptions and Proposition 6.4 imply that we have $F_n \rightarrow F$ uniformly on γ^* .

Now let $\epsilon > 0$ be arbitrary. Since $F_n \rightarrow F$ uniformly on γ^* , there exists $N \in \mathbb{N}$ such that

$$|F_n(z) - F(z)| \leq \frac{\epsilon}{\ell(\gamma) + 1},$$

whenever $z \in \gamma^*$, and $n \in \mathbb{N}$, $n \geq N$. Here $\ell(\gamma)$ is the length of γ .

Hence for all $n \geq N$ we have

$$\begin{aligned} \left| \int_{\gamma} F_n(z) dz - \int_{\gamma} F(z) dz \right| &= \left| \int_{\gamma} (F_n(z) - F(z)) dz \right| \\ &\leq \frac{\epsilon}{\ell(\gamma) + 1} \ell(\gamma) \quad (\text{by Lem. 4.9}) \\ &\leq \epsilon. \end{aligned}$$

The claim follows. $\square$

Remark 6.6. We will use the last statement also for function series. Let $U \subseteq \mathbb{C}$ be open, γ be a piecewise smooth path in U , and $f_n: U \rightarrow \mathbb{C}$ for $n \in \mathbb{N}$ be continuous functions for $n \in \mathbb{N}$. Suppose that $\sum_{n=1}^{\infty} f_n$ converges locally uniformly on U . Then we have

$$\begin{aligned} \int_{\gamma} \left(\sum_{k=1}^{\infty} f_k(z) \right) dz &= \int_{\gamma} \left(\lim_{n \rightarrow \infty} \sum_{k=1}^n f_k(z) \right) dz = \lim_{n \rightarrow \infty} \int_{\gamma} \left(\sum_{k=1}^n f_k(z) \right) dz \\ &= \lim_{n \rightarrow \infty} \sum_{k=1}^n \int_{\gamma} f_k(z) dz = \sum_{k=1}^{\infty} \int_{\gamma} f_k(z) dz. \end{aligned}$$

So in this situation the infinite series can be integrated term-by-term.

Definition 6.7 (Normal convergence). Let $U \subseteq \mathbb{C}$ be open, and $f_n: U \rightarrow \mathbb{C}$ for $n \in \mathbb{N}$ be a function. We say that the series $\sum_{n=1}^{\infty} f_n$ converges normally

on U if $\sum_{n=1}^{\infty} |f_n|$ converges locally uniformly on U , i.e., the sequence of partial sums of the last series converges locally uniformly on U to some limit function (which is necessarily the pointwise limit of this series).

Proposition 6.8. *Let $U \subseteq \mathbb{C}$ be open, and $f_n: U \rightarrow \mathbb{C}$ for $n \in \mathbb{N}$ be a function.*

(a) *Then the following conditions are equivalent:*

- (i) $\sum_{n=1}^{\infty} f_n$ converges normally on U ,
- (ii) for all compact sets $K \subseteq U$ and all $\epsilon > 0$ there exists $N \in \mathbb{N}$ with

$$\sum_{n=N}^{\infty} |f_n(z)| \leq \epsilon \quad \text{whenever } z \in K,$$

- (iii) for all $p \in U$ there exists a neighborhood $V \subseteq U$ of p such that for all $\epsilon > 0$ there exists $N \in \mathbb{N}$ with

$$\sum_{n=N}^{\infty} |f_n(z)| \leq \epsilon \quad \text{whenever } z \in V.$$

- (b) If $\sum_{n=1}^{\infty} f_n$ converges normally on U , then this series also converges locally uniformly on U .

- (c) Suppose $\sum_{n=1}^{\infty} f_n$ converges normally on U , and let $\varphi: \mathbb{N} \rightarrow \mathbb{N}$ be a bijection. Then $\sum_{n=1}^{\infty} f_{\varphi(n)}$ converges normally on U , and $\sum_{n=1}^{\infty} f_n = \sum_{n=1}^{\infty} f_{\varphi(n)}$.

The proof is left as an exercise.

Remark 6.9. Again it is often convenient to use some general infinite countable index set A to label the elements of an infinite function series. So suppose A is such a set, let $U \subseteq \mathbb{C}$ be open, and $f_\alpha: U \rightarrow \mathbb{C}$ be a function for $\alpha \in A$.

Then we say that $\sum_{\alpha \in A} f_\alpha$ *converges normally* if the infinite series $\sum_{n=1}^{\infty} f_{\varphi(n)}$ converges normally for some bijection $\varphi: \mathbb{N} \rightarrow A$. This converges behavior and the limit of the series $\sum_{n=1}^{\infty} f_{\varphi(n)}$ does not depend on the choice of the bijection φ .

Example 6.10. The series $\sum_{n \in \mathbb{Z}} \frac{1}{(z - n)^2}$ converges normally on $\mathbb{C} \setminus \mathbb{Z}$.

To see this, let $K \subseteq \mathbb{C} \setminus \mathbb{Z}$ be a compact set. Then K is bounded and so there exists $N \in \mathbb{N}$ such that $K \subseteq \overline{B}(0, N)$. Since finitely many terms do not affect the convergence behavior of a series, we may restrict ourselves to considering indices $n \in \mathbb{Z}$ such that $|n| \geq 2N$. Then for $z \in K$ and such n we have $|z| \leq N \leq |n|/2$. It follows that

$$\frac{1}{|z - n|^2} \leq \frac{1}{(|n| - |z|)^2} \leq \frac{1}{(|n| - |n|/2)^2} = \frac{4}{n^2}.$$

Now we have

$$\sum_{|n| \geq 2N} \frac{4}{n^2} = 8 \sum_{n=2N}^{\infty} \frac{1}{n^2} < \infty.$$

It follows from the M -test that the series $\sum_{|n| \geq 2N} \frac{1}{|z - n|^2}$ converges uniformly on K (no matter how we order the elements of the series). Hence $\sum_{n \in \mathbb{Z}} \frac{1}{|z - n|^2}$ converges uniformly on K (again, no matter how we order the terms of the series).

Since the compact set $K \subseteq \mathbb{C} \setminus \mathbb{Z}$ was arbitrary, it follows that $\sum_{n \in \mathbb{Z}} \frac{1}{(z - n)^2}$ converges normally on $\mathbb{C} \setminus \mathbb{Z}$ as claimed.

Definition 6.11 (Locally uniform convergence of an infinite product). Let $f_n: U \rightarrow \mathbb{C}$ for $n \in \mathbb{N}$ be functions defined on an open set $U \subseteq \mathbb{C}$ and suppose that the infinite product $\prod_{n=1}^{\infty} f_n(z)$ converges (properly) for each $z \in U$. We say that the infinite product *converges locally uniformly on U* if the sequence

of partial products converges locally uniformly on U to the limit function represented by the infinite product.

More precisely, if for $z \in U$ and $n \in \mathbb{N}$ we define $p_n(z) = \prod_{k=1}^n f_k(z)$ and $p(z) = \prod_{k=1}^{\infty} f_k(z)$, then the requirement is that $p_n \rightarrow p$ locally uniformly on U .

Proposition 6.12. *Let $U \subseteq \mathbb{C}$ be open, and $f_n: U \rightarrow \mathbb{C}$ for $n \in \mathbb{N}$ be a continuous function. If the series $\sum_{n=1}^{\infty} (f_n - 1)$ converges normally on U , then*

$\prod_{n=1}^{\infty} f_n(z)$ converges absolutely for each $z \in U$ and $\prod_{n=1}^{\infty} f_n$ converges locally uniformly on U .

Proof. We have $\prod_{n=1}^{\infty} f_n(z) = \prod_{n=1}^{\infty} (1 + (f_n(z) - 1))$ for $z \in U$. So our assumptions imply that $\prod_{n=1}^{\infty} f_n(z)$ converges absolutely for each $z \in U$.

To verify the second statement, for $n \in \mathbb{N}$ and $z \in U$ we set $p_n(z) := \prod_{k=1}^n f_k(z)$ and $p(z) := \prod_{k=1}^{\infty} f_k(z)$. We have to show that $p_n \rightarrow p$ locally uniformly on U , or, equivalently, that we have uniform convergence $p_n \rightarrow p$ on each compact subset K of U .

To see this, let $K \subseteq U$ be compact. Then by our assumptions the series $\sum_{n=1}^{\infty} |f_n - 1|$ converges uniformly on K to a non-negative continuous limit function L . Then L is bounded on K and so there exists $C_1 \geq 0$ such that $L(z) \leq C_1$ for $z \in K$. Then we have

$$\sum_{k=1}^n |f_k(z) - 1| \leq \sum_{k=1}^{\infty} |f_k(z) - 1| = L(z) \leq C_1$$

for all $n \in \mathbb{N}$ and $z \in K$. It follows that

$$\begin{aligned} |p_n(z)| &\leq 1 + |p_n(z) - 1| = 1 + \left| \prod_{k=1}^n (1 + (f_k(z)) - 1) - 1 \right| \\ &\leq \prod_{k=1}^n (1 + |f_k(z) - 1|) \quad (\text{see proof of Prop. 5.13 (a)}) \\ &\leq \exp\left(\sum_{k=1}^n |f_k(z) - 1|\right) \leq C_2 := e^{C_1} \end{aligned}$$

for all $n \in \mathbb{N}$ and $z \in U$. We conclude the the functions p_n are uniformly bounded on K .

Now let $\epsilon > 0$ be arbitrary. We may assume that $\epsilon \in (0, 1)$. Then $e^\epsilon - 1 \leq 3\epsilon$ (exercise!). Since $\sum_{n=1}^{\infty} |f_n - 1|$ converges uniformly on K , it follows that there exists $N \in \mathbb{N}$ such that for all $z \in K$,

$$\sum_{n=N}^{\infty} |f_n(z) - 1| \leq \epsilon.$$

Then for all $m, n \in \mathbb{N}$ with $m \geq n > N$ and $z \in K$,

$$\begin{aligned} |p_n(z) - p_m(z)| &= |p_{n-1}(z)| \cdot \left| \prod_{k=n}^m (1 + (f_k(z) - 1)) - 1 \right| \\ &\leq C_2 \left[\exp\left(\sum_{k=n}^m |f_k(z) - 1|\right) - 1 \right] \\ &\leq C_2(e^\epsilon - 1) \leq 3C_2\epsilon. \end{aligned}$$

Letting $m \rightarrow \infty$, we see that

$$|p_n(z) - p(z)| \leq 3C_2\epsilon$$

for $z \in K$ and $n > N$. Hence, $p_n \rightarrow p$ uniformly on K and the statement follows. $\square$

7 Power series

The most important series in complex analysis are *power series*.

Definition 7.1 (Power series). A series of the form

$$\sum_{n=0}^{\infty} a_n(z - z_0)^n$$

with fixed $z_0 \in \mathbb{C}$ and $a_n \in \mathbb{C}$ for $n \in \mathbb{N}_0$, is called a *(complex) power series (in z centered at z_0)*.

The convergence of the series depends on z .

Theorem 7.2. Let $\sum_{n=0}^{\infty} a_n(z - z_0)^n$ be a power series and define

$$R = \frac{1}{\limsup_{n \rightarrow \infty} \sqrt[n]{|a_n|}} \in [0, \infty]. \quad (7.1)$$

Then the power series $\left\{ \begin{array}{c} \text{converges} \\ \text{diverges} \end{array} \right\}$ for all $z \in \mathbb{C}$ with $\left\{ \begin{array}{c} |z - z_0| < R \\ |z - z_0| > R \end{array} \right\}$.

In (7.1) we use the conventions $1/0 = \infty$ and $1/\infty = 0$. For a point $z \in \mathbb{C}$ with $|z - z_0| = R$ both convergence or divergence may happen.

The value R is called the *radius of convergence* of the power series.

Proof. We will only consider the case $R \in (0, \infty)$. The cases $R = 0$ and $R = \infty$ are similar (and easier).

Let $z \in \mathbb{C}$ with $|z - z_0| > R$ be arbitrary. We want to show that the series diverges at z . Pick $R' \in \mathbb{R}$ with $R < R' < |z - z_0|$. Then

$$\limsup_{n \rightarrow \infty} \sqrt[n]{|a_n|} = \frac{1}{R} > \frac{1}{R'},$$

and so

$$\sqrt[n]{|a_n|} > \frac{1}{R'} \quad \text{for infinitely many } n \in \mathbb{N}.$$

Hence

$$|a_n| \cdot |z - z_0|^n > \frac{|z - z_0|^n}{(R')^n} \geq 1 \quad \text{for infinitely many } n \in \mathbb{N}.$$

This means that the terms in $\sum_{n=0}^{\infty} a_n(z - z_0)^n$ do *not* converge to 0, and so the series diverges.

Let $z \in \mathbb{C}$ with $|z - z_0| < R$ be arbitrary. Then

$$\begin{aligned} \limsup_{n \rightarrow \infty} \sqrt[n]{|a_n| \cdot |z - z_0|^n} &= \limsup_{n \rightarrow \infty} |z - z_0| \sqrt[n]{|a_n|} \\ &= |z - z_0| \limsup_{n \rightarrow \infty} \sqrt[n]{|a_n|} = \frac{|z - z_0|}{R} < 1, \end{aligned}$$

and so the power series converges by the root test. $\square$

Remark 7.3. One can show that the radius of convergence R of a power series $\sum_{n=0}^{\infty} a_n(z - z_0)^n$ is given by

$$R = \lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right| \in [0, \infty]$$

if the last limit exists (possibly as an improper limit with value ∞).

Actually, more can be said about the convergence behavior of a power series.

Lemma 7.4. *Let $\sum_{n=0}^{\infty} a_n(a - z_0)^n$ be a complex power series with positive radius R of convergence and define $D := B(z_0, R)$ ($=\mathbb{C}$ if $R = \infty$). Then the series converges normally and in particular locally uniformly on D .*

Note that in general, the power series does not converge uniformly on D .

Proof. Let $K \subseteq D$ be a compact set. Then there exists $r > 0$ with $r < R$ such that $K \subseteq B(z_0, r) \subseteq D$. Since $r < R$ we can find a constant $R' > 0$ with $r < R' < R$.

Then

$$1/R' > 1/R = \limsup_{n \rightarrow \infty} \sqrt[n]{|a_n|}$$

(here we set $1/R = 0$ if $R = \infty$). It follows that we have $|a_n| \leq (1/R')^n$ for all $n \in \mathbb{N}_0$ with at most finitely many exceptions. Therefore we can choose a sufficiently large constant $C \geq 1$ such that $|a_n| \leq C(1/R')^n$ for all $n \in \mathbb{N}_0$.

Now if $z \in K$, then $|z - z_0| \leq r$. It follows that for all $z \in K$ and $n \in \mathbb{N}_0$ we have

$$|a_n| \cdot |z - z_0|^n \leq C(r/R')^n.$$

Since $r/R' < 1$, the series $C \sum_{n=0}^{\infty} (r/R')^n$ converges (why?). Hence the series

$\sum_{n=0}^{\infty} |a_n| \cdot |z - z_0|^n$ converges uniformly for $z \in K$ by the M -test and we

conclude that $\sum_{n=0}^{\infty} a_n(a - z_0)^n$ converges normally on D . $\square$

Lemma 7.5. *If $\sum_{n=0}^{\infty} a_n(a - z_0)^n$ is a complex power series, then the series*

$$\sum_{n=1}^{\infty} n a_n (z - z_0)^{n-1} = \sum_{n=0}^{\infty} (n+1) a_{n+1} (z - z_0)^n$$

(obtained by “term-by-term differentiation”) has the same radius of convergence.

Proof. Note that $\sum_{n=1}^{\infty} n a_n (z - z_0)^{n-1}$ converges if and only if

$$(z - z_0) \sum_{n=1}^{\infty} n a_n (z - z_0)^{n-1} = \sum_{n=1}^{\infty} n a_n (z - z_0)^n$$

converges. So by Theorem 7.2 it suffices to show that

$$\limsup_{n \rightarrow \infty} \sqrt[n]{|a_n|} = \limsup_{n \rightarrow \infty} \sqrt[n]{n|a_n|}.$$

This follows from

$$\begin{aligned} \limsup_{n \rightarrow \infty} \sqrt[n]{|a_n|} &\leq \limsup_{n \rightarrow \infty} \sqrt[n]{n|a_n|} \\ &\leq \underbrace{\limsup_{n \rightarrow \infty} \sqrt[n]{n}}_{=1} \cdot \limsup_{n \rightarrow \infty} \sqrt[n]{|a_n|} \\ &= \limsup_{n \rightarrow \infty} \sqrt[n]{|a_n|}. \end{aligned}$$

$\square$

Lemma 7.6. (a) For all $w, z \in \mathbb{C}$, $w \neq z$, $n \in \mathbb{N}$, we have

$$\frac{w^n - z^n}{w - z} - nz^{n-1} = \left(\sum_{k=1}^{n-1} kz^{k-1}w^{n-k-1} \right) (w - z).$$

(b) If $\sum_{n=0}^{\infty} a_n(z - z_0)^n$ is a power series with radius of convergence R and $0 \leq r < R$, then $\sum_{n=1}^{\infty} n^2 |a_n| r^n$ converges.

Proof. (a)

$$\begin{aligned} (w - z) \sum_{k=1}^{n-1} kz^{k-1}w^{n-k-1} &= \sum_{k=1}^{n-1} kz^{k-1}w^{n-k} - \sum_{k=1}^{n-1} kz^k w^{n-k-1} \\ &= \sum_{k=0}^{n-2} (k+1)z^k w^{n-k-1} - \sum_{k=1}^{n-1} kz^k w^{n-k-1} \\ &= w^{n-1} - (n-1)z^{n-1} + \sum_{k=1}^{n-2} z^k w^{n-k-1} \\ &= \sum_{k=0}^{n-1} z^k w^{n-k-1} - nz^{n-1} \\ &= \frac{w^n - z^n}{w - z} - nz^{n-1}. \end{aligned}$$

In the last step we used that

$$\begin{aligned} \frac{w^n - z^n}{w - z} &= w^{n-1} + w^{n-2}z + \cdots + wz^{n-2} + z^{n-1} \\ &= \sum_{k=0}^{n-1} z^k w^{n-k-1}. \end{aligned}$$

(b) We apply the root test:

$$\limsup_{n \rightarrow \infty} \sqrt[n]{n^2 |a_n| r^n} \leq r \limsup_{n \rightarrow \infty} \sqrt[n]{|a_n|} \cdot \underbrace{\limsup_{n \rightarrow \infty} (\sqrt[n]{n})^2}_{=1} = \frac{r}{R} < 1. \quad \square$$

Theorem 7.7. Suppose that $\sum_{n=0}^{\infty} a_n(z - z_0)^n$ is a complex power series with radius of convergence $R > 0$. Let $D := B(z_0, R)$ ($= \mathbb{C}$ if $R = \infty$) and

$$f(z) = \sum_{n=0}^{\infty} a_n(z - z_0)^n \quad \text{for } z \in D.$$

Then f is holomorphic on D , and

$$f'(z) = \sum_{n=0}^{\infty} (n+1)a_{n+1}(z - z_0)^n \quad \text{for } z \in D. \quad (7.2)$$

Note that the last power series converges for $z \in D$ by Lemma 7.5.

The statement says that a function represented by a power series is holomorphic in its disk of convergence, and the derivative can be obtained by term-by-term differentiation of the power series.

Proof. Without loss of generality $z_0 = 0$. Fix $z \in D$ and choose $r > 0$ such that $|z| < r < R$. We denote by $g(z)$ the value of the power series in (7.2). We have to show that

$$\lim_{w \rightarrow z} \left(\frac{f(w) - f(z)}{w - z} - g(z) \right) = 0. \quad (7.3)$$

To see this consider $w \in D$ close to z , $w \neq z$. Then $|w| < r$, and so

$$\begin{aligned} \left| \frac{f(w) - f(z)}{w - z} - g(z) \right| &= \left| \sum_{n=1}^{\infty} a_n \left(\frac{w^n - z^n}{w - z} - nz^{n-1} \right) \right| \\ &\leq \sum_{n=1}^{\infty} |a_n| \cdot \left| \frac{w^n - z^n}{w - z} - nz^{n-1} \right| \\ &= |w - z| \sum_{n=1}^{\infty} \left| a_n \sum_{k=1}^{n-1} kz^{k-1}w^{n-k-1} \right| \quad (\text{by Lem. 7.6 (a)}) \\ &\leq |w - z| \sum_{n=1}^{\infty} \left(|a_n| \sum_{k=1}^{n-1} kr^{n-2} \right) \\ &= |w - z| \sum_{n=1}^{\infty} \frac{n(n-1)}{2} |a_n| r^{n-2} \\ &\leq \frac{|w - z|}{r^2} \sum_{n=1}^{\infty} n^2 |a_n| r^n \leq C|w - z| \quad (\text{by Lem. 7.6 (b)}), \end{aligned}$$

where $C > 0$ is independent of w . Letting $w \rightarrow z$, we see that (7.3) holds. $\square$

Corollary 7.8. *A function represented by a power series*

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$$

has derivatives of arbitrarily high order:

$$\begin{aligned} f^{(k)}(z) &= \sum_{n=k}^{\infty} n(n-1)\dots(n-k+1)a_n(z-z_0)^{n-k} \\ &= \sum_{n=0}^{\infty} (n+k)\dots(n+1)a_{n+k}(z-z_0)^n, \quad k \in \mathbb{N}_0. \end{aligned}$$

In particular,

$$f^{(k)}(z_0) = k!a_k,$$

and so

$$a_k = \frac{f^{(k)}(z_0)}{k!}.$$

The last equation says that the coefficients of a power series equal the Taylor coefficients of the functions that it represents.

Theorem 7.9. *The exponential function has the power series representation*

$$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!} \quad \text{for } z \in \mathbb{C}.$$

Proof. We know that this power series converges for all $z \in \mathbb{C}$ (see Example 5.6 (b)). Let $E: \mathbb{C} \rightarrow \mathbb{C}$ be the function represented by the power series. Then by Theorem 7.7 the function E is holomorphic in $\mathbb{C}$ and

$$E'(z) = \sum_{n=1}^{\infty} n \frac{z^{n-1}}{n!} = \sum_{n=0}^{\infty} \frac{z^n}{n!} = E(z)$$

for all $z \in \mathbb{C}$. We know that the exponential function $z \mapsto e^z$ is also holomorphic on $\mathbb{C}$, and $\frac{d}{dz}e^z = e^z$. Hence

$$\frac{d}{dz}(E(z)e^{-z}) = E'(z)e^{-z} - E(z)e^{-z} = E(z)e^{-z} - E(z)e^{-z} = 0,$$

and so by Theorem 3.5 the function $z \mapsto E(z)e^{-z}$ is constant on $\mathbb{C}$. Since

$$E(0)e^{-0} = 1 \cdot 1 = 1$$

it follows that

$$E(z)e^{-z} \equiv 1.$$

Hence

$$E(z) = E(z)e^0 = E(z)e^{-z}e^z = 1 \cdot e^z = e^z$$

for all $z \in \mathbb{C}$.

□

7.1 Exercises

Problem 7.1. Suppose $\sum_{n=0}^{\infty} a_n(z - z_0)^n$ is a power series with nonzero coefficients. Show that its radius of convergence R is given by

$$R = \lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right|$$

if the limit exists.

Problem 7.2. Find the radius of convergence of the power series $\sum_{n=1}^{\infty} a_n z^n$, where

- (a) $a_n = (2i)^n / (n + i)^2$,
- (b) $a_n = n^2 + 5in + 1$,
- (c) $a_n = n^{\log n}$.

Problem 7.3. Show that the series $\sum_{n=1}^{\infty} \frac{z^n}{n}$ converges for $z \in \mathbb{C}$ if and only if $|z| \leq 1$ and $z \neq 1$.

Problem 7.4. Suppose that the power series $f(z) = \sum_{n=0}^{\infty} a_n z^n$ and $g(z) = \sum_{n=0}^{\infty} b_n z^n$ have radii of convergence $R_1 > 0$ and $R_2 > 0$, respectively. For $n \in \mathbb{N}_0$ define

$$c_n = \sum_{k=0}^n a_k b_{n-k}.$$

Show that the power series $h(z) = \sum_{n=0}^{\infty} c_n z^n$ (the so-called *Cauchy product* of f and g) converges for all $|z| < R := \min\{R_1, R_2\}$ and that $h(z) = f(z)g(z)$ for all such z .

Problem 7.5. Let $R > 0$, and suppose that the power series $f(z) = \sum_{n=0}^{\infty} a_n z^n$ converges for all $z \in B := B(0, R)$.

(a) Show that the power series converges uniformly on each compact subset of B .

(b) For $n \in \mathbb{N}_0$ and $z \in B(0, R)$ define

$$S_n(z) := \left| \sum_{k=0}^n a_k z^k \right|^2.$$

Show that on each compact subset of B the function sequence $\{S_n\}$ converges uniformly to $|f|^2$.

(c) Show that for each $0 \leq r < R$ Parseval's formula is valid:

$$\frac{1}{2\pi} \int_0^{2\pi} |f(re^{it})|^2 dt = \sum_{n=0}^{\infty} |a_n|^2 r^{2n}.$$

Problem 7.6. (a) Show that the power series

$$\sum_{n=0}^{\infty} \frac{i^{n+1}}{(n+1)!} z^n$$

converges for all $z \in \mathbb{C}$.

(b) Let g be the function represented by the power series in (a). Find a simple expression for $g(z)$ if $z \neq 0$.

(c) For $r > 0$ let $\alpha_r: [0, 2] \rightarrow \mathbb{C}$ be the path defined by

$$\alpha_r(t) = \begin{cases} re^{i\pi t}, & t \in [0, 1], \\ (2t - 3)r, & t \in [1, 2]. \end{cases}$$

Show that $\int_{\alpha_r} g(z) dz = 0$. Hint: This follows from a general theorem that will be proved later, but show this directly.

(d) Show that $\lim_{r \rightarrow \infty} \int_{[-r, r]} g(z) dz = i\pi$.

(e) Show that $\int_0^{\infty} \frac{\sin x}{x} dx = \frac{\pi}{2}$.

Problem 7.7. Let $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$.

(a) Show that for each $t \in [0, 2\pi]$ the function P_t defined as

$$P_t(z) = \frac{e^{it} + z}{e^{it} - z} \quad \text{for } z \in \mathbb{D}$$

can be represented by a power series that is centered at $z_0 = 0$ and converges on $\mathbb{D}$.

(b) Let $u: \partial\mathbb{D} \rightarrow \mathbb{R}$ be a continuous function. Show that the function $f: \mathbb{D} \rightarrow \mathbb{C}$ defined as

$$f(z) = \frac{1}{2\pi} \int_0^{2\pi} \frac{e^{it} + z}{e^{it} - z} u(e^{it}) dt \quad \text{for } z \in \mathbb{D}$$

is holomorphic. Hint: Find a power series representation of f on $\mathbb{D}$.

Problem 7.8. Define $\gamma: [0, 2\pi] \rightarrow \mathbb{C}$ as $\gamma(t) = e^{it}$ for $t \in [0, 2\pi]$.

(a) Show that

$$\frac{1}{2\pi i} \int_{\gamma} \left(w - \frac{1}{w}\right)^n \frac{dw}{w} = \begin{cases} 0, & n \in \mathbb{N}_0, n \text{ odd}, \\ (-1)^{n/2} \binom{n}{n/2}, & n \in \mathbb{N}_0, n \text{ even}. \end{cases}$$

The *Bessel function* $J_0: \mathbb{C} \rightarrow \mathbb{C}$ is defined as

$$J_0(z) = \frac{1}{2\pi i} \int_{\gamma} \exp\left(\frac{z}{2} \left(w - \frac{1}{w}\right)\right) \frac{dw}{w}$$

for $z \in \mathbb{C}$.

(b) Show that J_0 is represented by the power series

$$J_0(z) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(n!)^2} \left(\frac{z}{2}\right)^{2n}$$

convergent for all $z \in \mathbb{C}$ and is hence an *entire function*, that is, holomorphic on $\mathbb{C}$.

Problem 7.9. (a) Show that the function J_0 defined in Problem 7.8 satisfies Bessel's differential equation

$$zJ_0''(z) + J_0'(z) + zJ_0(z) = 0$$

for $z \in \mathbb{C}$.

(b) Suppose $f(z) = \sum_{n=0}^{\infty} a_n z^n$ is a power series that converges for $z \in \mathbb{C}$ and the function f represented by this power series satisfies Bessel's differential equation. Show that if $f(0) = 1$ and $f'(0) = 0$, then $f = J_0$. Hint: Find a recursive relation that the coefficients a_n must satisfy.

Problem 7.10. The periodic oscillations of a thin membrane (such as a drum) can be modeled as follows: in rest position the membrane occupies a region U in the x - y plane. Its movement under periodic (undamped) oscillations is described by a function $u = u(x, y, t)$ measuring the (small) deflection of the membrane in the z -direction in a three-dimensional x - y - z coordinate system over the point $(x, y) \in U$ at time t . The motion of the membrane is then governed by the two-dimensional wave equation

$$u_{tt} = u_{xx} + u_{yy} \quad (7.4)$$

with the boundary condition $u = 0$ on ∂U (with a suitable unit of time depending on material constants of the membrane). Here we denote by a subscript differentiation with respect to the variables t, x, y .

(a) Suppose u is a solution of (7.4) of the form $u(x, y, t) = f(t)v(x, y)$ with sufficiently smooth functions f and v , where $f \neq 0$ is a periodic function with $f(0) = 0$. Show that then there exist constants $A \in \mathbb{R}$ and $\lambda > 0$ such that

$$u(x, y, t) = A \sin(\lambda t) v(x, y),$$

where

$$\Delta v + \lambda^2 v = 0. \quad (7.5)$$

(b) Suppose that the membrane is circular, say $U = \mathbb{D}$. Suppose that $u(x, y, t) = f(t)g(r)$ is a solution of (7.4), where f is as in (a) and g only depends on the radius r in polar coordinates in the x - y -plane. Show that g then satisfies the differential equation

$$r g''(r) + g'(r) + \lambda^2 r g(r) = 0. \quad (7.6)$$

(c) Find solutions g of (7.6) in terms of λ and the Bessel function J_0 (see Problem 7.9). Which condition on λ guarantees that $u(x, y, t) = f(t)g(r)$ as in (b) for one of these solutions g satisfies the boundary condition $u = 0$ on $\partial\mathbb{D}$?

8 Local Cauchy theorems

Remark 8.1. Cauchy's Integral Theorem says that if f is a holomorphic function on a region $\Omega \subseteq \mathbb{C}$ and γ is a loop in Ω such that all points “surrounded” by γ also belong to Ω , then $\int_{\gamma} f = 0$.

Since it requires some effort to make the phrase “points surrounded by γ ” mathematically precise, we will prove a preliminary version of this statement first.

8.2. Convex sets. A set $K \subseteq \mathbb{C}$ is called *convex* if for all $z, w \in K$ we have $[z, w] \subseteq K$. One can show that a set is convex if and only if for all $z_1, \dots, z_n \in K$, $n \in \mathbb{N}$, and all $\lambda_1, \dots, \lambda_n \in [0, 1]$ with $\lambda_1 + \dots + \lambda_n = 1$ the *convex combination* $\lambda_1 z_1 + \dots + \lambda_n z_n$ also belongs to K . If $M \subseteq \mathbb{C}$ is any set, then the *convex hull* $\text{co}(M)$ is the smallest convex set containing M ; it consists of all convex combinations of points in M .

A *closed triangle* Δ is the convex hull of three points $z_1, z_2, z_3 \in \mathbb{C}$ called the *vertices* of $\mathbb{C}$; so

$$\Delta = \text{co}(\{z_1, z_2, z_3\}).$$

We say that Δ is *oriented* if we specify a cyclic order of its vertices z_1, z_2, z_3 . We then write $\Delta = \Delta(z_1, z_2, z_3)$ to indicate the vertices and their cyclic order. The *oriented boundary* of $\Delta = \Delta(z_1, z_2, z_3)$ is given by

$$\partial\Delta := [z_1, z_2] \cup [z_2, z_3] \cup [z_3, z_1],$$

where the edges of Δ are traversed according to the cyclic order of the vertices. We write $\int_{\partial\Delta} f$ for the corresponding path integral of a function f defined on $\partial\Delta$.

Theorem 8.3 (Goursat's Lemma). *Let $\Omega \subseteq \mathbb{C}$ be an open set, $p \in \Omega$, and $f: \Omega \rightarrow \mathbb{C}$ be a continuous function that is holomorphic on $\Omega \setminus \{p\}$. If Δ is a closed oriented triangle contained in Ω , then*

$$\int_{\partial\Delta} f = 0.$$

Proof. Let $a, b, c \in \Omega$ be the vertices of Δ cyclically ordered according to the orientation of Δ . Then $\Delta = \Delta(a, b, c)$.

1. We first assume that $p \notin \Delta$. We will inductively define a nested sequence of closed oriented triangles Δ_n , $n \in \mathbb{N}_0$, in Ω . Let $\Delta_0 := \Delta$. We define $I := \int_{\partial\Delta} f$, and denote by L the length of $\partial\Delta$.

Let $\begin{Bmatrix} a' \\ b' \\ c' \end{Bmatrix}$ be the midpoint of the interval $\begin{Bmatrix} [b, c] \\ [c, a] \\ [a, c] \end{Bmatrix}$, and let $\tilde{\Delta}_1 := \Delta(a, c', b')$, $\tilde{\Delta}_2 := \Delta(b, a', c')$, $\tilde{\Delta}_3 := \Delta(c, b', a')$, $\tilde{\Delta}_4 := \Delta(a', b', c')$. Then

$$I = \int_{\partial\Delta} f = \sum_{k=1}^4 \int_{\partial\tilde{\Delta}_k} f,$$

and so

$$|I| = \left| \sum_{k=1}^4 \int_{\partial\tilde{\Delta}_k} f \right| \leq \sum_{k=1}^4 \left| \int_{\partial\tilde{\Delta}_k} f \right|.$$

Hence there exists $j \in \{1, 2, 3, 4\}$ such that

$$\left| \int_{\partial\tilde{\Delta}_j} f \right| \geq \frac{1}{4}|I|.$$

Define $\Delta_1 := \tilde{\Delta}_j$. Note that

$$\ell(\partial\Delta_1) = \frac{1}{2}\ell(\partial\Delta) = \frac{1}{2}L \text{ and } \text{diam}(\Delta_1) = \frac{1}{2}\text{diam}(\Delta) \leq \frac{1}{2}L.$$

Now we apply the same construction to Δ_1 to obtain another closed oriented triangle Δ_2 , then to Δ_2 to get Δ_3 , etc.

In this way, we obtain a nested sequence of closed oriented triangles $\Delta_0 \supseteq \Delta_1 \supseteq \Delta_2 \supseteq \dots$ in Ω such that

$$\left| \int_{\partial\Delta_n} f \right| \geq \frac{1}{4^n}|I|$$

for $n \in \mathbb{N}_0$. Moreover, we also have

$$\ell(\partial\Delta_n) = \frac{1}{2^n}L \text{ and } \text{diam}(\Delta_n) \leq \frac{1}{2^n}L$$

for $n \in \mathbb{N}_0$.

If $K_0 \supseteq K_1 \supseteq K_2 \supseteq \dots$ is a nested sequence of compact sets in $\mathbb{C}$ with $\text{diam}(K_n) \rightarrow 0$ as $n \rightarrow \infty$, then $\bigcap_{n \in \mathbb{N}_0} K_n$ consists of exactly one point (exercise!). This implies that there exists a unique point $z_0 \in \bigcap_{n \in \mathbb{N}_0} \Delta_n \subseteq \Delta \subseteq \Omega \setminus \{p\}$. Since $z_0 \in \Omega$ and $z_0 \neq p$, the function f is differentiable at z_0 .

Now let $\epsilon > 0$ be arbitrary. Then there exists $\delta > 0$ such that $B(z_0, \delta) \subseteq \Omega$ and

$$\left| \frac{f(z) - f(z_0)}{z - z_0} - f'(z_0) \right| \leq \epsilon, \quad (8.1)$$

whenever $0 < |z - z_0| < \delta$. We can find $n \in \mathbb{N}$ such that

$$\text{diam}(\Delta_n) \leq \frac{1}{2^n} L < \delta.$$

Then

$$\Delta_n \subseteq \overline{B}(z_0, L/2^n) \subseteq B(z_0, \delta) \subseteq \Omega,$$

and so by (8.1) we have

$$|f(z) - f(z_0) - f'(z_0)(z - z_0)| \leq \epsilon |z - z_0| \quad (8.2)$$

whenever $z \in \Delta_n$.

The functions $z \mapsto f(z_0)$ and $z \mapsto f'(z_0)(z - z_0)$ have the primitives $z \mapsto zf(z_0)$ and $z \mapsto \frac{1}{2}f'(z_0)(z - z_0)^2$, respectively. Hence by 4.7 (c) we have

$$\int_{\partial\Delta_n} f(z_0) dz = 0 = \int_{\partial\Delta_n} f'(z_0)(z - z_0) dz.$$

It follows from (8.2) that

$$\begin{aligned} \left| \int_{\partial\Delta_n} f(z) dz \right| &= \left| \int_{\partial\Delta_n} (f(z) - f(z_0) - f'(z_0)(z - z_0)) dz \right| \\ &\leq \ell(\partial\Delta_n) \sup_{z \in \partial\Delta_n} |f(z) - f(z_0) - f'(z_0)(z - z_0)| \\ &\leq \frac{1}{2^n} L \sup_{z \in \partial\Delta_n} \epsilon |z - z_0| \\ &\leq \epsilon \frac{1}{2^n} L \text{diam}(\Delta) \leq \epsilon L \frac{1}{2^n} \cdot L \frac{1}{2^n} = \epsilon L^2 \frac{1}{4^n}. \end{aligned}$$

This implies

$$|I| \leq 4^n \left| \int_{\partial\Delta_n} f(z) dz \right| \leq 4^n \epsilon L^2 \frac{1}{4^n} = \epsilon L^2.$$

Since $\epsilon > 0$ was arbitrary, we must have $|I| = 0$, and so $I = \int_{\partial\Delta} f = 0$ as claimed.

2. We now assume that p coincides with one of the vertices a, b, c of Δ , say $p = a$. Without loss of generality, the vertices a, b, c are distinct (otherwise, it is easy to see that the claim trivially holds). Then we may choose points $b' \in [a, b]$ and $c' \in [a, c]$ close to, but distinct from $a = p$.

Now let $\Delta_1 = \Delta(b', b, c)$, $\Delta_2 = \Delta(c, c', b')$, and $\Delta_3 = \Delta(a, b', c')$. Then

$$\int_{\partial\Delta} f = \int_{\partial\Delta_1} f + \int_{\partial\Delta_2} f + \int_{\partial\Delta_3} f.$$

Moreover, $\Delta_1, \Delta_2 \subseteq \Omega \setminus \{p\}$, and so

$$\int_{\partial\Delta_1} f = 0 = \int_{\partial\Delta_2} f$$

by the first part of the proof. Hence

$$\int_{\partial\Delta} f = \int_{\partial\Delta_3} f,$$

and so

$$\left| \int_{\partial\Delta} f \right| \leq \ell(\partial\Delta_3) \sup_{z \in \partial\Delta_3} |f(z)|.$$

Since f is continuous at p , this function is bounded in a neighborhood of p . Choosing b' and c' close enough to p , we can make $\ell(\partial\Delta_3)$ as small as we want, while $\sup_{z \in \partial\Delta_3} |f(z)|$ stays uniformly bounded. It follows that

$$\left| \int_{\partial\Delta} f \right| \leq \epsilon$$

for all $\epsilon > 0$, which implies $\int_{\partial\Delta} f = 0$, as claimed.

3. Finally, we prove the claim if $p \in \Delta$, but where p is not necessarily a vertex of Δ . We cut Δ into three triangles so that p becomes a vertex in each of them; namely, we define $\Delta_1 = \Delta(a, b, p)$, $\Delta_2 = \Delta(b, c, p)$, and $\Delta_3 = \Delta(c, a, p)$. Then

$$\int_{\partial\Delta} f = \int_{\partial\Delta_1} f + \int_{\partial\Delta_2} f + \int_{\partial\Delta_3} f,$$

and by the second part of the proof

$$\int_{\partial\Delta_1} f = \int_{\partial\Delta_2} f = \int_{\partial\Delta_3} f = 0.$$

It follows that $\int_{\partial\Delta} f = 0$.

The proof is complete. $\square$

Corollary 8.4 (Cauchy's Integral Theorem for convex sets). *Suppose $\Omega \subseteq \mathbb{C}$ is an open convex set, $p \in \Omega$, and $f: \Omega \rightarrow \mathbb{C}$ is a continuous function that is holomorphic on $\Omega \setminus \{p\}$. Then f has a primitive in Ω , i.e., there exists $F \in H(\Omega)$ such that $F' = f$.*

In particular,

$$\int_{\gamma} f = 0,$$

whenever γ is a piecewise smooth loop in Ω .

Proof. Fix $a \in \Omega$, and define

$$F(w) = \int_{[a,w]} f(z) dz$$

for $w \in \Omega$. Since Ω is convex, and so $[a, w] \subseteq \Omega$, and since f is continuous on Ω , the function F is well defined.

Let $w, w_0 \in \Omega$ be arbitrary. Then the triangle $\Delta = \Delta(a, w, w_0)$ lies in Ω . So it follows from Goursat's lemma that

$$\int_{\partial\Delta} f = \int_{[a,w]} f + \int_{[w,w_0]} f + \int_{[w_0,a]} f = 0.$$

Hence

$$F(w) - F(w_0) = \int_{[a,w]} f + \int_{[w_0,a]} f = - \int_{[w,w_0]} f = \int_{[w_0,w]} f.$$

So for $w \neq w_0$ we get

$$\begin{aligned} \left| \frac{F(w) - F(w_0)}{w - w_0} - f(w_0) \right| &= \\ &= \left| \frac{1}{w - w_0} \int_{[w_0,w]} f(z) dz - \frac{1}{w - w_0} \int_{[w_0,w]} f(w_0) dz \right| \\ &= \frac{1}{|w - w_0|} \left| \int_{[w_0,w]} (f(z) - f(w_0)) dz \right| \\ &\leq \sup_{z \in [w_0,w]} |f(z) - f(w_0)| \quad (\text{by Lem. 4.9}) \\ &\rightarrow 0 \quad \text{as } w \rightarrow w_0. \end{aligned}$$

The last relation follows from the continuity of f at w_0 .

So F is differentiable at w_0 , and $F'(w_0) = f(w_0)$. Since w_0 was arbitrary, we conclude that $F \in H(\Omega)$ and $F' = f$.

The statement about path integrals now follows from 4.7 (c). $\square$

Lemma 8.5. *Let $a \in \mathbb{C}$, $r > 0$, $\gamma(t) = a + re^{it}$ for $t \in [0, 2\pi]$, and $z_0 \in B(a, r)$. Then*

$$\int_{\gamma} \frac{dz}{z - z_0} = 2\pi i.$$

Proof. It is not hard, but tedious to evaluate this integral by reducing it to integrals over real-valued functions and applying a trigonometric substitution. We prefer to show the claim by using methods from complex analysis.

Note that for $z \in \mathbb{C}$ with $|z - a| = r$ and $z_0 \in B(a, r)$ we have

$$\begin{aligned} \frac{1}{z - z_0} &= \frac{1}{(z - a) - (z_0 - a)} = \frac{1}{z - a} \cdot \frac{1}{1 - \left(\frac{z_0 - a}{z - a}\right)} \\ &= \frac{1}{z - a} \sum_{n=0}^{\infty} \left(\frac{z_0 - a}{z - a}\right)^n = \sum_{n=0}^{\infty} \frac{(z_0 - a)^n}{(z - a)^{n+1}}. \end{aligned}$$

This sum converges uniformly in z on the circle $\gamma^* = \{z \in \mathbb{C} : |z - a| = r\}$ (to see this, use the Weierstrass M -test with

$$M_n := \frac{1}{r} \underbrace{\left(\frac{|z_0 - a|}{r}\right)^n}_{< 1}$$

for $n \in \mathbb{N}_0$).

Hence by Remark 6.6, we can integrate the infinite series term-by-term and obtain

$$\int_{\gamma} \frac{dz}{z - z_0} = \sum_{n=0}^{\infty} \int_{\gamma} \frac{(z_0 - a)^n}{(z - a)^{n+1}} dz.$$

Now by Example 4.5 we have

$$\int_{\gamma} \frac{dz}{(z - a)^{n+1}} = \int_{\gamma} (z - a)^{-n-1} dz = \begin{cases} 2\pi i & \text{for } n = 0, \\ 0 & \text{for } n \neq 0. \end{cases}$$

Hence

$$\int_{\gamma} \frac{dz}{z - z_0} = 2\pi i \cdot (z_0 - a)^0 = 2\pi i. \quad \square$$

Corollary 8.6 (Cauchy's Integral Formula. Version I). *Let $U \subseteq \mathbb{C}$ be an open set, $a \in U$, $r > 0$, $f \in H(U)$, and $\gamma(t) = a + re^{it}$ for $t \in [0, 2\pi]$. If $\overline{B}(a, r) \subseteq U$, then*

$$f(z_0) = \frac{1}{2\pi i} \int_{\gamma} \frac{f(z)}{z - z_0} dz$$

for all $z_0 \in B(a, r)$.

Proof. Suppose $\overline{B}(a, r) \subseteq U$, and let $z_0 \in B(a, r)$. Note that $\gamma^* \subseteq U$, and $z_0 \notin \gamma^*$. So $z \mapsto \frac{f(z) - f(z_0)}{z - z_0}$ is continuous on γ^* and we can integrate this function over γ^* .

Since $\overline{B}(a, r) \subseteq U$, and U is open, we can find a small number $\delta > 0$ such that for $R := r + \delta$ we have

$$\overline{B}(a, r) \subseteq \Omega := B(a, R) \subseteq U$$

(exercise!). Note that Ω is an open and convex set. Moreover, the function $g: \Omega \rightarrow \mathbb{C}$ defined by

$$g(z) = \begin{cases} \frac{f(z) - f(z_0)}{z - z_0} & \text{for } z \in \Omega \setminus \{z_0\}, \\ f'(z_0) & \text{for } z = z_0, \end{cases}$$

is continuous on Ω and holomorphic on $\Omega \setminus \{z_0\}$. Since $\gamma^* \subseteq \Omega$, Corollary 8.4 implies that

$$\begin{aligned} \int_{\gamma} \frac{f(z)}{z - z_0} dz - 2\pi i f(z_0) &= \\ \int_{\gamma} \frac{f(z)}{z - z_0} dz - f(z_0) \int_{\gamma} \frac{dz}{z - z_0} &\quad (\text{by Lem. 8.5}) \\ &= \int_{\gamma} \frac{f(z) - f(z_0)}{z - z_0} dz \\ &= \int_{\gamma} g(z) dz = 0. \end{aligned}$$

The statement follows. □

8.1 Exercises

Problem 8.1. Let f be a continuous function in $\mathbb{C}$, and $\{z_n\}$ and $\{w_n\}$ be convergent sequences in $\mathbb{C}$ with $z_n \rightarrow z$ and $w_n \rightarrow w$ as $n \rightarrow \infty$. Show that

$$\lim_{n \rightarrow \infty} \int_{[z_n, w_n]} f(z) dz = \int_{[z, w]} f(z) dz.$$

Problem 8.2. Show that $|e^z - e^w| \leq |z - w|$ whenever z and w are complex numbers with non-positive real part.

Problem 8.3. Let $\Omega \subseteq \mathbb{C}$ be a convex region and f be a zero-free holomorphic function on Ω .

(a) Show that f has a holomorphic logarithm, that is, there exists a function $L \in H(\Omega)$ such that

$$f(z) = \exp(L(z)) \text{ for all } z \in \Omega.$$

Hint: Do not use the function $w \mapsto \log(w)$ for $w \in \mathbb{C} \setminus \{0\}$ which we have not yet defined (and cannot be defined as a holomorphic function on $\mathbb{C} \setminus \{0\}$).

(b) Show that f has a holomorphic square root, that is, there exists a function $S \in H(\Omega)$ such that

$$f(z) = S(z)^2 \text{ for all } z \in \Omega.$$

9 Power series representations of holomorphic functions

Theorem 9.1. *Let $U \subseteq \mathbb{C}$ be open, and $f \in H(U)$. Then f has a local representation as a power series at every point of U ; more precisely, let $a \in U$ be arbitrary, $r > 0$ be such that $\overline{B}(a, r) \subseteq U$, and $z_0 \in B(a, r)$. Then there exists a power series $\sum_{n=0}^{\infty} a_n(z - z_0)^n$ that converges and satisfies*

$$f(z) = \sum_{n=0}^{\infty} a_n(z - z_0)^n$$

for all $z \in B(z_0, r - |z_0 - a|)$. Moreover, if $\gamma(t) = a + re^{it}$ for $t \in [0, 2\pi]$, then

$$a_n = \frac{1}{2\pi i} \int_{\gamma} \frac{f(w)}{(w - z_0)^{n+1}} dw \quad (9.1)$$

for $n \in \mathbb{N}_0$.

The formulation is slightly technical, but the most important case is if $z_0 = a$. Then the statement says that f can be expanded in a power series centered at $z_0 = a$ that converges in $B(a, r)$.

Proof. Under the given assumptions, fix $z \in B(z_0, r - |z_0 - a|)$. Then by Cauchy's Integral Formula (Corollary 8.6) we have

$$f(z) = \frac{1}{2\pi i} \int_{\gamma} \frac{f(w)}{w - z} dw.$$

Now for $w \in \partial B(a, r)$ we have

$$|w - z_0| \geq |w - a| - |z_0 - a| = r - |z_0 - a| > |z - z_0|.$$

So if we define $\rho := (r - |z_0 - a|) \leq |w - z_0|$, then $|z - z_0|/|w - z_0| \leq |z - z_0|/\rho < 1$ and so

$$\begin{aligned} \frac{1}{w - z} &= \frac{1}{(w - z_0) - (z - z_0)} = \frac{1}{(w - z_0)} \frac{1}{1 - \left(\frac{z - z_0}{w - z_0}\right)} \\ &= \sum_{n=0}^{\infty} \frac{(z - z_0)^n}{(w - z_0)^{n+1}}. \end{aligned}$$

It follows that

$$\frac{f(w)}{w - z} = \sum_{n=0}^{\infty} \frac{(z - z_0)^n}{(w - z_0)^{n+1}} f(w) \quad (9.2)$$

for all $w \in \partial B(z_0, r)$.

The series in (9.2) (for fixed z) converges uniformly for $w \in \partial B(a, r)$ by the Weierstrass M -test; indeed, since $\partial B(a, r)$ is a compact set and $|f|$ is continuous on $\partial B(a, r)$, there exists $K \geq 0$ such that

$$|f(w)| \leq K \quad \text{for } w \in \partial B(a, r).$$

So for $n \in \mathbb{N}_0$ we have

$$\left| \frac{(z - z_0)^n}{(w - z_0)^{n+1}} f(w) \right| \leq \frac{|z - z_0|^n}{\rho^{n+1}} K =: M_n,$$

and

$$\sum_{n=0}^{\infty} M_n = \frac{K}{\rho} \sum_{n=0}^{\infty} \frac{|z - z_0|^n}{\rho^n} = \frac{K}{\rho} \frac{1}{1 - \frac{|z - z_0|}{\rho}} = \frac{K}{\rho - |z - z_0|} < \infty.$$

Integrating the series in (9.2) term-by-term, we obtain

$$\begin{aligned} f(z) &= \frac{1}{2\pi i} \int_{\gamma} \frac{f(w)}{w - z} dw \\ &= \frac{1}{2\pi i} \int_{\gamma} \left(\sum_{n=0}^{\infty} \frac{(z - z_0)^n}{(w - z_0)^{n+1}} f(w) \right) dw \\ &= \sum_{n=0}^{\infty} \left(\frac{1}{2\pi i} \int_{\gamma} \frac{(z - z_0)^n}{(w - z_0)^{n+1}} f(w) dw \right) \\ &= \sum_{n=0}^{\infty} a_n (z - z_0)^n, \end{aligned} \quad (9.3)$$

where

$$a_n = \frac{1}{2\pi i} \int_{\gamma} \frac{f(w)}{(w - z_0)^{n+1}} dw \quad (9.4)$$

for $n \in \mathbb{N}_0$.

Since a_n is independent of z , and $z \in B(z_0, r)$ was arbitrary, we see that the series in (9.3) converges for all $z \in B(z_0, \rho) = B(z_0, r - |z_0 - a|)$ and represents f . The claim follows. $\square$

Corollary 9.2. *Let $U \subseteq \mathbb{C}$ be open, and $f \in H(U)$. Then $f' \in H(U)$.*

This corollary establishes the fact used in Section 3.

Proof. By Theorem 9.1 the function f has a local power series representation; so f' can also be locally represented by a power series and $f' \in H(U)$ as follows from Theorem 7.7. $\square$

Corollary 9.3. *Let $U \subseteq \mathbb{C}$ be open, and $f \in H(U)$. Then f has derivatives $f^{(n)}$ of arbitrary order $n \in \mathbb{N}$, and $f^{(n)} \in H(U)$.*

Holomorphic functions are indefinitely differentiable!

Corollary 9.4. *Let $U \subseteq \mathbb{C}$ be open, and $f \in H(U)$. Then at each point $z_0 \in U$, the function f is represented by its Taylor series, and so*

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} (z - z_0)^n.$$

This is valid for all $z \in \mathbb{C}$ with $|z - z_0| < \text{dist}(z, \partial U)$; in particular, if R is the radius of convergence of the power series, then $R \geq \text{dist}(z_0, \partial U)$.

In other words, the Taylor series of f converges and represents the function in the largest open disk centered at z_0 that is contained in U ; if $\partial U = \emptyset$ and so $U = \mathbb{C}$, this means that this Taylor series converges and represents the function for *all* $z \in \mathbb{C}$.

Proof. By Theorem 9.1 the function f can locally be represented by a power series centered at z_0 . By Corollary 7.8 the coefficients of this power series are uniquely determined, because they are the Taylor coefficients of f at z_0 . Since each point $z \in U$ with $|z - z_0| < \text{dist}(z_0, \partial U)$ lies in a disk $B(z_0, r)$ with $\overline{B}(z_0, r) \subseteq U$, the claim follows from Theorem 9.1. $\square$

Corollary 9.5 (Cauchy's Integral Formula for derivatives). *Let $U \subseteq \mathbb{C}$ be open, and $f \in H(U)$. Suppose that $a \in U$, $r > 0$, $\overline{B}(a, r) \subseteq U$, and define $\gamma(t) = a + re^{it}$ for $t \in [0, 2\pi]$. Then*

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \int_{\gamma} \frac{f(z)}{(z - z_0)^{n+1}} dz \quad (9.5)$$

for all $z_0 \in B(a, r)$ and $n \in \mathbb{N}_0$.

Proof. Equation (9.5) follows from formula (9.4) in the proof of Theorem 9.1, since $a_n = \frac{f^{(n)}(z_0)}{n!}$ is the n -th Taylor coefficient of f at z_0 . $\square$

Corollary 9.6. *Let $U \subseteq \mathbb{C}$ be open, and $f \in H(U)$. Suppose that $z_0 \in U$, $r > 0$, $\overline{B}(z_0, r) \subseteq U$, and $|f(z)| \leq M$ for all $z \in \partial B(z_0, r)$. Then for $n \in \mathbb{N}_0$ we have*

$$|f^{(n)}(z_0)| \leq \frac{n!M}{r^n} \quad (\text{Cauchy estimates}).$$

Proof. We apply Corollary 9.5 with $a = z_0$ and $\gamma(t) = z_0 + re^{it}$ for $t \in [0, 2\pi]$ to conclude that

$$\begin{aligned} |f^{(n)}(z_0)| &= \left| \frac{n!}{2\pi i} \int_{\gamma} \frac{f(z)}{(z - z_0)^{n+1}} dz \right| \\ &\leq \frac{n!}{2\pi} \ell(\gamma) \frac{M}{r^{n+1}} = \frac{n!}{2\pi} 2\pi r \frac{M}{r^{n+1}} = \frac{n!M}{r^n}. \end{aligned} \quad \square$$

Definition 9.7. A function $f: \mathbb{C} \rightarrow \mathbb{C}$ that is holomorphic on the whole complex plane $\mathbb{C}$ is called an *entire function*.

Corollary 9.8. *Let f be an entire function. Then at each point z_0 we can represent f by a power series*

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$$

that converges for all $z \in \mathbb{C}$.

Proof. This follows from Corollary 9.4. $\square$

9.1 Exercises

Problem 9.1. Show that a complex power series $\sum_{n=0}^{\infty} a_n(z - z_0)^n$ has a positive radius of convergence if and only if there exist constants $C > 0$ and $M > 0$ such that $|a_n| \leq CM^n$ for all $n \in \mathbb{N}_0$.

Problem 9.2. Let $\Omega \subseteq \mathbb{C}$ be a region, and a, b be holomorphic functions on Ω . Suppose that $f \in H(\Omega)$ satisfies the differential equation

$$f'' + af' + bf = 0. \quad (9.6)$$

Show that if there exists a point $z_0 \in \Omega$ such that

$$f(z_0) = f'(z_0) = 0,$$

then f vanishes identically on Ω .

Problem 9.3. Suppose we have the same setup as in Problem 9.2 and $z_0 \in \Omega$. Show for given values $f(z_0), f'(z_0) \in \mathbb{C}$, the equation (9.6) has a unique solution f that is holomorphic near z_0 . Hint: Assume that

$$f(z) = \sum_{n=0}^{\infty} c_n(z - z_0)^n.$$

Show that the coefficients c_n can be computed recursively. To establish convergence of the series near z_0 , prove an estimate of the form $|c_n| \leq CM^n$ by induction.

Problem 9.4. Let $f(z) = \sum_{k=0}^{\infty} a_k z^k$ be an entire function and define

$$P_n(z) = \sum_{k=0}^n \frac{n!}{(n-k)!} a_k z^k$$

for $n \in \mathbb{N}_0$. Show that then $P_n(z/n) \rightarrow f(z)$ locally uniformly for $z \in \mathbb{C}$.

10 Applications of Cauchy's theory

In this section, we collect various application of the Cauchy theory.

Theorem 10.1 (Liouville's Theorem). *Every bounded entire function is constant.*

Proof. Let $f \in H(\mathbb{C})$ and suppose that there exists $M \geq 0$ such that $|f(z)| \leq M$ for all $z \in \mathbb{C}$. Then f has a power series representation

$$f(z) = \sum_{n=0}^{\infty} a_n z^n$$

that converges for all $z \in \mathbb{C}$. We know that

$$a_n = \frac{f^{(n)}(0)}{n!}$$

for all $n \in \mathbb{N}_0$. The Cauchy estimates with $z_0 = 0$ are valid for all $n \in \mathbb{N}_0$ and all $r > 0$. Hence

$$|a_n| = \frac{|f^{(n)}(0)|}{n!} \leq \frac{M}{r^n}$$

for all $n \in \mathbb{N}$ and all $r > 0$. Letting $r \rightarrow \infty$ (for fixed $n \in \mathbb{N}$), we see that $a_n = 0$ for $n \in \mathbb{N}$. So

$$f(z) = \sum_{n=0}^{\infty} a_n z^n = a_0$$

for all $z \in \mathbb{C}$. This shows that f is a constant function. $\square$

Theorem 10.2 (Fundamental Theorem of Algebra). *Every non-constant complex polynomial has a root; more precisely, if $n \in \mathbb{N}$, $a_0, a_1, \dots, a_n \in \mathbb{C}$, $a_n \neq 0$, and $P: \mathbb{C} \rightarrow \mathbb{C}$,*

$$P(z) = a_0 + a_1 z + \dots + a_n z^n \tag{10.1}$$

for $z \in \mathbb{C}$, then there exists $z_0 \in \mathbb{C}$ such that $P(z_0) = 0$.

Proof. Let P be a polynomial as in (10.1). We argue by contradiction and assume that $P(z) \neq 0$ for all $z \in \mathbb{C}$. Then $f = 1/P$ is an entire function.

Claim: f is bounded.

Obviously, $|P(z)| \rightarrow \infty$ as $|z| \rightarrow \infty$. Hence there exists $R > 0$ such that

$$|P(z)| > |P(0)| \quad \text{whenever } z \in \mathbb{C}, |z| \geq R,$$

and so

$$|f(z)| < |f(0)| \quad \text{whenever } z \in \mathbb{C}, |z| \geq R. \quad (10.2)$$

Since f is continuous and $\overline{B}(0, R)$ is compact, there exists $K \geq 0$ such that

$$|f(z)| \leq K \quad \text{for all } z \in \overline{B}(0, R).$$

So

$$|f(z)| \leq \max\{K, |f(0)|\} \quad \text{for all } z \in \mathbb{C}.$$

The claim follows.

Liouville's Theorem now implies that f is constant. This is a contradiction, since $|f(R)| < |f(0)|$ by (10.2). $\square$

Corollary 10.3. *The field $\mathbb{C}$ is algebraically closed.*

Remark 10.4. If $P \neq 0$ is a polynomial as in (10.1), where $a_n \neq 0$, then $n \in \mathbb{N}_0$ is called the *degree* of P , denoted by $\deg(P)$. For the constant polynomial $P = 0$ the degree is undefined.

If P and Q are polynomials and $Q \neq 0$, then we can divide P by Q with remainder by the usual division algorithm. Hence there exist polynomials R and S such that

$$P = S \cdot Q + R,$$

where $\deg(R) < \deg(Q)$ or $R = 0$.

If z_0 is a root of $P \neq 0$, then we can apply this to $Q(z) := z - z_0$, and we obtain

$$P(z) = S(z)(z - z_0) + R(z) \quad \text{for } z \in \mathbb{C},$$

where $\deg(R) < \deg(Q) = 1$ or $R = 0$. So R must be a constant polynomial, and $0 = P(z_0) = R(z_0)$ shows that $R = 0$. We conclude that

$$P(z) = S(z)(z - z_0) \quad \text{for } z \in \mathbb{C}.$$

If z_0 is also a root of S , we can repeat this reasoning, etc. It follows that there exists $k \in \mathbb{N}$ and a polynomial T with $T(z_0) \neq 0$ such that

$$P(z) = (z - z_0)^k T(z) \quad \text{for } z \in \mathbb{C}.$$

The number k is uniquely determined and called the *multiplicity* of the root z_0 of P .

Corollary 10.5. *Every polynomial P of degree $n \in \mathbb{N}$ has precisely n roots counting multiplicities; more precisely, if $z_1, \dots, z_l \in \mathbb{C}$, $l \in \mathbb{N}$, are the distinct roots of P and $k_1, \dots, k_l \in \mathbb{N}$ their respective multiplicities, then $k_1 + \dots + k_l = n$.*

Moreover, there exists $a \in \mathbb{C}$, $a \neq 0$, such that

$$P(z) = a(z - z_1)^{k_1} \dots (z - z_m)^{k_m} \quad \text{for } z \in \mathbb{C}.$$

Proof (Outline). This follows from induction on n . For the induction step let P be a polynomial of degree n . Then P has a root z_0 by the Fundamental Theorem of Algebra. Then $P(z) = (z - z_0)Q(z)$ for $z \in \mathbb{C}$, where $\deg(Q) = n - 1$. Now we apply the induction hypothesis to Q . $\square$

Theorem 10.6 (Morera's Theorem). *Let $\Omega \subseteq \mathbb{C}$ be open, and $f: \Omega \rightarrow \mathbb{C}$ be continuous. Suppose that for every closed oriented triangle $\Delta \subseteq \Omega$ we have*

$$\int_{\partial\Delta} f = 0.$$

Then f is holomorphic on Ω .

This can be considered as a “converse” of Goursat's lemma.

Proof. It is enough to show that f is holomorphic on every open disk $B \subseteq \Omega$. If B is such a disk, fix $a \in B$ and define

$$F(w) = \int_{[a,w]} f(z) dz \quad \text{for } w \in B.$$

This makes sense, since B is convex. As in the proof of Corollary 8.4, one shows that $F \in H(B)$ and $F'(w) = f(w)$ for $w \in B$. So in B the function f is the derivative of a holomorphic function. Hence f is holomorphic itself (Corollary 9.2). $\square$

Theorem 10.7. *Let $U \subseteq \mathbb{C}$ be open, $F: U \rightarrow \mathbb{C}$, and $F_n \in H(U)$ for $n \in \mathbb{N}$. Suppose that $F_n \rightarrow f$ locally uniformly on U . Then $F \in H(U)$. Moreover, for each $k \in \mathbb{N}$ we have $F_n^{(k)} \rightarrow F^{(k)}$ locally uniformly on U .*

So locally uniform limits of holomorphic functions are holomorphic, and taking derivatives is compatible with taking locally uniform limits.

Proof. The function F is continuous on U as a locally uniform limit of continuous functions (Theorem 6.5). The holomorphicity of F now follows from Morera's Theorem; indeed, if $\Delta \subseteq U$ is an arbitrary closed oriented triangle in U , then $F_n \rightarrow F$ uniformly on $\partial\Delta$. Hence

$$\int_{\partial\Delta} F = \lim_{n \rightarrow \infty} \underbrace{\int_{\partial\Delta} F_n}_{=0} = 0.$$

For the second statement is enough to establish the case $k = 1$; the general case then follows from this by repeated application.

So let $z_0 \in U$ be arbitrary. Pick $r > 0$ such that $\overline{B}(z_0, r) \subseteq U$. Let $\gamma(t) = z_0 + re^{it}$ for $t \in [0, 2\pi]$. Then by Corollary 9.5 for all $z \in \overline{B}(z_0, r/2)$,

$$\begin{aligned} |F'_n(z) - F'(z)| &= \frac{1}{2\pi} \left| \int_{\gamma} \frac{F_n(w) - F(w)}{(w - z)^2} dw \right| \\ &\leq \frac{r}{(r/2)^2} \max_{w \in \partial B(z_0, r)} |F_n(w) - F(w)| \\ &= \frac{4}{r} \max_{w \in \partial B(z_0, r)} |F_n(w) - F(w)| \rightarrow 0 \quad \text{as } n \rightarrow \infty. \end{aligned}$$

Here the last limit relation follows from the fact that $F_n \rightarrow F$ uniformly on $\partial B(z_0, r)$. This estimate above now implies that $F'_n \rightarrow F'$ uniformly on $\overline{B}(z_0, r/2)$, and it follows that $F'_n \rightarrow F'$ locally uniformly on U . $\square$

Theorem 10.8. *Let $U \subseteq \mathbb{C}$ be open, and $f_n \in H(U)$ for $n \in \mathbb{N}$. If the series $\sum_{n=1}^{\infty} f_n$ converges normally on U , then the series $\sum_{n=1}^{\infty} f_n^{(k)}$ also converges normally on U for each $k \in \mathbb{N}$. Moreover, if $F(z) = \sum_{n=1}^{\infty} f_n(z)$ for $z \in U$, then $F \in H(U)$ and $F^{(k)}(z) = \sum_{n=1}^{\infty} f_n^{(k)}(z)$ for all $z \in U$ and $k \in \mathbb{N}$.*

So normal convergence of a series of holomorphic functions is preserved under differentiation of the terms. Moreover, derivatives of the function that the series represents can be obtained by term-by-term differentiation of the series.

Proof. It suffices to prove the first statement for $k = 1$. For this let $a \in U$ be arbitrary. Then we will find a neighborhood $V \subseteq U$ of a on which the series $\sum_{n=1}^{\infty} |f'_n|$ converges uniformly. We can find a radius $r > 0$ such that $\overline{B}(a, r) \subseteq U$. Now define $V := B(a, r/2)$.

In order to show the uniform convergence of $\sum_{n=1}^{\infty} |f'_n|$ on V , we want to apply Cauchy's Integral Formula for derivatives. So define $\gamma(t) = a + re^{it}$ for $t \in [0, 2\pi]$ and let $z_0 \in V = B(a, r/2)$ and $\ell, m \in \mathbb{N}$ with $m \geq \ell$ be arbitrary. Then for each $n = \ell, \dots, m$, we can find a number $\alpha_n \in [0, 2\pi]$ such that $|f'_n(z_0)| = e^{i\alpha_n} f'_n(z_0)$. We now apply Cauchy's integral formula for the function $g := \sum_{n=\ell}^m e^{i\alpha_n} f_n \in H(U)$ and its derivative g' at z_0 .

Note that for $z \in \gamma^* = \partial B(a, r)$ and $z_0 \in V = B(a, r/2)$ we have

$$|z - z_0| \geq |z - a| - |z_0 - a| \geq r - r/2 = r/2.$$

It follows that

$$\begin{aligned} \sum_{n=\ell}^m |f'_n(z_0)| &= \left| \sum_{k=n}^m e^{i\alpha_k} f'_k(z_0) \right| = |g'(z_0)| = \left| \frac{1}{2\pi i} \int_{\gamma} \frac{g(z)}{(z - z_0)^2} dz \right| \\ &\leq \frac{4}{r} \sup_{z \in \partial B(a, r)} |g(z)| \leq \frac{4}{r} \sup_{z \in \partial B(a, r)} \sum_{n=\ell}^m |f_n(z)|. \end{aligned}$$

Letting $m \rightarrow \infty$, we see that

$$\sum_{n=\ell}^{\infty} |f'_n(z_0)| \leq \frac{4}{r} \sup_{z \in \partial B(a, r)} \sum_{n=\ell}^{\infty} |f_n(z)|$$

for all $z_0 \in V$.

Now let $\epsilon > 0$ be arbitrary. Since by assumption $\sum_{n=1}^{\infty} |f_n|$ converges locally uniformly on U , and hence uniformly on the compact set $\partial B(a, r) \subseteq U$, we can choose $N \in \mathbb{N}$ so large such that

$$\sum_{n=N}^{\infty} |f_k(z)| \leq \frac{r}{4} \epsilon$$

for all $z \in \partial B(a, r)$. Combining this with the previous estimate, we see that then

$$\sum_{n=N}^{\infty} |f'_n(z_0)| \leq \epsilon$$

for all $z_0 \in V$. Then shows that the series $\sum_{n=1}^{\infty} |f'_n(z_0)|$ converges uniformly on the neighborhood V of a . Since $a \in U$ was arbitrary, we conclude that $\sum_{n=1}^{\infty} |f'_n|$ converges uniformly on U , and hence the series $\sum_{n=1}^{\infty} f'_n$ converges normally on U .

To see the second statement, note that the partial sums $F_n := \sum_{j=1}^n f_j$ of $\sum_{n=1}^{\infty} f_n$ are holomorphic functions on U and that we have locally uniform convergence $F_n \rightarrow F$ on U as follows from normal convergence of $\sum_{n=1}^{\infty} f_n$.

It then follows from Theorem 10.7 that F is holomorphic and that $F_n^{(k)} \rightarrow F^{(k)}$ locally uniformly on U for all $k \in \mathbb{N}$. But for fixed $k \in \mathbb{N}$, the function $F_n^{(k)}$ is the partial sum of the series $\sum_{n=1}^{\infty} f_n^{(k)}$. In other words, we have

$$\sum_{n=1}^{\infty} f_n^{(k)} = F^{(k)} \text{ and the statement follows.}$$

□

10.1 Exercises

Problem 10.1. Let f be a holomorphic function in the unit disk $\mathbb{D}$. Suppose that $|f(1/n)| \leq e^{-n}$ for all $n \in \mathbb{N}$, $n \geq 2$. Show that f vanishes identically.

Problem 10.2. Let f be a continuous function in $\mathbb{C}$ that is holomorphic on $\mathbb{C} \setminus \mathbb{R}$. Show that f is holomorphic on $\mathbb{C}$.

Problem 10.3. Let $U \subseteq \mathbb{C}$ be an open set, and $f: U \rightarrow \mathbb{C}$ be a holomorphic function. For $(z, w) \in U \times U \subseteq \mathbb{C}^2$ we define

$$g(z, w) := \begin{cases} \frac{f(z) - f(w)}{z - w}, & z \neq w, \\ f'(z), & z = w. \end{cases}$$

(a) Show that g is continuous on $U \times U$. Hint: Here we use the product topology on $U \times U$. Essentially, you have to show that if $\{z_n\}$ and $\{w_n\}$ are sequences in U with $z_n \rightarrow z \in U$ and $w_n \rightarrow w \in U$, then $g(z_n, w_n) \rightarrow g(z, w)$.

(b) Show that for fixed $w \in U$ the function $z \mapsto g(z, w)$ is holomorphic on U .

Problem 10.4. Let $U, V \subseteq \mathbb{C}$ be open, and $f: U \times V \rightarrow \mathbb{C}$ be a continuous function. Assume that for each $w \in V$ the function $z \mapsto f(z, w)$ is holomorphic on U , and denote by $\frac{\partial f}{\partial z}$ the derivative of this function (so as in real analysis we differentiate with respect to z with w fixed).

(a) Show that $\frac{\partial f}{\partial z}: U \times V \rightarrow \mathbb{C}$ is continuous.

(b) Show that if γ a piecewise smooth path in V , then

$$z \mapsto \int_{\gamma} f(z, w) dw$$

is a holomorphic function on U and its derivative is given by

$$\frac{d}{dz} \left(\int_{\gamma} f(z, w) dw \right) = \int_{\gamma} \frac{\partial f}{\partial z}(z, w) dw.$$

So under the given assumptions we can “differentiate under the integral sign”.

Assume in addition that the function f is also holomorphic in its second variable, i.e., for fixed $z \in U$ the map $w \mapsto f(z, w)$ is holomorphic on V .

(c) Show that for each $z \in U$ the map $w \mapsto \frac{\partial f}{\partial z}(z, w)$ is holomorphic on V .

Problem 10.5 (Riemann ζ -function). In the following, we consider the half-plane $H = \{z \in \mathbb{C} : \operatorname{Re}(z) > 1\}$.

(a) Show that the series

$$\zeta(z) = \sum_{n=1}^{\infty} \frac{1}{n^z}$$

converges normally for $z \in H$ and there represents a holomorphic function, the *Riemann ζ -function*.

(b) Let P be the set of all prime numbers. Show that the infinite product

$$\prod_{p \in P} \frac{1}{1 - \frac{1}{p^z}}$$

converges absolutely and locally uniformly for $z \in H$.

(c) Show that

$$\zeta(z) = \sum_{n=1}^{\infty} \frac{1}{n^z} = \prod_{p \in P} \frac{1}{1 - \frac{1}{p^z}}$$

for $z \in H$. Hint: Show that each partial product of the infinite product can be written in the form $\sum_{n \in A} \frac{1}{n^z}$ with $A \subseteq \mathbb{N}$.

Problem 10.6 (Eisenstein series). (a) Let $K \subseteq \mathbb{H} := \{\tau \in \mathbb{C} : \operatorname{Im}(\tau) > 0\}$ be compact. Show that then there exists $c_0 > 0$ such that

$$|x\tau + y| \geq c_0$$

for all $\tau \in K$ and $(x, y) \in \mathbb{R}^2$ with $x^2 + y^2 = 1$.

(b) Show that for each $k \in \mathbb{N}$, $k \geq 2$, the so-called *Eisenstein series*

$$G_{2k}(\tau) = \sum_{(n,m) \in \mathbb{Z} \times \mathbb{Z} \setminus \{(0,0)\}} \frac{1}{(n\tau + m)^{2k}}$$

converges normally for $\tau \in \mathbb{H}$ and there represents a holomorphic function.

(c) Let $k \in \mathbb{N}$, $k \geq 2$. Show that then for all $\tau \in \mathbb{H}$,

$$G_{2k}(\tau + 1) = G_{2k}(\tau) \text{ and } G_{2k}(-1/\tau) = \tau^{2k} G_{2k}(\tau).$$

11 Zeros of holomorphic functions

Definition 11.1. Let (X, d) be a metric space, and $A \subseteq X$.

- (a) A point $x \in X$ is called an *isolated point* of A if $x \in A$ and if there exists $\epsilon > 0$ such that $B(x, \epsilon) \cap A = \{x\}$.
- (b) A point $x \in X$ is called a *limit point* of A if for each $\epsilon > 0$ the ball $B(x, \epsilon)$ contains infinitely many distinct points of A .

Remark 11.2. (a) A point $x \in X$ is a limit point of A if and only if there exists a sequence $\{x_n\}$ of distinct elements in A such that $x = \lim_{n \rightarrow \infty} x_n$.

(b) Every point $x \in A$ is either an isolated point of A or a limit point of A .

(c) A set $K \subseteq X$ is compact if and only if every infinite set $A \subseteq K$ has a limit point in K .

Theorem 11.3. Let $\Omega \subseteq \mathbb{C}$ be a region, and $f \in H(\Omega)$. Define

$$Z(f) := \{a \in \Omega : f(a) = 0\} \quad (\text{the zero-set of } f).$$

Then either $Z(f) = \Omega$ (iff $f \equiv 0$), or $Z(f)$ is countable, consists of isolated points, and has no limit points in Ω .

In the latter case, for each $a \in Z(f)$ there exists $m \in \mathbb{N}$ (the multiplicity or order of the zero a), and $g \in H(\Omega)$ with $g(a) \neq 0$ such that

$$f(z) = (z - a)^m g(z) \quad \text{for } z \in \Omega. \quad (11.1)$$

Proof. Let $a \in Z(f)$, and fix $r > 0$ such that $B(a, r) \subseteq \Omega$. By Corollary 9.4 we can represent f by a power series

$$f(z) = \sum_{n=0}^{\infty} a_n (z - a)^n$$

for $z \in B(a, r)$. Here $a_0 = f(a) = 0$. Now either $a_n = 0$ for all $n \in \mathbb{N}$, or there exists a smallest $m \in \mathbb{N}$ such that $a_m \neq 0$.

In the first case, $B(a, r) \subseteq Z(f)$. In the second case, we define

$$g(z) := \begin{cases} \frac{1}{(z - a)^m} f(z) & \text{for } z \in \Omega \setminus \{a\}, \\ \sum_{n=0}^{\infty} a_{n+m} (z - a)^n & \text{for } z \in B(a, r). \end{cases}$$

The definitions for g agree in

$$(\Omega \setminus \{a\}) \cap B(a, r) = B(a, r) \setminus \{a\},$$

and show that $g \in H(\Omega)$. Moreover, $g(a) = a_m \neq 0$ and

$$f(z) = (z - a)^m g(z) \quad \text{for } z \in \Omega. \quad (11.2)$$

By continuity of g at a , there exists $\epsilon > 0$ such that $g(z) \neq 0$ for $z \in B(a, \epsilon) \subseteq \Omega$. Then (11.2) shows that $f(z) \neq 0$ for $z \in B(a, \epsilon) \setminus \{a\}$, and so a is an isolated point of $Z(f)$.

To summarize, either a whole neighborhood of $a \in Z(f)$ belongs to $Z(f)$ or a is an isolated point of $Z(f)$.

Now let A be the set of limit points of $Z(f)$ in Ω . We want to show that A is open and relatively closed in Ω .

If $a \in A$, then there exists a sequence $\{z_n\}$ of distinct points in $Z(f)$ such that $a = \lim_{n \rightarrow \infty} z_n$ (Remark 11.2 (a)). By continuity of f we then have

$$f(a) = \lim_{n \rightarrow \infty} f(z_n) = 0,$$

and so $a \in Z(f)$. Then a is not an isolated point of $Z(f)$, and so there exists $r > 0$ such that $B(a, r) \subseteq Z(f)$ by the first part of the proof. Then $B(a, r) \subseteq A$ which shows that A is an open set.

To establish that A is relatively closed in Ω , we verify that $\Omega \setminus A$ is open. To see this, let $b \in \Omega \setminus A$ be arbitrary. Then b is not a limit point of $Z(f)$, and so by the first part of the proof there exists $\epsilon > 0$ such that $B(b, \epsilon) \subseteq \Omega$, and $B(b, \epsilon)$ contains at most one point in $Z(f)$. Then $B(b, \epsilon) \subseteq \Omega \setminus A$.

We have shown that A is open and relatively closed in Ω . Since Ω is a region, we conclude that $A = \Omega$ or $A = \emptyset$. In the first case, $\Omega = A \subseteq Z(f) \subseteq \Omega$, and so $Z(f) = \Omega$.

In the second case, $A = \emptyset$ and so $Z(f)$ has no limit points in Ω . By the first part of the proof, we also see that $Z(f)$ consists of isolated points, and that at each point $a \in Z(f)$ the function f has a representation as in (11.2).

It remains to show that if $Z(f) \neq \Omega$, then $Z(f)$ is countable. The set Ω (as any open set in $\mathbb{C}$) has a *compact exhaustion*; namely, there are compact sets $K_n \subseteq \Omega$ for $n \in \mathbb{N}$ such that $K_n \subseteq K_{n+1}$ for $n \in \mathbb{N}$, and $\bigcup_{n \in \mathbb{N}} K_n = \Omega$.

For example, we can take

$$K_n = \{z \in \Omega : |z| \leq n \text{ and } \text{dist}(z, \partial\Omega) \geq 1/n\}.$$

Now $Z(f) \cap K_n$ must be a finite set for each $n \in \mathbb{N}$, for otherwise $Z(f)$ would have a limit point in K_n (see Remark 11.2 (c)), and hence in Ω . So

$$Z(f) = \bigcup_{n \in \mathbb{N}} (K_n \cap Z(f))$$

is a countable union of finite sets, which implies that $Z(f)$ is countable. $\square$

Example 11.4. (a) In Theorem 11.3 it is important that Ω is a region. For example, let $\Omega = B(0, 1) \cup B(1, 1)$ and

$$f(z) := \begin{cases} 0 & \text{for } z \in B(0, 1), \\ e^z & \text{for } z \in B(1, 1), \end{cases}$$

Then f is holomorphic on Ω . We have $Z(f) = B(0, 1) \neq \Omega$, but every point in $B(0, 1)$ is a limit point of $Z(f)$ in Ω .

(b) Let $f(z) = \exp(1/z) - 1$ for $z \in \Omega := \mathbb{C} \setminus \{0\}$. Then $f \in H(\Omega)$, and $Z(f) := \{1/(2\pi in) : n \in \mathbb{Z} \setminus \{0\}\}$. Then $Z(f) \neq \Omega$ has a limit point, namely 0, but this point does not lie in Ω .

Theorem 11.5 (Uniqueness Theorem). *Let $\Omega \subseteq \mathbb{C}$ be a region, and $f, g \in H(\Omega)$. If the set*

$$\{z \in \Omega : f(z) = g(z)\}$$

has a limit point in Ω , then $f = g$.

Proof. Consider $h = f - g \in H(\Omega)$. Then the set

$$\begin{aligned} Z(h) &= \{z \in \Omega : h(z) = 0\} \\ &= \{z \in \Omega : f(z) = g(z)\} \end{aligned}$$

has a limit point in Ω . So $Z(h) = \Omega$ by Theorem 11.3 which implies $h = f - g = 0$. Hence $f = g$. $\square$

We conclude this section with a statement that allows us to obtain derivatives of infinite function products.

Theorem 11.6. *Let $U \subseteq \mathbb{C}$ be a region, and $f_n : U \rightarrow \mathbb{C}$ for $n \in \mathbb{N}$ be a holomorphic function. Suppose the series $\sum_{n=1}^{\infty} (f_n - 1)$ converges normally on U and no function f_n , $n \in \mathbb{N}$, vanishes identically on U . Then the following statements are true:*

- (a) The infinite product $p(z) := \prod_{n=1}^{\infty} f_n(z)$ converges absolutely for each $z \in U$ and represents a holomorphic function p on U .
- (b) For $z_0 \in U$ we have $p(z_0) = 0$ if and only if $f_n(z_0) = 0$ for some $n \in \mathbb{N}$. Moreover, if $p(z_0) = 0$, then $f_n(z_0) = 0$ for only finitely many $n \in \mathbb{N}$, and the order of the zero of p at z_0 is the sum of the orders of the zeros of these finitely many functions f_n at z_0 .
- (c) The function p does not vanish identically on U .
- (d) Let $Z(p) := \{z_0 \in U : f(z_0) = 0\}$ be the zero set of p . Then for $z \in U \setminus Z(p)$ we have

$$\frac{p'(z)}{p(z)} = \sum_{n=1}^{\infty} \frac{f'_n(z)}{f_n(z)},$$

where the series converges normally on $U \setminus Z(p)$.

Proof. (a) Proposition 6.12 implies that the infinite product $p(z) = \prod_{n=1}^{\infty} f_n(z)$ converges absolutely for each $z \in U$. Moreover, if $p_n(z) := \prod_{k=1}^n f_k(z)$ for $z \in U$, then $p_n \rightarrow p$ locally uniformly on U . Since $p_n \in H(U)$ for each $n \in \mathbb{N}$, it follows that $p \in H(U)$.

(b) For each point $z_0 \in U$ there exists a neighborhood $V \subseteq U$ of z_0 such that $\sum_{n=1}^{\infty} |f_n - 1|$ converges uniformly on V . This in turn implies that there exists $N \in \mathbb{N}$ such that

$$\sum_{n=N+1}^{\infty} |f_n(z) - 1| < 1$$

for all $z \in V$. Then we have $f_n(z) \neq 0$ for all $z \in V$ and $n \geq N+1$. This in turn implies that

$$g(z) := \prod_{n=N+1}^{\infty} f_n(z) \neq 0$$

for all $z \in V$. Now by the argument in (a) we know that g is a holomorphic function on U . We also have

$$p(z) = \prod_{n=1}^{\infty} f_n(z) = f_1(z) \cdots f_N(z) \prod_{n=N+1}^{\infty} f_n(z) = f_1(z) \cdots f_N(z) \cdot g(z).$$

This shows that $p(z_0) = 0$ if and only if $f_1(z_0) \cdots f_N(z_0) = 0$. The statement now easily follows.

(c) Since for each $n \in \mathbb{N}$ the function f_n does not vanish identically, the set $Z(f_n)$ is countable. By (b) we have $Z(p) = \bigcup_{n \in \mathbb{N}} Z(f_n)$ and so $Z(p)$ is countable as well. Hence p cannot vanish identically on U .

(d) We have $p_n \rightarrow p$ and $p'_n \rightarrow p'$ locally uniformly on U . Now if $z \in U \setminus Z(p)$, then $p(z) \neq 0$ and $f_n(z) \neq 0$ for all $n \in \mathbb{N}$. In particular, $p_n(z) \neq 0$ for all $n \in \mathbb{N}$. This implies that

$$\sum_{k=1}^n \frac{f'_k(z)}{f_k(z)} = \frac{p'_n(z)}{p_n(z)} \rightarrow \frac{p'(z)}{p(z)}$$

as $n \rightarrow \infty$ for all $z \in U \setminus Z(p)$. In other words, we have pointwise convergence

$$\sum_{n=1}^{\infty} \frac{f'_k(z)}{f_k(z)} = \frac{p'(z)}{p(z)}$$

for all $z \in U \setminus Z(p)$.

To show that the series converges normally on $U \setminus Z(p)$, let $a \in U \setminus Z(p)$ be arbitrary. Since $Z(p)$ is a countable set with no limit point in U , we can find $r > 0$ such that $\overline{B}(a, r) \subseteq U \setminus Z(p)$. By our assumptions the series $\sum_{n=1}^{\infty} |f_n - 1|$ converges uniformly on $\overline{B}(a, r)$. In particular, there exists $N \in \mathbb{N}$ such that $|f_n(z) - 1| < 1/2$ and so $|f_n(z)| > 1/2$ for all $z \in \overline{B}(a, r)$ and $n \geq N$. Moreover, it easily follows from Cauchy's integral formula for derivatives that there exists a constant $C > 0$ such that

$$|f'_n(z)| = |(f_n - 1)'(z)| \leq C|f_n(z) - 1|$$

for all $z \in B(a, r/2)$ and $n \in \mathbb{N}$. Then for all $z \in B(a, r/2)$ and $n \geq N$ we have

$$\sum_{k=n}^{\infty} \left| \frac{f'_k(z)}{f_k(z)} \right| \leq 2C \sum_{k=n}^{\infty} |f_k(z) - 1|.$$

Now for large enough n the series on the right hand side is arbitrarily small uniformly on $B(a, r/2)$ by our assumptions. It follows that $\sum_{n=1}^{\infty} \left| \frac{f'_k}{f_k} \right|$ converges uniformly on $B(a, r/2)$. Since $a \in U \setminus Z(p)$ was arbitrary, this series converges locally uniformly on $U \setminus Z(p)$. Hence $\sum_{n=1}^{\infty} \frac{f'_k}{f_k}$ converges normally on U . $\square$

Example 11.7. The infinite product $p(z) = z \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2}\right)$ converges for each $z \in \mathbb{C}$ and represents an entire function p . It has zeros precisely at the points in $\mathbb{Z}$ and these zeros are all of the first order.

These statements immediately follow from Theorem 11.6 and the fact that the series $\sum_{n=1}^{\infty} \frac{z^2}{n^2}$ converges normally on $\mathbb{C}$.

12 The Maximum Modulus Theorem

Theorem 12.1 (Maximum Modulus Theorem. Version I). *Let $\Omega \subseteq \mathbb{C}$ be a region, and $f \in H(\Omega)$. If $|f|$ attains a local maximum at a point in Ω , then f is a constant function.*

Proof. Suppose f attains a local maximum at $a \in \Omega$. Then there exists $\epsilon > 0$ such that $B(a, \epsilon) \subseteq \Omega$ and

$$|f(a)| \geq |f(z)|$$

for all $z \in B(a, \epsilon)$. Replacing f with $e^{i\theta}f$ with suitable $\theta \in [0, 2\pi]$, we may assume $f(a) \geq 0$ without loss of generality. Then

$$f(a) = |f(a)| \geq |f(z)| \geq |\operatorname{Re}(f(z))|$$

for $z \in B(a, \epsilon)$.

Pick $r \in (0, \epsilon)$ and let $\gamma(t) = a + re^{it}$, $t \in [0, 2\pi]$. Then by Cauchy's Integral Formula we have

$$\begin{aligned} f(a) &= \frac{1}{2\pi i} \int_{\gamma} \frac{f(z)}{z - a} dz = \frac{1}{2\pi i} \int_0^{2\pi} \frac{f(a + re^{it})}{re^{it}} ire^{it} dt \\ &= \frac{1}{2\pi} \int_0^{2\pi} f(a + re^{it}) dt \quad (\text{Mean Value Property}) \end{aligned}$$

(the average of the function values on the circle γ^* is equal to the function value at the center of the circle!). Hence

$$\begin{aligned} f(a) &= \operatorname{Re}(f(a)) = \frac{1}{2\pi} \int_0^{2\pi} \operatorname{Re}(f(a + re^{it})) dt \\ &\leq \frac{1}{2\pi} \int_0^{2\pi} |f(a + re^{it})| dt \\ &\leq \frac{1}{2\pi} \int_0^{2\pi} |f(a)| dt = |f(a)| = f(a). \end{aligned}$$

So we must have equality in all these inequalities, which shows that

$$\begin{aligned} \int_0^{2\pi} (|f(a + re^{it})| - \operatorname{Re}(f(a + re^{it}))) dt &= 0, \\ \int_0^{2\pi} (|f(a)| - |f(a + re^{it})|) dt &= 0. \end{aligned}$$

Since the integrands in these integrals are non-negative continuous functions, we conclude that

$$\operatorname{Re} (f(a + re^{it})) = |f(a + re^{it})| = |f(a)| = f(a)$$

for all $t \in [0, 2\pi]$. In particular,

$$\operatorname{Im} (f(a + re^{it})) = 0,$$

and so

$$f(a + re^{it}) = \operatorname{Re} (f(a + re^{it})) = f(a)$$

for all $t \in [0, 2\pi]$. This shows that the function f takes the constant value $f(a)$ on the circle $\partial B(a, r)$. Hence $f \equiv f(a)$ as follows from the Uniqueness Theorem, and so f is a constant function. $\square$

We record two important facts that were derived in the proof of the last theorem.

Corollary 12.2. *Let $\Omega \subseteq \mathbb{C}$ be a region, $f \in H(\Omega)$, and $a \in \Omega$ and $r > 0$ be such that $\overline{B}(a, r) \subseteq \Omega$. Then we have:*

$$(a) \quad f(a) = \frac{1}{2\pi} \int_0^{2\pi} f(a + re^{it}) dt \quad (\text{Mean Value Property}),$$

$$(b) \quad |f(a)| \leq \max\{|f(a + re^{it})| : t \in [0, 2\pi]\} \text{ with equality if and only if } f \text{ is a constant function.}$$

Proof. In the proof of Theorem 12.1 statement (a) was proved explicitly, and statement (b) easily follows from the considerations in this proof. $\square$

Corollary 12.3. *Let $\Omega \subseteq \mathbb{C}$ be a region, $f \in H(\Omega)$, $a \in \Omega$, and $r > 0$ be such that $\overline{B}(a, r) \subseteq \Omega$. If $f(z) \neq 0$ for all $z \in B(a, r)$, then we have*

$$|f(a)| \geq \min\{|f(a + re^{it})| : t \in [0, 2\pi]\} \quad (12.1)$$

with equality if and only if f is a constant function.

Proof. If f has a zero on $\partial B(a, r)$, then the inequality is obvious with strict inequality.

In the other case, where f has no zero on $\partial B(a, r)$, we have $f(z) \neq 0$ for all $z \in \overline{B}(a, r)$ by our hypotheses. Hence there exists $R > r$ such that

$B(a, R) \subseteq \Omega$ and f is zero-free on $\tilde{\Omega} := B(a, R)$. For otherwise, we could find a sequence $\{z_n\}$ of zeros of f with $|z_n - a| \rightarrow r$. Passing to a subsequence if necessary, we may without loss of generality assume that $\{z_n\}$ converges, say $z_n \rightarrow w$. Then

$$|w - a| = \lim_{n \rightarrow \infty} |z_n - a| = r,$$

and so $w \in \partial B(a, r) \subseteq \Omega$. Moreover, the continuity of f shows that

$$f(w) = \lim_{n \rightarrow \infty} f(z_n) = 0.$$

This contradicts the fact that f has no zero on $\partial B(a, r)$.

The function $g := 1/f$ is then holomorphic on $\tilde{\Omega}$, and so by Corollary 12.2 (b) we have

$$|g(a)| \leq \max\{|g(a + re^{it})| : t \in [0, 2\pi]\}. \quad (12.2)$$

By taking reciprocals here, we get (12.1). If we have equality in (12.1), then we must have equality in (12.2). This implies that g is constant on $\tilde{\Omega}$. So $f = 1/g$ is constant on $\tilde{\Omega}$, and hence constant on Ω by the Uniqueness Theorem. $\square$

Theorem 12.4 (Maximum Modulus Theorem. Version II). *Let $\Omega \subseteq \mathbb{C}$ be a bounded region, and $f: \overline{\Omega} \rightarrow \mathbb{C}$ be a continuous function that is holomorphic on Ω . Then*

$$\max\{|f(z)| : z \in \overline{\Omega}\} = \max\{|f(z)| : z \in \partial\Omega\} \quad (12.3)$$

The theorem says that the maximum of $|f|$ on $\overline{\Omega}$ (which exists by continuity of $|f|$ and compactness of $\overline{\Omega}$) is attained at some point on the boundary $\partial\Omega$ of Ω . Note that $\partial\Omega \neq \emptyset$, because the only open sets $U \subseteq \mathbb{C}$ with $\partial U = \emptyset$ are $U = \mathbb{C}$ and $U = \emptyset$ (exercise!).

Proof. The maximum on the left-hand side of (12.3) is attained at some point $z_0 \in \overline{\Omega}$. If $z_0 \in \partial\Omega$, then (12.3) holds.

If $z_0 \in \Omega$, then f is constant on Ω by Theorem 12.1. Then f is also constant on $\overline{\Omega}$ by the continuity of f (exercise!). Since $\partial\Omega \neq \emptyset$, again (12.3) holds. $\square$

12.1 Exercises

Problem 12.1. Let $f: \mathbb{C} \rightarrow \mathbb{C}$ be an entire function and suppose that $f(\sqrt[n]{n}) \in \mathbb{R}$ for all $n \in \mathbb{N}$. Show that $f(\mathbb{R}) \subseteq \mathbb{R}$.

Problem 12.2. Let $f: \overline{\mathbb{D}} \rightarrow \mathbb{C}$ be a continuous function such that $f|_{\mathbb{D}}$ is holomorphic on $\mathbb{D}$. Suppose that $|f(e^{i\theta})| \leq 1$ for $\theta \in [0, \pi)$ and $|f(e^{i\theta})| \leq 1/4$ for $\theta \in [\pi, 2\pi)$. Show that $|f(0)| \leq 1/2$.

Problem 12.3. A function $B: \overline{\mathbb{D}} \rightarrow \mathbb{C}$ is called a (*finite*) *Blaschke product* if there exist $\theta \in \mathbb{R}$, $n \in \mathbb{N}$, and $a_1, \dots, a_n \in \mathbb{D}$ such that

$$B(z) = e^{i\theta} \prod_{k=1}^n \frac{z - a_k}{1 - \overline{a_k}z}$$

for $z \in \overline{\mathbb{D}}$.

(a) Show that every Blaschke product B is continuous on $\overline{\mathbb{D}}$, holomorphic on $\mathbb{D}$, and satisfies $|B(z)| = 1$ for $|z| = 1$.

(b) Show that conversely every non-constant continuous function $f: \overline{\mathbb{D}} \rightarrow \mathbb{C}$ that is holomorphic on $\mathbb{D}$ and satisfies $|f(z)| = 1$ for $|z| = 1$ is a Blaschke product.

Problem 12.4. A *holomorphic germ* P at a point $z_0 \in \mathbb{C}$ is a power series $P(z) = \sum_{n=0}^{\infty} a_n(z - z_0)^n$ centered at z_0 with positive radius of convergence.

Suppose we have a (continuous) path $\gamma: [0, 1] \rightarrow \mathbb{C}$ and germs P_t at $z_t := \gamma(t)$ for each $t \in [0, 1]$. We say that this family P_t , $t \in [0, 1]$, of germs is *compatible along* γ if for each $t_0 \in [0, 1]$ there exist a holomorphic function f in a neighborhood of z_{t_0} such that P_t is the power series expansion of f at z_t for all t sufficiently close to t_0 .

Let $z_0, z_1 \in \mathbb{C}$, $\gamma: [0, 1] \rightarrow \mathbb{C}$ be a path in $\mathbb{C}$ with $\gamma(0) = z_0$ and $\gamma(1) = z_1$, and Q_0 and Q_1 be germs at z_0 and z_1 , respectively. Then Q_1 is called the *analytic continuation* of Q_0 along γ if there exists a family of germs P_t , $t \in [0, 1]$, that is compatible along γ with $P_0 = Q_0$ and $P_1 = Q_1$.

Show that if the analytic continuation Q_1 of a germ Q_0 along a given path γ exists, then it is uniquely determined.

Problem 12.5. Let $\Omega \subseteq \mathbb{C}$ be a region, a, b be holomorphic functions on Ω .

(a) Show that for each $z_0 \in \Omega$ there exists a constant $r_0 > 0$ with the following property: if f is any holomorphic germ at z_0 (see Problem 12.4)

that satisfies the differential equation

$$f'' + af' + bf = 0 \quad (12.4)$$

near z_0 , then the power series representing f has radius of convergence $R \geq r_0$.

(b) Let f be a holomorphic germ at a point $z_0 \in \Omega$ and suppose that f satisfies the equation (12.4) locally near z_0 . Show that if $\gamma: [0, 1] \rightarrow \Omega$ is a path in Ω , then f has an analytic continuation along γ (see Problem 12.4) and that this analytic continuation produces a holomorphic germ g at $z_1 := \gamma(1)$ that satisfies the differential equation (12.4) (with $f = g$) locally near z_1 .

13 The Open Mapping Theorem

Theorem 13.1. *Let $\Omega \subseteq \mathbb{C}$ be open, $\varphi \in H(\Omega)$, and $z_0 \in \Omega$. If $\varphi'(z_0) \neq 0$, then φ is locally injective near z_0 with a holomorphic inverse map.*

More precisely, there exists an open neighborhood $V \subseteq \Omega$ of z_0 such that

- (i) φ is injective on V and $\varphi'(z) \neq 0$ for $z \in V$,
- (ii) $W = \varphi(V)$ is an open set,
- (iii) the inverse map $\psi := (\varphi|_V)^{-1}$ is holomorphic on W .

Proof. Since $\varphi \in H(\Omega)$, we have $\varphi' \in H(\Omega)$ by Corollary 9.2. In particular, φ' is continuous at z_0 . Hence there exists $\delta > 0$ such that $\overline{B}(z_0, \delta) \subseteq \Omega$ and

$$|\varphi'(z) - \varphi'(z_0)| \leq \frac{1}{2}|\varphi'(z_0)| \quad \text{for all } z \in \overline{B}(z_0, \delta). \quad (13.1)$$

Define $V := B(z_0, \delta)$. Then V is an open convex neighborhood of z_0 .

For $z_1, z_2 \in V$ we have $[z_1, z_2] \subseteq V$, and so

$$\begin{aligned} |\varphi(z_1) - \varphi(z_2)| &= \left| \int_{[z_1, z_2]} \varphi'(z) dz \right| \\ &= \left| \int_{[z_1, z_2]} \varphi'(z_0) dz + \int_{[z_1, z_2]} (\varphi'(z) - \varphi'(z_0)) dz \right| \\ &\geq \left| \int_{[z_1, z_2]} \varphi'(z_0) dz \right| - \left| \int_{[z_1, z_2]} (\varphi'(z) - \varphi'(z_0)) dz \right| \quad (13.2) \\ &\geq |z_1 - z_2| \cdot |\varphi'(z_0)| - |z_1 - z_2| \cdot \max_{z \in [z_1, z_2]} |\varphi'(z) - \varphi'(z_0)| \\ &\geq \frac{1}{2}|\varphi'(z_0)| \cdot |z_1 - z_2| \quad \text{by (13.1).} \end{aligned}$$

This inequality shows that φ is injective on V .

Moreover, inequality (13.1) implies that

$$\begin{aligned} |\varphi'(z)| &= |\varphi'(z_0) + (\varphi'(z) - \varphi'(z_0))| \\ &\geq |\varphi'(z_0)| - |\varphi'(z) - \varphi'(z_0)| \\ &\geq \frac{1}{2}|\varphi'(z_0)| \end{aligned}$$

for $z \in V$. Hence $\varphi'(z) \neq 0$ for all $z \in V$. We have established (i).

In order to prove (ii), let $w \in W = \varphi(V)$ be arbitrary. Then there exists $a \in V$ such that $w = \varphi(a)$. Pick $r > 0$ such that $\overline{B}(a, r) \subseteq V$. Then by (13.2) we have

$$|\varphi(a + re^{it}) - \varphi(a)| \geq \frac{1}{2}|\varphi'(z_0)|r \quad \text{for } t \in [0, 2\pi]. \quad (13.3)$$

Let $\epsilon := \frac{1}{6}|\varphi'(z_0)|r > 0$. We claim that $B(w, \epsilon) \subseteq W$. Indeed, if $u \in B(w, \epsilon)$ is arbitrary, then

$$\begin{aligned} |\varphi(a) - u| &= |w - u| < \epsilon \leq 2\epsilon - |\varphi(a) - u| \\ &< \min_{t \in [0, 2\pi]} |\varphi(a + re^{it}) - \varphi(a)| - |\varphi(a) - u| \quad (\text{by (13.3)}) \\ &\leq \min_{t \in [0, 2\pi]} |\varphi(a + re^{it}) - u|. \end{aligned}$$

By Corollary 12.3 this is only possible if $\varphi - u$ has a zero in $B(a, r)$. Hence there exists $z \in B(a, r) \subseteq V$ such that $u = \varphi(z)$. This implies that $u \in \varphi(V)$, and so $B(a, \epsilon) \subseteq W$. It follows that W is open.

To prove (iii) note that (13.2) can be rewritten as

$$|\psi(w_1) - \psi(w_2)| \leq \frac{2}{|\varphi'(z_0)|} \cdot |w_1 - w_2| \quad \text{for } w_1, w_2 \in W.$$

This shows that $\psi: W \rightarrow V$ is continuous on W .

In order to prove that $\psi \in H(W)$, let $w \in W$ be arbitrary. Then there exists a unique point $z \in V$ with $\varphi(z) = w$. We claim that

$$\psi'(w) = \frac{1}{\varphi'(z)}. \quad (13.4)$$

Note that $\varphi'(z) \neq 0$ by (i). To prove (13.4), we have to show that if $\{w_n\}$ is an arbitrary sequence in $W \setminus \{w\}$ with $w_n \rightarrow w$, then

$$\lim_{n \rightarrow \infty} \frac{\psi(w_n) - \psi(w)}{w_n - w} = \frac{1}{\varphi'(z)}.$$

If $z_n := \varphi^{-1}(w_n)$ for $n \in \mathbb{N}$, then $z_n \rightarrow \varphi^{-1}(w) = z$ by the continuity of ψ . Therefore,

$$\lim_{n \rightarrow \infty} \frac{\psi(w_n) - \psi(w)}{w_n - w} = \lim_{n \rightarrow \infty} \frac{z_n - z}{\varphi(z_n) - \varphi(z)} = \frac{1}{\varphi'(z)}.$$

It follows that $\psi \in H(W)$. The proof is complete. $\square$

Example 13.2. Let $m \in \mathbb{N}$ and $\pi_m: \mathbb{C} \rightarrow \mathbb{C}$ be defined as $\pi_m(z) = z^m$ for $z \in \mathbb{C}$ (the m -th power function). Then every point $w = Re^{i\beta} \neq 0$, $R > 0$, $\beta \in \mathbb{R}$, has the m distinct preimages

$$z_k = R^{1/m} e^{i\beta/m + 2\pi i k/m}, \quad k \in \{0, \dots, m-1\}.$$

This is also true for the point 0 if we count its only preimage 0 with multiplicity m . So every point has exactly m preimages counting multiplicities and π_m is “ m -to-1”.

As the next theorem shows, a similar statement is true locally for every non-constant holomorphic function.

Theorem 13.3. Let $\Omega \subseteq \mathbb{C}$ be open, $f \in H(\Omega)$ be non-constant near $z_0 \in \Omega$, and $w_0 = f(z_0)$. Let $m \in \mathbb{N}$ be the order of the zero of the function $g := f - w_0$ at z_0 . Then there exists an open neighborhood $V \subseteq \Omega$ of z_0 , and $\varphi \in H(V)$ such that

- (i) $g(z) = f(z) - w_0 = \varphi(z)^m$ for $z \in V$,
- (ii) φ is a bijective map from V onto a disk $B(0, r)$ with $r > 0$, $\varphi(z_0) = 0$, and $\varphi'(z) \neq 0$ for $z \in V$.

If we define $U = B(w_0, r^m)$, $\psi(w) = w - w_0$, and let π_m be the m -th power function as in Example 13.2, then we get a commutative diagram

$$\begin{array}{ccc} V & \xrightarrow{f} & U \\ \varphi \downarrow & & \downarrow \psi \\ B(0, r) & \xrightarrow{\pi_m} & B(0, r^m). \end{array}$$

Here the maps φ and ψ can be considered as bijective holomorphic “changes of coordinates”. So near z_0 the map f behaves in the same way as π_m and is locally m -to-1.

Proof. Without loss of generality, Ω is a convex open neighborhood of z_0 such that $f(z) - w_0 \neq 0$ for $z \in \Omega \setminus \{z_0\}$ (this is based on Theorem 11.3). Then

$$f(z) - w_0 = (z - z_0)^m \cdot h(z) \quad \text{for } z \in \Omega,$$

where $h \in H(\Omega)$ and $h(z) \neq 0$ for all $z \in \Omega$ (see Theorem 11.3).

We have $h'/h \in H(\Omega)$, and so h'/h has a primitive $k \in H(\Omega)$ by Corollary 8.4 which means $k' = h'/h$.

Then

$$(he^{-k})' = h'e^{-k} - hk'e^{-k} = h'e^{-k} - h'e^{-k} = 0,$$

and so he^{-k} is a constant function. Then $he^{-k} \equiv h(z_0)e^{-k(z_0)} \neq 0$. There exists $c \in \mathbb{C}$ such that $e^c = h(z_0)e^{-k(z_0)} = he^{-k}$. Define $l := k + c$. Then $l \in H(\Omega)$ and

$$e^l = e^k e^c = e^k he^{-k} = h.$$

Define $\varphi(z) := (z - z_0)e^{l(z)/m}$ for $z \in \Omega$. Then $\varphi \in H(\Omega)$ and $g = f - w_0 = \varphi^m$.

Moreover, $\varphi(z_0) = 0$ and $\varphi'(z_0) = e^{l(z_0)/m} \neq 0$. By Theorem 13.1 we can choose an open neighborhood $\tilde{V}$ of z_0 such that $\varphi|_{\tilde{V}}$ is a bijection onto some open neighborhood W of $\varphi(z_0) = 0$ and such that $\varphi'(z) \neq 0$ for $z \in \tilde{V}$. Now pick $r > 0$ such that $B(0, r) \subseteq W$ and let $V := \varphi^{-1}(B(0, r))$. Then V and φ (restricted to V) have the desired properties. $\square$

Remark 13.4. The number m in Theorem 13.3 is called the *local degree* of f at z_0 , written $m = \deg_f(z_0)$. Note that

$$f'(z_0) = m\varphi'(z_0) \cdot \varphi(z_0)^{m-1} \begin{cases} \neq 0 & \text{for } m = 1, \\ = 0 & \text{for } m \geq 2. \end{cases}$$

So $m = \deg_f(z_0) \geq 2$ if and only if $f'(z_0) = 0$.

The zeros of f' are called the *critical points* of f . Each image point of a critical point is called a *critical value* of f . The critical points of f are precisely the points where the local degree is greater than 1. If f is a non-constant holomorphic function on a region Ω , then the set of critical points of f consists of isolated points and has no limit point in Ω as follows from the Uniqueness Theorem.

Corollary 13.5 (Open Mapping Theorem). *Let $\Omega \subseteq \mathbb{C}$ be a region, and $f \in H(\Omega)$ be non-constant. Then f is an open map, i.e., the image $f(U)$ of every open set $U \subseteq \Omega$ is open. Moreover, the image $f(U)$ of every region $U \subseteq \Omega$ is a region.*

Proof. The first part follows from Theorem 13.3. So if $U \subseteq \Omega$ is a region, then $f(U)$ is open. Moreover, $f(U)$ is also connected as the continuous image of a connected set. Hence $f(U)$ is a region. $\square$

Corollary 13.6. *Let $\Omega \subseteq \mathbb{C}$ be a region, and $f: \Omega \rightarrow \mathbb{C}$ be a holomorphic function that is a bijection onto its image $\tilde{\Omega} := f(\Omega)$ (i.e., $f: \Omega \rightarrow \tilde{\Omega}$ is a bijection). Then $f'(z) \neq 0$ for all $z \in \Omega$, the inverse map $g := f^{-1}: \tilde{\Omega} \rightarrow \Omega$ is holomorphic on the open set $\tilde{\Omega}$, and we have*

$$g'(w) = \frac{1}{f'(g(w))} \quad \text{for all } w \in \tilde{\Omega}. \quad (13.5)$$

Proof. Since f is a bijection, the local degree of f must be equal to 1 at each point of Ω . Hence $f'(z) \neq 0$ for all $z \in \Omega$ by Remark 13.4. By Corollary 13.5 the set $\tilde{\Omega}$ is open, and $g = f^{-1}$ is holomorphic on $\tilde{\Omega}$ by Theorem 13.1 (iii).

Since $f(g(w)) = w$ for all $w \in \tilde{\Omega}$, we conclude from the chain rule that

$$f'(g(w)) \cdot g'(w) = 1$$

for $w \in \tilde{\Omega}$. So (13.5) follows. □

13.1 Exercises

Problem 13.1. Let $\Omega \subseteq \mathbb{C}$ be open, and $f \in H(\Omega)$. Show that for every point $z_0 \in \Omega$ there exist distinct points $z_1, z_2 \in \Omega$ such that

$$f'(z_0) = \frac{f(z_1) - f(z_2)}{z_1 - z_2}.$$

Problem 13.2. Suppose that the function f is holomorphic on $\mathbb{D}$ and injective on $\mathbb{D} \setminus \{0\}$. Show that f is injective on $\mathbb{D}$.

14 Elementary functions

14.1. Trigonometric functions. How can we define $\sin z$ and $\cos z$ for $z \in \mathbb{C}$ so that

- (i) the definitions agree with the usual definition for $z \in \mathbb{R}$,
- (ii) $\sin$ and $\cos$ are holomorphic on $\mathbb{C}$?

By the Uniqueness Theorem this is possible in at most one way, and by Corollary 9.8 the functions $\sin$ and $\cos$ should have power series representations converging on $\mathbb{C}$.

So we define

$$\sin z = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n+1}}{(2n+1)!} \quad \text{and} \quad \cos z = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n}}{(2n)!}$$

for $z \in \mathbb{C}$. These series converge for all $z \in \mathbb{C}$ by the ratio test; so by Theorem 7.7 the functions $\sin$ and $\cos$ are holomorphic on $\mathbb{C}$. Moreover, with these definitions we get the usual functions for $z \in \mathbb{R}$ by the well-known power series representations for $\sin$ and $\cos$ on $\mathbb{R}$.

All the standard trigonometric identities are also valid for complex arguments. For example, we have

$$\sin^2 z + \cos^2 z = 1 \quad \text{for } z \in \mathbb{C}.$$

Indeed, the functions on both sides of this equation are holomorphic functions of z on $\mathbb{C}$. They are the same for $z \in \mathbb{R}$. So by the Uniqueness Theorem they are the same for *all* $z \in \mathbb{C}$.

Similarly, we have

$$\sin(z + w) = \sin z \cos w + \cos z \sin w \tag{14.1}$$

for $z, w \in \mathbb{C}$.

Indeed, we know that (14.1) holds for $z, w \in \mathbb{R}$. Fix $w \in \mathbb{R}$. Then

$$z \mapsto \sin(z + w) \quad \text{and} \quad z \mapsto \sin z \cos w + \cos z \sin w$$

are holomorphic functions on $\mathbb{C}$ that are identical on $\mathbb{R}$. So they have the same values for all $z \in \mathbb{C}$. We conclude that (14.1) holds for all $z \in \mathbb{C}$, $w \in \mathbb{R}$.

Now fix $z \in \mathbb{C}$ and consider the right-hand and the left-hand side of (14.1) as holomorphic functions of w on $\mathbb{C}$. They agree for $w \in \mathbb{R}$, and so they agree for all $w \in \mathbb{C}$ by the uniqueness theorem. So (14.1) holds for all $z, w \in \mathbb{C}$.

We have

$$\sin z = \frac{e^{iz} - e^{-iz}}{2i} \quad \text{and} \quad \cos z = \frac{e^{iz} + e^{-iz}}{2} \quad \text{for } z \in \mathbb{C}. \quad (14.2)$$

Again we note that both sides in these equations are holomorphic functions of z on $\mathbb{C}$, and that the functions agree for $z \in \mathbb{R}$. For example, for $z = t \in \mathbb{R}$ we have

$$\frac{e^{it} - e^{-it}}{2i} = \frac{(\cos t + i \sin t) - (\cos t - i \sin t)}{2i} = \sin t.$$

So the equations (14.2) are valid for all $z \in \mathbb{C}$.

An immediate consequence of (14.2) is the identity

$$e^{iz} = \cos z + i \sin z \quad \text{for } z \in \mathbb{C} \quad (\text{Euler's formula}).$$

What are the zeros of the functions $\sin$ and $\cos$? Note that

$$\begin{aligned} \sin z = 0 &\Leftrightarrow e^{iz} - e^{-iz} = 0 \\ &\Leftrightarrow e^{2iz} = 1 \\ &\Leftrightarrow 2iz = 2\pi i k \text{ for some } k \in \mathbb{Z} \\ &\Leftrightarrow z = \pi k \text{ for some } k \in \mathbb{Z}. \end{aligned}$$

Since $\cos z = \sin(\frac{\pi}{2} - z)$ for $z \in \mathbb{C}$, we have

$$\cos z = 0 \Leftrightarrow z = \frac{\pi}{2} + k\pi \text{ for some } k \in \mathbb{Z}.$$

We can compute $\sin z$ and $\cos z$ for $z = x + iy$, $x, y \in \mathbb{R}$, by using the addition formulas. First note that

$$\cos(iy) = \frac{e^{-y} + e^y}{2} = \cosh y, \quad \sin(iy) = \frac{e^{-y} - e^y}{2i} = i \sinh y.$$

Hence

$$\begin{aligned} \cos(x + iy) &= \cos x \cos(iy) - \sin x \sin(iy) \\ &= \cos x \cosh y - i \sin x \sinh y, \end{aligned}$$

and similarly

$$\sin(x + iy) = \sin x \cosh y + i \cos x \sinh y.$$

The functions $\tan$ and $\cot$ are defined in the usual way. So

$$\tan z := \frac{\sin z}{\cos z}$$

is a holomorphic function of z on $\mathbb{C} \setminus \{k\pi + \pi/2 : k \in \mathbb{Z}\}$ and

$$\cot z := \frac{\cos z}{\sin z}$$

is a holomorphic function of z on $\mathbb{C} \setminus \{k\pi : k \in \mathbb{Z}\}$.

14.2. Logarithms. Every solution of the equation $z = e^w$ is called a *logarithm* of z . A number $z \in \mathbb{C}$ has a logarithm if and only if $z \neq 0$. Let $z = re^{i\alpha}$, $r > 0$, $\alpha \in \mathbb{R}$. Then

$$z = e^{\log r + i\alpha} = e^w$$

iff $w = \log r + i\alpha + 2\pi ik$, $k \in \mathbb{Z}$. So every $z \in \mathbb{C} \setminus \{0\}$ has infinitely many logarithms.

A continuous function $L: \Omega \rightarrow \mathbb{C}$ defined on a region $\Omega \subseteq \mathbb{C}$ is called a *branch of the logarithm* if $e^{L(z)} = z$ for all $z \in \Omega$. Note that $0 \notin \Omega$ if a branch of the logarithm exists, but this condition is not sufficient for the existence of the branch of the logarithm on a region Ω . For example, if $\Omega = \mathbb{C} \setminus \{0\}$, then no branch of the logarithm exists on Ω .

If $L_1, L_2: \Omega \rightarrow \mathbb{C}$ are two branches of the logarithm, then $e^{L_1(z)} = e^{L_2(z)}$, and so $e^{L_1(z) - L_2(z)} = 1$ for all $z \in \Omega$. It follows that

$$(L_1 - L_2)(\Omega) \subseteq \{2\pi ik : k \in \mathbb{Z}\}.$$

Since $(L_1 - L_2)(\Omega)$ is a connected set, we conclude that there exists $k \in \mathbb{Z}$ such that

$$L_1(z) = L_2(z) + 2\pi ik$$

for all $z \in \Omega$. So the branches of the logarithm in a region are obtained from one branch (if it exists) by adding $2\pi ik$, $k \in \mathbb{Z}$. In particular, knowing the function value of a branch at one point determines the branch uniquely.

The exponential function maps the strip

$$S := \{w \in \mathbb{C} : -\pi < \operatorname{Im}(w) < \pi\}$$

bijectionally onto the slit plane $\mathbb{C} \setminus (-\infty, 0]$. By Corollary 13.6 the function $\exp|_S$ has a holomorphic inverse function denoted by $\log$ and called the *principal branch of the logarithm*. It is the unique branch of the logarithm on $\Omega = \mathbb{C} \setminus (-\infty, 0]$ with $\log 1 = 0$.

By Corollary 13.6 we have

$$\frac{d}{dz} \log z = \frac{1}{e^{\log z}} = \frac{1}{z}.$$

The expression $\log z$ agrees with the usual definition for $z > 0$. Note that $\log$ is discontinuous along the “branch cut” $(-\infty, 0]$. Indeed, if for fixed $r > 0$ the points $z_1 = re^{i\alpha_1}$ and $z_2 = re^{i\alpha_2}$ with $\alpha_1, \alpha_2 \in (-\pi, \pi)$ approach $-r$ from the upper and lower half-planes, respectively, then $\alpha_1 \rightarrow \pi$ and $\alpha_2 \rightarrow -\pi$. So

$$\log(z_1) = \log r + i\alpha_1 \rightarrow \log r + i\pi$$

and

$$\log(z_2) = \log r + i\alpha_2 \rightarrow \log r - i\pi.$$

This shows that the values of the principal branch of the logarithm differ by $2\pi i$ on “different sides” of the branch cut.

Note that in general we have

$$\log(z_1 z_2) \neq \log(z_1) + \log(z_2)$$

if $\log$ is the principal branch of the logarithm. For example, if $z_1 = z_2 = i \in \mathbb{C} \setminus (-\infty, 0]$, then $z_1 z_2 = i^2 = -1$ is not even contained in the domain of definition $\mathbb{C} \setminus (-\infty, 0]$ of the principal branch.

14.3. Roots and powers. If $\Omega \subseteq \mathbb{C}$ is a region where a branch L of the logarithm exists, we define for $a \in \mathbb{C}$ and $z \in \Omega$,

$$z^a := \exp(aL(z)).$$

Note that if $\tilde{L}$ is another branch of the logarithm, then $\tilde{L} = L + 2\pi i k$, $k \in \mathbb{Z}$, and so

$$\exp(a\tilde{L}(z)) = \exp(aL(z) + a2\pi i k) = \exp(aL(z)) \exp(a2\pi i k)$$

for $z \in \Omega$. So different branches of the power function z^a differ by a multiplicative factor. If $a \in \mathbb{Z}$, then $\exp(a2\pi i k) = 1$, and so there exists only one branch. So for $a = n \in \mathbb{Z}$, $z \neq 0$,

$$z^a = \exp(aL(z)) = (\exp(L(z)))^a = z^n$$

agrees with the power function as defined in Section 1.

Unless otherwise stated, z^a will denote the *principal branch* defined on $\mathbb{C} \setminus (-\infty, 0]$ as

$$z^a := \exp(a \log(z)),$$

where $\log$ is the principal branch of the logarithm. This branch is normalized so that

$$1^a = \exp(a \log(1)) = \exp(0) = 1,$$

and agrees with the usual definition of the power function for $z > 0$, $a \in \mathbb{R}$.

Note that

$$\begin{aligned} \frac{d}{dz} z^a &= \frac{d}{dz} \exp(a \log(z)) = a \frac{1}{z} \exp(a \log(z)) \\ &= a \exp(-\log(z)) \exp(a \log(z)) = a \exp((a-1) \log(z)) = az^{a-1} \end{aligned}$$

for $z \in \mathbb{C} \setminus (-\infty, 0]$.

If $z = re^{i\alpha}$, $r > 0$, $\alpha \in (-\pi, \pi)$, we have

$$z^a = \exp(a \log(re^{i\alpha})) = \exp(a(\log r + i\alpha)).$$

For $a > 0$ this means that

$$z^a = \exp(a(\log r + i\alpha)) = r^a e^{ia\alpha}.$$

In particular, a sector

$$S = \{re^{i\alpha} : R > 0, \varphi_1 < \alpha < \varphi_2\},$$

where $-\pi \leq \varphi_1 < \varphi_2 \leq \pi$ and whose opening angle $\varphi_2 - \varphi_1$ is sufficiently small, is mapped by $z \mapsto z^a$ onto the sector

$$S' = \{\rho e^{i\beta} : \rho > 0, a\varphi_1 < \beta < a\varphi_2\}$$

whose opening angle differs from the opening angle of S by the factor a .

If $a = 1/n$, where $n \in \mathbb{N}$, we also use the notation

$$\sqrt[n]{z} := z^{1/n} = z^a.$$

For the principal branch we get

$$\sqrt[n]{re^{i\alpha}} = r^{1/n} e^{i\alpha/n}$$

for $z = re^{i\alpha}$, $r > 0$, $\alpha \in (-\pi, \pi)$.

Note that if $r > 0$ is fixed and $z_1 = re^{i\alpha}$ approaches the point $-r$ on the branch cut $(-\infty, 0]$ from the upper half-plane, then $\alpha \rightarrow \pi$ and so $\sqrt[n]{z_1} = \sqrt[n]{re^{i\alpha}} \rightarrow r^{1/n}e^{i\pi n}$. Similarly, if $z_2 = re^{i\alpha}$ approaches the point $-r$ on the branch cut from the lower half-plane, then $\alpha \rightarrow -\pi$ and so $\sqrt[n]{z_2} = \sqrt[n]{re^{i\alpha}} \rightarrow r^{1/n}e^{-i\pi n}$.

In particular, if $n = 2$, then the values of $\sqrt{z}$ for $z = -r$ on “different sides of the branch cut” are $\sqrt{r}i$ and $-\sqrt{r}i$. So these values differ by a factor -1 .

14.4. Riemann surfaces spread over the plane. How can we get rid of the artificial branch cut of the logarithm? We cannot define the logarithm as a “sensible” (for example, continuous) function on $\mathbb{C} \setminus \{0\}$. Indeed, each $z = re^{i\alpha} \in \mathbb{C} \setminus \{0\}$ as infinitely many logarithms

$$\log r + i\alpha + 2\pi ik, \quad k \in \mathbb{Z}, \quad (14.3)$$

and these values cannot be patched up to a continuous function on $\mathbb{C} \setminus \{0\}$. So the logarithm is a “multi-valued” function. To get a well-defined function in the usual sense, one introduces a surface with infinitely many layers (roughly speaking, one layer for each $k \in \mathbb{Z}$ in (14.3)) so that on this surface the logarithm is a well-defined and continuous function.

One way to do this is as follows. Let $S = \{(z, w) \in \mathbb{C}^2 : z = \exp(w)\}$. Then we may think of S as a “covering surface” of $\mathbb{C} \setminus \{0\}$ where each point $(z, w) \in S$ lies “above” the point z . The logarithm is then defined as the map $(z, w) \in S \mapsto w$.

Another way to visualize this *Riemann surface spread over $\mathbb{C} \setminus \{0\}$* is by the following gluing procedure. For $k \in \mathbb{Z}$ let

$$P_k = \{re^{i\alpha} : r > 0, 2\pi i(k-1) \leq \alpha \leq 2\pi i(k+1)\}$$

be a copy of punctured complex plane, where the points in P_k with $\alpha = 2\pi i(k \pm 1)$ on different sides of the branch cut $(-\infty, 0]$ are considered as distinct. Then the Riemann surface S of the logarithm is obtained by gluing together (i.e., identifying) each point $re^{i2\pi i(k+1)}$, $r > 0$, on the upper side of the branch cut of P_k with the same point on lower side of the branch cut of P_{k+1} . On a more intuitive level, this surface has the shape of a “bi-infinite spiral staircase” lying above $\mathbb{C} \setminus \{0\}$.

The logarithm on this surface S is defined as follows. If $z \in S$, then $z \in re^{i\alpha} \in P_k$ for some $k \in \mathbb{Z}$, where $2\pi i(k-1) \leq \alpha \leq 2\pi i(k+1)$. Then we set $\log(z) := \log r + i\alpha$. This is a well-defined function on S .

14.5. The Riemann surface of $\sqrt{z}$. The square-root function has two branches on $\mathbb{C} \setminus (-\infty, 0]$ that differ by the factor ± 1 . To get a well-defined square-root function we define a Riemann surface $S = \{(z, w) \in \mathbb{C}^2 : z = w^2\}$. Then by the projection map $(z, w) \in S \mapsto z$, this surface is spread over the complex plane $\mathbb{C}$. On S the square-root function is given by the map $(z, w) \in S \mapsto w$.

Similarly as for the logarithm, one can obtain S from a gluing procedure also as follows. We take two copies P_+ and P_- of the complex plane with a slit along the negative real axis $(-\infty, 0]$ as branch cut. In each copy we consider the points $re^{\pm i\pi}$, $r > 0$, on different sides of the branch cut as distinct. Then we glue P_+ and P_- together “cross-wise” so that the point $re^{+i\pi} \in P_+$ is identified with $re^{-i\pi} \in P_-$, and $re^{-i\pi} \in P_+$ is identified with $re^{+i\pi} \in P_-$. We also identify $0 \in P_+$ with $0 \in P_-$. Each point $z \in S$ can be considered as lying above its corresponding point $z \in P_+$ or $z \in P_-$ considered as a point in the complex plane. In this way, S is a Riemann surface spread over $\mathbb{C}$ with a natural projection map $S \rightarrow \mathbb{C}$ that is locally injective away from the so-called *branch point* $0 \in S$.

The surface S cannot be realized as an embedded surface in $\mathbb{R}^3$, but if one wants a model of S in $\mathbb{R}^3$ one has to allow a “self-intersection” of S along the branch cut.

If $z \in S$ is arbitrary, then the point z lies in one of the copies P_+ or P_- (or both). If $z \in P_+$ or $z \in P_-$ and $z = re^{i\alpha}$, where $r \geq 0$ and $-\pi \leq \alpha \leq \pi$, we define $\sqrt{z} = \sqrt{r}e^{i\alpha/2}$ or $\sqrt{z} = -\sqrt{r}e^{i\alpha/2}$, respectively. This gives a well-defined square-root function on S .

14.1 Exercises

Problem 14.1. The purpose of this problem is to outline a purely analytic theory of the functions $\sin$ and $\cos$ that does not rely on any previous results about trigonometric functions from real analysis.

For this we define

$$S(z) = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n+1}}{(2n+1)!} \quad \text{and} \quad C(z) = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n}}{(2n)!}$$

for $z \in \mathbb{C}$. Then S and C are entire functions (why?).

- (a) Show that $S(-z) = -S(z)$ and $C(-z) = C(z)$ for $z \in \mathbb{C}$.
- (b) Show that $S'(z) = C(z)$ and $C'(z) = -S(z)$ for $z \in \mathbb{C}$.
- (c) Show that

$$\begin{aligned} S(z+w) &= S(z)C(w) + C(z)S(w), \\ C(z+w) &= C(z)C(w) - S(z)S(w) \end{aligned}$$

for $z, w \in \mathbb{C}$. Hint: Use Problem 9.2.

- (d) Show that $S(z)^2 + C(z)^2 = 1$ for $z \in \mathbb{C}$.
- (e) Show that on the interval $[0, 2]$ the function C takes real values and is strictly decreasing. Hint: Investigate the sign of $S(x)$ for $x \in (0, 2]$.
- (f) Show that there exists a unique number $\omega \in (0, 2)$ such that $C(\omega) = 0$.
- (g) Show that for the number ω in (f) we have $S(\omega) = 1$,

$$\begin{aligned} S(2\omega) &= 0, & C(2\omega) &= -1, \\ S(4\omega) &= 0, & C(4\omega) &= 1. \end{aligned}$$

- (h) Show that $S(z + 4\omega) = S(z)$ and $C(z + 4\omega) = C(z)$ for $z \in \mathbb{C}$.
- (i) Show that for each $u \in \mathbb{C}$ with $|u| = 1$, $\operatorname{Re}(u) \geq 0$, and $\operatorname{Im}(u) \geq 0$ there exists precisely one value $t \in [0, \omega]$ with $C(t) + iS(t) = u$.
- (j) We define a path $\gamma: [0, \omega] \rightarrow \mathbb{C}$ as $\gamma(t) = C(t) + iS(t)$ for $t \in [0, \omega]$. Show that for the length $\ell(\gamma)$ of γ we have $\ell(\gamma) = \omega$.

Note that we can use these results to give an analytic definition of π : by (j) the length of a quarter of the unit circle is equal to ω . So if we define π to be the length of a semicircle of radius 1, then $\pi = 2\omega$.

Problem 14.2. (a) Consider the function $f(z) := \frac{z}{e^z - 1}$ that is holomorphic in $z \neq 0$ near 0. Show that if we set $f(0) := 1$, then f is holomorphic in a neighborhood of 0.

(b) By (a) the function f can be expanded in a power series centered at the point 0. What is the radius of convergence of this power series?

(c) If we write the power series expansion of f near 0 in the form

$$f(z) = \sum_{n=0}^{\infty} \frac{B_n}{n!} z^n,$$

then the number B_n is called the n th *Bernoulli number*. Show that $B_0 = 1$, and that

$$B_n = -\frac{1}{n+1} \sum_{k=0}^{n-1} \binom{n+1}{k} B_k \quad \text{for } n \in \mathbb{N}.$$

(d) Show that $B_{2n+1} = 0$ for $n \in \mathbb{N}$.

(e) Show that

$$z \cot z = \sum_{n=0}^{\infty} (-1)^n \frac{2^{2n} B_{2n}}{(2n)!} z^{2n} \quad \text{for } 0 < |z| < \pi.$$

(f) Show that

$$\tan z = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{2^{2n} (2^{2n} - 1) B_{2n}}{(2n)!} z^{2n-1} \quad \text{for } |z| < \frac{\pi}{2}.$$

Problem 14.3. Let $\alpha \in \mathbb{C}$. We define the generalized binomial coefficient as $\binom{\alpha}{0} = 1$ and as

$$\binom{\alpha}{n} = \frac{\alpha(\alpha-1) \cdots (\alpha-n+1)}{n!}$$

for $n \in \mathbb{N}$.

Show that then

$$(1+z)^\alpha = \sum_{n=0}^{\infty} \binom{\alpha}{n} z^n$$

for $z \in \mathbb{D}$, where $(1+z)^\alpha = \exp(\alpha \log(z))$ with the principal branch of the logarithm.

Problem 14.4. We consider the path $\gamma: [0, 1] \rightarrow \mathbb{C}$ defined as $\gamma(t) = e^{2\pi it}$ for $t \in [0, 1]$ and the germ (see Problem 12.4)

$$Q_0(z) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} (z-1)^n$$

at $\gamma(0) = 1$. Show that Q_0 has an analytic continuation along the path γ to a (unique) germ Q_1 at $\gamma(1) = 1$. Determine Q_1 explicitly.

Problem 14.5. By the solution of Problem 7.10 (c) the zeros of the Bessel function J_0 (see Problem 7.8) are important. The purpose of this problem is to show that each of them lies on the real axis.

(a) Suppose $a \in \mathbb{C}$ and $J_0(a) = 0$. Show that then the function $f(z) := J_0(az)$ satisfies

$$(zf'(z))' = -a^2zf(z)$$

for $z \in \mathbb{C}$.

(b) Suppose $b \in \mathbb{C}$ also satisfies $J_0(b) = 0$ and define $g(z) := J_0(bz)$ for $z \in \mathbb{C}$. Find a holomorphic function H expressed in terms of f and g such that

$$H'(z) = (a^2 - b^2)zf(z)g(z)$$

for $z \in \mathbb{C}$.

(c) Show that if $a, b \in \mathbb{C}$ and $J_0(a) = J_0(b) = 0$, then

$$(a^2 - b^2) \int_0^1 tJ_0(at)J_0(bt) dt = 0.$$

(d) Show that if $J_0(a) = 0$, then $a \in \mathbb{R}$. Hint: Use (c) to first show that $a^2 = \bar{a}^2$.

Problem 14.6. The purpose of this problem is to show that J_0 has infinitely many zeros on the positive real axis.

For this we consider the function

$$u(z) = \sqrt{z}J_0(z)$$

defined for $z \in \mathbb{C} \setminus (-\infty, 0]$, where $\sqrt{z}$ denotes the principal branch of the square root function.

(a) Show that

$$u''(z) + u(z) = -\frac{1}{4z^2}u(z)$$

for $z \in \mathbb{C} \setminus (-\infty, 0]$.

(b) Find a primitive of the function $f(z) = \sin(z)(u''(z) + u(z))$ expressed in term of u , u' , and trigonometric functions.

(c) Show that for each $n \in \mathbb{N}$ we have

$$\int_{2n\pi}^{(2n+1)\pi} \frac{u(t) \sin(t)}{4t^2} dt = -[u(2n\pi) + u((2n+1)\pi)].$$

(d) Show that for each $n \in \mathbb{N}$ the function J_0 has at least one zero in the interval $(2n\pi, (2n+1)\pi)$.

Problem 14.7. The purpose of this problem is to establish some basic properties of the Γ -function.

(a) Let $H := \{z \in \mathbb{C} : \operatorname{Re}(z) > 0\}$ be the open right half-plane. Show that the improper integral

$$\Gamma(z) := \int_0^\infty t^{z-1} e^{-t} dt$$

converges for each $z \in H$. Here $t^{z-1} = e^{(z-1)\log t}$ for $t > 0$. Hint: You have to show the existence of

$$\lim_{\delta \rightarrow 0^+} \int_\delta^1 t^{z-1} e^{-t} dt \quad \text{and} \quad \lim_{R \rightarrow +\infty} \int_1^R t^{z-1} e^{-t} dt.$$

(b) For $n \in \mathbb{N}$ and $z \in H$ define

$$\Gamma_n(z) = \int_{1/n}^n t^{z-1} e^{-t} dt.$$

Show that Γ_n is holomorphic on H and that $\Gamma_n \rightarrow \Gamma$ uniformly on each compact subset of H .

(c) Show that Γ is a holomorphic function on H and that it satisfies the functional equation $\Gamma(z+1) = z\Gamma(z)$ for $z \in H$.

(d) Show that $\Gamma(1/2) = \sqrt{\pi}$ and $\Gamma(n) = (n-1)!$ for $n \in \mathbb{N}$.

(e) Show that Γ has a unique extension to a holomorphic function on $U := \mathbb{C} \setminus \{-n : n \in \mathbb{N}_0\}$. Why does the functional equation $\Gamma(z+1) = z\Gamma(z)$ remain valid for all $z \in U$?

Problem 14.8 (Logarithmic derivative). Suppose f is a holomorphic function defined locally near a point $z_0 \in \mathbb{C}$ (that is, in a (small) region containing z_0). Suppose $w_0 := f(z_0) \neq 0$. Then the *logarithmic derivative* $\mathcal{L}(f)$ of f is defined as

$$\mathcal{L}(f)(z) = \frac{f'(z)}{f(z)}$$

for z near z_0 .

(a) Let L be any branch of the logarithm defined near $w_0 = f(z_0) \neq 0$. Show that then

$$\mathcal{L}(f)(z) = \frac{d}{dz} L(f(z))$$

for z near z_0 .

This shows that while $\log(f(z))$ is not well-defined, because it depends on the branch of the logarithm, the logarithmic derivative

$$\frac{d}{dz} \log(f(z)) = \frac{f'(z)}{f(z)} \quad (14.4)$$

is well-defined locally near z_0 , since it is independent of the choice of branch. For this reason, we will often use the notation $\frac{d}{dz} \log(f(z))$ for the logarithmic derivative of f .

(b) If g is another function holomorphic near z_0 with $g(z_0) \neq 0$, then

$$\mathcal{L}(fg) = \mathcal{L}(f) + \mathcal{L}(g)$$

near z_0 . Justify this by considering branches of $\log(fg)$, $\log(f)$, and $\log(g)$ (instead of using direct computation).

(c) Suppose h is a holomorphic function defined near $u_0 \in \mathbb{C}$ such that $z_0 = h(u_0)$. Then we have the chain rule

$$\mathcal{L}(f \circ h) = (\mathcal{L}(f) \circ h)h'$$

locally near u_0 . Justify this by considering a branch of $\log(f \circ h)$.

Problem 14.9 (Schwarzian derivative). Let f be a holomorphic function near $z_0 \in \mathbb{C}$ such that $f'(z_0) \neq 0$. Then its *Schwarzian derivative* $\mathcal{S}(f)$ is defined as

$$\mathcal{S}(f)(z) := \frac{d^2}{dz^2} \log(f'(z)) - \frac{1}{2} \left(\frac{d}{dz} \log(f'(z)) \right)^2 \quad (14.5)$$

for z near z_0 . Here we use the notation established in (14.4).

(a) Find an explicit expression of $\mathcal{S}(f)$ in terms of f' , f'' , and f''' .

(b) Suppose h is a holomorphic function defined near $u_0 \in \mathbb{C}$ such that $z_0 = h(u_0)$ and $h'(u_0) \neq 0$. Show that then the chain rule

$$\mathcal{S}(f \circ h) = (\mathcal{S}(f) \circ h)(h')^2 + \mathcal{S}(h) \quad (14.6)$$

holds locally near u_0 .

15 The Riemann sphere

15.1. The extended complex plane. We want to compactify $\mathbb{C}$ by adding a point at infinity denoted by ∞ . Then $\widehat{\mathbb{C}} := \mathbb{C} \cup \{\infty\}$ is called the *extended complex plane*. One can visualize $\widehat{\mathbb{C}}$ as the *Riemann sphere* by *stereographic projection*.

To do this, let $S = \{(u, v, w) \in \mathbb{R}^3 : u^2 + v^2 + w^2 = 1\}$ be the unit sphere in $\mathbb{R}^3$. We consider the complex plane as a subset of $\mathbb{R}^3$ by the identification $z = x + iy \in \mathbb{C} \cong (x, y, 0) \in \mathbb{R}^3$, where $x, y \in \mathbb{R}$. Then $\mathbb{C}$ cut the sphere S along its equator. Let $N = (0, 0, 1)$ be the “north pole” of S . Then one defines a projection map $\pi: S \setminus \{N\} \rightarrow \mathbb{C}$ as follows. If $Z \in S \setminus \{N\}$ is arbitrary, then we join N and Z by a line L . The line L intersects the complex plane $\mathbb{C} \subseteq \mathbb{R}^3$ in a unique point z . We define $\pi(Z) := z$. This *stereographic projection* π gives a bijection of $S \setminus \{N\}$ onto $\mathbb{C}$.

It is easy to find an explicit formula for the map π and its inverse. If $Z = (x_1, x_2, x_3) \in S \setminus \{N\}$ is arbitrary, then the line L passing through Z and N is given by

$$L = \{t(x_1, x_2, x_3) + (1 - t)(0, 0, 1) \in \mathbb{R}^3 : t \in \mathbb{R}\}.$$

The line L meets $\mathbb{C}$ at a point z determined by the equation

$$tx_3 + (1 - t) = 0.$$

Hence $t = 1/(1 - x_3)$. Inserting this back into the parametrization of L gives

$$Z = (x_1, x_2, x_3) \in S \setminus \{N\} \mapsto \pi(Z) = z = \frac{x_1 + ix_2}{1 - x_3} \in \mathbb{C}.$$

Conversely, let $z = x + iy \in \mathbb{C}$ be arbitrary, where $x, y \in \mathbb{R}$. The line L' passing through $N = (0, 0, 1)$ and $(x, y, 0) \cong z$ is given by

$$L' = \{t(x, y, 0) + (1 - t)(0, 0, 1) \in \mathbb{R}^3 : t \in \mathbb{R}\}.$$

A point on L' different from N lies on S iff

$$\begin{aligned} t^2x^2 + t^2y^2 + (1 - t)^2 &= t^2|z|^2 + (1 - t)^2 = 1 \\ \Rightarrow t^2(|z|^2 + 1) &= 2t \\ \Rightarrow t &= \frac{2}{|z|^2 + 1} \quad (t = 0 \text{ corresponds to } N). \end{aligned}$$

Hence

$$z = x + iy \in \mathbb{C} \mapsto \pi^{-1}(z) = \left(\frac{2x}{|z|^2 + 1}, \frac{2y}{|z|^2 + 1}, \frac{|z|^2 - 1}{|z|^2 + 1} \right) \in S \setminus \{N\}.$$

If we make $|z|$ larger and larger, then the corresponding point $Z = \pi^{-1}(z)$ on S gets closer and closer to N . So it is natural to extend π to a bijection from S onto $\widehat{\mathbb{C}}$ by setting $\pi(N) = \infty$. In this way, we can identify $\widehat{\mathbb{C}}$ with S by stereographic projection and refers to $\widehat{\mathbb{C}}$ as the *Riemann sphere*.

Let $z, w \in \mathbb{C}$ be points in the complex plane, and $Z = \pi^{-1}(z)$ and $W = \pi^{-1}(w)$. Then for the Euclidean distance of Z and W we get

$$\begin{aligned} |Z - W|^2 &= \left| \frac{2z}{|z|^2 + 1} - \frac{2w}{|w|^2 + 1} \right|^2 + \left(\frac{|z|^2 - 1}{|z|^2 + 1} - \frac{|w|^2 - 1}{|w|^2 + 1} \right)^2 \\ &= \frac{4|z(|w|^2 + 1) - w(|z|^2 + 1)|^2 + 4(|z|^2 - |w|^2)^2}{(|z|^2 + 1)^2(|w|^2 + 1)^2} \\ &= -4 \frac{z\bar{w} + w\bar{z}}{(|z|^2 + 1)(|w|^2 + 1)} + 4 \frac{|z|^2(|w|^2 + 1)^2 + |w|^2(|z|^2 + 1)^2 + (|z|^2 - |w|^2)^2}{(|z|^2 + 1)^2(|w|^2 + 1)^2} \\ &= -4 \frac{z\bar{w} + w\bar{z}}{(|z|^2 + 1)(|w|^2 + 1)} + 4 \frac{(|z|^2 + |w|^2)(|z|^2 + 1)(|w|^2 + 1)}{(|z|^2 + 1)^2(|w|^2 + 1)^2} \\ &= 4 \frac{|z|^2 + |w|^2 - z\bar{w} - w\bar{z}}{(|z|^2 + 1)(|w|^2 + 1)} = 4 \frac{|z - w|^2}{(|z|^2 + 1)(|w|^2 + 1)}, \end{aligned}$$

and so

$$|Z - W| = \frac{2|z - w|}{\sqrt{1 + |z|^2} \sqrt{1 + |w|^2}}.$$

Moreover,

$$|Z - N|^2 = \frac{4|z|^2}{(|z|^2 + 1)^2} + \frac{4}{(|z|^2 + 1)^2} = \frac{4}{(|z|^2 + 1)},$$

and so

$$|Z - N| = \frac{2}{\sqrt{1 + |z|^2}}.$$

15.2. The chordal metric. We define a distance function on $\widehat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$ as follows. For $z, w \in \mathbb{C} \subseteq \widehat{\mathbb{C}}$ we define

$$d(z, w) := \frac{2|z - w|}{\sqrt{1 + |z|^2} \sqrt{1 + |w|^2}},$$

$$d(z, \infty) = d(\infty, z) := \frac{2}{\sqrt{1 + |z|^2}},$$

and let $d(\infty, \infty) = 0$.

As we have seen above, d corresponds to the Euclidean metric on S by stereographic projection. This shows that d is a metric on $\widehat{\mathbb{C}}$, called the *chordal metric*. Unless otherwise stated, $\widehat{\mathbb{C}}$ will always carry this metric.

Stereographic projection is an isometry between S equipped with the (restriction of the) Euclidean metric on $\mathbb{R}^3$ and $\widehat{\mathbb{C}}$ equipped with the chordal metric. In particular, S and $\widehat{\mathbb{C}}$ are homeomorphic (i.e., there exists a continuous bijection between these spaces with continuous inverse). So the topological and metric properties of S and $\widehat{\mathbb{C}}$ are the same. In particular, $\widehat{\mathbb{C}}$ is a compact metric space.

Remark 15.3. It is not hard to show that $\mathbb{C}$ equipped with the restriction of the chordal metric and $\mathbb{C}$ equipped with the usual Euclidean metric have the same topology (i.e., the same open sets). So chordal and Euclidean metrics are *equivalent metrics*.

In particular, if $\{z_n\}$ is a sequence in $\mathbb{C}$ and $z \in \mathbb{C}$, then $z_n \rightarrow z$ with respect to the chordal metric if and only if $z_n \rightarrow z$ with respect to the Euclidean metric.

We have $z_n \rightarrow \infty \in \widehat{\mathbb{C}}$ (convergence in the metric space $\widehat{\mathbb{C}}$) iff

$$d(z_n, \infty) = \frac{2}{\sqrt{1 + |z_n|^2}} \rightarrow 0$$

iff $|z_n| \rightarrow +\infty$ (in the sense of real analysis).

Definition 15.4. Let X be a topological Hausdorff space (e.g., a metric space), and $n \in \mathbb{N}$. Then X is called a (*topological*) *n-manifold* if every point $x \in X$ has an open neighborhood that is homeomorphic to an open subset of $\mathbb{R}^n$.

A *chart* on an n -manifold X is a homeomorphism $\varphi: U \rightarrow V$ of an open subset $U \subseteq X$ onto an open subset $V \subseteq \mathbb{R}^n$. An *atlas* on X is a family $\mathcal{A} = \{\varphi_i: U_i \rightarrow V_i : i \in I\}$ of charts on X such that $X = \bigcup_{i \in I} U_i$, i.e., the chart neighborhoods of charts in the family cover the whole manifold X .

Recall that a topological space X is called *Hausdorff* if for all points $x, y \in X$, $x \neq y$, there exists open sets $U, V \subseteq X$ with $x \in U$, $y \in V$, and $U \cap V = \emptyset$.

Our definition of an n -manifold is slightly non-standard as one usually requires that an n -manifold should have a *countable basis* for its topology (i.e., there exists a countable set $\mathcal{B}$ of open sets in X such that every open set in U is a union of sets in $\mathcal{B}$).

We are mostly interested in the case $n = 2$. A 2-manifold is also called a *surface*. If we make the identification $\mathbb{R}^2 \cong \mathbb{C}$, then we can think of a chart on a surface as a map into $\mathbb{C}$.

Example 15.5. The unit sphere $S \subseteq \mathbb{R}^3$ (and hence also $\widehat{\mathbb{C}}$) is a topological 2-manifold. To see this, it is enough to exhibit an atlas of charts on S . It consists of two charts, namely the map $\varphi_1: S \setminus \{n\} \rightarrow \mathbb{R}^2$ given by stereographic projection from the north pole $n = (0, 0, 1)$, and the map $\varphi_2: S \setminus \{s\} \rightarrow \mathbb{R}^2$ given by stereographic projection from the south pole $s = (0, 0, -1)$ (defined in the obvious way).

Similarly, one can define an atlas on $\widehat{\mathbb{C}}$ given by two charts ψ_1 and ψ_2 . Here $\psi_1: U_1 := \mathbb{C} \subseteq \widehat{\mathbb{C}} \rightarrow \mathbb{C}$, $\psi_1(z) = z$ for $z \in \mathbb{C}$, and $\psi_2: U_2 := \mathbb{C} \setminus \{0\} \cup \{\infty\} \subseteq \widehat{\mathbb{C}} \rightarrow \mathbb{C}$,

$$\psi_2(z) = \begin{cases} 1/z & \text{for } z \in \mathbb{C} \setminus \{0\}, \\ 0 & \text{for } z = \infty. \end{cases}$$

Definition 15.6. Let $U, V \subseteq \mathbb{C}$ be open sets. A map $f: U \rightarrow V$ is called a *biholomorphism* if f is a bijection, f is holomorphic on U , and f^{-1} is holomorphic on V .

Two charts $\varphi_1: U_1 \rightarrow V_1 \subseteq \mathbb{C}$ and $\varphi_2: U_2 \rightarrow V_2$ on a 2-manifold X are called *holomorphically compatible* if $U_1 \cap U_2 = \emptyset$, or if $U_1 \cap U_2 \neq \emptyset$ and the map

$$\varphi_2 \circ \varphi_1^{-1}|_{\varphi_1(U_1 \cap U_2)}: \varphi_1(U_1 \cap U_2) \rightarrow \varphi_2(U_1 \cap U_2)$$

is a biholomorphism.

Example 15.7. Let $\psi_1: U_1 \rightarrow \mathbb{C}$ and $\psi_2: U_2 \rightarrow \mathbb{C}$ be the charts on $\widehat{\mathbb{C}}$ as in Example 15.5. Then ψ_1 and ψ_2 are holomorphically compatible. Indeed, $U_1 = \mathbb{C}$ and $U_2 = \mathbb{C} \setminus \{0\} \cup \{\infty\}$, and so $U_1 \cap U_2 = \mathbb{C} \setminus \{0\}$. Then $W := \psi_1(U_1 \cap U_2) = \psi_2(U_1 \cap U_2) = \mathbb{C} \setminus \{0\}$. Moreover, $f := \psi_2 \circ \psi_1^{-1}|_W: W \rightarrow W$ is given by $f(z) = 1/z$ for $z \in W$. This is a holomorphic bijection with holomorphic inverse $z \mapsto f^{-1}(z) = 1/z$.

Definition 15.8. An atlas $\mathcal{A} = \{\varphi_i: U_i \rightarrow V_i \subseteq \mathbb{C} : i \in I\}$ on a 2-manifold X is called a *complex atlas* if the charts in $\mathcal{A}$ are pairwise holomorphically compatible.

Two complex atlases $\mathcal{A}_1$ and $\mathcal{A}_2$ on X are called *analytically equivalent* if each chart in $\mathcal{A}_1$ is holomorphically compatible with each chart in $\mathcal{A}_2$. This defines an equivalence relation for atlases. A *complex structure* on X is an equivalence class of analytically equivalent atlases on X . A *Riemann surface* is a topological 2-manifold equipped with a complex structure.

Remark 15.9. One should think of the complex structure on a Riemann surface as being represented by some fixed complex atlas. A *complex chart* on a Riemann surface is a chart that is holomorphically compatible with all the charts in one atlas (and hence in all atlases) that define the complex structure. Often it is convenient to add complex charts to a given atlas as needed, for example if one wants to make the chart neighborhoods smaller: if $\varphi: U \rightarrow V$ is a chart in a complex atlas $\mathcal{A}$, and $U' \subseteq U$ is open, then $\varphi|_{U'}: U' \rightarrow \varphi(U')$ is a chart that is holomorphically compatible with each chart in $\mathcal{A}$.

Definition 15.10. Let X be a Riemann surface, and $f: X \rightarrow \mathbb{C}$. Then f is called a *holomorphic function* if for every complex chart $\varphi: U \rightarrow V \subseteq \mathbb{C}$ on X the function $f \circ \varphi^{-1}: V \rightarrow \mathbb{C}$ is a holomorphic function on the open set $V \subseteq \mathbb{C}$.

Definition 15.11. Let X and Y be Riemann surfaces, and $f: X \rightarrow Y$ be continuous. Then f is called a *holomorphic map* (between X and Y) if for every complex chart $\varphi: U \rightarrow V \subseteq \mathbb{C}$ on X and every complex chart $\psi: U' \rightarrow V' \subseteq \mathbb{C}$ on Y with $f(U) \subseteq U'$ the function $\psi \circ f \circ \varphi^{-1}: V \rightarrow V'$ is holomorphic.

Note that the function $\psi \circ f \circ \varphi^{-1}$ is defined on the open set $V \subseteq \mathbb{C}$ and takes values in $V' \subseteq \mathbb{C}$. So it makes sense to require the holomorphicity of this function (as defined in Section 3).

Example 15.12. The atlas $\mathcal{A}$ consisting of the charts ψ_1 and ψ_2 on the Riemann sphere $\widehat{\mathbb{C}}$ as defined Example 15.5 is a complex atlas (see Example 15.7). In the following, we always think of $\widehat{\mathbb{C}}$ as a Riemann surface equipped with the complex structure represented by the complex atlas $\mathcal{A}$.

How do we verify that a given continuous map $f: \widehat{\mathbb{C}} \rightarrow \widehat{\mathbb{C}}$ is holomorphic? We check this locally near each point $z_0 \in \widehat{\mathbb{C}}$. Let $w_0 := f(z_0)$. The basic idea is this: whenever the point ∞ is involved, we send it to 0 by the chart ψ_2 . This leads to several cases.

Case 1: $z_0, w_0 \in \mathbb{C}$. In this case, $z \mapsto f(z)$ is just a complex-valued function near z_0 , and we check its holomorphicity as usual.

Case 2: $z_0 = \infty, w_0 \in \mathbb{C}$. We check $z \mapsto g(z) := f(1/z)$ for holomorphicity near 0.

Here it is understood that $g(0) := f(\infty)$, and we make similar definitions in the remaining two cases.

Case 3: $z_0 \in \mathbb{C}, w_0 = \infty$. We check $z \mapsto 1/f(z)$ for holomorphicity near $z_0 \in \mathbb{C}$.

Case 4: $z_0 = w_0 = \infty$. We check $z \mapsto 1/f(1/z)$ for holomorphicity near 0.

Example 15.13. Every rational function R can be considered as a holomorphic map $R: \widehat{\mathbb{C}} \rightarrow \widehat{\mathbb{C}}$. To see this, let $R = P/Q$, where P and Q are polynomials with no common roots and $Q \neq 0$. We write

$$\begin{aligned} P(z) &= a_0 + a_1z + \cdots + a_nz^n, \quad a_n \neq 0, \\ Q(z) &= b_0 + b_1z + \cdots + b_kz^k, \quad b_k \neq 0. \end{aligned}$$

Then R is given by $R(z) = P(z)/Q(z)$ for $z \in \mathbb{C}$ if $Q(z) \neq 0$. We extend R continuously to a map $R: \widehat{\mathbb{C}} \rightarrow \widehat{\mathbb{C}}$. To do this, we set

$$R(z) := \lim_{w \rightarrow z} R(w) = \infty$$

for $z \in \mathbb{C}$ if $Q(z) = 0$ (note that then $P(z) \neq 0$) and

$$R(\infty) := \lim_{w \rightarrow \infty} R(w) = \begin{cases} 0 & \text{if } n < k, \\ a_n/b_k & \text{if } n = k, \\ \infty & \text{if } n > k. \end{cases}$$

Then $R(z)$ is defined for all $z \in \widehat{\mathbb{C}}$, and $R: \widehat{\mathbb{C}} \rightarrow \widehat{\mathbb{C}}$ is continuous.

Moreover, $R: \widehat{\mathbb{C}} \rightarrow \widehat{\mathbb{C}}$ is a holomorphic map as can be seen by the case analysis as in Example 15.12.

For example, suppose $z_0 \in \mathbb{C}$ and $Q(z_0) = 0$. Then $R(z_0) = \infty$, and so we have to investigate the holomorphicity of

$$z \mapsto \frac{1}{R(z)} = \frac{Q(z)}{P(z)} \tag{15.1}$$

near z_0 . Since P and Q have no common roots, we have $P(z_0) \neq 0$ and so $P(z) \neq 0$ for z near z_0 . The holomorphicity of the function in (15.1) near z_0 follows.

15.1 Exercises

Problem 15.1. Let X be a topological 2-manifold, and $\mathcal{A} = \{\phi_i : U_i \rightarrow V_i : i \in I\}$ be a complex atlas on X , i.e., an atlas of holomorphically compatible charts.

(a) Suppose that $\psi_1 : M_1 \rightarrow N_1$ and $\psi_2 : M_2 \rightarrow N_2$ are two charts on X that are holomorphically compatible with each chart in $\mathcal{A}$. Show that ψ_1 and ψ_2 are holomorphically compatible.

(b) Let $\mathcal{B}$ be the set of all charts on X that are holomorphically compatible with each chart in $\mathcal{A}$. Show that $\mathcal{B}$ is a maximal complex atlas containing $\mathcal{A}$, i.e., $\mathcal{B}$ is a complex atlas containing $\mathcal{A}$, and every chart that is holomorphically compatible with all the charts in $\mathcal{A}$ is contained in $\mathcal{B}$.

(c) Suppose that $\mathcal{A}'$ is another complex atlas on X , and let $\mathcal{B}'$ be the associated maximal complex atlas as defined in (b). Show that $\mathcal{A}$ and $\mathcal{A}'$ are analytically equivalent if and only if $\mathcal{B} = \mathcal{B}'$.

(d) Show that analytic equivalence (see Definition 15.8) is an equivalence relation on the set of all complex atlases $\mathcal{A}$ on X .

Problem 15.2. Let X and Y be Riemann surfaces, and $f : X \rightarrow Y$ be a continuous map. Recall that f is called *holomorphic* if for each complex chart $\varphi : U_1 \rightarrow V_1$ on X and each complex chart $\psi : U_2 \rightarrow V_2$ on Y with $\varphi(U_1) \subseteq U_2$ the function $\psi \circ f \circ \varphi^{-1} : V_1 \rightarrow V_2 \subseteq \mathbb{C}$ is holomorphic on the open set $V_1 \subseteq \mathbb{C}$.

Show that this is equivalent to the following condition: for each $x \in X$ there *exists* a complex chart $\varphi : U_1 \rightarrow V_1$ on X , and a complex chart $\psi : U_2 \rightarrow V_2$ on Y with $x \in U_1$ and $\varphi(U_1) \subseteq U_2$ such that $\psi \circ f \circ \varphi^{-1} : V_1 \rightarrow V_2$ is holomorphic on V_1 .

Problem 15.3. Let X and Y be connected Riemann surfaces, and $f : X \rightarrow Y$ and $g : X \rightarrow Y$ be holomorphic maps. Suppose that the set

$$A = \{x \in X : f(x) = g(x)\}$$

has a limit point in X . Show that $f = g$.

Problem 15.4. Let Γ be a *rank-2 lattice* in $\mathbb{C}$, that is, a set of the form

$$\Gamma = \{n\omega_1 + k\omega_2 : k, n \in \mathbb{Z}\},$$

where $\omega_1, \omega_2 \in \mathbb{C} \setminus \{0\}$ are linearly independent over $\mathbb{R}$.

(a) For $z, w \in \mathbb{C}$ we write $z \sim w$ iff $z - w \in \Gamma$. Show that this defines an equivalence relation on $\mathbb{C}$.

In the following, we denote by $[z] := \{w \in \mathbb{C} : z \sim w\}$ the equivalence class of a point $z \in \mathbb{C}$, by $\mathbb{C}/\Gamma := \{[z] : z \in \mathbb{C}\}$ the corresponding *quotient space* (that is, the set of all equivalence classes), and by $\pi: \mathbb{C} \rightarrow \mathbb{C}/\Gamma$ the *quotient map* defined as $\pi(z) := [z] \in \mathbb{C}/\Gamma$ for $z \in \mathbb{C}$.

(b) We declare a set $U \subseteq \mathbb{C}/\Gamma$ to be *open* iff $\pi^{-1}(U)$ is open in $\mathbb{C}$. Show that the family of all open sets in $\mathbb{C}/\Gamma$ is a topology on $\mathbb{C}/\Gamma$, called the *quotient topology*.

In the following, we equip $\mathbb{C}/\Gamma$ with this quotient topology.

(c) Show that the quotient map $\pi: \mathbb{C} \rightarrow \mathbb{C}/\Gamma$ is continuous and open.

(d) Show that there exists a number $r > 0$ such that for each $z_0 \in \mathbb{C}$, the restriction $\pi|_{B(z_0, r)}$ of the quotient map π is a homeomorphism of the disk $B(z_0, r) = \{z \in \mathbb{C} : |z - z_0| < r\}$ onto its image in $\mathbb{C}/\Gamma$.

(e) Show that $\mathbb{C}/\Gamma$ is a Hausdorff space.

(f) Let $P := \{u\omega_1 + v\omega_2 : u, v \in [0, 1]\}$. Show that $\pi(P) = \mathbb{C}/\Gamma$ and conclude that $\mathbb{C}/\Gamma$ is compact.

(g) Show that $\mathbb{C}/\Gamma$ is a surface (that is, a (topological) 2-manifold).

(h) Show that on a more intuitive level the surface $\mathbb{C}/\Gamma$ can be obtained by “gluing” parts of the boundary of the set P (as in (f)) together and that in this way one can see that $\mathbb{C}/\Gamma$ has the “shape” of a torus.

(i) Show that there exists a unique complex structure on the torus $\mathbb{C}/\Gamma$ such that if we equip $\mathbb{C}/\Gamma$ with this complex structure, then $\mathbb{C}/\Gamma$ a Riemann surface (called a *complex torus*) and $\pi: \mathbb{C} \rightarrow \mathbb{C}/\Gamma$ is a holomorphic map.

Hint: Use (d) to find a suitable complex atlas $\mathcal{A}$ obtained from local inverses of π . Show that if $\mathcal{A}'$ is another complex atlas on $\mathbb{C}/\Gamma$ such that the map π becomes holomorphic if we equip $\mathbb{C}/\Gamma$ with the complex structure represented by $\mathcal{A}'$, then $\mathcal{A}'$ is analytically equivalent to $\mathcal{A}$.

16 Isolated singularities

Definition 16.1. Let $U \subseteq \mathbb{C}$ be open, and $a \in U$. If $f \in H(U \setminus \{a\})$, then f is said to have an *isolated singularity* at a . The singularity is called:

- (i) *removable* if the function f can be extended to a such that the extended function is holomorphic on U , i.e., if there exists $g \in H(U)$ such that $f = g|_{U \setminus \{a\}}$,
- (ii) a *pole* if $\lim_{z \rightarrow a} |f(z)| = \infty$,
- (iii) an *essential singularity* if it is neither removable nor a pole.

So each isolated singularity is either removable, a pole, or an essential singularity. These cases are mutually exclusive.

Example 16.2. The following functions f are defined and holomorphic on $\mathbb{C} \setminus \{0\}$; so they have 0 as an isolated singularity.

- (a) $f(z) = (e^z - 1)/z$. Then 0 is a removable singularity for f ; indeed,

$$f(z) = \frac{e^z - 1}{z} = \sum_{n=1}^{\infty} \frac{z^{n-1}}{n!} = \sum_{n=0}^{\infty} \frac{z^n}{(n+1)!}.$$

The last power series converges for all $z \in \mathbb{C}$, and so it represents an entire function that gives a holomorphic extension of f to the whole complex plane.

- (b) $f(z) = 1/z^2$. Then 0 is a pole of f , because $|f(z)| = \frac{1}{|z|^2} \rightarrow \infty$ as $z \rightarrow 0$.

- (c) $f(z) = e^{1/z}$. Then 0 is an essential singularity of f ; indeed, if $z_n := 1/n$, then $z_n \rightarrow 0$ and $f(z_n) = e^n \rightarrow \infty$ as $n \rightarrow \infty$; so 0 is not removable. But 0 is not a pole of f either: if $z'_n = -1/n$, then $z'_n \rightarrow 0$ and $f(z'_n) = e^{-n} \rightarrow 0$ as $n \rightarrow \infty$.

Theorem 16.3. Let $U \subseteq \mathbb{C}$ be open, $a \in U$, and $f \in H(U \setminus \{a\})$. Then the isolated singularity a of f is removable if and only if f is bounded near a , i.e., there exists $r > 0$ and $M \geq 0$ such that $B(a, r) \subseteq U$ and $|f(z)| \leq M$ for all $z \in B(a, r) \setminus \{a\}$.

Proof. “ $\Rightarrow$ ” This is the easy implication. If a is removable, then f has a holomorphic, and in particular continuous, extension to a ; so $\lim_{z \rightarrow a} f(z)$ exists which implies that f is bounded near a .

“ $\Leftarrow$ ” Suppose that f is bounded near a . Define

$$g(z) = \begin{cases} 0 & \text{for } z = a, \\ (z - a)^2 f(z) & \text{for } z \in U \setminus \{a\}. \end{cases}$$

Then

$$\lim_{z \rightarrow a} \frac{g(z) - g(a)}{z - a} = \lim_{z \rightarrow a} f(z)(z - a) = 0.$$

So g is differentiable at $a = 0$, and $g'(a) = 0$. Obviously, g is also differentiable at each point $z \in U \setminus \{a\}$. Hence $g \in H(U)$, and so g has a power series expansion

$$g(z) = \sum_{n=0}^{\infty} c_n (z - a)^n$$

near a . Here $c_n = g^{(n)}(a)/n!$ are the Taylor coefficients of g at a . Since $g(a) = 0 = g'(a)$, we have $c_0 = c_1 = 0$. This implies that

$$f(z) = \sum_{n=2}^{\infty} c_n (z - a)^{n-2} = \sum_{n=0}^{\infty} c_{n+2} (z - a)^n$$

is represented by a power series near a . This shows that f can be holomorphically extended to a by setting $f(a) := c_2$. $\square$

Theorem 16.4. *Let $U \subseteq \mathbb{C}$ be open, $a \in U$, and $f \in H(U \setminus \{a\})$. If a is a pole of f , then there exists a unique number $m \in \mathbb{N}$ (called the order or multiplicity of the pole), and a unique function $g \in H(U)$ with $g(a) \neq 0$ such that*

$$f(z) = \frac{1}{(z - a)^m} g(z) \quad \text{for } z \in U \setminus \{a\}. \quad (16.1)$$

If f has a pole of order m at a , then there exist a unique function $h \in H(U)$ and unique numbers $A_1, \dots, A_m \in \mathbb{C}$, $A_m \neq 0$, such that

$$f(z) = \frac{A_m}{(z - a)^m} + \dots + \frac{A_1}{(z - a)} + h(z) \quad \text{for } z \in U \setminus \{a\}. \quad (16.2)$$

The expression

$$P(z) := \frac{A_m}{(z-a)^m} + \cdots + \frac{A_1}{(z-a)}$$

is called the *principal* or *singular part* of the pole a of f .

Proof. Since $|f(z)| \rightarrow \infty$ as $z \rightarrow a$, the function f has no zeros near a . So the function $z \mapsto 1/f(z)$ is defined in a punctured neighborhood of a and holomorphic there; since

$$\lim_{z \rightarrow a} \frac{1}{f(z)} = 0,$$

this function has a removable singularity at a by Theorem 16.3. So there exists a function u that is holomorphic near a such that $u(a) = 0$, and $u = 1/f$ in a punctured neighborhood of a .

Let $m \in \mathbb{N}$ be the order of the zero at a (note that $u \not\equiv 0$); then there exists a function v that is holomorphic near a such that $v(a) \neq 0$ and

$$u(z) = (z-a)^m v(z) \quad \text{for } z \text{ near } a.$$

Then

$$f(z) = \frac{1}{(z-a)^m v(z)} \quad \text{for } z \text{ near } a, z \neq a.$$

If we define

$$g(z) = \begin{cases} 1/v(z) & \text{for } z \text{ near } a, \\ f(z)(z-a)^m & \text{for } z \text{ elsewhere in } U, \end{cases}$$

then $g \in H(U)$, $g(a) = 1/v(a) \neq 0$, and (16.1) is true.

The function g has a power series representation

$$g(z) = \sum_{n=0}^{\infty} a_n (z-a)^n = a_0 + \cdots + a_{m-1} (z-a)^{m-1} + (z-a)^m h(z)$$

near a , where $a_0 = g(a) \neq 0$, and $h \in H(U)$. Dividing by $(z-a)^m$ we obtain (16.2) with $A_k = a_{m-k}$ for $k = 1, \dots, m$.

The uniqueness statements are straightforward to prove. For example, suppose that

$$\frac{A_m}{(z-a)^m} + \cdots + \frac{A_1}{(z-a)} + h(z) = f(z) = \frac{B_m}{(z-a)^m} + \cdots + \frac{B_1}{(z-a)} + \tilde{h}(z)$$

for z near a , $z \neq a$, where $B_1, \dots, B_m \in \mathbb{C}$, and $\tilde{h} \in H(U)$. Then one successively proves $A_m = B_m, \dots, A_1 = B_1$, $h = \tilde{h}$; indeed, letting $z \rightarrow a$ in the identity

$$\begin{aligned} A_m - B_m &= (B_{m-1} - A_{m-1})(z - a) + \dots \\ &\quad + (B_1 - A_1)(z - a)^{m-1} + (h(z) - \tilde{h}(z))(z - a)^m, \end{aligned}$$

we obtain $A_m - B_m = 0$, etc. $\square$

Definition 16.5. Let $U \subseteq \mathbb{C}$ be open. A complex-valued function f is called *meromorphic* on U if there exists a set $A \subseteq U$ such that

- (i) A has no limit points in the set U and so consists of isolated points,
- (ii) $f \in H(U \setminus A)$,
- (iii) each point in A is a pole of f .

So a meromorphic function on U is a function that is holomorphic except for isolated singularities that are poles (we may always assume that removable singularities are absent). The possibility $A = \emptyset$ is allowed in the definition; so every holomorphic function on U is also meromorphic.

One can extend a meromorphic function to a holomorphic map $f: U \rightarrow \hat{\mathbb{C}}$ by setting $f(a) = \infty$ for a pole a ; it is easy to see that in this way the meromorphic functions f on U correspond precisely to the holomorphic maps $f: U \rightarrow \hat{\mathbb{C}}$ that are not identically equal to ∞ on any component of U .

Remark 16.6. If U is an open subset of a Riemann surface, $a \in U$, and $f: U \setminus \{a\} \rightarrow \mathbb{C}$ is a holomorphic function, then one calls a an *isolated singularity* of f . One defines a removable singularity, a pole, or an essential singularity in the same way as in Definition 16.1. By using local charts near a one can actually treat this situation in the same way as for functions with isolated singularities in $\mathbb{C}$, and prove analogs of Theorems 16.3 and 16.4.

An important special case is when f is a holomorphic function defined in a punctured neighborhood of ∞ on the Riemann sphere. Then ∞ is an isolated singularity of f ; its type is the same as the type of the isolated singularity 0 of the map $z \mapsto f(1/z)$. For example, if f is a non-constant polynomial, then ∞ is a pole of f whose order is equal to the degree of the polynomial.

Example 16.7 (Rational functions). Let R be a non-constant rational function written as

$$R(z) = \frac{P(z)}{Q(z)}$$

where P and Q are polynomials, $P, Q \neq 0$. Every zero a of Q is an isolated singularity of R . We can write

$$P(z) = (z - a)^k \tilde{P}(z) \quad \text{and} \quad Q(z) = (z - a)^l \tilde{Q}(z),$$

where $k \in \mathbb{N}_0$, $l \in \mathbb{N}$, $\tilde{P}$ and $\tilde{Q}$ are polynomials, and $\tilde{P}(a), \tilde{Q}(a) \neq 0$. Then

$$\frac{P(z)}{Q(z)} = (z - a)^{k-l} \frac{\tilde{P}(z)}{\tilde{Q}(z)},$$

and so a is removable if $k \geq l$, and a pole if $k < l$. By dividing out common factors of P and Q one can assume that R has no removable singularities. Then the only singularities of R are possible poles $a \in \mathbb{C}$ (which are always present unless R is a polynomial). If P and Q have no common factor, then the zeros a of Q are precisely the poles of R . In particular, every rational function is a meromorphic function on $\mathbb{C}$.

One can obtain a *partial fraction decomposition* of R as follows. By dividing P by Q with remainder, we can write R in the form

$$R(z) = \frac{P(z)}{Q(z)} = S(z) + \frac{T(z)}{Q(z)},$$

where S and T are polynomials, and $\deg(T) < \deg(Q)$ or $T = 0$. Let us assume that $T \neq 0$, because otherwise R is a polynomial, and that T and Q have no common zeros (otherwise we can cancel common factors). Then the zeros a of Q are precisely the poles of T/Q . Let p_a denote the principal part of the pole of T/Q at a ; then the function

$$z \mapsto \frac{T(z)}{Q(z)} - \sum_{a \text{ zero of } Q} p_a(z)$$

is holomorphic on $\mathbb{C}$ except for finitely many removable singularities corresponding to the zeros of Q ; so one can consider it as an entire function. Moreover, it tends to 0 as $|z| \rightarrow \infty$. Hence by Liouville's Theorem this function must be identical 0.

The conclusion is that every rational function R can be written as

$$R(z) = S(z) + \sum_{k=1}^n \sum_{l=1}^{m_k} \frac{A_{kl}}{(z - a_k)^l} \quad (\text{partial fraction decomposition}),$$

where S is a polynomial, $n \in \mathbb{N}_0$, $a_1, \dots, a_n \in \mathbb{C}$ are the poles of R with their respective orders $m_1, \dots, m_n \in \mathbb{N}$, and the numbers $A_{kl} \in \mathbb{C}$, $l = 1, \dots, m_k$, are coefficients associated with the principal part of each pole a_k , $k = 1, \dots, n$.

Similar considerations show that every holomorphic map $R: \widehat{\mathbb{C}} \rightarrow \widehat{\mathbb{C}}$ is a rational function (including constant functions, possibly identical equal to $\infty \in \widehat{\mathbb{C}}$). To see this, one may assume that R is non-constant, and that $R(\infty) = 0$ (otherwise one considers R composed with a suitable Möbius transformation). Then the Uniqueness Theorem implies that $R^{-1}(\infty)$ is a finite subset of $\mathbb{C}$. Each point in $R^{-1}(\infty)$ is a pole of R . Then one subtracts principal parts, applies Liouville's Theorem, etc.

Theorem 16.8 (Casorati–Weierstrass). *Let $U \subseteq \mathbb{C}$ be open, $a \in U$, and $f \in H(U \setminus \{a\})$. Suppose that a is an essential singularity of f . Then for all $\delta > 0$ such that $B(a, \delta) \subseteq U$, the set $f(B(a, \delta) \setminus \{a\})$ is dense in $\mathbb{C}$, i.e., $\overline{f(B(a, \delta) \setminus \{a\})} = \mathbb{C}$.*

Proof. Suppose not. Then there exists $\delta > 0$ such that $B(a, \delta) \subseteq U$ and $A := \overline{f(B(a, \delta) \setminus \{a\})} \neq \mathbb{C}$. Then the set $\mathbb{C} \setminus A$ is open and non-empty, and so we can pick a point $w \in \mathbb{C} \setminus A$. Moreover, there exists $\epsilon > 0$ such that

$$|f(z) - w| \geq \epsilon \quad \text{for all } z \in B(a, \delta) \setminus \{a\}.$$

Now define

$$g(z) = \frac{1}{f(z) - w} \quad \text{for } z \in B(a, \delta) \setminus \{a\}.$$

Then $g \in H(B(a, \delta) \setminus \{a\})$ and $|g| \leq 1/\epsilon$. By Theorem 16.3 the function g has a removable singularity at a ; so the definition of g can be extended to a such that $g \in H(B(a, \delta))$. We can write g in the form

$$g(z) = (z - a)^m \tilde{g}(z),$$

where $m \in \mathbb{N}_0$, $\tilde{g} \in H(B(a, \delta))$, and $\tilde{g}(a) \neq 0$. Then

$$f(z) = w + \frac{1}{g(z)} = w + \frac{1}{(z - a)^m \tilde{g}(z)}$$

for z near a . This shows that a is a removable singularity (if $m = 0$) or a pole (if $m > 0$) of f , contradicting our hypothesis that a is an essential singularity. $\square$

Remark 16.9. (a) Another version of the Casorati–Weierstrass Theorem is the following statement: if f is a non-constant entire function, then $f(\mathbb{C})$ is dense in $\mathbb{C}$. The proof is very similar. Again one argues by contradiction, and defines an auxiliary function g as in the previous proof. Then one invokes Liouville’s Theorem to conclude that g is constant, and hence also f . This gives the desired contradiction.

Alternatively, one can analyze the singularity at $a = \infty$ by considering $z \mapsto h(z) = f(1/z)$ which has an isolated singularity at 0; this singularity can only be a pole or an essential singularity. In the first case f must be a polynomial, and we have $f(\mathbb{C}) = \mathbb{C}$ as follows from the Fundamental Theorem of Algebra. In the second case, where 0 is an essential singularity of h , the statement $\overline{f(\mathbb{C})} = \mathbb{C}$ follows from the above version of the Casorati–Weierstrass Theorem.

(b) One can actually prove much stronger statements: if f is as in Theorem 16.8, then there exists $w_0 \in \mathbb{C}$ such that for all small $\delta > 0$ we have $\mathbb{C} \setminus \{w_0\} \subseteq f(B(a, \delta) \setminus \{a\})$. So in every neighborhood of the essential singularity a the function f attains every value with at most one exception. This is known as Picard’s “Big” Theorem.

Similarly, if f is a non-constant entire function, then there exists $w_0 \in \mathbb{C}$ such that $\mathbb{C} \setminus \{w_0\} \subseteq f(\mathbb{C})$, and so f attains every value with at most one exception (Picard’s “Little” Theorem).

These theorems will be proved later in this course.

Example 16.10. We consider the function

$$f(z) = \frac{\pi^2}{\sin^2(\pi z)}.$$

Note that $\sin(\pi z) = 0$ if and only if $z \in \mathbb{Z}$. It follows that f is a holomorphic function on $\mathbb{C} \setminus \mathbb{Z}$. It obviously satisfies $f(z + 1) = f(z)$ for $z \in \mathbb{C} \setminus \mathbb{Z}$.

Actually, f is a meromorphic function on $\mathbb{C}$ with a second order pole at each point $n \in \mathbb{Z}$ (and no other poles). To see this, note that for $n \in \mathbb{Z}$ and

$z \in \mathbb{C}$ near n we have

$$\begin{aligned} f(z) &= \frac{\pi^2}{\sin^2(\pi z)} = \frac{\pi^2}{\sin^2(\pi(z-n))} \\ &= \frac{1}{(z-n)^2 P(z-n)}, \end{aligned}$$

where P is a power series centered at 0 representing an even function that is holomorphic near 0 and satisfies $P(0) = 1$. Then $1/P$ is a function of the same type and it follows that

$$f(z) = \frac{1}{(z-n)^2} (1 + O((z-n)^2)) = \frac{1}{(z-n)^2} + h(z),$$

where h is holomorphic for z near n . So f has a second order pole at each point $z = n$ with principal part $\frac{1}{(z-n)^2}$ and no other poles.

It is tempting to try to obtain an analog of the partial fraction decomposition of rational functions for the meromorphic function f . This suggests that we should compare f with $g(z) = \sum_{n \in \mathbb{Z}} \frac{1}{(z-n)^2}$. Now the latter series converges normally on $\mathbb{C} \setminus \mathbb{Z}$. Moreover, $s = f - g$ has a removable singularity at each point $z = n$, and hence s can be considered as an entire function.

Note that for each $z \in \mathbb{C} \setminus \mathbb{Z}$, we have

$$g(z+1) = \sum_{n \in \mathbb{Z}} \frac{1}{((z+1)-n)^2} = \sum_{n \in \mathbb{Z}} \frac{1}{(z-(n-1))^2} = g(z).$$

Here we used the fact that $n \mapsto n-1$ is a bijection on $\mathbb{Z}$. So g is 1-periodic as the function f . This implies that $s = f - g$ is 1-periodic.

Now consider the strip $S := \{x + iy \in \mathbb{C} : -1/2 \leq x \leq 1/2, y \in \mathbb{R}\}$. Let $z \in S$ be of the form $z = x + iy$ with $x \in [-1/2, 1/2]$ and $y \in \mathbb{R}$ with $|y| > 0$.

Then we have

$$\begin{aligned}
|g(z)| &\leq \sum_{n \in \mathbb{Z}} \frac{1}{|z - n|^2} = \sum_{n \in \mathbb{Z}} \frac{1}{(n - x)^2 + y^2} = \sum_{n \in \mathbb{Z}} \frac{1}{n^2 - 2xn + 1 + y^2} \\
&\leq \frac{1}{1 + y^2} + \sum_{n \in \mathbb{Z} \setminus \{0\}} \frac{1}{n^2 - |n| + 1 + y^2} \\
&\leq \frac{3}{y^2} + 2 \sum_{n=2}^{\infty} \frac{1}{(n - 1/2)^2 + y^2} \\
&\leq \frac{3}{y^2} + 2 \sum_{n=2}^{\infty} \int_{n-3/2}^{n-1/2} \frac{du}{u^2 + y^2} \leq \frac{3}{y^2} + \int_{-\infty}^{\infty} \frac{du}{u^2 + y^2} \\
&= \frac{3}{y^2} + \frac{1}{|y|} \int_{-\infty}^{\infty} \frac{dv}{v^2 + 1} = \frac{3}{y^2} + \frac{\pi}{|y|}.
\end{aligned}$$

We conclude that $g(z) \rightarrow 0$ as $|\operatorname{Im}(z)| \rightarrow \infty$ for $z \in S$.

Similarly, for $z = x + iy \in S$ with $x \in [-1/2, 1/2]$ and $y \in \mathbb{R}$, $|y| > 0$, we have

$$\begin{aligned}
|f(z)| &= \frac{\pi^2}{|\sin^2(\pi z)|} = \frac{4\pi^2}{|e^{i\pi z} - e^{-i\pi z}|^2} \\
&\leq \frac{4\pi^2}{(e^{\pi|y|} - e^{-\pi|y|})^2}.
\end{aligned}$$

This shows that we also have $f(z) \rightarrow 0$ as $|\operatorname{Im}(z)| \rightarrow \infty$ for $z \in S$.

We conclude that $s = f - g$ is an entire function that is 1-periodic and satisfies $s(z) \rightarrow 0$ as $|\operatorname{Im}(z)| \rightarrow \infty$ for $z \in S$. It now easily follows from Liouville's Theorem that s is the constant function 0.

We have shown that

$$\frac{\pi^2}{\sin^2(\pi z)} = \sum_{n \in \mathbb{Z}} \frac{1}{(z - n)^2} \tag{16.3}$$

for $z \in \mathbb{C} \setminus \mathbb{Z}$.

Example 16.11. We consider the function

$$f(z) = \pi \cot(\pi z) = \pi \frac{\cos(\pi z)}{\sin(\pi z)}.$$

This is a meromorphic function on $\mathbb{C}$ with first order poles at each point $n \in \mathbb{Z}$ (and no other poles). To see this, note that for $n \in \mathbb{Z}$ and $z \in \mathbb{C}$ near n we have

$$\begin{aligned} f(z) &= \pi \frac{\cos(\pi z)}{\sin(\pi z)} = \pi \frac{\cos(\pi(z-n))}{\sin(\pi(z-n))} \\ &= \frac{\pi + O((z-n)^2)}{\pi(z-n) + O((z-n)^3)} = \frac{1}{z-n} \cdot \frac{1 + O((z-n)^2)}{1 + O((z-n)^2)} \\ &= \frac{1}{z-n} + h(z), \end{aligned}$$

where h is a holomorphic function near $z = n$ with $h(n) = 0$. So the principal part of the pole of f at $z = n$ is given by $\frac{1}{z-n}$ and we see that this is a first order pole.

Again is tempting to try obtain an analog of the partial fraction decomposition of rational functions for the meromorphic function f . This suggests that we should compare f with $\sum_{n \in \mathbb{Z}} \frac{1}{z-n}$. The problem is that the latter series does not converge. Instead, we consider

$$g(z) = \frac{1}{z} + \sum_{k \in \mathbb{Z} \setminus \{0\}} \left(\frac{1}{z-k} + \frac{1}{k} \right)$$

for $z \in \mathbb{C} \setminus \mathbb{Z}$. It is easy to see that this series converges normally on $\mathbb{C} \setminus \mathbb{Z}$ (exercise!) and hence represents a holomorphic function on $\mathbb{C} \setminus \mathbb{Z}$. We can obtain the derivative of g by term-by-term differentiation of the series. So we have

$$g'(z) = -\frac{1}{z^2} - \sum_{k \in \mathbb{Z} \setminus \{0\}} \frac{1}{(z-k)^2} = -\sum_{n \in \mathbb{Z}} \frac{1}{(z-n)^2} = -\frac{\pi^2}{\sin^2(\pi z)}.$$

But we also have $f'(z) = -\frac{\pi^2}{\sin^2(\pi z)}$. It follows that $f - g$ is a constant function on $\mathbb{C} \setminus \mathbb{Z}$, say $f - g = c \in \mathbb{C}$. To see what this constant is, we rewrite the last identity as

$$\pi \cot(\pi z) - \frac{1}{z} = c + \sum_{k \in \mathbb{Z} \setminus \{0\}} \left(\frac{1}{z-k} + \frac{1}{k} \right)$$

As $z \rightarrow 0$ the function on the left has the limit 0 (exercise!) while the right hand side evaluates to c at $z = 0$. This implies that $c = 0$. We conclude that

$$\pi \cot(\pi z) = \frac{1}{z} + \sum_{n \in \mathbb{Z} \setminus \{0\}} \left(\frac{1}{z-n} + \frac{1}{n} \right) \quad (16.4)$$

for $z \in \mathbb{C} \setminus \mathbb{Z}$. Since the latter series converges normally on $\mathbb{C} \setminus \mathbb{Z}$, we can combine the terms in any way. In particular, if we combine the terms for $n \in \mathbb{N}$ and $-n$, we obtain

$$\left(\frac{1}{z-n} + \frac{1}{n} \right) + \left(\frac{1}{z+n} - \frac{1}{n} \right) = \frac{1}{z-n} + \frac{1}{z+n} = \frac{2z}{z^2 - n^2}.$$

It follows that

$$\pi \cot(\pi z) = \frac{1}{z} + \sum_{n \in \mathbb{N}} \frac{2z}{z^2 - n^2} \quad (16.5)$$

for $z \in \mathbb{C} \setminus \mathbb{Z}$.

These identities have some interesting consequences if we combine them with the power series expansion

$$z \cot(z) = \sum_{n=0}^{\infty} (-1)^n \frac{2^{2n} B_{2n}}{(2n)!} z^{2n}$$

derived in Problem 14.2 (e). This expansion is valid near 0 and the coefficients B_{2n} are the Bernoulli numbers. In particular, $B_0 = 1$ and $B_2 = 1/6$. This identity together with (16.4) leads to

$$\sum_{n \in \mathbb{Z} \setminus \{0\}} \left(\frac{1}{z-n} + \frac{1}{n} \right) = \frac{\pi z \cot(\pi z) - 1}{z} = \sum_{n=0}^{\infty} (-1)^n \frac{2^{2n} B_{2n} \pi^{2n}}{(2n)!} z^{2n-1}. \quad (16.6)$$

The series on the left hand side converges normally near 0 and hence represents a holomorphic function near 0. We can obtain its derivative $f^{(2k-1)}$ for $k \in \mathbb{N}$ by term-by-term differentiation of the series. In this way we see that

$$f^{(2k-1)}(z) = -(2k-1)! \sum_{n \in \mathbb{Z} \setminus \{0\}} \frac{1}{(z-n)^{2k}}$$

for z near 0. In particular,

$$f^{(2k-1)}(0) = -2(2k-1)! \sum_{n=1}^{\infty} \frac{1}{n^{2k}} = -2(2k-1)! \zeta(2k),$$

where ζ is the Riemann ζ -function. On the other hand, we can read off $f^{(2k-1)}(0)$ from the power series expansion of f in (16.6) to see that

$$f^{(2k-1)}(0) = (2k-1)!(-1)^k \frac{2^{2k} B_{2k} \pi^{2k}}{(2k)!}$$

Equating these two expressions for $f^{(2k-1)}(0)$ we conclude that

$$\zeta(2k) = \sum_{n=1}^{\infty} \frac{1}{n^{2k}} = (-1)^{k-1} \frac{2^{2k-1} B_{2k} \pi^{2k}}{(2k)!} \quad (16.7)$$

for each $k \in \mathbb{N}$.

Example 16.12. We consider the infinite product $p(z) = z \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2}\right)$ for $z \in \mathbb{C}$. It easily follows from Theorem 11.6 that this infinite product converges absolutely for each $z \in \mathbb{C}$ and locally uniformly on $\mathbb{C}$. Hence it represents an entire function p . This function p has zeros at each point $z = n \in \mathbb{Z}$ and no other zeros.

To see what this function p is, we compute its logarithmic derivative according to Theorem 11.6 (d) as

$$\frac{p'(z)}{p(z)} = \frac{1}{z} + \sum_{n \in \mathbb{N}} \frac{2z}{z^2 - n^2}$$

for $z \in \mathbb{C} \setminus \mathbb{Z}$. As we have seen, the last expression is equal to $\pi \cot(\pi z)$ (see (16.5)). Now the function $\sin(\pi z)$ also has its logarithmic derivative equal to $\pi \cot(\pi z)$. If two holomorphic functions have the same logarithmic derivatives, then they must be the same up to a multiplication constant (exercise!).

It follows that there exists a constant $c \in \mathbb{C}$ such that

$$\sin(\pi z) = cz \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2}\right)$$

for $z \in \mathbb{C}$. To determine this constant, we rewrite this identity as

$$\frac{\sin(\pi z)}{z} = c \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2}\right)$$

for $z \in \mathbb{C} \setminus \{0\}$. Now as $z \rightarrow 0$, the right left hand side has the limit π , while the right hand side evaluates to c at $z = 0$. Hence $c = \pi$ and we obtain the identity

$$\sin(\pi z) = \pi z \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2}\right) \quad (16.8)$$

for $z \in \mathbb{C}$.

16.1 Exercises

Problem 16.1. Let $\Gamma \subseteq \mathbb{C}$ be a rank-2 lattice in $\mathbb{C}$ (see Problem 5.2).

(a) Show that the series

$$\wp(z) = \frac{1}{z^2} + \sum_{\gamma \in \Gamma \setminus \{0\}} \left(\frac{1}{(z - \gamma)^2} - \frac{1}{\gamma^2} \right)$$

converges normally on $\mathbb{C} \setminus \Gamma$.

(b) Show that $\wp$ represents a meromorphic function on $\mathbb{C}$ (called the *Weierstrass $\wp$ -function* for the given lattice Γ) with a second order pole at each point $\gamma \in \Gamma$ with principal part given by $\frac{1}{(z - \gamma)^2}$ and no other poles.

(c) Show that $\wp$ is an even function, that is, $\wp(-z) = \wp(z)$ for all $z \in \mathbb{C}$.

(d) Show that $\wp'(z + \gamma) = \wp'(z)$ for all $\gamma \in \Gamma$ and $z \in \mathbb{C}$.

(e) Show that $\wp(z + \gamma) = \wp(z)$ for all $\gamma \in \Gamma$ and $z \in \mathbb{C}$.

Problem 16.2. Let $\Gamma \subseteq \mathbb{C}$ be a rank-2 lattice in $\mathbb{C}$. Show that then the associated associated Weierstrass $\wp$ -function has the representation

$$\wp(z) = \frac{1}{z^2} + \sum_{n=1}^{\infty} (2n+1)G_{2n+2}z^{2n} \quad (16.9)$$

for $z \in \mathbb{C}$ sufficiently close to 0. Here G_{2k} for $k \in \mathbb{N}$, $k \geq 2$, is the *homogeneous Eisenstein series* defined as

$$G_{2k} := G_{2k}(\Gamma) = \sum_{\gamma \in \Gamma \setminus \{0\}} \frac{1}{\gamma^{2k}}. \quad (16.10)$$

Problem 16.3. Let $\Gamma \subseteq \mathbb{C}$ be a rank-2 lattice in $\mathbb{C}$ and define

$$g_2 := g_2(\Gamma) = 60G_4(\Gamma) \text{ and } g_3 := g_3(\Gamma) = 140G_6(\Gamma).$$

where G_4 and G_6 are the homogeneous Eisenstein series. Show that then the associated Weierstrass $\wp$ -function satisfies the differential equation

$$\wp'(z)^2 = 4\wp(z)^3 - g_2\wp(z) - g_3, \quad z \in \mathbb{C}.$$

Hint: Show first that an entire function f satisfying $f(z + \gamma) = f(z)$ for all $z \in \mathbb{C}$ and $\gamma \in \Gamma$ is constant. Then consider local expansions near 0 and near other poles of $\wp$.

Problem 16.4. Let the setup be as in Problem 16.3.

- (a) Show that $\wp''(z) = 6\wp(z)^2 - \frac{1}{2}g_2$ for $z \in \mathbb{C}$.
- (b) Show that we have $G_8 = \frac{3}{7}G_4^2$ and $G_{10} = \frac{5}{11}G_4G_6$ for the Eisenstein series associated with the given lattice Γ .
- (c) More generally, show that for each $k \in \mathbb{N}$, $k \geq 2$, there exists a polynomial P_k of two variables such that $G_{2k} = P_k(G_4, G_6)$.
- (d) Suppose $\Gamma, \Gamma' \subseteq \mathbb{C}$ are two rank-2 lattices that for the associated Eisenstein series we have $G_4(\Gamma) = G_4(\Gamma')$ and $G_6(\Gamma) = G_6(\Gamma')$. Show that then $\Gamma = \Gamma'$.

17 Möbius transformations

Definition 17.1. A rational map $R: \widehat{\mathbb{C}} \rightarrow \widehat{\mathbb{C}}$ of the form

$$R(z) = \frac{az + b}{cz + d} \quad \text{with } a, b, c, d \in \mathbb{C}, ad - bc \neq 0, \quad (17.1)$$

is called a *Möbius transformation*. The set of all Möbius transformations is denoted by Möb .

The condition $ad - bc \neq 0$ in (17.1) ensures that we cannot cancel the numerator against denominator in the formula for $R(z)$. Note that $ad - bc$ is the determinant of the coefficient matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ associated with R .

17.2. Composition of Möbius transformations. The composition of two Möbius transformations is again a Möbius transformation. This operation corresponds to multiplication of the associated coefficient matrices. Indeed, suppose that $S, T \in \text{Möb}$, where

$$S(w) = \frac{a_{11}w + a_{12}}{a_{21}w + a_{22}}, \quad \text{and} \quad w = T(z) = \frac{b_{11}z + b_{12}}{b_{21}z + b_{22}}.$$

Let

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix}$$

be the corresponding coefficient matrices. Then $S \circ T$ is also a Möbius transformation, because

$$\begin{aligned} (S \circ T)(z) &= S(T(z)) = \frac{a_{11} \frac{b_{11}z + b_{12}}{b_{21}z + b_{22}} + a_{12}}{a_{21} \frac{b_{11}z + b_{12}}{b_{21}z + b_{22}} + a_{22}} \\ &= \frac{(a_{11}b_{11} + a_{12}b_{21})z + a_{11}b_{12} + a_{12}b_{22}}{(a_{21}b_{11} + a_{22}b_{21})z + a_{21}b_{12} + a_{22}b_{22}}, \end{aligned}$$

and its associated coefficient matrix is

$$\begin{pmatrix} a_{11}b_{11} + a_{12}b_{21} & a_{11}b_{12} + a_{12}b_{22} \\ a_{21}b_{11} + a_{22}b_{21} & a_{21}b_{12} + a_{22}b_{22} \end{pmatrix} = AB,$$

which is the product of A and B .

17.3. Möb is a group. The set Möb of Möbius transformations with the composition given by composition $\circ$ of maps is a group. Indeed:

1. If $S, T \in \text{Möb}$, then $S \circ T \in \text{Möb}$.
2. Associativity of $\circ$ is clear, because composition of maps is associative.
3. The identity on $\widehat{\mathbb{C}}$ is the unit element in Möb; note that $\text{id}_{\widehat{\mathbb{C}}} \in \text{Möb}$, because

$$\text{id}_{\widehat{\mathbb{C}}}(z) = z = \frac{1 \cdot z + 0}{0 \cdot z + 1}.$$

4. If $S \in \text{Möb}$ with associated coefficient matrix A , then the Möbius transformation with associated coefficient matrix A^{-1} (it exists, because $\det(A) \neq 0$) is the inverse S^{-1} of S .

In particular, every Möbius transformation S is a biholomorphism from $\widehat{\mathbb{C}}$ onto $\widehat{\mathbb{C}}$ (i.e., S is a bijection, and both maps S and S^{-1} are holomorphic).

17.4. Generators of Möb. The group Möb is *generated* by translations, dilations, rotations, and the *inversion*. This means that every $R \in \text{Möb}$ can be written as a composition of maps of these types. These maps are Möbius transformation of the following forms:

$$\begin{aligned} S(z) &= z + a, & a \in \mathbb{C}, & \quad (\text{translation}), \\ S(z) &= \lambda z, & \lambda > 0, & \quad (\text{dilation}), \\ S(z) &= e^{i\theta} z, & \theta \in \mathbb{R}, & \quad (\text{rotation}), \\ S(z) &= 1/z & & \quad (\text{inversion}). \end{aligned}$$

To see that Möb is generated by these maps, let R be a Möbius transformation of the form

$$R(z) = \frac{az + b}{cz + d}.$$

Assume $c \neq 0$ (the case $c = 0$ is similar and easier). By dividing the numerator in the fraction representing R by the denominator, we can find $A, B \in \mathbb{C}$, $B \neq 0$, such that

$$R(z) = A + \frac{B}{cz + d}.$$

Then the map $z \mapsto R(z)$ can be written as a composition of generators as above in the following way:

$$\begin{aligned} z &\xrightarrow{\text{dilation}} |c|z \xrightarrow{\text{rotation}} cz \xrightarrow{\text{translation}} cz + d \xrightarrow{\text{inversion}} \frac{1}{cz + d} \\ &\xrightarrow{\text{dilation+rotation}} \frac{B}{cz + d} \xrightarrow{\text{translation}} A + \frac{B}{cz + d} = R(z). \end{aligned}$$

17.5. Fixed points of Möbius transformations. Every Möbius transformation $R \neq \text{id}_{\widehat{\mathbb{C}}}$ has precisely one or two fixed points. To see this, let R be a Möbius transformation of the form

$$R(z) = \frac{az + b}{cz + d}.$$

We consider the fixed point equation

$$z = \frac{az + b}{cz + d}. \quad (17.2)$$

This leads to two cases.

Case 1: $c = 0$. Then $d \neq 0$ and (17.2) is equivalent to

$$\frac{a}{d}z + \frac{b}{d} = z \Leftrightarrow \left(\frac{a}{d} - 1\right)z + \frac{b}{d} = 0.$$

This equation has exactly one solution $z \in \mathbb{C}$ if $a/d - 1 \neq 0$, and no solution $z \in \mathbb{C}$ if $a/d - 1 = 0$ and $b/d \neq 0$. Note that if $a/d - 1 = 0$ and $b/d = 0$, then $a = d$ and $b = c = 0$. This is impossible as $R \neq \text{id}_{\widehat{\mathbb{C}}}$.

So R has no or exactly one fixed point $z \in \mathbb{C}$. Since $c = 0$, we must have $a \neq 0$, and so $R(\infty) = \infty$. Hence R has precisely one or two fixed points in $\widehat{\mathbb{C}}$.

Case 2: $c \neq 0$. Then $R(\infty) = a/c \neq \infty$. So ∞ is not a fixed point of R . The fixed point equation (17.2) is equivalent to

$$cz^2 + (d - a)z + b = 0.$$

Since $c \neq 0$, this equation has precisely one or two solutions $z \in \mathbb{C}$. It follows that R has precisely one or two fixed points in $\widehat{\mathbb{C}}$.

17.6. Möbius transformations are uniquely determined by images of three distinct points. Let $S, T \in \text{Möb}$, and $z_1, z_2, z_3 \in \widehat{\mathbb{C}}$ be three distinct points. If $S(z_k) = T(z_k)$ for $k = 1, 2, 3$, then $S = T$.

Indeed, if $S(z_k) = T(z_k)$, then $(T^{-1} \circ S)(z_k) = z_k$ for $k = 1, 2, 3$. So the Möbius transformation $T^{-1} \circ S$ has three distinct fixed points. By what we have seen in Subsection 17.5, this is only possible if $T^{-1} \circ S = \text{id}_{\widehat{\mathbb{C}}}$. Hence $S = T$.

17.7. Cross-ratios. If $z_1, z_2, z_3, z_4 \in \widehat{\mathbb{C}}$ are four distinct points, we define their *cross-ratio* as

$$(z_1, z_2, z_3, z_4) := \frac{z_1 - z_3}{z_1 - z_4} : \frac{z_2 - z_3}{z_2 - z_4} = \frac{(z_1 - z_3)(z_2 - z_4)}{(z_1 - z_4)(z_2 - z_3)}.$$

This has to be interpreted as an appropriate limit if one of the points is equal to $\infty \in \widehat{\mathbb{C}}$; for example,

$$(\infty, z_2, z_3, z_4) := \lim_{z \rightarrow \infty} (z, z_2, z_3, z_4) = \frac{z_2 - z_4}{z_2 - z_3}.$$

Note that the map

$$z \mapsto (z, z_2, z_3, z_4) = \frac{(z - z_3)(z_2 - z_4)}{(z - z_4)(z_2 - z_3)}$$

is a Möbius transformation. The values of this Möbius transformation for $z = z_2, z_3, z_4$ have to be interpreted as appropriate limits. Hence this Möbius transformation maps z_2 to 1, z_3 to 0, and z_4 to ∞ .

17.8. Möbius transformations preserve cross-ratios. If $S \in \text{Möb}$ and $z_1, z_2, z_3, z_4 \in \widehat{\mathbb{C}}$ are distinct points, then

$$(S(z_1), S(z_2), S(z_3), S(z_4)) = (z_1, z_2, z_3, z_4).$$

Proof. The maps $z \mapsto (z, z_2, z_3, z_4)$ and $z \mapsto (S(z), S(z_2), S(z_3), S(z_4))$ are Möbius transformations. The first maps z_2 to 1, z_3 to 0, and z_4 to ∞ . The second map has the same behavior as is indicated by the following table:

$$\begin{array}{llll} z & \mapsto w = S(z) & \mapsto (w, S(z_2), S(z_3), S(z_4)), \\ z_2 & \mapsto S(z_2) & \mapsto 1, \\ z_3 & \mapsto S(z_3) & \mapsto 0, \\ z_4 & \mapsto S(z_4) & \mapsto \infty. \end{array}$$

So these Möbius transformations agree on z_2, z_3, z_4 . It follows that they are the same (see 17.6), and so give the same value for $z = z_1$. $\square$

17.9. Möb acts triply transitive on $\widehat{\mathbb{C}}$. If $z_1, z_2, z_3 \in \widehat{\mathbb{C}}$ are three distinct points, and $w_1, w_2, w_3 \in \widehat{\mathbb{C}}$ are three distinct points, then there exists a unique Möbius transformation S with $S(z_k) = w_k$ for $k = 1, 2, 3$.

Proof. Uniqueness follows from 17.6.

To establish existence, let $U, V \in \text{Möb}$ be given by

$$U(z) = (z, z_1, z_2, z_3) \quad \text{and} \quad V(z) = (z, w_1, w_2, w_3)$$

for $z \in \widehat{\mathbb{C}}$. These maps have mapping properties as indicated by the following table:

U	V
$z_1 \mapsto 1,$	$w_1 \mapsto 1,$
$z_2 \mapsto 0,$	$w_2 \mapsto 0,$
$z_3 \mapsto \infty,$	$w_3 \mapsto \infty.$

Then $S := V^{-1} \circ U \in \text{Möb}$ has the desired mapping behavior:

U	V^{-1}
$z_1 \mapsto 1$	$\mapsto w_1,$
$z_2 \mapsto 0$	$\mapsto w_2,$
$z_3 \mapsto \infty$	$\mapsto w_3.$

□

The map S in the last proof satisfies $V(S(z)) = U(z)$, i.e.,

$$(S(z), w_2, w_3, w_4) = (z, z_2, z_3, z_4).$$

This equation can be used to compute $S(z)$ explicitly.

Example 17.10. How to find the Möbius transformation S that maps $1, i, -i$, to $-1, 1, \infty$, respectively?

Solution: The map $z \mapsto w = S(z)$ satisfies

$$(z, 1, i, -i) = (w, -1, 1, \infty),$$

or equivalently,

$$\frac{(z-i)(1+i)}{(z+i)(1-i)} = \frac{“(w-1)(-1-\infty)”}{(w-\infty)(-1-1)} = \frac{1-w}{2}.$$

Of course, the term in the middle enclosed in “...” has a purely symbolic meaning and has to be interpreted as an appropriate limit given by the last term. Solving for w gives (after some computation)

$$w = S(z) = \frac{-(1+2i)z + (-1+2i)}{z+1}.$$

17.11. Circles in $\widehat{\mathbb{C}}$. A *circle* (in $\widehat{\mathbb{C}}$) is either a Euclidean circle in $\mathbb{C}$ in the usual sense or a line (“circle through ∞ ”). Three distinct points in $\widehat{\mathbb{C}}$ determine a unique circle on which they lie.

The circles in $\widehat{\mathbb{C}}$ are precisely the curves whose points $z \in \mathbb{C}$ can be characterized by an equation of the form

$$\alpha z\bar{z} + \beta z + \bar{\beta}\bar{z} + \gamma = 0, \quad (17.3)$$

where $\alpha, \gamma \in \mathbb{R}$, $\beta \in \mathbb{C}$, and $|\beta|^2 > \alpha\gamma$.

To see this, note that every equation of the form (17.3) represents a circle. Indeed, there are two cases.

Case 1. $\alpha = 0$. Then $|\beta|^2 > \alpha\gamma = 0$, and so $\beta \neq 0$. Moreover, if $\beta = a+ib$ and $z = x+iy$, where $a, b, x, y \in \mathbb{R}$, then a and b cannot both be equal to 0, and (17.3) is equivalent to

$$2ax - 2by + \gamma = 0.$$

This equation represents a line.

Case 2. $\alpha \neq 0$. Then $\alpha = 1$ without loss of generality, and (17.3) can be written as

$$\begin{aligned} 0 &= z\bar{z} + \beta z + \bar{\beta}\bar{z} + \gamma = (z + \bar{\beta})(\bar{z} + \beta) + \gamma - |\beta|^2 \\ &\Leftrightarrow |z - (-\bar{\beta})|^2 = |\beta|^2 - \gamma > 0. \end{aligned}$$

This represents a circle of radius $R = (|\beta|^2 - \gamma)^{1/2}$ centered at $-\bar{\beta}$.

Conversely, every equation for a circle or a line in $\mathbb{C}$ can be given the form (17.3) (exercise!).

17.12. Möbius transformations map circles to circles. If $S \in \text{Möb}$, and $C \subseteq \widehat{\mathbb{C}}$ is a circle, then $S(C) \subseteq \widehat{\mathbb{C}}$ is a circle.

Proof. Since we know that the group Möb is generated by translations, dilations, rotations, and the inversion $z \mapsto I(z) := 1/z$ (see 17.4), it is enough to show the statement for these types of maps.

Since the assertion is clear for translations, dilations, and rotations, it suffices to prove it for the inversion I . Let $C \subseteq \widehat{\mathbb{C}}$ be a circle, and suppose it is represented by an equation of the form (17.3). Then

$$\begin{aligned} z \in C &\Leftrightarrow \alpha z\bar{z} + \beta z + \bar{\beta}\bar{z} + \gamma = 0 \\ &\Leftrightarrow \alpha + \frac{\beta}{\bar{z}} + \frac{\bar{\beta}}{z} + \frac{\gamma}{z\bar{z}} = 0 \\ &\Leftrightarrow \alpha + \bar{\beta}w + \beta\bar{w} + \gamma w\bar{w} = 0 \quad \text{with } w = I(z) = 1/z, \\ &\Leftrightarrow w \in \tilde{C}, \end{aligned}$$

where $\tilde{C} \subseteq \widehat{\mathbb{C}}$ is the circle represented by the equation

$$\gamma w\bar{w} + \bar{\beta}w + \beta\bar{w} + \alpha = 0.$$

This is equivalent to $I(C) = \tilde{C}$. □

17.13. Cross-ratios and points on a circle. Four distinct points z_1, z_2, z_3, z_4 in $\widehat{\mathbb{C}}$ lie on a circle if and only if $(z_1, z_2, z_3, z_4) \in \mathbb{R}$.

Proof. Let C be the unique circle with $z_2, z_3, z_4 \in C$, and $S \in \text{Möb}$ be given by $S(z) = (z, z_2, z_3, z_4)$. Then $S(C)$ is a circle containing $S(z_2) = 1$, $S(z_3) = 0$, and $S(z_4) = \infty$. Hence $S(C) = \mathbb{R} \cup \{\infty\} =: \widehat{\mathbb{R}}$. Note that for $u \in \mathbb{C}$ we have

$$(u, 1, 0, \infty) = \frac{(u-0)(1-\infty)}{(u-\infty)(1-0)} = u.$$

Using this and the invariance of cross-ratios under Möbius transformations, we see that z_1, z_2, z_3, z_4 lie on a circle iff $z_1 \in C$ iff $S(z_1) \in S(C) = \widehat{\mathbb{R}}$ iff

$$S(z_1) = (S(z_1), 1, 0, \infty) = (S(z_1), S(z_2), S(z_3), S(z_4)) \in \mathbb{R}$$

iff $(z_1, z_2, z_3, z_4) \in \mathbb{R}$. □

Lemma 17.14. Let $S \in \text{Möb}$. Then $S(\widehat{\mathbb{R}}) = \widehat{\mathbb{R}}$ if and only if S has the form

$$S(z) = \frac{az + b}{cz + d}, \quad \text{where } a, b, c, d \in \mathbb{R}, \quad ad - bc \neq 0. \quad (17.4)$$

Proof. $\Leftarrow$: If S can be written in the form (17.4), then S maps point in $\widehat{\mathbb{R}}$ to points in $\widehat{\mathbb{R}}$. This is only possible if $S(\widehat{\mathbb{R}}) = \widehat{\mathbb{R}}$.

$\Rightarrow$: Conversely, suppose S can be written as in (17.4), let $z \in \widehat{\mathbb{C}}$, and $w = S(z)$. Then

$$z = (z, 1, 0, \infty) = (w, S(1), S(0), S(\infty)) = \frac{\alpha w + \beta}{\gamma w + \delta},$$

where $\alpha, \beta, \gamma, \delta \in \mathbb{C}$, $\alpha\delta - \beta\gamma \neq 0$. Since $S(1), S(0), S(\infty) \in \widehat{\mathbb{R}}$, we actually have $\alpha, \beta, \gamma, \delta \in \mathbb{R}$. Solving this for $w = S(z)$ in terms of z , we get an expression for $S(z)$ as in (17.4). $\square$

Lemma 17.15. *Let $S \in \text{Möb}$ with $S(\widehat{\mathbb{R}}) = \widehat{\mathbb{R}}$. Then $S(\bar{z}) = \overline{S(z)}$ for $z \in \widehat{\mathbb{C}}$ (here $\infty := \infty$).*

Proof. This immediately follows from Lemma 17.14. $\square$

17.16. Points symmetric with respect to a circle. Let $C \subseteq \widehat{\mathbb{C}}$ be a circle, and $z, z^* \in \widehat{\mathbb{C}}$. We say that z and z^* are *symmetric with respect to C* if the following condition is true: Pick $U \in \text{Möb}$ such that $U(C) = \widehat{\mathbb{R}}$. Then the requirement is that

$$U(z^*) = \overline{U(z)} \quad \text{which is equivalent to} \quad U(z) = \overline{U(z^*)}. \quad (17.5)$$

Note that a Möbius transformation U with $U(C) = \widehat{\mathbb{R}}$ exists, because we can send three distinct points on C to $0, 1, \infty$ by a Möbius transformation. Then $U(C) = \widehat{\mathbb{R}}$.

Note also that condition (17.5) is independent of the Möbius transformation we choose. To see this, suppose that V is another Möbius transformation with $V(C) = \widehat{\mathbb{R}}$. If $S := V \circ U^{-1} \in \text{Möb}$, then $S(\widehat{\mathbb{R}}) = \widehat{\mathbb{R}}$, and so $S(\bar{w}) = \overline{S(w)}$ for all $w \in \widehat{\mathbb{C}}$. So if (17.5) is true, then we also have

$$\begin{aligned} V(z^*) &= (V \circ U^{-1})(U(z^*)) = S(\overline{U(z)}) \\ &= \overline{S(U(z))} = \overline{V(z)}. \end{aligned}$$

17.17. The symmetry principle. Let $S \in \text{Möb}$, and z and z^* be points in $\widehat{\mathbb{C}}$ that are symmetric with respect to a circle $C \subseteq \widehat{\mathbb{C}}$. Then $w = S(z)$ and $w^* = S(z^*)$ are symmetric with respect to the circle $\tilde{C} = S(C)$.

Proof. Pick $U \in \text{Möb}$ such that $U(C) = \widehat{\mathbb{R}}$. Then $U(z^*) = \overline{U(z)}$. Then for $W := U \circ S^{-1} \in \text{Möb}$ we have $W(\tilde{C}) = U(C) = \widehat{\mathbb{R}}$. Moreover,

$$\begin{aligned} W(w^*) &= (U \circ S^{-1})(S(z^*)) = U(z^*) \\ &= \overline{U(z)} = \overline{(U \circ S^{-1})(S(z))} = \overline{W(w)}. \end{aligned}$$

Hence w and w^* are symmetric with respect to $\tilde{C}$. $\square$

Lemma 17.18. *Let $C \subseteq \widehat{\mathbb{C}}$ be a circle, and $z_1, z_2, z_3 \in C$ be distinct points. Then $z \in \widehat{\mathbb{C}}$ and $z^* \in \widehat{\mathbb{C}}$ are symmetric with respect to C if and only if*

$$(z^*, z_1, z_2, z_3) = \overline{(z, z_1, z_2, z_3)}. \quad (17.6)$$

Proof. Note that $u \mapsto (u, z_1, z_2, z_3)$ is a Möbius transformation that maps C onto $\widehat{\mathbb{R}}$ (cf. 17.13). So (17.6) corresponds to (17.5). $\square$

17.19. Geometric description of symmetry with respect to a circle. Let $C \subseteq \widehat{\mathbb{C}}$ be a circle, and z and z^* be points that are symmetric with respect to C .

Case 1: C is a line L . Then we can find a Möbius transformation U that is a composition of a rotation and a translation that maps L to $\widehat{\mathbb{R}}$. If $w = U(z)$ and $w^* = U(z^*)$, then $w^* = \bar{w}$, and so w^* is the image of w under reflection in $\mathbb{R}$. Since U that is a composition of a rotation and a translation, this reflections corresponds to the reflection in the line L . It follows that z^* is the image of z under reflection in L .

Since we can reverse the argument, it follows that two points z^* and z are symmetric with respect to the line $L = C$ if and only if z^* is the image of z under reflection in L .

Case 2: $C \subseteq \mathbb{C}$ is a Euclidean circle. Then a point $w \in C$ satisfies an equation of the type

$$|w - a| = R \Leftrightarrow (w - a)\overline{(w - a)} = R^2 \Leftrightarrow \overline{w - a} = \frac{R^2}{w - a}, \quad (17.7)$$

where $a \in \mathbb{C}$ is the center and $R > 0$ is the radius of C .

We pick three distinct points $z_1, z_2, z_3 \in C$. Then from Lemma 17.18 and

the invariance of cross-ratios under Möbius transformations we conclude that

$$\begin{aligned}
 (z^*, z_1, z_2, z_3) &= \overline{(z, z_1, z_2, z_3)} = (\bar{z}, \bar{z}_1, \bar{z}_2, \bar{z}_3) \\
 &= (\bar{z} - \bar{a}, \bar{z}_1 - \bar{a}, \bar{z}_2 - \bar{a}, \bar{z}_3 - \bar{a}) \quad (\text{by applying } u \mapsto u - \bar{a}) \\
 &= \left(\frac{\bar{z} - \bar{a}}{R^2}, \frac{\bar{z}_1 - \bar{a}}{R^2}, \frac{\bar{z}_2 - \bar{a}}{R^2}, \frac{\bar{z}_3 - \bar{a}}{R^2} \right) \quad (\text{by (17.7)}) \\
 &= \left(\frac{R^2}{\bar{z} - \bar{a}}, z_1 - a, z_2 - a, z_3 - a \right) \quad (\text{by applying } u \mapsto R^2/u) \\
 &= \left(a + \frac{R^2}{z - a}, z_1, z_2, z_3 \right) \quad (\text{by applying } u \mapsto u + a).
 \end{aligned}$$

If we consider the cross-ratio as a function of the first argument with the other arguments fixed, then it is a Möbius transformation and hence a bijection. So we conclude that

$$z^* = a + \frac{R^2}{\bar{z} - \bar{a}} \quad (\text{where } z^* = \infty \text{ if } z = a, \text{ and } z^* = a \text{ if } z = \infty). \quad (17.8)$$

This equation defines a map $z \mapsto z^*$ on $\widehat{\mathbb{C}}$ called the *reflection* in the circle C .

If $z = a + re^{it}$ with $r > 0$ and $t \in \mathbb{R}$, then

$$z^* = a + \frac{R^2}{re^{-it}} = a + \frac{R^2}{r}e^{it}.$$

This implies that z and z^* lie on the ray $S = \{a + \rho e^{it} : \rho > 0\}$ emanating from a , and we have

$$|z - a| \cdot |z^* - a| = R^2.$$

Note that the fixed points of the reflection in the circle C are precisely the points in C .

Again one can reverse the argument, and our discussion shows that two points z and z^* are symmetric with respect to C if and only if z^* is the image of z under the reflection in C .

17.20. Oriented circles. We say that a circle $C \subseteq \widehat{\mathbb{C}}$ is *oriented* if we specify an ordered triple (z_1, z_2, z_3) of distinct points on C . Given an orientation on C we define the *right side* of C as the set

$$\{z \in \widehat{\mathbb{C}} : \text{Im}(z, z_1, z_2, z_3) > 0\}$$

and the *left side* of C as the set

$$\{z \in \widehat{\mathbb{C}} : \operatorname{Im}(z, z_1, z_2, z_3) < 0\}.$$

Note that $\operatorname{Im}(z, z_1, z_2, z_3) = 0$ if and only if $z \in C$. This implies that the left and the right side of C form the two connected components of the complement of C in $\widehat{\mathbb{C}}$.

17.21. Möbius transformations are orientation-preserving. Let $S \in \operatorname{Möb}$ and $C \subseteq \widehat{\mathbb{C}}$ be a circle. Let $\tilde{C} = S(C)$ be the image of C under S . Suppose C is oriented by the three distinct points $z_1, z_2, z_3 \in C$. We orient $\tilde{C}$ by the image points $S(z_1), S(z_2), S(z_3) \in \tilde{C}$. Then S takes the left side of C to the left side of $\tilde{C}$, and consequently the right side of C to the right side of $\tilde{C}$.

Indeed, a point $z \in \widehat{\mathbb{C}}$ belongs to the left side of C if and only if

$$\operatorname{Im}(z, z_1, z_2, z_3) = \operatorname{Im}(S(z), S(z_1), S(z_2), S(z_3)) < 0.$$

This is true if and only if $S(z)$ belongs to the left side of $\tilde{C}$.

17.22. Conformal maps. Let $U, V \subseteq \widehat{\mathbb{C}}$ be open, and $f: U \rightarrow V$ be a map. Then f is called a *conformal map* (from U onto V) if f is a bijection, f is holomorphic on U , and f^{-1} is holomorphic on V . In other words, a conformal map from U onto V is the same as a biholomorphism from U onto V .

If $f: U \rightarrow V$ is a conformal map, and $g = f^{-1}$, then $g(f(z)) = z$ for all $z \in U$. In particular, if $U, V \subseteq \mathbb{C}$, then

$$g'(f(z)) \cdot f'(z) = 1$$

and so $f'(z) \neq 0$ for all $z \in U$. So the derivative of a conformal map does not vanish anywhere (this also follows from Theorem 13.3).

17.23. Conformal maps preserve angles between curves. Let $\Omega \subseteq \mathbb{C}$ be open, and $\gamma_1, \gamma_2: [a, b] \rightarrow \Omega$ be two smooth paths with common initial point $p = \gamma_1(a) = \gamma_2(a)$. If $\gamma_1'(a), \gamma_2'(a) \neq 0$, then we define the *angle between γ_1 and γ_2 at p* as

$$\angle_p(\gamma_1, \gamma_2) := \arg \gamma_2'(a) - \arg \gamma_1'(a).$$

This is equal to the angle between the tangent vectors of γ_1 and γ_2 at p . Intuitively, $\angle_p(\gamma_1, \gamma_2)$ is the angle of a rotation at p that sends the direction of $\gamma_1'(a)$ into the direction of $\gamma_2'(a)$. Note that the order of the paths γ_1 and γ_2 is important here.

Now let $f: \Omega \rightarrow \mathbb{C}$ be a holomorphic function with $f'(p) \neq 0$. Define $q := f(p)$ and $\sigma_k := f \circ \gamma_k$, $k = 1, 2$. Then

$$\angle_q(\sigma_1, \sigma_2) = \angle_p(\gamma_1, \gamma_2). \quad (17.9)$$

To see this, assume that $f'(p) = re^{i\alpha}$. Then

$$\sigma_k'(a) = f'(\gamma_k(a)) \cdot \gamma_k'(a) = f'(p) \cdot \gamma_k'(a) = re^{i\alpha} \gamma_k'(a),$$

and so

$$\arg \sigma_k'(a) = \alpha + \arg \gamma_k'(a)$$

for $k = 1, 2$. Hence

$$\arg \sigma_2'(a) - \arg \sigma_1'(a) = \arg \gamma_2'(a) - \arg \gamma_1'(a)$$

and (17.9) follows.

17.24. Angles between curves on Riemann surfaces. One can define angles between paths on a Riemann surface X similarly as in Subsection 17.23 by using local charts. Since transition maps between charts are conformal maps and hence angle-preserving, this leads to a well-defined notion.

In particular, if γ_1 and γ_2 are suitable path in $\widehat{\mathbb{C}}$ with initial point ∞ , then

$$\angle_\infty(\gamma_1, \gamma_2) := \angle_0(I \circ \gamma_1, I \circ \gamma_2),$$

where $I: \widehat{\mathbb{C}} \rightarrow \widehat{\mathbb{C}}$, $I(z) = 1/z$ is the inversion.

17.25. Möbius transformations preserve angles between paths. Let $S \in \text{Möb}$, and γ_1, γ_2 be smooth paths in $\widehat{\mathbb{C}}$ with initial point $p \in \widehat{\mathbb{C}}$. Let $q = S(p)$. Then

$$\angle_p(\gamma_1, \gamma_2) = \angle_q(S \circ \gamma_1, S \circ \gamma_2).$$

To see this, assume that

$$S(z) = \frac{az + b}{cz + d} \quad \text{with } a, b, c, d \in \mathbb{C}, ad - bc \neq 0.$$

We consider several cases similar to the cases in Example 15.12.

Case 1: $p, q \in \mathbb{C}$. By the considerations in 17.23 it suffices to show that $S'(p) \neq 0$. This is true, since S is a biholomorphism, or more explicitly,

$$S'(p) = \frac{a(cp + d) - c(ap + b)}{(cp + d)^2} = \frac{ad - bc}{(cp + d)^2} \neq 0.$$

Note that $cp + d \neq 0$, because $q = S(p) \in \mathbb{C}$.

Case 2: $p = \infty, q \in \mathbb{C}$. Then we consider $z \mapsto S(1/z) =: T(z)$, and have to check whether $T'(0) \neq 0$. Since $T \in \text{Möb}$, and $T(0) = S(\infty) = q \in \mathbb{C}$, this follows from the considerations in Case 1.

There are two more cases, namely where $p \in \mathbb{C}, q = \infty$, and where $p = q = \infty$. They can be treated in a similar way as in Case 2.

Example 17.26. Let

$$S(z) = \frac{1 - z}{1 + z}.$$

What is the set $S(\mathbb{D})$?

Solution: S has real coefficients; so $S(\widehat{\mathbb{R}}) = \widehat{\mathbb{R}}$ (see 17.14). We have $-1, 1 \in \partial\mathbb{D}$, and $S(-1) = \infty$ and $S(1) = 0$; so $S(\partial\mathbb{D})$ is a line L . Since $\partial\mathbb{D}$ meets $\widehat{\mathbb{R}}$ orthogonally at 1, this line L meets $S(\widehat{\mathbb{R}}) = \widehat{\mathbb{R}}$ orthogonally at $S(1) = 0$ (see 17.25). It follows that L is the imaginary axis. Since $S(0) = 1$, we conclude that $S(\mathbb{D}) = \{w \in \mathbb{C} : \operatorname{Re} w > 0\}$ is the open right half-plane.

17.1 Exercises

Problem 17.1. We call two Möbius transformations S and T (*Möbius*) *conjugate* if there exists $U \in \text{Möb}$ such that

$$T = U \circ S \circ U^{-1}. \quad (17.10)$$

If $S \in \text{Möb}$ and $z_0 \in \mathbb{C}$ is a (finite) fixed point of S , then we call $\lambda_S(z_0) := S'(z_0)$ the *multiplier* of z_0 (for given S).

(a) Show that multipliers are invariant under conjugation in the following sense: Suppose $S, T, U \in \text{Möb}$ satisfy (17.10), $z_0 \in \mathbb{C}$ is a fixed point of S , and $w_1 := U(z_0) \in \mathbb{C}$. Then w_1 is a fixed point of T and we have $\lambda_T(w_1) = \lambda_S(z_0)$.

(b) If $S \in \text{Möb}$ and $z_0 = \infty$ is a fixed point of S , then (a) suggests that we define the multiplier $\lambda_S(\infty)$ as

$$\lambda_S(\infty) := \lambda_{\tilde{S}}(0),$$

where $\tilde{S} = I \circ S \circ I^{-1}$ with $I(z) = 1/z$. Show that with this definition a statement as in (a) is true under the more general assumption $z_0, w_0 \in \hat{\mathbb{C}}$.

Problem 17.2. Let $S \in \text{Möb}$ and $S \neq \text{id}_{\hat{\mathbb{C}}}$. Show that then S is conjugate to a Möbius transformation T of precisely one of the following types:

$$T(z) = \begin{cases} z + 1, & \text{(parabolic type),} \\ \lambda z, & \lambda \in \partial\mathbb{D} \setminus \{1\}, \quad \text{(elliptic type),} \\ \lambda z, & \lambda \in (0, \infty) \setminus \{1\}, \quad \text{(hyperbolic type),} \\ \lambda z, & \lambda \in \mathbb{C} \setminus (\partial\mathbb{D} \cup (0, \infty)), \quad \text{(loxodromic type).} \end{cases}$$

Hint: Consider the fixed points of S and their multipliers (see Problem 17.1).

Note: According to these types, S is called *parabolic*, *elliptic*, *hyperbolic*, or *loxodromic*. In the literature, hyperbolic type is sometimes considered as a subcase of loxodromic type.

Problem 17.3. Let $\text{GL}_2(\mathbb{C})$ be the set of all 2×2 -matrices A with complex coefficients and determinant $\det(A) \neq 0$.

(a) Show that $\text{GL}_2(\mathbb{C})$ is a group with matrix multiplication as a composition on $\text{GL}_2(\mathbb{C})$.

(b) If $A \in \text{GL}_2(\mathbb{C})$, we denote by $S_A \in \text{Möb}$ the Möbius transformation with coefficient matrix A . Show that $A \in \text{GL}_2(\mathbb{C}) \mapsto S_A \in \text{Möb}$ is a surjective group homomorphism.

(c) Show that for $A, B \in \text{GL}_2(\mathbb{C})$ we have $S_A = S_B$ if and only if there exists $\lambda \in \mathbb{C} \setminus \{0\}$ such that $A = \lambda B$.

Note: Based on (b) and (c) one can consider Möb as a “projective version” of $\text{GL}_2(\mathbb{C})$, where two elements are considered the same when they are equal up to a non-zero multiplicative factor. For this reason one often denotes the group Möb as $\text{PGL}_2(\mathbb{C})$, where P indicates this “projective” equivalence.

(d) Let $\text{SL}_2(\mathbb{C})$ be the set of all 2×2 -matrices A with complex coefficients and $\det(A) = 1$. Show that $\text{SL}_2(\mathbb{C})$ is a subgroup of $\text{GL}_2(\mathbb{C})$ that maps onto Möb under the homomorphism in (b).

Note: Based on this fact one also uses the notation $\text{PSL}_2(\mathbb{C})$ for Möb.

(e) What is the *kernel* of the homomorphism $\text{SL}_2(\mathbb{C}) \rightarrow \text{Möb}$, i.e., which elements in $\text{SL}_2(\mathbb{C})$ are mapped to the identity element $\text{id}_{\widehat{\mathbb{C}}} \in \text{Möb}$?

Problem 17.4. Let

$$S(z) = \frac{az + b}{cz + d},$$

be a Möbius transformation, where $a, b, c, d \in \mathbb{C}$ and $ad - bc = 1$. Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ be the coefficient matrix of S and denote by $\text{tr}(A) := a + d$

its *trace*. Show that then S is $\begin{cases} \text{parabolic iff } \text{tr}(A)^2 = 4, \\ \text{elliptic iff } \text{tr}(A)^2 \in [0, 4), \\ \text{hyperbolic iff } \text{tr}(A)^2 \in (4, +\infty), \\ \text{loxodromic iff } \text{tr}(A)^2 \notin [0, +\infty). \end{cases}$

Here these types correspond to the classification given in Problem 17.2.

Problem 17.5. If $S \in \text{Möb}$, then we have the identity

$$S'(z)S'(w) = \left(\frac{S(z) - S(w)}{z - w} \right)^2$$

for all $z, w \in \mathbb{C}$, $z \neq w$. Note that if here $S(z) = \infty$ or $S(w) = \infty$, then one has to interpret the identity as a suitable limit.

(a) Show the identity by direct computation. Hint: you may assume that for the coefficient matrix A of S we have $\det(A) = 1$.

(b) Derive the identity from the fact that S preserves cross-ratios by passing to a suitable limit.

Problem 17.6. Let C_1 and C_2 be two disjoint circles in $\widehat{\mathbb{C}}$.

(a) Show that then there exist a unique pair of distinct points $z, z^* \in \widehat{\mathbb{C}}$ such that z and z^* are symmetric with respect to both circles C_1 and C_2 . Hint: Reduce to the case in which one of the circles is the unit circle $\partial\mathbb{D}$ and the other circle is contained in $\mathbb{D}$ and centered at a point in the interval $[0, 1)$.

(b) Show that there exists $S \in \text{Möb}$ such that both $S(C_1)$ and $S(C_2)$ are circles in $\mathbb{C}$ centered at 0.

Problem 17.7. Let $T \in \text{Möb}$ and f be a holomorphic function on some region in $\mathbb{C}$.

(a) Show that $\mathcal{S}(T) = 0$ for the Schwarzian derivative of T (see Problem 14.9).

(b) Show that

$$\mathcal{S}(T \circ f) = \mathcal{S}(f)$$

wherever both sides are defined.

(c) Show that

$$\mathcal{S}(f \circ T) = (\mathcal{S}(f) \circ T) \cdot (T')^2$$

wherever both sides are defined.

Problem 17.8 (The modular group). The *modular group* is defined to be the group $\Gamma = \text{PSL}_2(\mathbb{Z})$ consisting of all Möbius transformations of the form

$$S(z) = \frac{az + b}{cz + d},$$

where $a, b, c, d \in \mathbb{Z}$ and $ad - bc = 1$.

(a) Show that Γ is indeed a group under composition of Möbius transformations.

(b) Let $T(z) = -1/z$ and $U(z) = z + 1$. Then $U, T \in \Gamma$. Show that T and U generate Γ , that is, each $S \in \Gamma$ can be written as a finite composition of Möbius transformations, where each of these Möbius transformations is equal to $T = T^{-1}$, U or U^{-1} .

Hint: Show that if $S \in \Gamma$ is represented by the coefficient matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$,

where $a, b, c, d \in \mathbb{Z}$ and $ad - bc = 1$, then $T \circ S$ is represented by $\begin{pmatrix} -c & -d \\ a & b \end{pmatrix}$

and $U^k \circ S$ for $k \in \mathbb{Z}$ by $\begin{pmatrix} a+kc & b+kd \\ c & d \end{pmatrix}$. Minimize $|c|$ by repeated applications of T and powers of U .

18 Schwarz's Lemma

Theorem 18.1 (Schwarz's Lemma). *Let $f: \mathbb{D} \rightarrow \mathbb{C}$ be a holomorphic function satisfying $f(0) = 0$ and $|f(z)| \leq 1$ for $z \in \mathbb{D}$. Then*

$$|f(z)| \leq |z| \quad \text{for all } z \in \mathbb{D}, \quad (18.1)$$

and

$$|f'(0)| \leq 1. \quad (18.2)$$

If there is a point $z \in \mathbb{D} \setminus \{0\}$ for which we have equality in (18.1), or if we have equality in (18.2), then there exists $\theta \in \mathbb{R}$ such that

$$f(z) = e^{i\theta} z \quad \text{for all } z \in \mathbb{D}.$$

Proof. Since f has a zero at 0, we can write this function in the form $f(z) = zh(z)$ for $z \in \mathbb{D}$, where $h: \mathbb{D} \rightarrow \mathbb{C}$ is holomorphic (see Theorem 11.3).

Claim: $|h(z)| \leq 1$ for all $z \in \mathbb{D}$.

To see this, let $z_0 \in \mathbb{D}$, and consider an arbitrary number $r > 0$ with $|z_0| < r < 1$. Then h is continuous on $\overline{B}(0, r) \subseteq \mathbb{D}$ and holomorphic on $B(0, r)$. By the Maximum Modulus Principle (see Theorem 12.4), we have

$$|h(z_0)| \leq \max_{z \in \partial B(0, r)} |h(z)| = \max_{z \in \partial B(0, r)} \frac{|f(z)|}{|z|} \leq \frac{1}{r}.$$

Letting $r \rightarrow 1$, we conclude $|h(z_0)| \leq 1$, and the claim follows.

The claim implies that

$$|f(z)| = |zh(z)| \leq |z|$$

for $z \in \mathbb{D}$, and

$$|f'(0)| = |h(0)| \leq 1.$$

So we have established (18.1) and (18.2).

If we have equality in (18.1) for some $z = z_0 \in \mathbb{D} \setminus \{0\}$, then $|h(z_0)| = 1$, and so h must be constant by the Maximum Modulus Principle (see Theorem 12.1). This is also true if $|f'(0)| = 1 = |h(0)|$.

In both cases, $h \equiv e^{i\theta}$ for some $\theta \in \mathbb{R}$, and so $f(z) = e^{i\theta} z$ for $z \in \mathbb{D}$. $\square$

18.2. The automorphisms of a region. Let $\Omega \subseteq \widehat{\mathbb{C}}$ be a region. A conformal map from Ω onto Ω is called a (*conformal*) *automorphism* of Ω . The set of all automorphisms of Ω is denoted by $\text{Aut}(\Omega)$. This is a group under compositions of maps, the so-called *automorphism group* of Ω .

Theorem 18.3 (Automorphisms of the unit disk). *A map $f: \mathbb{D} \rightarrow \mathbb{C}$ is a conformal automorphism of $\mathbb{D}$ if and only if f is a Möbius transformation of the form*

$$f(z) = e^{i\theta} \frac{z - a}{1 - \bar{a}z}, \quad (18.3)$$

where $\theta \in \mathbb{R}$, $a \in \mathbb{D}$.

By this theorem any point $z_0 \in \mathbb{D}$ can be sent to 0 by a suitable Möbius transformation $f \in \text{Aut}(\mathbb{D})$ (take $a = z_0$ in (18.3)). The rotation factor $e^{i\theta}$ can be used to send a second point $w_0 \in \mathbb{D}$ to the positive real axis, and so into $[0, 1)$. In other words, if $z_0, w_0 \in \mathbb{D}$, then there exists $f \in \text{Aut}(\mathbb{D})$ such that $f(z_0) = 0$ and $f(w_0) \in [0, 1)$.

Proof of Theorem 18.3. “ $\Leftarrow$ ” Every $f \in \text{Möb}$ of the form (18.3) is an automorphism of $\mathbb{D}$; indeed,

$$|f(e^{it})| = \left| e^{i\theta} \frac{e^{it} - a}{1 - \bar{a}e^{it}} \right| = \frac{|1 - ae^{-it}|}{|1 - ae^{-it}|} = 1$$

for $t \in \mathbb{R}$. Hence $f(\partial\mathbb{D}) = \partial\mathbb{D}$. Since $f(0) = -e^{i\theta}a \in \mathbb{D}$, we have $f(\mathbb{D}) = \mathbb{D}$. So f maps $\mathbb{D}$ onto $\mathbb{D}$ bijectively. Since f and the inverse of f are Möbius transformations, and hence holomorphic, it follows that f is an automorphism of $\mathbb{D}$.

“ $\Rightarrow$ ” Let $f \in \text{Aut}(\mathbb{D})$ be arbitrary. Then there exists $a \in \mathbb{D}$ such that $f(a) = 0$. Define

$$g(z) = \frac{z + a}{1 + \bar{a}z}.$$

Then $g \in \text{Aut}(\mathbb{D})$ by the first part of the proof. Let $\varphi = f \circ g$. Then $\varphi \in \text{Aut}(\mathbb{D})$, and

$$\varphi(0) = f(g(0)) = f(a) = 0.$$

We also have $\psi := \varphi^{-1} \in \text{Aut}(\mathbb{D})$, and $\psi(0) = 0$. Schwarz's Lemma gives

$$|\varphi'(0)| \leq 1 \quad \text{and} \quad |\psi'(0)| \leq 1.$$

On the other hand, $\psi(\varphi(z)) = z$ for $z \in \mathbb{D}$, and so

$$|\psi'(0)| \cdot |\varphi'(0)| = 1.$$

It follows that

$$|\psi'(0)| = 1 = |\varphi'(0)|,$$

and so by the second part Schwarz's Lemma there exists $\theta \in \mathbb{R}$ such that

$$\varphi(z) = f(h(z)) = e^{i\theta} z$$

for $z \in \mathbb{D}$. Computation gives

$$g^{-1}(z) = \frac{z - a}{1 - \bar{a}z},$$

which implies that

$$f(z) = f(g(g^{-1}(z))) = e^{i\theta} g^{-1}(z) = e^{i\theta} \frac{z - a}{1 - \bar{a}z}.$$

This shows that f can be written as in (18.3). $\square$

18.4. Hyperbolic geometry. Based on Theorem 18.3 one can show that

$$\frac{2|S'(z)|}{1 - |S(z)|^2} = \frac{2}{1 - |z|^2} \quad (18.4)$$

for all $S \in \text{Aut}(\mathbb{D})$ and $z \in \mathbb{D}$ (exercise!).

If $\gamma: [a, b] \rightarrow \mathbb{D}$ is a piecewise smooth path, one defines its *hyperbolic length* as

$$\ell_h(\gamma) = \int_{\gamma} \frac{2|dz|}{1 - |z|^2} := \int_a^b \frac{2|\gamma'(t)|}{1 - |\gamma(t)|^2} dt.$$

Then it follows from (18.4) that

$$\ell_h(S \circ \gamma) = \ell_h(\gamma) \quad (18.5)$$

for all $S \in \text{Aut}(\mathbb{D})$, i.e., the hyperbolic length of a path is invariant under automorphisms of $\mathbb{D}$.

The *hyperbolic metric* on $\mathbb{D}$ is defined by

$$d_h(z, w) := \inf_{\gamma} \ell_h(\gamma) \quad \text{for } z, w \in \mathbb{D}, \quad (18.6)$$

where the infimum is taken over all piecewise smooth paths in $\mathbb{D}$ with endpoints z and w . By (18.5) each $S \in \text{Aut}(\mathbb{D})$ is an isometry of $\mathbb{D}$ equipped with hyperbolic metric.

If $z = 0$ and $w = u \in \mathbb{D}$, then one can show that the (parametrized) segment $[0, u]$ is the unique *hyperbolic geodesic segment*, i.e., the unique (up

to reparametrization) piecewise smooth path γ for which the infimum in (18.6) is attained. Hence

$$\begin{aligned} d_h(0, u) &= \int_0^{|u|} \frac{2dt}{1-t^2} = \int_0^{|u|} \left(\frac{1}{1+t} + \frac{1}{1-t} \right) dt \\ &= (\log(1+t) - \log(1-t)) \Big|_0^{|u|} \\ &= \log \left(\frac{1+|u|}{1-|u|} \right) = 2 \operatorname{artanh} |u|. \end{aligned} \quad (18.7)$$

Using this, the fact that every $S \in \operatorname{Aut}(\mathbb{D})$ is a hyperbolic isometry, and the remark after Theorem 18.3, one easily obtains the explicit expression

$$d_h(z, w) = \log \left(\frac{1 + \left| \frac{z-w}{1-\bar{z}w} \right|}{1 - \left| \frac{z-w}{1-\bar{z}w} \right|} \right) = 2 \operatorname{artanh} \left| \frac{z-w}{1-\bar{z}w} \right| \quad (18.8)$$

for the hyperbolic distance between two points $z, w \in \mathbb{D}$.

The hyperbolic geodesic segment Σ joining two points $z, w \in \mathbb{D}$ is the image of an interval of the form $[0, r]$, $0 \leq r < 1$, under a suitable map $S \in \operatorname{Aut}(\mathbb{D})$. Since $\widehat{\mathbb{R}} \supseteq [0, r]$ meets the unit circle orthogonally and S preserves angles, Σ is contained in a unique circle C that passes through z and w , and meets $\partial\mathbb{D}$ orthogonally. Then Σ is given by the subarc of C in $\mathbb{D}$ with endpoints z and w .

The unit circle $\mathbb{D}$ equipped with the hyperbolic metric is a model of the hyperbolic plane in non-Euclidean geometry, called the *Poincaré model* of the hyperbolic plane.

Theorem 18.5 (Schwarz–Pick Lemma). *Let $f: \mathbb{D} \rightarrow \mathbb{C}$ be a holomorphic function with $f(\mathbb{D}) \subseteq \mathbb{D}$. Then we have*

$$\left| \frac{f(z_1) - f(z_2)}{1 - \overline{f(z_1)}f(z_2)} \right| \leq \left| \frac{z_1 - z_2}{1 - \bar{z}_1 z_2} \right| \quad (18.9)$$

for all $z_1, z_2 \in \mathbb{D}$, and

$$\frac{|f'(z)|}{1 - |f(z)|^2} \leq \frac{1}{1 - |z|^2} \quad (18.10)$$

for all $z \in \mathbb{D}$.

If we have equality in (18.9) for some points $z_1, z_2 \in \mathbb{D}$, $z_1 \neq z_2$, or if we have equality in (18.10) for some $z \in \mathbb{D}$, then $f \in \operatorname{Aut}(\mathbb{D})$.

Remark 18.6. If we define

$$B(z_1, z_2) := \left| \frac{z_1 - z_2}{1 - \overline{z_1}z_2} \right|,$$

then (see (18.8)),

$$d_h(z_1, z_2) = \log \left(\frac{1 + B(z_1, z_2)}{1 - B(z_1, z_2)} \right).$$

Since the function $r \mapsto \log \left(\frac{1+r}{1-r} \right)$ is strictly increasing on $[0, 1)$, inequality (18.9) is equivalent to

$$d_h(f(z_1), f(z_2)) \leq d_h(z_1, z_2).$$

So a holomorphic map $f: \mathbb{D} \rightarrow \mathbb{D}$ does not increase hyperbolic distances of points.

If $f \in \text{Aut}(\mathbb{D})$, then f is a hyperbolic isometry and we have equality in (18.9) for *all* $z_1, z_2 \in \mathbb{D}$, and also in (18.10) for *all* $z \in \mathbb{D}$ by (18.4).

Proof of Theorem 18.5 (Outline). The idea is to reduce the statements to corresponding statements in Schwarz's Lemma.

Let $z_1, z_2 \in \mathbb{D}$ be arbitrary, and define $w_1 = f(z_1)$, $w_2 = f(z_2)$,

$$S(z) = \frac{z - z_1}{1 - \overline{z_1}z} \quad \text{and} \quad T(w) = \frac{w - w_1}{1 - \overline{w_1}w}.$$

Then $S, T \in \text{Aut}(\mathbb{D})$, $S(z_1) = 0$, and $T(w_1) = 0$.

Define $g = T \circ f \circ S^{-1}$. Then $g \in H(\mathbb{D})$, $g(0) = 0$, and $g(\mathbb{D}) \subseteq \mathbb{D}$. So Schwarz's Lemma gives

$$|g(u)| \leq |u| \tag{18.11}$$

for all $u \in \mathbb{D}$, in particular for $u = S(z_2)$. This translates into (18.9). Similarly, we have

$$|g'(0)| \leq 1, \tag{18.12}$$

which after some computation gives (18.12) (for $z = z_1$).

In case of (non-trivial) equality in (18.9), or of equality in (18.10), we have equality in (18.11) or (18.12). Then $g(z) \equiv e^{i\theta}z$ for some $\theta \in \mathbb{R}$, and so $g \in \text{Aut}(\mathbb{D})$. Then $f = T^{-1} \circ g \circ S \in \text{Aut}(\mathbb{D})$. $\square$

18.7. Conformal metrics. Let $U \subseteq \mathbb{C}$ be a region, and $\rho: U \rightarrow [0, \infty)$ be a continuous function such that $\rho^{-1}(0)$ consists of isolated points in U (the “singularities” of ρ). Given such a *conformal density*, one defines the ρ -length of a piecewise smooth path $\gamma: [a, b] \rightarrow U$ by

$$\ell_\rho(\gamma) = \int_\gamma \rho(z) |dz| := \int_a^b \rho(\gamma(t)) |\gamma'(t)| dt,$$

and an associated *conformal metric* given by

$$d_\rho(z, w) = \inf_\gamma \ell_\rho(\gamma) \quad \text{for } z, w \in U,$$

where the infimum is taken over all piecewise smooth paths γ in U joining z and w .

Then d_ρ is a metric on U that induces the standard topology (to prove this one needs that $\rho^{-1}(0)$ consists of isolated points in U). One says that this metric is given by the (symbolic) *length element*

$$ds = \rho(z) |dz|.$$

For example, the hyperbolic metric on $\mathbb{D}$ is induced by the length element

$$ds = \rho_h(z) |dz|,$$

where

$$\rho_h(z) := \frac{2}{1 - |z|^2}, \quad z \in \mathbb{D},$$

is the *hyperbolic density*.

18.8. Pull-backs of conformal metrics. Let $U, V \subseteq \mathbb{C}$ be regions, $f: U \rightarrow V$ be a non-constant holomorphic map, and $ds = \rho(w) |dw|$ be a conformal metric on V . Here we use w to denote points in V , and z for points in U . Then the *pull-back* $f^*(ds)$ of ds by the map f (given by $z \mapsto w = f(z)$) is the conformal metric on U with length element

$$\begin{aligned} f^*(ds) &= \rho(w) |dw| \\ &= \rho(f(z)) \left| \frac{dw}{dz} \right| |dz| \\ &= \rho(f(z)) |f'(z)| |dz|. \end{aligned}$$

Of course, this “computation” is purely symbolic and is intended to motivate the definition of the length element in the last line. Note that this pull-back operation is functorial in the sense that

$$(g \circ f)^*(ds) = f^*(g^*(ds))$$

for compositions of holomorphic maps (this immediately follows from the chain rule).

If the map f is conformal, then f is an isometry between V equipped with the conformal metric given by ds and U equipped with $f^*(ds)$.

If we have two conformal metrics

$$ds_1 = \rho_1(z) |dz| \quad \text{and} \quad ds_2 = \rho_2(z) |dz|$$

on U , we write $ds_1 \leq ds_2$ iff $\rho_1 \leq \rho_2$. Then the Schwarz–Pick Lemma is equivalent to the statement

$$f^*(ds_h) \leq ds_h, \tag{18.13}$$

where $ds_f = \rho_h(z) |dz|$ is the hyperbolic metric on $\mathbb{D}$, and $f: \mathbb{D} \rightarrow \mathbb{D}$ is a holomorphic map; indeed, (18.13) is the same as (18.10), from which one can derive (18.11) by integration.

Example 18.9. (a) Let

$$w = f(z) = \frac{1 + iz}{1 - iz}.$$

Then f is a Möbius transformation that maps the upper half-plane $H = \{z \in \mathbb{C} : \operatorname{Im} z > 0\}$ onto $\mathbb{D}$. If we pull-back the hyperbolic metric $ds_h = \rho_h(w) |dw|$ on $\mathbb{D}$ by f to H , then we get

$$f^*(ds_h) = \frac{|dz|}{\operatorname{Im} z}.$$

This conformal metric on H gives an isometric model of the hyperbolic plane, called the *upper half-plane model*.

(b) One can represent the unit sphere in $\mathbb{R}^3$ equipped with the spherical metric d_s (given by geodesic length between points) as a conformal metric on $\hat{\mathbb{C}}$, namely by

$$ds = \frac{2|dz|}{1 + |z|^2}$$

(suitably interpreted at $z = \infty$). One can show that then

$$d_s(z, w) = 2 \arctan \left| \frac{z - w}{1 + \bar{z}w} \right|$$

for $z, w \in \widehat{\mathbb{C}}$. Note that this expression is very similar to (18.8).

18.1 Exercises

Problem 18.1. Let $\text{Aut}(\mathbb{D})$ be the set of all conformal automorphisms of the unit disk $\mathbb{D}$, i.e., the set of all biholomorphisms $S: \mathbb{D} \rightarrow \mathbb{D}$.

(a) Show that if $S \in \text{Aut}(\mathbb{D})$, then

$$\frac{|S'(z)|}{1 - |S(z)|^2} = \frac{1}{1 - |z|^2} \text{ for } z \in \mathbb{D}.$$

(b) Let $\gamma: [a, b] \rightarrow \mathbb{D}$ be a piecewise smooth path in $\mathbb{D}$. Define its *hyperbolic length* as

$$\ell_h(\gamma) := \int_{\gamma} \frac{2|dz|}{1 - |z|^2} = \int_a^b \frac{2|\gamma'(t)|}{1 - |\gamma(t)|^2} dt.$$

Show that if $S \in \text{Aut}(\mathbb{D})$, then

$$\ell_h(S \circ \gamma) = \ell_h(\gamma).$$

(c) Define the *hyperbolic distance* between two points $z, w \in \mathbb{D}$ as

$$d_h(z, w) := \inf_{\gamma} \ell_h(\gamma),$$

where the infimum is taken over all piecewise smooth paths γ in $\mathbb{D}$ with endpoints z and w . Show that d_h is a metric on $\mathbb{D}$, and that each $S \in \text{Aut}(\mathbb{D})$ is an isometry on the space $(\mathbb{D}, d_h)$, i.e., we have

$$d_h(S(z), S(w)) = d_h(z, w) \text{ for all } z, w \in \mathbb{D}.$$

Problem 18.2. (a) Show that a circle C intersects the unit circle $\partial\mathbb{D}$ orthogonally if and only if either C is a line passing through 0 or $C = \{z \in \mathbb{C} : |z - a| = r\}$, where $a \in \mathbb{C}$, $|a| > 1$, and $r = \sqrt{|a|^2 - 1}$.

(b) Suppose $z, w \in \mathbb{D}$ are distinct points. Show that there exists a unique map $S \in \text{Aut}(\mathbb{D})$ such that $S(z) = 0$ and $S(w) \in (0, 1)$. Express $S(w)$ in terms of z and w .

(c) Show that there exists a unique circle C that passes through z and w and intersects the unit circle $\partial\mathbb{D}$ orthogonally.

Problem 18.3. Let $r \in (0, 1)$ and $\gamma: [0, 1] \rightarrow \mathbb{D}$ be a piecewise smooth path in $\mathbb{D}$ with $\gamma(0) = r$ and $\gamma(1) = 0$.

(a) Define

$$s(t) := \int_0^t |\gamma'(u)| du \text{ for } t \in [0, 1].$$

Show that

$$|\gamma(t)| \geq \max\{0, r - s(t)\} \text{ for } t \in [0, 1].$$

(b) Show that

$$\ell_h(\gamma) \geq \int_0^r \frac{2ds}{1 - (r - s)^2}.$$

(c) Show that the hyperbolic distance between 0 and r is equal to the hyperbolic length of the line segment joining 0 and r . Compute this length explicitly.

Problem 18.4. Let $z, w \in \mathbb{D}$ be two distinct points, C the unique circle that passes through z and w and intersects the unit circle $\partial\mathbb{D}$ orthogonally, and A the unique subarc of C contained in $\mathbb{D}$ with endpoints z and w .

(a) Find an explicit expression for the hyperbolic distance $d_h(z, w)$.

(b) Show that A is a *hyperbolic geodesic segment* connecting z and w , i.e., a path in $\mathbb{D}$ with endpoints z and w whose hyperbolic length is equal to the hyperbolic distance of z and w .

(c) The circle C intersects $\partial\mathbb{D}$ in two points u, v . Suppose the notation is chosen such that the points z, w, u, v are in cyclic order on C . Show that

$$d_h(z, w) = \log(z, w, u, v).$$

Problem 18.5. Let $\Omega \subseteq \mathbb{C}$ be a region, and $\varrho: \Omega \rightarrow (0, \infty)$ be a C^2 -smooth conformal density. The *Gaussian curvature* $\kappa_\varrho(z)$ of the underlying conformal metric $ds = \varrho(z)|dz|$ at $z \in \Omega$ is defined as

$$\kappa_\varrho(z) = -\frac{(\Delta \log \varrho)(z)}{\varrho(z)^2}.$$

(a) Show that if $\varrho_h: \mathbb{D} \rightarrow (0, \infty)$ is the hyperbolic density on $\mathbb{D}$ given by

$$\varrho_h(z) = \frac{2}{1 - |z|^2}, \quad z \in \mathbb{D},$$

then $\kappa_{\varrho_h}(z) = -1$ for all $z \in \mathbb{D}$.

(b) Show the following invariance property of the Gaussian curvature under conformal change of coordinates: Suppose $f: \tilde{\Omega} \rightarrow \Omega$ is a conformal map between two regions $\Omega, \tilde{\Omega} \subseteq \mathbb{C}$, $\varrho: \Omega \rightarrow (0, \infty)$ a C^2 -smooth conformal density, and $\tilde{\varrho}$ the pull-back of ϱ by f , i.e.,

$$\tilde{\varrho}(z) = \varrho(f(z))|f'(z)|, \quad z \in \tilde{\Omega}.$$

Then

$$\kappa_{\tilde{\varrho}}(z) = \kappa_{\varrho}(f(z))$$

for all $z \in \tilde{\Omega}$. Hint: Express the Laplace operator Δ in terms of z - and $\bar{z}$ -derivatives.

Problem 18.6. Prove the following version of the *Ahlfors–Schwarz Lemma*: If $\varrho: \mathbb{D} \rightarrow (0, \infty)$ is a C^2 -smooth conformal density with Gaussian curvature satisfying $\kappa_{\varrho}(z) \leq -1$ for each $z \in \mathbb{D}$, then $\rho(z) \leq \rho_h(z)$ for $z \in \mathbb{D}$, where ρ_h is the hyperbolic density on $\mathbb{D}$.

Hint: First assume that ϱ is bounded on $\mathbb{D}$. Argue by contradiction and investigate a point where $\log \varrho - \log \rho_h$ attains a positive maximum.

Problem 18.7. (a) Suppose $L: \mathbb{D} \rightarrow \mathbb{C}$ is a holomorphic function such that $L(0) = 1$ and $\operatorname{Re}(L(z)) > 0$ for all $z \in \mathbb{D}$. Show that $|L'(0)| \leq 2$.

(b) Suppose $f: \mathbb{D} \rightarrow \mathbb{C}$ is a holomorphic function with $0 < |f(z)| < 1$ for all $z \in \mathbb{D}$. Show that $|f'(0)| \leq 2/e$.

19 Winding numbers

Remark 19.1. We want to define an integer that counts how often a loop $\gamma: [0, 1] \rightarrow \mathbb{C}$ winds around a given point $z \in \mathbb{C}$, say $z = 0$. This only makes sense if $0 \notin \gamma^*$. The idea is to look at the total change of the polar angle if we move along the loop γ . For this we want to write γ in the form $\gamma(t) = e^{\alpha(t)}$, $t \in [0, 1]$, where $\alpha: [0, 1] \rightarrow \mathbb{C}$ is continuous. Then the imaginary part of α corresponds to the polar angle of points on γ , and we define the winding number of γ around 0 as

$$\text{winding number} = \frac{1}{2\pi}(\text{Im } \alpha(1) - \text{Im } \alpha(0)). \quad (19.1)$$

The first problem that one has to address here is the existence of α . We will study this problem in greater generality based on the following definition.

Let Z be a metric space, and $f: Z \rightarrow \mathbb{C}^*$ be a continuous map. A continuous map $\tilde{f}: Z \rightarrow \mathbb{C}$ is called a *lift* of f (under the exponential map) if $f = \exp \circ \tilde{f}$, i.e., if the diagram

$$\begin{array}{ccc} & & \mathbb{C} \\ & \nearrow \tilde{f} & \downarrow \exp \\ Z & \xrightarrow{f} & \mathbb{C}^* \end{array}$$

commutes. So finding a continuous function α as above is the same as finding a lift of the loop γ .

Remark 19.2. The map $\exp: \mathbb{C} \rightarrow \mathbb{C}^*$ is an example of a covering map. By definition a continuous map $p: X \rightarrow Y$ between topological spaces X and Y is a *covering map* if each point $y \in Y$ has an open neighborhood V with the following property: there exists an index set I such that $p^{-1}(V)$ can be written as a union

$$p^{-1}(V) = \bigcup_{i \in I} U_i$$

of pairwise disjoint open sets $U_i \subseteq X$ such that $p|_{U_i}$ is a homeomorphism from U_i onto V for each $i \in I$. Such a neighborhood V is called a *fundamental neighborhood* or an *evenly covered neighborhood* of the point y .

To see that $\exp: \mathbb{C} \rightarrow \mathbb{C}^*$ is indeed a covering map, let $w_0 \in \mathbb{C}^*$ be arbitrary. Then $w_0 = re^{i\alpha}$ for some $r > 0$ and $\alpha \in \mathbb{R}$. If we define

$$V = \{\rho e^{it} : \rho > 0, \alpha - \pi < t < \alpha + \pi\} = \mathbb{C} \setminus \{\rho e^{i(\alpha+\pi)} : \rho \geq 0\},$$

then V is an evenly covered neighborhood of w_0 . Indeed, V is an open set containing w_0 , and so a neighborhood of w_0 . Moreover,

$$\exp^{-1}(V) = \bigcup_{k \in \mathbb{Z}} U_k,$$

where

$$U_k = \{z \in \mathbb{C} : \alpha + (2k - 1)\pi < \operatorname{Im} z < \alpha + (2k + 1)\pi\}$$

for $k \in \mathbb{Z}$. The sets U_k , $k \in \mathbb{Z}$, are open and pairwise disjoint, and for each $k \in \mathbb{Z}$ the restriction $\exp|_{U_k}: U_k \rightarrow V$ is a holomorphic bijection of U_k onto V . Then $\exp|_{U_k}$ has a holomorphic, and in particular continuous inverse which shows that $\exp|_{U_k}$ is a homeomorphism of U_k onto V .

Note that $V' = B(w_0, |w_0|) \subseteq V$ is also an evenly covered neighborhood of w_0 .

Proposition 19.3. *Let Z be a compact and connected metric space, and $f: Z \rightarrow \mathbb{C}^*$ be a continuous map.*

- (a) *Suppose that $\tilde{f}_1$ and $\tilde{f}_2$ are lifts of f and that there exists a point $z_0 \in Z$ such that $\tilde{f}_1(z_0) = \tilde{f}_2(z_0)$. Then $\tilde{f}_1 = \tilde{f}_2$.*
- (b) *Let $z_0 \in Z$, define $w_0 = f(z_0)$, and suppose that*

$$\operatorname{diam}(f(Z)) < \operatorname{dist}(0, f(Z)).$$

If $u_0 \in \exp^{-1}(w_0)$, then there exists a unique lift $\tilde{f}$ of f with $\tilde{f}(z_0) = u_0$.

The first statement says that under the given assumptions, lifts are uniquely determined by the image of one point.

Note that in (b) we have $\operatorname{dist}(0, f(Z)) > 0$, because $f(Z)$ is a compact subset of $\mathbb{C}^*$. The statement in (b) essentially says that lifts exist if the image $f(Z)$ is sufficiently small (depending on $\operatorname{dist}(0, f(Z))$).

Proof. (a) We have

$$\exp(\tilde{f}_1 - \tilde{f}_2) = \exp(\tilde{f}_1) / \exp(\tilde{f}_2) = f / f = 1.$$

Since Z is connected, we conclude $\tilde{f}_1 - \tilde{f}_2 \equiv \text{const.}$ (see Subsection 14.2 for a similar reasoning). By assumption $\tilde{f}_1(z_0) - \tilde{f}_2(z_0) = 0$, and so $\tilde{f}_1 = \tilde{f}_2$.

(b) Since $u_0 \in \exp^{-1}(w_0)$ and $V = B(w_0, |w_0|)$ is an evenly covered neighborhood of w_0 for the covering map $\exp: \mathbb{C} \rightarrow \mathbb{C}^*$, there exists a local

inverse of $\exp$ on V , i.e., a branch L of the logarithm on V such that $L(w_0) = u_0$. Note that for each $z \in Z$ we have

$$\begin{aligned} |f(z) - w_0| &= |f(z) - f(z_0)| \leq \text{diam}(f(Z)) \\ &< \text{dist}(0, f(Z)) \leq |f(z_0)| = |w_0|. \end{aligned}$$

Hence $f(Z) \subseteq B(w_0, |w_0|)$. So $\tilde{f} := L \circ f$ is defined. It is immediate that $\tilde{f}$ is a lift of f with $\tilde{f}(z_0) = u_0$. Moreover, by (a) this lift is unique. $\square$

Lemma 19.4. *Let X and Y be metric spaces, and $f: X \rightarrow Y$ be a map. Suppose that there are finitely many closed sets $F_1, \dots, F_n \subseteq X$ such that $f|_{F_k}$ is continuous for each $k \in \{1, \dots, n\}$ and $X = \bigcup_{k=1}^n F_k$. Then f is continuous.*

Proof. From the assumptions it easily follows that if $A \subseteq Y$ is closed, then $f^{-1}(A)$ is also closed. Hence f is continuous (see Proposition 2.22). $\square$

Theorem 19.5 (Existence of lifts). (a) *Let $\gamma: [0, 1] \rightarrow \mathbb{C}^*$ be a continuous path, and $u_0 \in \exp^{-1}(\gamma(0))$. Then there exists a unique lift $\tilde{\gamma}: [0, 1] \rightarrow \mathbb{C}$ of γ with $\tilde{\gamma}(0) = u_0$. If γ is piecewise smooth, then $\tilde{\gamma}$ is also piecewise smooth.*

(b) *Let $H: [0, 1] \times [0, 1] \rightarrow \mathbb{C}^*$ be continuous and $u_0 \in \exp^{-1}(H(0, 0))$. Then there exists a unique lift $\tilde{H}: [0, 1] \times [0, 1] \rightarrow \mathbb{C}$ of H such that $\tilde{H}(0, 0) = u_0$.*

The first part of (a) is true in greater generality for covering maps and called the “path lifting property” of such maps. The statement in (b) can be considered as a special case of the “Homotopy Lifting Theorem” for covering maps.

Proof (Outline). (a) One can find a fine partition of $[0, 1]$ given by points $t_0 = 0 < t_1 < \dots < t_n = 1$ such that each of the paths $\gamma_k := \gamma|_{[t_{k-1}, t_k]}$ has small enough image so that we can apply Proposition 19.3 (b). Then we successively lift the paths $\gamma_1, \dots, \gamma_n$ to paths $\tilde{\gamma}_1, \dots, \tilde{\gamma}_n$. We can do this in such a way that $\tilde{\gamma}_1(0) = u_0$, and so that the endpoint $\tilde{\gamma}_k(t_k)$ of a lift $\tilde{\gamma}_k$ coincides with the initial point $\tilde{\gamma}_{k+1}(t_k)$ of the next lift $\tilde{\gamma}_{k+1}$.

By Lemma 19.4 these lifts $\tilde{\gamma}_k$ paste together to a continuous map $\tilde{\gamma}: [0, 1] \rightarrow \mathbb{C}$ such that $\gamma|_{[t_{k-1}, t_k]} = \tilde{\gamma}_k$. Then $\exp \circ \tilde{\gamma} = \gamma$ and $\tilde{\gamma}(0) = \tilde{\gamma}_1(0) = u_0$, and

so $\tilde{\gamma}$ is a lift of γ with the desired properties. It is unique as follows from Proposition 19.3 (a).

If γ is piecewise smooth, then we can choose the partition so that the paths γ_k are (C^1) -smooth. Since each lift $\tilde{\gamma}_k$ can be obtained by composing a branch of the logarithm with γ_k , the lift $\tilde{\gamma}_k$ is also smooth. Hence $\tilde{\gamma}$ is piecewise smooth.

(b) The basic idea for the proof is the same as in (a). We subdivide $[0, 1] \times [0, 1]$ into small squares Q so that we can apply Proposition 19.3 (b) to each map $H|_Q$. We can ensure that $(0, 0)$ is mapped to u_0 by a suitable lift. Moreover, based on Proposition 19.3 (a) one can obtain lifts of the maps $H|_Q$ so they match along common boundary points of squares Q . Again by Lemma 19.4 these lifts will paste together to a lift $\tilde{H}$ of H with $\tilde{H}(0, 0) = u_0$. Uniqueness of $\tilde{H}$ follows from Proposition 19.3 (a). The details are left as an exercise. $\square$

Definition 19.6 (Index of a loop with respect to a point). Let $\gamma: [0, 1] \rightarrow \mathbb{C}$ be a loop with $0 \notin \gamma^*$. Then we define the *index of γ with respect to 0* (“the winding number of γ around 0”), denoted by $\text{ind}_\gamma(0)$, as follows: Pick a lift $\tilde{\gamma}: [0, 1] \rightarrow \mathbb{C}$ of γ . Then

$$\text{ind}_\gamma(0) := \frac{1}{2\pi i}(\tilde{\gamma}(1) - \tilde{\gamma}(0)). \quad (19.2)$$

More generally, the *index of γ with respect to a point $w \in \mathbb{C} \setminus \gamma^*$* (“the winding number of γ around w ”), is defined as

$$\text{ind}_\gamma(w) := \text{ind}_{\gamma-w}(0).$$

Remark 19.7. (a) Note that a lift $\tilde{\gamma}$ as in the previous definition exists by Theorem 19.5 (a). Moreover, $\text{ind}_\gamma(0)$ is well-defined. Indeed, if $\tilde{\gamma}_1$ and $\tilde{\gamma}_2$ are two lifts of γ , then

$$\exp(\tilde{\gamma}_1 - \tilde{\gamma}_2) = \exp(\tilde{\gamma}_1) / \exp(\tilde{\gamma}_2) = \gamma / \gamma = 1,$$

and so $\tilde{\gamma}_1 - \tilde{\gamma}_2 = \text{const.}$ Hence $\tilde{\gamma}_1 = \tilde{\gamma}_2 + \text{const.}$, which implies

$$\tilde{\gamma}_1(1) - \tilde{\gamma}_1(0) = \tilde{\gamma}_2(1) - \tilde{\gamma}_2(0).$$

(b) The index is an integer. Indeed, since γ is a loop, we have

$$\exp(\tilde{\gamma}(0)) = \gamma(0) = \gamma(1) = \exp(\tilde{\gamma}(1)).$$

Hence $\tilde{\gamma}(1) = \tilde{\gamma}(0) + 2\pi ik$ for some $k \in \mathbb{Z}$, and

$$\text{ind}_\gamma(0) = \frac{1}{2\pi i}(\tilde{\gamma}(1) - \tilde{\gamma}(0)) = k \in \mathbb{Z}.$$

(c) The definition given in (19.2) agrees with a similar definition based on (19.1) with $\alpha = \tilde{\gamma}$. To see this note that

$$\text{Re}(\tilde{\gamma}(1)) = \log |\gamma(1)| = \log |\gamma(0)| = \text{Re}(\tilde{\gamma}(0)).$$

(d) In the previous discussion the assumption that the loops are defined on $[0, 1]$ was of course just for convenience. One defines winding numbers for loops defined on arbitrary compact subintervals of $\mathbb{R}$ in essentially the same way as for loops on $[0, 1]$.

Theorem 19.8. *Let $\gamma: [0, 1] \rightarrow \mathbb{C}$ be a piecewise smooth loop and $w \in \mathbb{C} \setminus \gamma^*$. Then*

$$\text{ind}_\gamma(w) = \frac{1}{2\pi i} \int_\gamma \frac{dz}{z - w}.$$

Proof. By Theorem 19.5 (a) the path $\gamma - w$ has a piecewise smooth lift $\tilde{\gamma}$. Then $\gamma(t) - w = \exp(\tilde{\gamma}(t))$ for $t \in [0, 1]$. This implies that

$$\begin{aligned} \frac{1}{2\pi i} \int_\gamma \frac{dz}{z - w} &= \frac{1}{2\pi i} \int_0^1 \frac{\gamma'(t)}{\exp(\tilde{\gamma}(t))} dt = \frac{1}{2\pi i} \int_0^1 \tilde{\gamma}'(t) dt \\ &= \frac{1}{2\pi i}(\tilde{\gamma}(1) - \tilde{\gamma}(0)) = \text{ind}_\gamma(0). \end{aligned} \quad \square$$

Proposition 19.9. *Let $\alpha, \gamma_1, \gamma_2: [0, 1] \rightarrow \mathbb{C}$ be loops, and assume $\gamma_1(1) = \gamma_2(0)$.*

(a) *Define $\beta: [0, 1] \rightarrow \mathbb{C}$ by $\beta(t) = \alpha(1 - t)$ for $t \in [0, 1]$ (so β is the path opposite to α). If $z \in \mathbb{C} \setminus \alpha^* = \mathbb{C} \setminus \beta^*$, then*

$$\text{ind}_\beta(z) = -\text{ind}_\alpha(z).$$

(b) *Define $\gamma: [0, 2] \rightarrow \mathbb{C}$ by*

$$\gamma(t) = \begin{cases} \gamma_1(t) & \text{for } t \in [0, 1], \\ \gamma_2(t - 1) & \text{for } t \in [1, 2]. \end{cases}$$

If $z \in \mathbb{C}$ and $z \notin \gamma^ = \gamma_1^* \cup \gamma_2^*$, then*

$$\text{ind}_\gamma(z) = \text{ind}_{\gamma_1}(z) + \text{ind}_{\gamma_2}(z).$$

Proof. Without loss of generality $z = 0$.

(a) Let $\tilde{\alpha}: [0, 1] \rightarrow \mathbb{C}$ be a lift of α . Then $\tilde{\beta}: [0, 1] \rightarrow \mathbb{C}$, $\tilde{\beta}(t) := \tilde{\alpha}(1 - t)$, $t \in [0, 1]$, is a lift of $\tilde{\beta}$, and so

$$\text{ind}_{\beta}(0) = \frac{1}{2\pi i}(\tilde{\beta}(1) - \tilde{\beta}(0)) = -\frac{1}{2\pi i}(\tilde{\alpha}(1) - \tilde{\alpha}(0)) = -\text{ind}_{\alpha}(z).$$

(b) First note that γ is a loop, because

$$\gamma(0) = \gamma_1(0) = \gamma_1(1) = \gamma_2(0) = \gamma_2(1) = \gamma(2).$$

Let $\tilde{\gamma}_1$ be a lift of γ_1 . Then

$$\tilde{\gamma}_1(1) \in \exp^{-1}(\gamma_1(1)) = \exp^{-1}(\gamma_2(0)),$$

and so there exists a lift $\tilde{\gamma}_2: [0, 1] \rightarrow \mathbb{C}$ of γ_2 such that $\tilde{\gamma}_2(0) = \tilde{\gamma}_1(1)$. Define

$$\tilde{\gamma}(t) = \begin{cases} \tilde{\gamma}_1(t) & \text{for } t \in [0, 1], \\ \tilde{\gamma}_2(t - 1) & \text{for } t \in [1, 2]. \end{cases}$$

Then $\tilde{\gamma}$ is a lift of γ , and so

$$\begin{aligned} \text{ind}_{\gamma}(0) &= \frac{1}{2\pi i}(\tilde{\gamma}(2) - \tilde{\gamma}(0)) = \frac{1}{2\pi i}(\tilde{\gamma}(2) - \tilde{\gamma}(1) + \tilde{\gamma}(1) - \tilde{\gamma}(0)) \\ &= \frac{1}{2\pi i}(\tilde{\gamma}_2(1) - \tilde{\gamma}_2(0)) + \frac{1}{2\pi i}(\tilde{\gamma}_1(1) - \tilde{\gamma}_1(0)) \\ &= \text{ind}_{\gamma_1}(0) + \text{ind}_{\gamma_2}(0). \end{aligned} \quad \square$$

Definition 19.10. (a) Let $U \subseteq \mathbb{C}$ and $\gamma_0, \gamma_1: [0, 1] \rightarrow U$ be loops in U . We say that γ_0 and γ_1 are *(loop-)homotopic in U* , written $\gamma_0 \sim \gamma_1$ in U , if there exists a continuous map $H: [0, 1] \times [0, 1] \rightarrow U$ such that $H(\cdot, 0) = \gamma_0$, $H(\cdot, 1) = \gamma_1$, and such that $H_t := H(\cdot, t)$ is a loop for each $t \in [0, 1]$ (i.e., $H(0, t) = H(1, t)$ for all $t \in [0, 1]$).

(b) We say that a region $U \subseteq \mathbb{C}$ is *simply connected* if every loop γ in U is *null-homotopic in U* , i.e., γ is loop-homotopic in U to a constant path (written $\gamma \sim 0$ in U).

A map H as in (a) is called a *loop-homotopy* or simply a *homotopy* between the loops γ_0 and γ_1 .

Remark 19.11. (a) If $\Omega \subseteq \mathbb{C}$ is a convex region, then it is simply connected. To see this, let $\gamma_0: [0, 1] \rightarrow \Omega$ be a loop in Ω . Pick a point $a \in \Omega$, and define

$$H(s, t) := (1 - t)\gamma_0(s) + ta$$

for $s, t \in [0, 1] \times [0, 1]$. Then H is a loop-homotopy in Ω between γ_0 and the constant path $s \mapsto \gamma_1(s) := a$.

(b) One can show that a region $\Omega \subseteq \mathbb{C}$ is simply connected if and only if $\widehat{\mathbb{C}} \setminus \Omega$ is connected. The proof is surprisingly difficult and will be given later.

Theorem 19.12. *Let $\gamma_0, \gamma_1: [0, 1] \rightarrow \mathbb{C}$ be paths and $z \in \mathbb{C} \setminus (\gamma_0^* \cup \gamma_1^*)$. Then γ_0 and γ_1 are homotopic in $\mathbb{C} \setminus \{z\}$ if and only if*

$$\text{ind}_{\gamma_0}(z) = \text{ind}_{\gamma_1}(z).$$

Proof. Without loss of generality we may assume that $z = 0$.

“ $\Rightarrow$ ” Suppose that there exists a homotopy $H: [0, 1] \times [0, 1] \rightarrow \mathbb{C}^*$ between γ_0 and γ_1 in $\mathbb{C}^* = \mathbb{C} \setminus \{z\}$. Then $H(\cdot, 0) = \gamma_0$, $H(\cdot, 1) = \gamma_1$, and $\gamma_t := H(\cdot, t)$ is a loop for each $t \in [0, 1]$. By Theorem 19.5 (b) there exists a lift $\tilde{H}: [0, 1] \times [0, 1] \rightarrow \mathbb{C}$ of H . Then $\tilde{\gamma}_t := \tilde{H}(\cdot, t)$ is a lift of γ_t for each $t \in [0, 1]$. Hence

$$\text{ind}_{\gamma_t}(0) = \frac{1}{2\pi i}(\tilde{\gamma}_t(1) - \tilde{\gamma}_t(0)) = \frac{1}{2\pi i}(\tilde{H}(1, t) - \tilde{H}(0, t)).$$

Since this is a continuous function of t with values in $\mathbb{Z}$, it must be constant on $[0, 1]$. It follows that $\text{ind}_{\gamma_0}(0) = \text{ind}_{\gamma_1}(0)$ as desired.

“ $\Leftarrow$ ” Suppose that $\text{ind}_{\gamma_0}(0) = \text{ind}_{\gamma_1}(0) =: k \in \mathbb{Z}$. Let $\tilde{\gamma}_0: [0, 1] \rightarrow \mathbb{C}$ be a lift of γ_0 and $\tilde{\gamma}_1: [0, 1] \rightarrow \mathbb{C}$ be a lift of γ_1 . Then

$$\tilde{\gamma}_0(1) - \tilde{\gamma}_0(0) = \tilde{\gamma}_1(1) - \tilde{\gamma}_1(0) = 2\pi i k.$$

Define $H: [0, 1] \times [0, 1] \rightarrow \mathbb{C}^*$ by

$$H(s, t) := \exp((1 - t)\tilde{\gamma}_0(s) + t\tilde{\gamma}_1(s)), \quad s, t \in [0, 1].$$

Then H is a homotopy between γ_0 and γ_1 in $\mathbb{C}^*$; indeed, H is a continuous map with image in $\mathbb{C}^*$, $H(\cdot, 0) = \exp(\tilde{\gamma}_0) = \gamma_0$, and $H(\cdot, 1) = \exp(\tilde{\gamma}_1) = \gamma_1$. Moreover, $H(0, t) = H(1, t)$ for all $t \in [0, 1]$, because

$$\begin{aligned} \frac{H(1, t)}{H(0, t)} &= \exp((1 - t)(\tilde{\gamma}_0(1) - \tilde{\gamma}_0(0)) + t(\tilde{\gamma}_1(1) - \tilde{\gamma}_1(0))) \\ &= \exp((1 - t)2\pi i k + t2\pi i k) = \exp(2\pi i k) = 1. \end{aligned}$$

Hence $\gamma_0 \sim \gamma_1$ in $\mathbb{C}^*$. □

Corollary 19.13. *Let $\Omega \subseteq \mathbb{C}$ be a region. If $\gamma: [0, 1] \rightarrow \Omega$ a loop that is null-homotopic in Ω , then*

$$\text{ind}_\gamma(z) = 0 \quad \text{for all } z \in \mathbb{C} \setminus \Omega.$$

Note that the hypothesis of this implication, and hence also the conclusion, is true for all loops γ in a simply connected region Ω .

Proof. Let γ be as in the statement and $z \in \mathbb{C} \setminus \Omega$ be arbitrary. Then γ is homotopic to constant path α in Ω , and hence also in $\mathbb{C} \setminus \{z\} \supseteq \Omega$. Then by Theorem 19.12 we have $\text{ind}_\gamma(z) = \text{ind}_\alpha(z) = 0$, where the last equality immediately follows from the definitions. $\square$

Theorem 19.14. *Let $\gamma: [0, 1] \rightarrow \mathbb{C}$ be a loop. Then the function $z \mapsto \text{ind}_\gamma(z)$ is constant on each connected component of $\mathbb{C} \setminus \gamma^*$, and equal to 0 on the unique unbounded component of $\mathbb{C} \setminus \gamma^*$.*

Proof. The set $\mathbb{C} \setminus \gamma^*$ is open, and so a disjoint union of regions, the connected components of $\mathbb{C} \setminus \gamma^*$. Let $D \subseteq \mathbb{C}$ be a sufficiently large open disk with $\gamma^* \subseteq D$. Then $\mathbb{C} \setminus D \subseteq \mathbb{C} \setminus \gamma^*$ is connected and must lie in one of these regions which is necessarily unbounded. All the other regions are contained in D and hence bounded. So there exists precisely one of the connected components of $\mathbb{C} \setminus \gamma^*$ that is unbounded.

Suppose the region Ω is one of the components of $\mathbb{C} \setminus \gamma^*$, and let $z_0, z_1 \in \Omega$ be arbitrary. Then there exists a path $\alpha: [0, 1] \rightarrow \Omega$ with $\alpha(0) = z_0$ and $\alpha(1) = z_1$ (see Theorem 2.31). Note that then $\alpha^* \cap \gamma^* = \emptyset$. Hence

$$(s, t) \in [0, 1] \times [0, 1] \mapsto H(s, t) := \gamma(s) - \alpha(t)$$

defines a homotopy between $\gamma - z_0$ and $\gamma - z_1$ in $\mathbb{C}^*$. So by Theorem 19.12 we have

$$\text{ind}_\gamma(z_0) = \text{ind}_{\gamma - z_0}(0) = \text{ind}_{\gamma - z_1}(0) = \text{ind}_\gamma(z_1).$$

This shows that the map $z \mapsto \text{ind}_\gamma(z)$ is constant on each component of $\mathbb{C} \setminus \gamma^*$.

As above, let $D \subseteq \mathbb{C}$ be an open disk with $\gamma^* \subseteq D$. Then D is convex, and so also simply connected. By Corollary 19.13 we have $\text{ind}_\gamma(z) = 0$ for all $z \in \mathbb{C} \setminus D$. By what we have seen, this implies that $\text{ind}_\gamma(z) = 0$ whenever z lies in the unique unbounded component of $\mathbb{C} \setminus \gamma^*$. $\square$

Remark 19.15. Apart from its applications in complex analysis, the winding number is a very powerful device for establishing some advanced statements in plane topology, for example the *Jordan Curve Theorem*. By definition a *Jordan curve* J in $\mathbb{C}$ (or $\widehat{\mathbb{C}}$) is a set homeomorphic to the unit circle $\partial\mathbb{D}$. The Jordan Curve Theorem says that for a Jordan curve $J \subseteq \mathbb{C}$ the set $\mathbb{C} \setminus J$ has precisely one bounded component U and one unbounded component V . Moreover, we have $\partial U = \partial V = J$.

A proof is outlined in Problems 19.8–19.12. Along the way a bunch of other interesting facts in plane topology are established.

19.1 Exercises

Problem 19.1. Provide the details for the proof of Theorem 19.5 (b): if $H: Q := [0, 1] \times [0, 1] \rightarrow \mathbb{C}^*$ is a continuous map and $u_0 \in \exp^{-1}(H(0, 0))$, then there exists a unique continuous map $\tilde{H}: Q \rightarrow \mathbb{C}$ such that $H = \exp \circ \tilde{H}$ and $\tilde{H}(0, 0) = u_0$.

Problem 19.2. The purpose of this problem is to develop a geometric method that allows the determination of winding numbers in many cases.

(a) Let $f: [0, 1] \rightarrow \mathbb{R}$ be a piecewise C^1 -smooth function with $f(0), f(1) \in \mathbb{Z}$, and assume that for all $t \in [0, 1]$ with $f(t) \in \mathbb{Z}$, the derivative $f'(t)$ exists and $f'(t) \neq 0$ (this condition essentially means that the graph of f in the x - y -plane meets each line $y = n \in \mathbb{Z}$ transversally).

A point $t \in [0, 1]$ with $f(t) \in \mathbb{Z}$ is called an *up-crossing* if $f'(t) > 0$, and a *down-crossing* if $f'(t) < 0$. Let U be the total number of up-crossings and D be the number of down-crossing, where the up- or down-crossings at $t = 0$ and $t = 1$ only contribute with $1/2$. Show that U and D are finite, and that

$$f(1) - f(0) = U - D.$$

(b) Let R be the positive real axis, $H_+ \subseteq \mathbb{C}$ be the open upper half-plane, and $H_- \subseteq \mathbb{C}$ be the open lower half-plane. Let $\gamma: [0, 1] \rightarrow \mathbb{C}^*$ be a piecewise smooth loop, and assume that $\gamma(0), \gamma(1) \notin R$ and that for all $t \in (0, 1)$ with $\gamma(t) \in R$, the derivative $\gamma'(t)$ exists and $\gamma'(t) \notin \mathbb{R}$ (which implies $\gamma'(t) \in H_+ \cup H_-$). A point $t \in (0, 1)$ with $\gamma(t) \in R$ is called an *up-crossing* if $\gamma'(t) \in H_+$, and a *down-crossing* if $\gamma'(t) \in H_-$. Let U be the total number of up-crossings and D be the number of down-crossing. Show that U and D are finite, and that

$$\text{ind}_\gamma(0) = U - D.$$

Problem 19.3. The purpose of this problem is to prove the following generalization of the Fundamental Theorem of Algebra: Let $f: \mathbb{C} \rightarrow \mathbb{C}$ be a continuous function. Suppose that there exist $n \in \mathbb{Z} \setminus \{0\}$ and $c \in \mathbb{C}^*$ such that

$$\lim_{|z| \rightarrow \infty} \frac{f(z)}{z^n} = c.$$

Then f has a zero in $\mathbb{C}$, i.e., there exists $z_0 \in \mathbb{C}$ such that $f(z_0) = 0$. Complete the following steps for the proof:

(a) Suppose $\alpha, \beta: [0, 1] \rightarrow \mathbb{C}$ are loops, and $w \in \mathbb{C} \setminus \alpha^*$. Show that if

$$|\alpha(s) - \beta(s)| < |\alpha(s) - w| \text{ for all } s \in [0, 1],$$

then $w \in \mathbb{C} \setminus \beta^*$, and we have

$$\text{ind}_\alpha(w) = \text{ind}_\beta(w).$$

(b) Suppose f is as above. For $R \geq 0$ we define the path $\gamma_R: [0, 1] \rightarrow \mathbb{C}$ by

$$\gamma_R(s) = f(Re^{2\pi is}) \text{ for } s \in [0, 1].$$

Use (a) to show that

$$\text{ind}_{\gamma_R}(0) = n$$

for large R .

(c) Use (b) to show that f has a zero in $\mathbb{C}$.

Problem 19.4. (a) Let $f: \overline{\mathbb{D}} \rightarrow \mathbb{C}$ be a continuous map, and $\gamma(s) := f(e^{2\pi is})$ for $s \in [0, 1]$. Show that if $\gamma(s) \neq 0$ for $s \in [0, 1]$ and $\text{ind}_\gamma(0) \neq 0$, then f has a zero in $\mathbb{D}$, i.e., there exists $z_0 \in \mathbb{D}$ such that $f(z_0) = 0$.

(b) Let $f: \overline{\mathbb{D}} \rightarrow \mathbb{C}$ be a continuous map such that

$$\frac{f(z)}{z} \in \mathbb{C} \setminus (-\infty, 0] \text{ for all } z \in \partial\mathbb{D}.$$

Show that f has a zero in $\mathbb{D}$.

(c) Use (b) to prove the following 2-dimensional version of *Brouwer's Fixed Point Theorem*: if $g: \overline{\mathbb{D}} \rightarrow \overline{\mathbb{D}}$ is a continuous map, then g has a fixed point, i.e., there exists $z_0 \in \overline{\mathbb{D}}$ such that $g(z_0) = z_0$.

Problem 19.5. Prove or disprove: If $f: \overline{\mathbb{D}} \rightarrow \mathbb{C}$ is continuous and $f(z) = z$ for all $z \in \partial\mathbb{D}$, then $\overline{\mathbb{D}} \subseteq f(\overline{\mathbb{D}})$.

Problem 19.6. (a) Let $f: \overline{\mathbb{D}} \rightarrow \mathbb{C}$ be a continuous map, and

$$A_k = \{e^{iu} : \pi(2k-1)/4 \leq u \leq \pi(2k+1)/4\} \subseteq \partial\mathbb{D}$$

for $k \in \{0, 1, 2, 3\}$. Suppose that $\text{Re}(i^k f(z)) > 0$ whenever $z \in A_k$, $k \in \{0, 1, 2, 3\}$. Show that f has a zero in $\mathbb{D}$.

Hint: First find a simple function f that satisfies the hypotheses.

(b) Consider the square $Q := [-1, 1] \times [-1, 1] \subseteq \mathbb{R}^2 \cong \mathbb{C}$, and let $\alpha: [-1, 1] \rightarrow Q$ and $\beta: [-1, 1] \rightarrow Q$ be paths in Q . Suppose that α joins the left and the right side of Q , and that β connects the bottom and top side of Q (i.e., $\operatorname{Re} \alpha(\pm 1) = \pm 1$ and $\operatorname{Im} \beta(\pm 1) = \pm 1$). Show that α and β have a point in common, i.e., $\alpha^* \cap \beta^* \neq \emptyset$.

Problem 19.7. (a) Let $f: \partial\mathbb{D} \rightarrow \mathbb{C}^*$ be a continuous map, and define $\gamma(s) = f(e^{2\pi i s})$ for $s \in [0, 1]$. Show that if $f(-z) = -f(z)$ for all $z \in \partial\mathbb{D}$, then $\operatorname{ind}_\gamma(0) \neq 0$.

(b) Prove the 2-dimensional version of the *Borsuk–Ulam Theorem*: if $g: \mathbb{S}^2 \rightarrow \mathbb{C}$ is a continuous map on the unit sphere $\mathbb{S}^2 \subseteq \mathbb{R}^3$, then there exists a pair of antipodal points on $\mathbb{S}^2$ with the same image under g ; in other words, there exists a point $p \in \mathbb{S}^2$ such that $g(p) = g(-p)$.

Hint: It is convenient to use complex coordinates and make the identification $\mathbb{R}^3 \cong \mathbb{C} \times \mathbb{R}$ and $\mathbb{S}^2 = \{(z, h) \in \mathbb{C} \times \mathbb{R} : |z|^2 + h^2 = 1\}$. Argue by contradiction, and consider $z \in \partial\mathbb{D} \mapsto f(z) := g(z, 0) - g(-z, 0)$.

Problem 19.8. A *retraction* of a topological space X onto a subset $M \subseteq X$ is a continuous map $f: X \rightarrow M$ with $f|_M = \operatorname{id}_M$.

(a) Let $K \neq \emptyset$ be a convex and compact subset of $\mathbb{C}$. Show that for each $p \in \mathbb{C}$ there exists a unique point $\tilde{p} \in K$ such that $|p - \tilde{p}| = \operatorname{dist}(p, K)$, and that the map $f: \mathbb{C} \rightarrow K$, $p \mapsto f(p) := \tilde{p}$ is a retraction of $\mathbb{C}$ onto K .

(b) By Brouwer's Fixed Point Theorem (see Problem 19.4) we know that every continuous map $f: \overline{\mathbb{D}} \rightarrow \overline{\mathbb{D}}$ has a fixed point. Use this and (a) to prove the following more general version of this theorem: every continuous map $g: K \rightarrow K$ on a convex and compact subset $K \neq \emptyset$ of $\mathbb{C}$ has a fixed point.

(c) Use (b) to give another proof of the fact stated in Problem 19.6 (b).

Hint: We can regard Q as the closed unit ball in the metric on $\mathbb{R}^2 \cong \mathbb{C}$ induced by the ℓ^∞ -norm given by $\|z\|_\infty := \max\{|\operatorname{Re} z|, |\operatorname{Im} z|\}$ for $z \in \mathbb{C}$. Argue by contradiction and consider the map

$$(s, t) \in Q \mapsto \frac{\overline{\beta(t)} - \overline{\alpha(s)}}{\|\beta(t) - \alpha(s)\|_\infty} \in Q.$$

Problem 19.9. Let $K \subseteq \mathbb{C}$ be a compact.

(a) Prove the following version of *Urysohn's Lemma*: if $K' \subseteq \mathbb{C}$ is another compact set with $K \cap K' = \emptyset$, there exists a continuous function $g: \mathbb{C} \rightarrow [0, 1]$ such that $g = 1$ on K and $g = 0$ on K' .

Hint: Consider $\text{dist}(z, K)$ for $z \in \mathbb{C}$.

(b) Prove the following version of *Tietze's Extension Theorem*: if $f: K \rightarrow [-1, 1]$ is continuous, then there exists a continuous map $F: \mathbb{C} \rightarrow [-1, 1]$ such that $F|_K = f$.

Hint: Use (a) to first find a continuous function $F_0: \mathbb{C} \rightarrow [-1/3, 1/3]$ such that $|f - F_0| \leq 2/3$ on K .

Problem 19.10. (a) Use Brouwer's Fixed Point Theorem to show that no closed disk $D \subseteq \mathbb{C}$ with non-empty interior can be retracted onto its boundary.

(b) A set $\alpha \subseteq \widehat{\mathbb{C}}$ is called a (*Jordan*) *arc* if there exists a homeomorphism $\varphi: [0, 1] \rightarrow \alpha$. Show that if α is an arc in $\mathbb{C}$, then there exists a retraction $f: \mathbb{C} \rightarrow \alpha$. Hint: Use a suitable version of Tietze's Extension Theorem (see Problem 19.9).

(c) A set $J \subseteq \widehat{\mathbb{C}}$ is called a *Jordan curve* if there exists a homeomorphism $\varphi: \partial\mathbb{D} \rightarrow J$. Let J be a Jordan curve in $\mathbb{C}$, and suppose $U \subseteq \mathbb{C}$ is a bounded component of $\mathbb{C} \setminus J$. Show that then $\partial U = J$.

Hint: Argue by contradiction. Show that then there exists an arc α such that $\partial U \subseteq \alpha \subseteq J$. Use this and (b) to find a retraction of $\mathbb{C}$ onto the complement of U , and derive a contradiction from (a).

Problem 19.11. Let $J \subseteq \mathbb{C}$ be a Jordan curve. The purpose of this problem is to show that the set $\mathbb{C} \setminus J$ has precisely one bounded component U .

(a) Show that we can make the following assumptions without loss of generality: J is contained in the square $Q = [-1, 1] \times [-1, 1] \subseteq \mathbb{R}^2 \cong \mathbb{C}$. Moreover, J has precisely one point p in common with the left side of Q , one point q in common with the right side of Q , and no other common points with ∂Q .

Hint: Use an extremality property to find candidates for the points p and q , and use these points to construct a suitable rectangle R containing J . Argue that we may assume that $R = Q$.

(b) Let $p, q \in J \subseteq Q$ be as in (a). Then there exist two arcs $\alpha, \beta \subseteq J$ with endpoints p and q with $\alpha \cup \beta = J$ and $\alpha \cap \beta = \{p, q\}$ (why?).

If we move from $-i$ to i along the imaginary axis, then we must meet J (why?). Say we meet J for the first time in a point $a \in \alpha$. Show that then the last point b where we meet J must belong to β .

(c) Show that there exist points a' and b' on the imaginary axis with $a' \in \alpha$, $b' \in \beta$, and $\operatorname{Im} a' < \operatorname{Im} b'$ such that the open line segment (a', b') has no point in common with J .

(d) Let $z_0 = \frac{1}{2}(a' + b') \in \mathbb{C} \setminus J$ be the midpoint of a' and b' . Show that then z_0 does not lie in the unbounded component of $\mathbb{C} \setminus J$.

Hint: Argue by contradiction. Then we can join z_0 to a point $w_0 \in \partial Q$ by a path in Q that does not meet J . Analyze the various possibilities for the location of w_0 , and derive a contradiction in all cases based on Problem 19.6 (c). For example, if w_0 lies on the top side of Q , then we can join $-i$ to w_0 by a path that does not meet β .

(e) Part (d) shows that $\mathbb{C} \setminus J$ has *at least* one bounded component. Using Problem 19.10 and Problem 19.6 (c) show that $\mathbb{C} \setminus J$ can have *at most* one bounded component and derive the desired statement.

Hint: Suppose U is the bounded component of $\mathbb{C} \setminus J$ containing z_0 , and $V \neq U$ is another bounded component of $\mathbb{C} \setminus J$. There exists a path γ in Q joining $-i$ and i that does not meet $V \cup \{p, q\}$ (why?). On the other hand, there are points $p' \in V$ and $q' \in V$ close to p and q , respectively. Use this to construct a path in Q joining p and q that does not meet γ .

Problem 19.12 (Jordan Curve Theorem). (a) Let $J \subseteq \mathbb{C}$ be a Jordan curve. Show that then $\mathbb{C} \setminus J$ has precisely one bounded component, called the *interior (component)* of J , denoted by $\mathcal{I}(J)$, and precisely one unbounded component, called the *exterior (component)* of J , denoted by $\mathcal{O}(J)$. Moreover, we have

$$\partial \mathcal{I}(J) = J = \partial \mathcal{O}(J).$$

(b) Let $J \subseteq \widehat{\mathbb{C}}$ be a Jordan curve. Show that then $\widehat{\mathbb{C}} \setminus J$ has precisely two components D_1 and D_2 and we have

$$\partial D_1 = J = \partial D_2.$$

Hint: First show that $J \neq \widehat{\mathbb{C}}$.

Problem 19.13. Let $U \subseteq \mathbb{C}$ be a bounded open set, and assume that $J := \partial U$ is a Jordan curve. Show that then U is equal to the interior $\mathcal{I}(J)$ of J .

20 Global Cauchy theorems

Proposition 20.1. Let $[a, b], [c, d] \subseteq \mathbb{R}$ be compact intervals, and $F: [a, b] \times [c, d] \rightarrow \mathbb{C}$ be a continuous function. Then we have

$$\int_a^b \left(\int_c^d F(x, y) dy \right) dx = \int_c^d \left(\int_a^b F(x, y) dx \right) dy. \quad (20.1)$$

Proof. This is a special case of Fubini's well-known theorem in Measure Theory. One can easily prove this special case directly as follows.

Uniform continuity of F on $R := [a, b] \times [c, d]$ implies that the functions

$$x \mapsto \int_c^d F(x, y) dy \quad \text{and} \quad y \mapsto \int_a^b F(x, y) dx$$

are continuous on $[a, b]$ and $[c, d]$, respectively. So the integrals in (20.1) exist.

To establish the equality in (20.1) one first notes that this is true for a constant function F . The case of an arbitrary continuous function F can easily be derived from this: one subdivides R into small rectangles, and approximates the integrals over F in (20.1) by sums of integrals of constant functions over the rectangles in the subdivision of R . $\square$

Corollary 20.2. Let $\alpha, \beta: [0, 1] \rightarrow \mathbb{C}$ be piecewise smooth paths, and $G: \alpha^* \times \beta^* \rightarrow \mathbb{C}$ be continuous. Then we have

$$\int_{\alpha} \left(\int_{\beta} G(z, w) dw \right) dz = \int_{\beta} \left(\int_{\alpha} G(z, w) dz \right) dw.$$

Proof. By breaking α and β up into smaller paths if necessary, one may in addition assume that α and β are smooth, and so have continuous derivatives α' and β' . The statement then immediately follows from Proposition 20.1 applied to the function

$$(x, y) \in [0, 1] \times [0, 1] \mapsto F(x, y) := G(\alpha(x), \beta(y))\alpha'(x)\beta'(y). \quad \square$$

Proposition 20.3. Let $\gamma: [0, 1] \rightarrow \mathbb{C}$ be a piecewise smooth path, and $f: \gamma^* \rightarrow \mathbb{C}$ be a continuous function. If we define

$$F(z) = \int_{\gamma} \frac{f(w)}{w - z} dw \quad \text{for } z \in U := \mathbb{C} \setminus \gamma^*,$$

then F is holomorphic on U , and $\lim_{|z| \rightarrow \infty} |F(z)| = 0$.

Proof. Note that for fixed $z \in \mathbb{C} \setminus \gamma^*$, the function

$$w \mapsto \frac{f(w)}{w - z}$$

is continuous on γ^* ; so $F(z)$ is defined.

To establish that F is holomorphic, we want to apply Morera's Theorem (see Theorem 10.6). For this we first show that F is continuous on U .

To see this, let $z_0 \in U$ be arbitrary. Then there exists $r > 0$ such that $|z_0 - w| \geq r$ for all $w \in \gamma^*$. Let $z \in B(z_0, r/2) \subseteq U$ be arbitrary. Then

$$|z - w| \geq |w - z_0| - |z - z_0| \geq r - r/2 = r/2$$

for $w \in \gamma^*$, and so

$$\begin{aligned} |F(z) - F(z_0)| &\leq \left| \int_{\gamma} f(w) \left(\frac{1}{w - z} - \frac{1}{w - z_0} \right) dw \right| \\ &\leq \ell(\gamma) \max_{w \in \gamma^*} |f(w)| \frac{|z - z_0|}{|w - z| \cdot |w - z_0|} \\ &\leq |z - z_0| \underbrace{\frac{2}{r^2} \ell(\gamma) \max_{w \in \gamma^*} |f(w)|}_{=: M} = M|z - z_0|. \end{aligned}$$

This inequality implies that F is continuous at z_0 .

To verify the other hypothesis in Morera's Theorem, let Δ be an arbitrary oriented triangle with $\Delta \subseteq U = \mathbb{C} \setminus \gamma^*$. If $w \in \gamma^*$, then $w \notin \Delta$, and so w lies in the unbounded component of $\mathbb{C} \setminus \partial\Delta$. So by Theorem 19.14 we have

$$\text{ind}_{\partial\Delta}(w) = 0 \quad \text{for } w \in \gamma^*, \quad (20.2)$$

where we consider $\partial\Delta$ as a loop (see Subsection 8.2).

Note that the function

$$(z, w) \in \partial\Delta \times \gamma^* \mapsto \frac{f(w)}{w - z}$$

is continuous. Hence by Corollary 20.2 we have

$$\begin{aligned} \int_{\partial\Delta} F(z) dz &= \int_{\partial\Delta} \left(\int_{\gamma} \frac{f(w)}{w - z} dw \right) dz = \int_{\gamma} \left(\int_{\partial\Delta} \frac{f(w)}{w - z} dz \right) dw \\ &= - \int_{\gamma} f(w) \left(\int_{\partial\Delta} \frac{dz}{z - w} \right) dw \\ &= -2\pi i \int_{\gamma} f(w) \text{ind}_{\partial\Delta}(w) dw = 0 \quad (\text{by Thm. 19.8 and (20.2)}). \end{aligned}$$

The holomorphicity of F on U follows.

Finally, we have

$$|F(z)| = \left| \int_{\gamma} \frac{f(w)}{w - z} dw \right| \leq \max_{w \in \gamma^*} |f(w)| \cdot \frac{\ell(\gamma)}{\text{dist}(z, \gamma^*)} \rightarrow 0 \quad \text{as } |z| \rightarrow \infty. \quad \square$$

20.4. Chains and cycles. Let $\Omega \subseteq \mathbb{C}$ be an open set. A *chain* Γ in Ω is a collection $\gamma_1, \dots, \gamma_n$ of paths in Ω where each of the paths γ_k has an integer $m_k \in \mathbb{Z}$ attached as a weight. We write symbolically

$$\Gamma = m_1 \gamma_1 + \dots + m_n \gamma_n. \quad (20.3)$$

The chain Γ is called a *cycle* in Ω if each of its paths $\gamma_1, \dots, \gamma_n$ is a loop. A chain or a cycle is called *piecewise smooth* if each of its paths has this property. If Γ is as in (20.3), then we define the *support* of Γ as

$$\Gamma^* := \gamma_1^* \cup \dots \cup \gamma_n^* \subseteq \mathbb{C}.$$

If Γ is piecewise smooth chain, and $f: \Gamma^* \rightarrow \mathbb{C}$ is continuous, we define

$$\int_{\Gamma} f(z) dz = m_1 \int_{\gamma_1} f(z) dz + \dots + m_n \int_{\gamma_n} f(z) dz.$$

If Γ is a cycle and $z \in \mathbb{C} \setminus \Gamma^*$, we define the *index of Γ with respect to the point z* as

$$\text{ind}_{\Gamma}(z) = m_1 \text{ind}_{\gamma_1}(z) + \dots + m_n \text{ind}_{\gamma_n}(z).$$

Definition 20.5. Let Ω be an open subset of $\mathbb{C}$. A cycle Γ in Ω is called *null-homologous in Ω* if $\text{ind}_{\Gamma}(z) = 0$ for all $z \in \mathbb{C} \setminus \Omega$.

Example 20.6. If γ is a null-homotopic loop in an open set $\Omega \subseteq \mathbb{C}$, then the cycle $\Gamma = \gamma$ is null-homologous in Ω . This follows from Corollary 19.13. In particular, every loop in a simply connected region is null-homologous.

More generally, if $\Gamma = m_1 \gamma_1 + \dots + m_n \gamma_n$ is any cycle in a simply connected region $\Omega \subseteq \mathbb{C}$, then Γ is null-homologous in Ω ; indeed, by Corollary 19.13 for all $z \in \mathbb{C} \setminus \Omega$ we have

$$\text{ind}_{\Gamma}(z) = m_1 \underbrace{\text{ind}_{\gamma_1}(z)}_{=0} + \dots + m_n \underbrace{\text{ind}_{\gamma_n}(z)}_{=0} = 0.$$

Theorem 20.7 (Cauchy's Theorem). *Let $\Omega \subseteq \mathbb{C}$ be open, $f: \Omega \rightarrow \mathbb{C}$ be a holomorphic function, and Γ be a piecewise smooth cycle that is null-homologous in Ω (i.e., $\text{ind}_\Gamma(z) = 0$ for all $z \in \mathbb{C} \setminus \Omega$).*

Then

$$\text{ind}_\Gamma(z) \cdot f(z) = \frac{1}{2\pi i} \int_\Gamma \frac{f(w)}{w - z} dw \quad (\text{Cauchy's Integral Formula}) \quad (20.4)$$

for all $z \in \Omega \setminus \Gamma^*$, and

$$\int_\Gamma f(z) dz = 0 \quad (\text{Cauchy's Integral Theorem}). \quad (20.5)$$

Proof. We will prove (20.4) first. Define $g: \Omega \times \Omega \rightarrow \mathbb{C}$ for $(z, w) \in \Omega \times \Omega$ as

$$g(z, w) = \begin{cases} \frac{f(w) - f(z)}{w - z} & \text{if } z \neq w, \\ f'(z) & \text{if } z = w. \end{cases}$$

Then g is continuous on $\Omega \times \Omega$ (exercise!), and so we can define

$$h(z) = \frac{1}{2\pi i} \int_\Gamma g(z, w) dw \quad \text{for } z \in \Omega.$$

Then (20.4) is equivalent to the statement

$$h(z) = 0 \quad \text{for all } z \in \Omega \setminus \Gamma^*. \quad (20.6)$$

To see this, we first want to show that h is holomorphic on Ω . Again we apply Morera's Theorem. First note that h is continuous on Ω as easily follows from the uniform continuity of g on compact subsets of $\Omega \times \Omega$ (exercise!). If Δ is an arbitrary closed oriented triangle in Ω , then by Corollary 20.2 we have

$$\begin{aligned} \int_{\partial\Delta} h(z) dz &= \frac{1}{2\pi i} \int_{\partial\Delta} \left(\int_\Gamma g(z, w) dw \right) dz \\ &= \frac{1}{2\pi i} \int_\Gamma \underbrace{\left(\int_{\partial\Delta} g(z, w) dz \right)}_{=0} dw = 0. \end{aligned}$$

Here we used Goursat's Lemma (see Theorem 8.3) and the fact that the function $z \mapsto g(z, w)$ is holomorphic on Ω for each fixed $w \in \Omega$ (exercise!). We conclude that $h \in H(\Omega)$.

Let $\tilde{\Omega} \subseteq \mathbb{C} \setminus \Gamma^*$ be the set of all $z \in \mathbb{C} \setminus \Gamma^*$ for which $\text{ind}_{\Gamma}(z) = 0$. The set $\tilde{\Omega}$ is open, because $z \mapsto \text{ind}_{\Gamma}(z)$ is a locally constant function (as follows from Theorem 19.14). Moreover, $\mathbb{C} \setminus \Omega \subseteq \tilde{\Omega}$ by our assumption that Γ is null-homologous in Ω . Define

$$\tilde{h}(z) = \frac{1}{2\pi i} \int_{\Gamma} \frac{f(w)}{w - z} dw \quad \text{for } z \in \tilde{\Omega}.$$

Then $h \in H(\tilde{\Omega})$ by Proposition 20.3, and

$$h(z) = \tilde{h}(z) \quad \text{for all } z \in \Omega \cap \tilde{\Omega}.$$

So if we define

$$F(z) = \begin{cases} h(z) & \text{for } z \in \Omega, \\ \tilde{h}(z) & \text{for } z \in \tilde{\Omega}, \end{cases}$$

then h is well-defined, and F is holomorphic on $\Omega \cup \tilde{\Omega} = \mathbb{C}$. So F is an entire function. Note that a point $z \in \mathbb{C}$ with $|z|$ sufficiently large lies in the unbounded component of Γ^* . Then $z \in \tilde{\Omega}$ by Theorem 19.14, and so by Proposition 20.3 we have

$$|F(z)| = |\tilde{h}(z)| = \left| \frac{1}{2\pi i} \int_{\Gamma} \frac{f(w)}{w - z} dz \right| \rightarrow 0 \quad \text{as } |z| \rightarrow \infty.$$

As follows from Liouville's Theorem, we must have $F \equiv 0$; so we have (20.6) which implies (20.4).

One easily derives Cauchy's Integral Theorem from this by applying (20.4) to the function $g \in H(\Omega)$ given by $g(z) = (z - z_0)f(z)$, where $z \in \Omega$ and z_0 is a fixed point in $\Omega \setminus \Gamma^*$; indeed,

$$\int_{\Gamma} f(w) dw = \int_{\Gamma} \frac{g(w)}{w - z_0} dw = \text{ind}_{\Gamma}(z_0) \cdot g(z_0) = 0.$$

□

Corollary 20.8. *Let $\Omega \subseteq \mathbb{C}$ be a simply connected region, $f \in H(\Omega)$, and γ be a piecewise smooth loop in Ω . Then*

$$\int_{\gamma} f(z) dz = 0. \quad (20.7)$$

Moreover, if α and β are two piecewise smooth paths in Ω with the same endpoints, then

$$\int_{\alpha} f(z) dz = \int_{\beta} f(z) dz. \quad (20.8)$$

Proof. The loop γ is null-homotopic in Ω , and so null-homologous in Ω . Hence (20.7) follows from Cauchy's Integral Theorem. If $\tilde{\beta}$ is the path opposite to β , then we get a loop σ by first running through α and then through $\tilde{\beta}$. So by the first part of the proof we have

$$0 = \int_{\sigma} f = \int_{\alpha} f + \int_{\tilde{\beta}} f = \int_{\alpha} f - \int_{\beta} f. \quad \square$$

Corollary 20.9. *Let $\Omega \subseteq \mathbb{C}$ be a simply connected region.*

- (a) *Every holomorphic function $f: \Omega \rightarrow \mathbb{C}$ has a primitive $F \in H(\Omega)$.*
- (b) *Every zero-free holomorphic function $f: \Omega \rightarrow \mathbb{C}$ has a holomorphic logarithm $L \in H(\Omega)$ (i.e., $f = e^L$) (and so a holomorphic square root $S = e^{L/2}$ and other n -th roots).*

Proof. (a) Fix $a \in \Omega$. If $z \in \Omega$ is arbitrary, then there exists a piecewise smooth path α in Ω joining a and z . Define

$$F(z) = \int_{\alpha} f(w) dw.$$

Then F is well-defined by Corollary 20.8. If z' is close to z , then we have

$$F(z') = \int_{\alpha} f(w) dw + \int_{[z, z']} f(w) dw,$$

and so

$$\frac{F(z') - F(z)}{z' - z} = \frac{1}{z' - z} \int_{[z, z']} f(w) dw.$$

Similarly as in the proof of Corollary 8.4 this implies that $F'(z)$ exists and $F'(z) = f(z)$.

(b) One can construct a logarithm L of f by considering a primitive of $f'/f \in H(\Omega)$. $\square$

Remark 20.10. There are other important facts about simply connected regions in $\mathbb{C}$; for example, we have the Riemann Mapping Theorem: if $\Omega \subseteq \mathbb{C}$ is a simply connected region with $\Omega \neq \mathbb{C}$, then there exists a conformal map from Ω onto $\mathbb{D}$. We will prove this later in this course.

20.1 Exercises

Problem 20.1. Fix $k \in \mathbb{Z}$.

(a) Show that

$$f(\tau) = \int_{-\infty}^{\infty} \exp(i\pi\tau z^2 - 2\pi i k z) dz := \lim_{R \rightarrow \infty} \int_{-R}^R \exp(i\pi\tau z^2 - 2\pi i k z) dz$$

is a holomorphic function of $\tau \in \mathbb{H} := \{u \in \mathbb{C} : \text{Im}(u) > 0\}$. Hint: First establish convergence of the integral.

(b) Prove that $f(it) = \frac{1}{\sqrt{t}} e^{-\pi k^2/t}$ for $t > 0$. Hint: First show that

$$\lim_{R \rightarrow \infty} \int_{-R}^R \exp(-\pi t(z + ik/t)^2) dz = \lim_{R \rightarrow \infty} \int_{-R}^R \exp(-\pi t z^2) dz.$$

(c) The function $\tau \in \mathbb{H} \mapsto -i\tau$ is zero-free on the simply connected region $\mathbb{H}$ and hence has a holomorphic square root uniquely $\sqrt{-i\tau}$ uniquely determined up to sign (why?). We choose this sign so that $\sqrt{-i\tau}$ takes the value $+1$ at $\tau = i$. Show that then

$$f(\tau) = \int_{-\infty}^{\infty} \exp(i\pi\tau z^2 - 2\pi i k z) dz = \frac{1}{\sqrt{-i\tau}} e^{-i\pi k^2/\tau}$$

for $\tau \in \mathbb{H}$.

Problem 20.2. Let $U \subseteq \mathbb{C}$ be a region, $f \in H(U)$, and Γ be a piecewise smooth cycle that is null-homologous in U . Show that if $|f(z)| \leq 1$ for all $z \in \Gamma^*$, then $|f(z_0)| \leq 1$ for all $z_0 \in U$ with $\text{ind}_{\Gamma}(z_0) \neq 0$.

21 Laurent series

Definition 21.1 (Doubly infinite series). If $c_n \in \mathbb{C}$ are numbers indexed by $n \in \mathbb{Z}$, then we associate with them a *doubly infinite series*, denoted by

$\sum_{n=-\infty}^{\infty} c_n$ or $\sum_{n \in \mathbb{Z}} c_n$. The series is said to *converge* if the infinite series $\sum_{n=0}^{\infty} c_n$ and $\sum_{n=-\infty}^{-1} c_n := \sum_{n=1}^{\infty} c_{-n}$ both converge. In this case, $\sum_{n=-\infty}^{\infty} c_n$ also denotes the limit $\sum_{n=0}^{\infty} c_n + \sum_{n=-\infty}^{-1} c_n$, i.e., $\sum_{n=-\infty}^{\infty} c_n = \lim_{N \rightarrow \infty} \sum_{n=0}^N c_n + \lim_{M \rightarrow \infty} \sum_{n=-M}^{-1} c_n$.

The series $\sum_{n=-\infty}^{\infty} c_n$ is said to be *absolutely convergent* if the series $\sum_{n \in \mathbb{Z}} c_n$ with the countable index set $\mathbb{Z}$ has this property (see Remark 5.7).

Remark 21.2. If $\sum_{n=-\infty}^{\infty} c_n$ converges, then $c_{\pm n} \rightarrow 0$ as $n \rightarrow \infty$, and $\sum_{n=-\infty}^{\infty} c_n = \lim_{N \rightarrow \infty} \sum_{n=-N}^N c_n$. Note that the existence of the last limit does *not* imply the

convergence of $\sum_{n=-\infty}^{\infty} c_n$. For example, $\sum_{n=-\infty}^{\infty} n$ does not converge, but the

limit $\lim_{N \rightarrow \infty} \sum_{n=-N}^N n = 0$ exists.

Definition 21.3 (Laurent series). A *Laurent series* is a doubly infinite series of the form $\sum_{n=-\infty}^{\infty} a_n(z - z_0)^n$, where $a_n \in \mathbb{C}$ for $n \in \mathbb{Z}$ are the *coefficients* of the series and $z_0 \in \mathbb{C}$ its *center*, and $z \in \mathbb{C}$ is a variable point on which the convergence of the Laurent series will depend in general.

We are mostly interested in *normal convergence* of a Laurent series

$$\sum_{n=-\infty}^{\infty} a_n(z - z_0)^n = \sum_{n \in \mathbb{Z}} a_n(z - z_0)^n$$

in z (see Definition 6.7).

Remark 21.4. For every Laurent series $\sum_{n=-\infty}^{\infty} a_n(z - z_0)^n$ there exists an *annulus*

$$A = \{z \in \mathbb{C} : R_1 < |z - z_0| < R_2\},$$

where $R_1, R_2 \in [0, \infty]$, such that the series converges normally on A and represents a holomorphic series on A , and diverges (i.e., does not converge) in the exterior of A ($= \mathbb{C} \setminus \overline{A}$). The set A may be empty, and for $z \in \partial A$ the series may converge or diverge.

To see this, note that

$$\sum_{n=-\infty}^{\infty} a_n(z - z_0)^n = \underbrace{\sum_{n=1}^{\infty} a_{-n} \left(\frac{1}{z - z_0} \right)^n}_{f_1(z)} + \underbrace{\sum_{n=0}^{\infty} a_n(z - z_0)^n}_{f_2(z)}.$$

Let $R_1 := 1/\tilde{R}_1 \in [0, \infty]$, where $\tilde{R}_1 \in [0, \infty]$ is the radius of convergence of the power series $\sum_{n=1}^{\infty} a_{-n}w^n$. Then the series $f_1(z)$ converges normally on the set of points z satisfying the restriction

$$|w| = \frac{1}{|z - z_0|} < \tilde{R}_1 \Leftrightarrow |z - z_0| > \frac{1}{\tilde{R}_1} = R_1;$$

similarly, $f_1(z)$ diverges if $|z - z_0| < R_1$.

If $R_2 \in [0, \infty]$ denotes the radius of convergence of $f_2(z)$, then we see that the Laurent series converges normally on the annulus A as defined above, and diverges in the exterior of A . Moreover, on A both $f_1(z)$ and $f_2(z)$ are holomorphic, which implies that the Laurent series represents a holomorphic function on A .

The Laurent series converges normally on A and so uniformly on compact subsets of A . In particular, if γ is a piecewise smooth path such that the compact set γ^* is contained in A , then the path integral of the Laurent series over the path γ can be obtained by integrating term-by-term.

Theorem 21.5. Let $R_1, R_2 \in [0, \infty]$, $R_1 < R_2$, and

$$A := \{z \in \mathbb{C} : R_1 < |z - z_0| < R_2\}.$$

If $f \in H(A)$, then there exists a Laurent series converging normally on A such that

$$f(z) = \sum_{n=-\infty}^{\infty} a_n(z - z_0)^n \quad \text{for } z \in A. \quad (21.1)$$

Let $R_1 < r < R_2$, and $\gamma(t) = z_0 + re^{it}$, $t \in [0, 2\pi]$. If f has a Laurent series representation as in (21.1), then

$$a_n = \frac{1}{2\pi i} \int_{\gamma} \frac{f(w)}{(w - z_0)^{n+1}} dw \quad \text{for } n \in \mathbb{Z}. \quad (21.2)$$

This formula for the coefficients implies that the Laurent series in (21.1) is uniquely determined. So every holomorphic function f on an annulus A has a unique Laurent series representation converging on A .

In particular, if z_0 is an isolated singularity of f , then there exists a unique Laurent series representation of f in a punctured neighborhood of z_0 .

Proof. Uniqueness: If f has a representation as in (21.1), then the Laurent series converges uniformly on the compact subset γ^* of A . Hence for all $n \in \mathbb{Z}$, we can evaluate the integral in (21.2) by integrating term-by-term:

$$\begin{aligned} \frac{1}{2\pi i} \int_{\gamma} \frac{f(w)}{(w - z_0)^{n+1}} dw &= \frac{1}{2\pi i} \int_{\gamma} \left(\sum_{k=-\infty}^{\infty} a_k(w - z_0)^{k-n-1} \right) dw \\ &= \frac{1}{2\pi i} \sum_{k=-\infty}^{\infty} a_k \underbrace{\int_{\gamma} (w - z_0)^{k-n-1} dw}_{=0 \text{ for } k \neq n} \\ &= \frac{1}{2\pi i} a_n \int_{\gamma} (w - z_0)^{-1} dw = \frac{1}{2\pi i} a_n \cdot 2\pi i = a_n. \end{aligned}$$

Existence: If we define a_n as in (21.2), then the Laurent series in (21.1) converges normally for $z \in A$, and represents f on A . To see this, let $z \in A$ be arbitrary. We can choose $r_1, r_2 \in (0, \infty)$ with

$$R_1 < r_1 < |z - z_0| < r_2 < R_2,$$

and define $\gamma_1(t) = z_0 + r_1 e^{it}$ and $\gamma_2(t) = z_0 + r_2 e^{it}$ for $t \in [0, 2\pi]$. We define the cycle Γ as $\Gamma = \gamma_2 - \gamma_1$. Then $\text{ind}_{\Gamma}(u) = 0$ for $u \in \mathbb{C} \setminus A$, and so Γ is

null-homologous in A . Moreover, $\text{ind}_\Gamma(z) = 1$, and so by Cauchy's Integral Formula we have

$$\begin{aligned} f(z) &= \frac{1}{2\pi i} \int_\Gamma \frac{f(w)}{w-z} dw \\ &= \underbrace{\frac{1}{2\pi i} \int_{\gamma_2} \frac{f(w)}{w-z} dw}_{=: f_2(z)} - \underbrace{\frac{1}{2\pi i} \int_{\gamma_1} \frac{f(w)}{w-z} dw}_{=: f_1(z)}. \end{aligned} \quad (21.3)$$

In the same way as in the proof of Theorem 9.1, for $w \in \gamma_2^*$ one writes

$$\begin{aligned} \frac{f(w)}{w-z} &= \frac{f(w)}{(w-z_0) - (z-z_0)} = \frac{f(w)}{(w-w_0)} \cdot \frac{1}{1 - \frac{(z-z_0)}{(w-w_0)}} \\ &= \sum_{n=0}^{\infty} (z-z_0)^n \cdot \frac{f(w)}{(w-w_0)^{n+1}}. \end{aligned}$$

Note that the convergence of the last series is uniform for $w \in \gamma_2^*$; this follows from

$$\frac{|z-z_0|}{|w-w_0|} = \frac{|z-z_0|}{r_2} < 1 \quad \text{for } w \in \gamma_2^*$$

and the Weierstrass M -test. By integrating term-by-term we conclude that

$$f_2(z) = \frac{1}{2\pi i} \int_{\gamma_2} \frac{f(w)}{w-z} dw = \sum_{n=0}^{\infty} b_n (z-z_0)^n, \quad (21.4)$$

where

$$b_n = \frac{1}{2\pi i} \int_{\gamma_2} \frac{f(w)}{(w-z_0)^{n+1}} dw \quad \text{for } n \in \mathbb{N}.$$

Actually, it is easy to see that in a sufficiently small neighborhood of z we have actually normal convergence of this series (exercise!).

To show that $b_n = a_n$, we want to shift the integration to γ in these integrals. Note that the cycle $\gamma_2 - \gamma$ is null-homologous in A , and that the function $w \mapsto \frac{f(w)}{(w-z_0)^{n+1}}$ is holomorphic on A for each $n \in \mathbb{N}$. So Cauchy's Integral Theorem implies

$$0 = \int_{\gamma_2 - \gamma} \frac{f(w)}{(w-z_0)^{n+1}} dw = 2\pi i (b_n - a_n).$$

So from (21.4) we conclude that

$$f_2(z) = \sum_{n=0}^{\infty} a_n(z - z_0)^n. \quad (21.5)$$

In a similar way one shows that

$$f_1(z) = - \sum_{n=-\infty}^{-1} a_n(z - z_0)^n. \quad (21.6)$$

Indeed, for $w \in \gamma_1^*$ we have

$$\begin{aligned} \frac{f(w)}{w - z} &= \frac{f(w)}{(w - z_0) - (z - z_0)} = - \frac{f(w)}{(z - z_0)} \cdot \frac{1}{1 - \frac{(w - z_0)}{(z - z_0)}} \\ &= - \sum_{n=0}^{\infty} (w - z_0)^n \cdot \frac{f(w)}{(z - z_0)^{n+1}} \\ &= - \sum_{n=-\infty}^{-1} (z - z_0)^n \cdot \frac{f(w)}{(w - z_0)^{n+1}}, \end{aligned}$$

and the convergence is uniform on γ_1^* . Integrating term-by-term and shifting the path γ_1 in the integrals to γ based on a similar argument as before, one obtains (21.6) and one can again easily see that the latter series converges normally in a sufficiently small neighborhood of z .

By combining (21.6), (21.5), and (21.3), we obtain (21.1) with normal convergence of the series on A . $\square$

Theorem 21.6. *Let $U \subseteq \mathbb{C}$ be open, $z_0 \in U$, $f \in H(U \setminus \{z_0\})$, and*

$$f(z) = \sum_{n=-\infty}^{\infty} a_n(z - z_0)^n$$

be the unique Laurent series expansion of f converging normally in a punctured neighborhood of z_0 . Then z_0 is

- (i) *removable if and only if $a_{-n} = 0$ for all $n \in \mathbb{N}$,*
- (ii) *a pole of order $m \in \mathbb{N}$ if and only if $a_{-m} \neq 0$ and $a_{-n} = 0$ for all $n \in \mathbb{N}$ with $n \geq m + 1$,*

(iii) *an essential singularity if and only if $a_{-n} \neq 0$ for infinitely many $n \in \mathbb{N}$.*

In other words, the nature of the isolated singularity can be determined from considering the *singular* or *principal part* $\sum_{n=-\infty}^{-1} a_n(z-z_0)^n$ of the isolated singularity z_0 . For a pole this notion of principal part agrees with our previous definition.

Proof. (i) z_0 is removable iff f has a power series expansion near z_0 iff the negative powers of $z - z_0$ in the Laurent series expansion of f are equal to 0 (by the uniqueness of the Laurent series) iff $a_{-n} = 0$ for all $n \in \mathbb{N}$.

(ii) z_0 is a pole of order $m \in \mathbb{N}$ iff

$$f(z) = \frac{1}{(z - z_0)^m} \sum_{n=0}^{\infty} b_n(z - z_0)^n \quad \text{near } z_0,$$

where the power series is convergent near z_0 and $b_0 \neq 0$ iff

$$f(z) = \sum_{n=-m}^{\infty} a_n(z - z_0)^n \quad \text{near } z_0, \text{ where } a_{-m} \neq 0.$$

(iii) z_0 is an essential singularity iff z_0 is neither removable nor a pole iff $a_{-n} \neq 0$ for infinitely many $n \in \mathbb{N}$. $\square$

Example 21.7. Let $f(z) = \frac{1}{(z-2)(z-3)}$. Then $f \in H(\mathbb{C} \setminus \{2, 3\})$. What is the Laurent series expansion of f on $A := \{1 < |z-1| < 2\} \subseteq \mathbb{C} \setminus \{2, 3\}$?

We first find the partial fraction decomposition of f which must have the form

$$\frac{1}{(z-2)(z-3)} = \frac{A}{z-2} + \frac{B}{z-3}.$$

By multiplying with the denominator of the left-hand side we obtain

$$1 = A(z-3) + B(z-2) = (A+B)z - (3A+2B).$$

By comparing coefficients we get

$$A + B = 0 \quad \text{and} \quad 3A + 2B = -1$$

which implies $A = -1$ and $B = 1$. So for $z \in A$ we have

$$\begin{aligned} f(z) &= \frac{1}{z-3} - \frac{1}{z-2} = \frac{1}{(z-1)-2} - \frac{1}{(z-1)-1} \\ &= -\frac{1}{2} \cdot \frac{1}{1 - \frac{(z-1)}{2}} - \frac{1}{(z-1)} \cdot \frac{1}{1 - \frac{1}{(z-1)}} \\ &= -\frac{1}{2} \sum_{n=0}^{\infty} \frac{1}{2^n} (z-1)^n - \sum_{n=1}^{\infty} \frac{1}{(z-1)^n} = \sum_{n=-\infty}^{\infty} a_n (z-1)^n, \end{aligned}$$

where

$$a_n = \begin{cases} -1/2^{n+1} & \text{for } n \geq 0, \\ -1 & \text{for } n < 0. \end{cases}$$

Theorem 21.8 (Fourier expansion of a periodic holomorphic functions). *Let $S = \{u \in \mathbb{C} : \alpha < \text{Im}(u) < \beta\}$, where $\alpha \in [-\infty, +\infty)$, $\beta \in (-\infty, +\infty]$, and $\alpha < \beta$. Suppose that f is a holomorphic function on S such that $f(u+1) = f(u)$ for all $u \in S$.*

Then there exist unique coefficients $a_n \in \mathbb{C}$ for $n \in \mathbb{Z}$ such that

$$f(u) = \sum_{n \in \mathbb{Z}} a_n e^{2\pi i n u}$$

for $u \in S$, where the latter series converges normally on S . Actually, for each $\alpha < t < \beta$ and $k \in \mathbb{Z}$, we have

$$a_k = \int_{[it, 1+it]} f(u) e^{-2\pi i k u} du.$$

Proof. Define $a := e^{-2\pi\alpha}$ and $b := e^{-2\pi\beta}$, where we set $e^{-\infty} := 0$ and $e^{+\infty} := +\infty$. We consider the annulus $A := \{z \in \mathbb{C} : b < |z| < a\}$. Note that then for $u \in \mathbb{C}$ we have $e^{2\pi i u} \in A$ if and only if $u \in S$.

We claim that there exists a holomorphic function $g: A \rightarrow \mathbb{C}$ such that

$$f(u) = g(e^{2\pi i u}) \tag{21.7}$$

for $u \in S$. Indeed, if $w \in A \in \mathbb{C} \setminus \{0\}$, then there exists $u \in S$ such that $w = e^{2\pi i u}$. We set $g(w) := f(u)$. It is easy to see that $g: A \rightarrow \mathbb{C}$ is well-defined, since f is 1-periodic. Moreover, near w we can find holomorphic branch L of the logarithm. Then for w' close to w we have $w' = e^{L(w')}$ and so

$g(w') = f(\frac{1}{2\pi i}L(w'))$. This local representation shows that g is holomorphic near each point in A and hence holomorphic on A .

The function g has a Laurent series representation of the form

$$g(z) = \sum_{n \in \mathbb{Z}} a_n z^n,$$

where $a_n \in \mathbb{C}$ for $n \in \mathbb{Z}$ and the series converges normally on A . It then easily follows from (21.7) that

$$f(u) = \sum_{n \in \mathbb{Z}} a_n e^{2\pi i n u}, \quad (21.8)$$

where the latter series converges normally for $u \in S$.

The coefficients a_n here are uniquely determined. Indeed, if $\alpha < t < \beta$, then the series (21.8) converges absolutely and uniformly on the set $[it, 1+it]$; so if $k \in \mathbb{Z}$, then we can multiply it by $e^{-2\pi i k u}$ and integrate term-by-term to obtain

$$\begin{aligned} \int_{[it, 1+it]} f(u) e^{-2\pi i k u} du &= \int_{[it, 1+it]} \left(\sum_{n \in \mathbb{Z}} a_n e^{2\pi i (n-k)u} \right) du \\ &= \sum_{n \in \mathbb{Z}} a_n \underbrace{\left(\int_{[it, 1+it]} e^{2\pi i (n-k)u} du \right)}_{=0 \text{ unless } n=k} = a_k. \end{aligned}$$

□

21.1 Exercises

Problem 21.1. (a) Let $k \in \mathbb{N}_0$. Show that

$$\frac{1}{(1-z)^{k+1}} = \sum_{n=0}^{\infty} \binom{n+k}{n} z^n \text{ for } |z| < 1.$$

(b) Find the Laurent series expansion of the function

$$f(z) = \frac{z^2 - z + 1}{(z-1)(z-2)^2}$$

converging in

(i) $A_1 = \{z \in \mathbb{C} : 0 < |z-1| < 1\},$

(ii) $A_2 = \{z \in \mathbb{C} : 1 < |z-1|\}.$

Problem 21.2. Show that every holomorphic map $f: \widehat{\mathbb{C}} \rightarrow \widehat{\mathbb{C}}$ is a rational function (including constant functions $f = c \in \widehat{\mathbb{C}}$).

Problem 21.3. Let $U \subseteq \mathbb{C}$ be open and $V = \{w \in \mathbb{C} : r < |w - w_0| < R\}$ be an annulus (where $w_0 \in \mathbb{C}$, $r, R \in [0, \infty]$, and $r < R$). Suppose that $f: U \times V \rightarrow \mathbb{C}$ is a continuous map such that $z \mapsto f(z, w)$ is holomorphic on U for each fixed $w \in V$ and $w \mapsto f(z, w)$ is holomorphic on V for each fixed $z \in U$ (a deep theorem due to Hartogs says that the last two conditions imply the continuity of f , but we will not use this).

For $z \in U$ let

$$f(z, w) = \sum_{n=-\infty}^{\infty} a_n(z)(w - w_0)^n$$

be Laurent expansion of $f(z, \cdot)$ on V .

(a) Show that each coefficient $a_n(z)$ considered as a function of z is holomorphic on U .

(b) Show that for $z \in U$ the Laurent series expansion of the holomorphic function $\frac{\partial f}{\partial z}(z, \cdot)$ (see Problem 10.4 (c)) on V is given by

$$\frac{\partial f}{\partial z}(z, w) = \sum_{n=-\infty}^{\infty} a'_n(z)(w - w_0)^n.$$

Problem 21.4. Fix $u \in \mathbb{C}$ and consider the function

$$f_u(z) := \exp\left(\frac{u}{2}\left(z - \frac{1}{z}\right)\right).$$

Then f_u is holomorphic for $z \neq 0$ and has an isolated singularity at 0. Let

$$f_u(z) = \sum_{n=-\infty}^{\infty} J_n(u) z^n$$

be the Laurent series expansion of f_u on $\mathbb{C} \setminus \{0\}$. The coefficient $J_n(u)$, considered as a function of u , is called the n -th *Bessel function*.

(a) Show that

$$J_n(u) = \frac{1}{\pi} \int_0^\pi \cos(u \sin \theta - n\theta) d\theta \quad \text{for } u \in \mathbb{C}, n \in \mathbb{Z}.$$

(b) Show that $J_{-n}(u) = (-1)^n J_n(u)$ for $u \in \mathbb{C}, n \in \mathbb{Z}$.

(c) Show that J_n is an entire function for each $n \in \mathbb{Z}$.

(d) Show that $J_n(u) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!(k+n)!} \left(\frac{u}{2}\right)^{n+2k}$ for $u \in \mathbb{C}, n \in \mathbb{N}_0$.

Problem 21.5. For $z \in \mathbb{C}$ and $n \in \mathbb{Z}$ show following identities for Bessel functions:

$$(a) \quad 2nJ_n(z) = zJ_{n-1}(z) + zJ_{n+1}(z),$$

$$(b) \quad zJ'_n(z) = -nJ_n(z) + zJ_{n-1}(z),$$

$$(c) \quad zJ'_n(z) = nJ_n(z) - zJ_{n+1}(z),$$

$$(d) \quad z^2 J''_n(z) + zJ'_n(z) + (z^2 - n^2)J_n(z) = 0 \text{ (Bessel's Differential Equation).}$$

Problem 21.6 (Functional equation of the Jacobi θ -function). (a) Fix $\tau \in \mathbb{C}$ with $\text{Im}(\tau) > 0$. Show that then

$$f_\tau(u) := \sum_{n \in \mathbb{Z}} e^{i\pi\tau(u+n)^2}$$

is a holomorphic function of $u \in \mathbb{C}$.

(b) The entire function f_τ is 1-periodic (why?) and hence has a Fourier series representation of the form

$$f_\tau(u) = \sum_{n \in \mathbb{Z}} a_n e^{2\pi i n u},$$

where the series converges normally on $\mathbb{C}$. Show that

$$a_k = \frac{1}{\sqrt{-i\tau}} e^{-i\pi k^2/\tau}$$

for $k \in \mathbb{Z}$. Here and below the branch of $\sqrt{-i\tau}$ is determined as in Problem 20.1 (c).

(c) The *Jacobi θ -function* is defined as

$$\theta(\tau) = \sum_{n \in \mathbb{Z}} e^{i\pi n^2 \tau} \quad (21.9)$$

for $\tau \in \mathbb{C}$ with $\text{Im}(\tau) > 0$. Show that θ is a holomorphic function on the upper half-plane $\mathbb{H} = \{\tau \in \mathbb{C} : \text{Im}(\tau) > 0\}$.

(d) Show that $\theta(-1/\tau) = \sqrt{-i\tau} \theta(\tau)$ for $\tau \in \mathbb{H}$.

Problem 21.7. Let $k \in \mathbb{N}$, $k \geq 2$. We know that the Eisenstein series $G_{2k}(\tau)$ is a holomorphic and 1-periodic function of $\tau \in \mathbb{H} = \{\tau \in \mathbb{C} : \text{Im}(\tau) > 0\}$ (see Problem 10.6). According to Theorem 21.8 the function G_{2k} has a Fourier series expansion of the form

$$G_{2k}(\tau) = \sum_{n \in \mathbb{Z}} a_n q^n$$

for $\tau \in \mathbb{H}$, where here and in the following we set $q := e^{2\pi i \tau}$. The purpose of this problem is to find the coefficients a_n explicitly.

(a) Show that

$$\pi \cot(\pi \tau) = -i\pi - 2\pi i \sum_{n=1}^{\infty} q^n$$

for $\tau \in \mathbb{H}$ with normal convergence of the series on $\mathbb{H}$.

(b) Show that

$$\sum_{n \in \mathbb{Z}} \frac{1}{(\tau - n)^\ell} = \frac{(-1)^\ell (2\pi i)^\ell}{(\ell - 1)!} \sum_{n=1}^{\infty} n^{\ell-1} q^n$$

for $\tau \in \mathbb{H}$, $\ell \in \mathbb{N}$, $\ell \geq 2$, with normal convergence of both series on $\mathbb{H}$.

(c) Show that

$$G_{2k}(\tau) = 2\zeta(2k) + 2 \frac{(2\pi i)^{2k}}{(2k-1)!} \sum_{(m,n) \in \mathbb{N} \times \mathbb{N}} n^{2k-1} q^{mn}$$

for $\tau \in \mathbb{H}$ with normal convergence of the series on $\mathbb{H}$. Here ζ is the Riemann ζ -function.

(d) For $n \in \mathbb{N}$ and $\ell \in \mathbb{N}_0$ we denote by $\sigma_\ell(n)$ the sum of the ℓ -th powers of all divisors $d \in \mathbb{N}$ of n , that is,

$$\sigma_\ell(n) := \sum_{d \text{ divides } n} d^\ell.$$

Show that

$$G_{2k}(\tau) = 2\zeta(2k) + 2 \frac{(2\pi i)^{2k}}{(2k-1)!} \sum_{n=1}^{\infty} \sigma_{2k-1}(n) q^n \quad (21.10)$$

for $\tau \in \mathbb{H}$ with normal convergence of the series on $\mathbb{H}$.

Problem 21.8. We use the notation from Problem 21.7. For $k \in \mathbb{N}$, $k \geq 2$, the *normalized* Eisenstein series E_{2k} is defined as $E_{2k} := G_{2k}/(2\zeta(2k))$. In this way, the leading coefficient of the Fourier expansion of E_{2k} corresponding to (21.10) is equal to 1.

(a) Show that

$$E_4(\tau) = 1 + 240 \sum_{n=1}^{\infty} \sigma_3(n) q^n, \quad \text{and} \quad E_8(\tau) = 1 + 480 \sum_{n=1}^{\infty} \sigma_7(n) q^n$$

for $\tau \in \mathbb{H}$.

(b) Show that $E_8(\tau) = E_4(\tau)^2$ for $\tau \in \mathbb{H}$.

(c) Show that

$$\sigma_7(n) = \sigma_3(n) + 120 \sum_{\ell=1}^{n-1} \sigma_3(\ell) \sigma_3(n-\ell)$$

for $n \in \mathbb{N}$.

Problem 21.9. Fix $k \in \mathbb{N}$ and denote by $\Gamma = \text{PSL}_2(\mathbb{Z})$ the modular group (see Problem 17.8). Let $A \in \Gamma$ and suppose that $A(z) = \frac{az+b}{cz+d}$, where $a, b, c, d \in \mathbb{Z}$ and $ad - bc = 1$.

We define an operation $|_k A$ defined on functions $f: \mathbb{H} \rightarrow \mathbb{C}$ on the upper half-plane $\mathbb{H} = \{z \in \mathbb{C} : \text{Im}(z) > 0\}$ by setting

$$(f|_k A)(z) := (cz + d)^{-2k} f\left(\frac{az+b}{cz+d}\right), \quad z \in \mathbb{H}.$$

(a) Show that if $f: \mathbb{H} \rightarrow \mathbb{C}$ is a function and $A, B \in \Gamma$, then

$$(f|_k A)|_k B = f|_k (A \circ B).$$

(b) Show that a function $f: \mathbb{H} \rightarrow \mathbb{C}$ satisfies

$$f\left(\frac{az+b}{cz+d}\right) = (cz+d)^{2k} f(z)$$

for all $z \in \mathbb{H}$ and all $a, b, c, d \in \mathbb{Z}$ with $ad - bc = 1$ if and only if

$$f(-1/z) = z^{2k} f(z) \text{ and } f(z+1) = f(z)$$

for all $z \in \mathbb{H}$.

22 The Residue Theorem

Definition 22.1. Let $U \subseteq \mathbb{C}$ be open, $z_0 \in U$, $f \in H(U \setminus \{z_0\})$, and

$$f(z) = \sum_{n=-\infty}^{\infty} a_n(z - z_0)^n$$

be the unique Laurent series expansion of f converging in a punctured neighborhood of z_0 . Then the coefficient a_{-1} is called the *residue of f at z_0* and denoted by $\text{Res}(f, z_0)$.

Proposition 22.2. Let $U \subseteq \mathbb{C}$ be open, $z_0 \in U$, and $f \in H(U \setminus \{z_0\})$. If f has a pole of order at most $m \in \mathbb{N}$ at z_0 , then

$$\text{Res}(f, z_0) = \frac{1}{(m-1)!} \frac{d^{m-1}}{dz^{m-1}} \left((z - z_0)^m f(z) \right) \Big|_{z=z_0}. \quad (22.1)$$

Proof. Near z_0 we have $f(z) = \frac{1}{(z - z_0)^m} g(z)$, where g is holomorphic; in particular, $z \mapsto (z - z_0)^m f(z) = g(z)$ has a holomorphic extension to $z = z_0$, and it is understood that one uses this holomorphic extension to evaluate the right-hand side of (22.1).

Now near z_0 the function g has a power series representation

$$g(z) = \sum_{n=0}^{\infty} b_n(z - z_0)^n;$$

$$\text{so } \text{Res}(f, z_0) = b_{m-1} = \frac{1}{(m-1)!} g^{(m-1)}(z_0). \quad \square$$

Theorem 22.3 (Residue Theorem). *Let $\Omega \subseteq \mathbb{C}$ be an open set, $a_1, \dots, a_n \in \Omega$ be distinct points, and $f \in H(\Omega \setminus \{a_1, \dots, a_n\})$. Suppose Γ is a piecewise smooth cycle in Ω that is null-homologous in Ω and for which $a_1, \dots, a_n \notin \Gamma^*$. Then*

$$\int_{\Gamma} f(z) dz = 2\pi i \sum_{k=1}^m \text{ind}_{\Gamma}(a_k) \cdot \text{Res}(f, a_k). \quad (22.2)$$

Proof. We can find a small number $r > 0$ such that the disks $\overline{B}(a_k, r)$, $k = 1, \dots, m$, are pairwise disjoint, do not meet Γ^* , and lie in Ω . Define $\gamma_k(t) = a_k + re^{it}$ for $t \in [0, 2\pi]$, and

$$\tilde{\Gamma} = \Gamma - \text{ind}_{\Gamma}(a_1)\gamma_1 - \dots - \text{ind}_{\Gamma}(a_m)\gamma_m.$$

Then $\tilde{\Gamma}$ is a piecewise smooth cycle in $\tilde{\Omega} := \Omega \setminus \{a_1, \dots, a_m\}$. For $z \in \mathbb{C} \setminus \Omega$ we have

$$\text{ind}_{\tilde{\Gamma}}(z) = \text{ind}_{\Gamma}(z) = 0;$$

moreover, if $k \in \{1, \dots, m\}$, then

$$\text{ind}_{\tilde{\Gamma}}(a_k) = \text{ind}_{\Gamma}(a_k) - \underbrace{\text{ind}_{\Gamma}(a_k) \cdot \text{ind}_{\gamma_k}(a_k)}_{=1} = 0.$$

Hence $\tilde{\Gamma}$ is null-homologous in $\tilde{\Omega}$. Since f is holomorphic on $\tilde{\Omega}$, it follows from Cauchy's Integral Theorem that

$$\int_{\tilde{\Gamma}} f(z) dz = \int_{\Gamma} f(z) dz - \sum_{k=1}^m \text{ind}_{\Gamma}(a_k) \cdot \int_{\gamma_k} f(z) dz = 0. \quad (22.3)$$

Now if $k \in \{1, \dots, m\}$ is fixed and $\sum_{n=-\infty}^{\infty} c_n(z - a_k)^n$ is the Laurent series representation of f near a_k , then this Laurent series converges uniformly for $z \in \gamma_k^*$. Hence

$$\int_{\gamma_k} f(z) dz = \sum_{n=-\infty}^{\infty} c_n \int_{\gamma_k} (z - a_k)^n dz = 2\pi i c_{-1} = 2\pi i \text{Res}(f, a_k).$$

From this and (22.3), the relation (22.2) follows. $\square$

Remark 22.4. In the proof of Theorem 20.7 we derived Cauchy's Integral Theorem from Cauchy's Integral Formula. In the previous argument we used Cauchy's Integral Theorem for the proof of the Residue Theorem. One can complete the circle by noting that the Residue Theorem implies Cauchy's Integral Formula. Actually, if the assumptions are as in Theorem 20.7, then

$$\text{ind}_{\Gamma}(z) \frac{f^{(n)}(z)}{n!} = \frac{1}{2\pi i} \int_{\Gamma} \frac{f(w)}{(w - z)^{n+1}} dw \quad (22.4)$$

for all $z \in \Omega \setminus \Gamma^*$ and $n \in \mathbb{N}_0$. This immediately follows from the Residue Theorem by noting that for $n \in \mathbb{N}_0$ the function

$$g_n(w) := \frac{f(w)}{(w - z)^{n+1}}$$

is holomorphic on $H(U \setminus \{z\})$ and has a pole of order at most $n + 1$ at z ; so

$$\operatorname{Res}(g_n, z) = \frac{1}{n!} \frac{d^n}{dw^n} \left((w - z)^{n+1} g_n(w) \right) \Big|_{w=z} = \frac{f^{(n)}(z)}{n!}.$$

The Residue Theorem is a very powerful tool for the computation of certain integrals.

Example 22.5. $I := \int_0^{2\pi} \frac{dt}{10 + 8 \cos t}$. To compute this integral, we let $z = \gamma(t) := e^{it}$ for $t \in [0, 2\pi]$, and convert I into an path integral over the loop γ . So we want to perform the usual substitution backwards. Here (symbolically)

$$dz = ie^{it} dt = iz dt \quad \text{and so} \quad dt = \frac{dz}{iz}.$$

Moreover, $\cos t = \frac{1}{2}(e^{it} + e^{-it}) = \frac{1}{2}\left(z + \frac{1}{z}\right)$, which gives

$$I = \int_0^{2\pi} \frac{dt}{10 + 8 \cos t} = \int_{\gamma} \frac{1}{10 + 4(z + 1/z)} \cdot \frac{dz}{iz} = \frac{1}{i} \int_{\gamma} \frac{dz}{4z^2 + 10z + 4}.$$

Now

$$4z^2 + 10z + 4 = 4(z + 1/2)(z + 2).$$

The function $f(z) := \frac{1}{(z + 1/2)(z + 2)}$ is holomorphic on $\mathbb{C}$ except for two poles of order 1 at $-1/2$ and -2 . So by the Residue Theorem,

$$\begin{aligned} I &= \frac{1}{4i} \int_{\gamma} \frac{dz}{(z + 1/2)(z + 2)} \\ &= \frac{2\pi i}{4i} \left(\underbrace{\operatorname{ind}_{\gamma}(-1/2)}_{=1} \cdot \operatorname{Res}(f, -1/2) + \underbrace{\operatorname{ind}_{\gamma}(-2)}_{=0} \cdot \operatorname{Res}(f, -2) \right) \\ &= \frac{\pi}{2} \cdot \frac{1}{z + 2} \Big|_{z=-1/2} = \frac{\pi}{2} \cdot \frac{1}{3/2} = \frac{\pi}{3}. \end{aligned}$$

Example 22.6.

$$I := \int_0^\infty \frac{dx}{(x^2 + 1)(x^2 + 4)} = \lim_{R \rightarrow \infty} \int_0^R \frac{dx}{(x^2 + 1)(x^2 + 4)}.$$

Then I is a convergent improper integral, and so the last limit exists (which also follows from the argument below).

For fixed $R > 0$ define $\alpha_R(t) = t$ for $t \in [-R, R]$, $\beta_R(t) = Re^{it}$ for $t \in [0, \pi]$, and $\gamma_R = \alpha_R + \beta_R$ (where this chain is considered as a loop obtained from concatenating α_R and β_R). Note that

$$(z^2 + 1)(z^2 + 4) = (z + i)(z - i)(z + 2i)(z - 2i).$$

So the rational function $f(z) = \frac{1}{(z^2 + 1)(z^2 + 4)}$ has four poles of order 1 at $\pm i$ and $\pm 2i$. For $R > 2$ we have

$$\text{ind}_{\gamma_R}(i) = \text{ind}_{\gamma_R}(2i) = 1 \quad \text{and} \quad \text{ind}_{\gamma_R}(-i) = \text{ind}_{\gamma_R}(-2i) = 0.$$

The Residue Theorem implies that

$$\begin{aligned} \int_{\gamma_R} f(z) dz &= 2\pi i \text{Res}(f, i) + 2\pi i \text{Res}(f, 2i) \\ &= 2\pi i \cdot \frac{1}{(z + i)(z^2 + 4)} \Big|_{z=i} + 2\pi i \cdot \frac{1}{(z^2 + 1)(z + 2i)} \Big|_{z=2i} \\ &= \frac{2\pi i}{2i \cdot 3} + \frac{2\pi i}{-3 \cdot 4i} = \frac{\pi}{3} - \frac{\pi}{6} = \frac{\pi}{6}. \end{aligned}$$

Now

$$\int_{\alpha_R} f(z) dz = \int_{-R}^R \frac{dx}{(x^2 + 1)(x^2 + 4)} = 2 \int_0^R \frac{dx}{(x^2 + 1)(x^2 + 4)},$$

and

$$\begin{aligned} \left| \int_{\beta_R} f(z) dz \right| &\leq \ell(\beta_R) \cdot \max_{z \in \beta_R^*} \frac{1}{|z^2 + 1| \cdot |z^2 + 4|} \\ &\leq \frac{\pi R}{(R^2 - 1)(R^2 - 4)} \rightarrow 0 \quad \text{as } R \rightarrow \infty. \end{aligned}$$

It follows that

$$\begin{aligned}
 I &= \lim_{R \rightarrow \infty} \int_0^R \frac{dx}{(x^2 + 1)(x^2 + 4)} = \frac{1}{2} \lim_{R \rightarrow \infty} \int_{\alpha_R} f(z) dz \\
 &= \frac{1}{2} \lim_{R \rightarrow \infty} \left(\int_{\gamma_R} f(z) dz - \int_{\beta_R} f(z) dz \right) \\
 &= \frac{\pi}{12} - \frac{1}{2} \lim_{R \rightarrow \infty} \int_{\beta_R} f(z) dz = \frac{\pi}{12}.
 \end{aligned}$$

Example 22.7.

$$I := \int_0^\infty \frac{dx}{\sqrt{x}(x+1)(x+2)}.$$

This is an improper integral for both integral limits 0 and ∞ . It is easy to see that it converges. One can compute I by making the substitution $x = u^2$ and using the method from Example 22.6.

We want to evaluate I based on a different method that applies to a larger class of integrals, for example the integrals obtained by replacing $\sqrt{x}$ in the integrand by a power x^α , $\alpha \in (0, 1)$. The main point here is that the square root function cannot be extended to holomorphic function on $\mathbb{C}^*$.

We let $\sqrt{z}$ be the branch of the square root with branch cut along the non-negative real axis, normalized so that $\sqrt{-1} = i$. Then $\sqrt{re^{it}} = \sqrt{r}e^{it/2}$ for $r > 0$, $t \in (0, 2\pi)$.

Fix $R > 2$ and two small numbers $\epsilon > 0$ and $\delta > 0$. We define $\gamma_1(t) = e^{i\delta}t$ for $t \in [\epsilon, R]$, $\gamma_2(t) = Re^{it}$ for $t \in [\delta, 2\pi - \delta]$, $\gamma_3(t) = e^{-i\delta}(R - t)$ for $t \in [0, R - \epsilon]$, and $\gamma_4(t) = \epsilon e^{-it}$ for $t \in [\delta, 2\pi - \delta]$. We consider the piecewise smooth loop $\gamma := \gamma_1 + \gamma_2 + \gamma_3 + \gamma_4$ obtained from concatenation of these paths. The paths depend on the parameters ϵ , δ , and R , but we suppress this in our notation.

Let

$$f(z) := \frac{1}{\sqrt{z}(z+1)(z+2)} \quad \text{for } z \in \Omega \setminus \{-1, -2\},$$

where $\Omega := \mathbb{C} \setminus [0, \infty)$. Then f is holomorphic on $\Omega \setminus \{-1, -2\}$ and has simple poles at -1 and -2 . Moreover, $\text{ind}_\gamma(z) = 0$ for $z \in [0, \infty)$ and so γ is

null-homologous in Ω . We have $\text{ind}_\gamma(-1) = \text{ind}_\gamma(-2) = 1$, and

$$\begin{aligned}\text{Res}(f, -1) &= \frac{1}{\sqrt{z}(z+2)} \Big|_{z=-1} = \frac{1}{\sqrt{-1} \cdot 1} = \frac{1}{i}, \\ \text{Res}(f, -2) &= \frac{1}{\sqrt{z}(z+1)} \Big|_{z=-2} = \frac{1}{\sqrt{-2} \cdot (-1)} = -\frac{1}{\sqrt{2}i}.\end{aligned}$$

So by the Residue Theorem,

$$\int_\gamma f(z) dz = 2\pi i (\text{Res}(f, -1) + \text{Res}(f, -2)) = 2\pi \left(1 - \frac{1}{\sqrt{2}}\right).$$

If $x > 0$ and z approaches x from the upper or lower half-plane, then $\sqrt{z} \rightarrow \sqrt{x}$ or $\sqrt{z} \rightarrow -\sqrt{x}$, respectively. From this one can derive that

$$\lim_{\delta, \epsilon \rightarrow 0} \int_{\gamma_1} f(z) dz = \int_0^R \frac{dx}{\sqrt{x}(x+1)(x+2)},$$

and

$$\lim_{\delta, \epsilon \rightarrow 0} \int_{\gamma_3} f(z) dz = \int_R^0 \frac{dx}{-\sqrt{x}(x+1)(x+2)} = \int_0^R \frac{dx}{\sqrt{x}(x+1)(x+2)}.$$

Moreover,

$$\left| \int_{\gamma_4} f(z) dz \right| \leq \frac{2\pi\epsilon}{\sqrt{\epsilon}(1-\epsilon)(2-\epsilon)} \rightarrow 0 \quad \text{as } \epsilon \rightarrow 0,$$

and

$$\left| \int_{\gamma_2} f(z) dz \right| \leq \frac{2\pi R}{\sqrt{R}(R-1)(R-2)} \rightarrow 0 \quad \text{as } R \rightarrow 0.$$

It follows that

$$\begin{aligned}2\pi \left(1 - \frac{1}{\sqrt{2}}\right) &= \int_\gamma f = \lim_{R \rightarrow \infty} \left(\lim_{\delta, \epsilon \rightarrow 0} \left(\int_{\gamma_1} f + \int_{\gamma_2} f + \int_{\gamma_3} f + \int_{\gamma_4} f \right) \right) \\ &= \lim_{R \rightarrow \infty} 2 \int_0^R \frac{dx}{\sqrt{x}(x+1)(x+2)} \\ &= \int_0^\infty \frac{dx}{\sqrt{x}(x+1)(x+2)} = 2I,\end{aligned}$$

and so $I = \pi \left(1 - \frac{1}{\sqrt{2}}\right)$.

Theorem 22.8 (Argument Principle). *Let $U \subseteq \mathbb{C}$ be a region, and f be a non-constant meromorphic function on U . Let $\gamma: [0, 1] \rightarrow U$ be a piecewise smooth loop in U that is null-homologous in U and satisfies $\text{ind}_\gamma(z) \in \{0, 1\}$ for each $z \in U \setminus \gamma^*$; in addition assume that there is no zero or pole of f on γ^* .*

Let $\Omega := \{z \in U \setminus \gamma^ : \text{ind}_\gamma(z) = 1\}$, and N_f and P_f be the number of zeros and poles of f in Ω , respectively, accounted according to multiplicity.*

Then N_f and P_f are finite, and

$$\text{ind}_\alpha(0) = \frac{1}{2\pi i} \int_\gamma \frac{f'(z)}{f(z)} dz = N_f - P_f,$$

where $\alpha := f \circ \gamma$.

Proof. First note that

$$\text{ind}_\alpha(0) = \frac{1}{2\pi i} \int_\alpha \frac{dw}{w} = \frac{1}{2\pi i} \int_0^1 \frac{f'(\gamma(t)) \cdot \gamma'(t)}{f(\gamma(t))} dt = \frac{1}{2\pi i} \int_\gamma \frac{f'(z)}{f(z)} dz.$$

If a is a zero of f of order $m \in \mathbb{N}$, then there exists a function g holomorphic near a such that $g(a) \neq 0$ and

$$f(z) = (z - a)^m g(z) \quad \text{near } a.$$

Then $f'(z) = m(z - a)^{m-1}g(z) + (z - a)^m g'(z)$, and so

$$\frac{f'(z)}{f(z)} = \frac{m}{z - a} + \frac{g'(z)}{g(z)} \quad \text{near } a.$$

Hence f'/f has a simple pole at a , and

$$\text{Res}(f'/f, a) = m = \text{multiplicity of the zero } a.$$

Similarly, if a is a pole of f , then there exists a function g holomorphic near a such that $g(a) \neq 0$ and

$$f(z) = \frac{1}{(z - a)^m} g(z) \quad \text{near } a.$$

Then

$$f'(z) = -\frac{m}{(z - a)^{m+1}} g(z) + \frac{1}{(z - a)^m} g'(z),$$

and so

$$\frac{f'(z)}{f(z)} = \frac{-m}{z-a} + \frac{g'(z)}{g(z)} \quad \text{near } a.$$

Hence f'/f has a simple pole at a , and

$$\text{Res}(f'/f, a) = -m = -\text{multiplicity of the pole } a.$$

The set $\Omega = \{z \in U \setminus \gamma^* : \text{ind}_\gamma(z) = 1\} \subseteq U$ is open as follows from Theorem 19.14. Let A be the set of zeros and poles of f in Ω , and B be the set of the remaining zeros and poles of f in U . Both A and B consists of isolated points. If $\tilde{U} := U \setminus B$, then $\tilde{U}$ is an open set and γ is null-homologous in $\tilde{U}$.

Note that the set A is finite (which implies that N_f and P_f are finite); indeed, A is a bounded set, because it must be disjoint from the unbounded component of $\mathbb{C} \setminus \gamma^*$; so if A were infinite, then it had a limit point w . Then necessarily $w \in \partial U \subseteq \mathbb{C} \setminus U$, and so $\text{ind}_\gamma(w) = 0$, because γ is null-homologous in U . Then $\text{ind}_\gamma(a) = 0$ for points $a \in A$ close to w by Theorem 19.14. This is impossible as $\text{ind}_\gamma(a) = 1$ for $a \in A$.

Since $f'/f \in H(\tilde{U} \setminus A)$ we can apply the Residue Theorem and obtain

$$\frac{1}{2\pi i} \int_\gamma \frac{f'(z)}{f(z)} dz = \sum_{a \in A} \text{Res}(f'/f, a) = N_f - P_f. \quad \square$$

Theorem 22.9 (Rouché's Theorem). *Let $U \subseteq \mathbb{C}$ be a region, and f and g be non-constant holomorphic functions on U . Let $\gamma: [0, 1] \rightarrow U$ be a piecewise smooth path that is null-homologous in U and satisfies $\text{ind}_\gamma(z) \in \{0, 1\}$ for each $z \in U \setminus \gamma^*$; in addition assume that there is no zero of f on γ^* .*

Let $\Omega = \{z \in U \setminus \gamma^ : \text{ind}_\gamma(z) = 1\}$, and N_f and N_g be the numbers of zeros of f and g in Ω , respectively, counted according to multiplicity.*

If

$$|g(z) - f(z)| < |f(z)| \quad \text{for all } z \in \gamma^*, \quad (22.5)$$

then $N_f = N_g$.

So if the functions f and g are sufficiently close on γ^* , then they have the same number of zeros in Ω .

Proof. Note that (22.5) implies that $g(z) \neq 0$ for $z \in \gamma^*$. Hence we can apply the Argument Principle to both f and g and conclude $N_f = \text{ind}_{f \circ \gamma}(0)$ and

$N_g = \text{ind}_{g \circ \gamma}(0)$. The statement will follow from the homotopy invariance of winding numbers if we can show that the loops $f \circ \gamma$ and $g \circ \gamma$ are homotopic in $\mathbb{C}^*$.

We define $H: [0, 1] \times [0, 1] \rightarrow \mathbb{C}^*$ by

$$H(s, t) = f(\gamma(s)) + t(g(\gamma(s)) - f(\gamma(s))) \quad \text{for } s, t \in [0, 1].$$

Then H is continuous. By (22.5) we have

$$|H(s, t)| \geq |f(\gamma(s))| - |g(\gamma(s)) - f(\gamma(s))| > 0$$

for all $s, t \in [0, 1]$. So H maps into $\mathbb{C}^*$. Since γ is a loop, and so $\gamma(0) = \gamma(1)$, we have $H(0, t) = H(1, t)$ for all $t \in [0, 1]$. So $H(\cdot, t)$ is a loop for all $t \in [0, 1]$. Since $H(\cdot, 0) = f \circ \gamma$ and $H(\cdot, 1) = g \circ \gamma$, the loops $f \circ \gamma$ and $g \circ \gamma$ are indeed homotopic in $\mathbb{C}^*$ as desired. $\square$

Example 22.10. (a) Consider the polynomial $g(z) = z^9 - 2z^6 + z^4 - 9z^2 + 3$. Then one can use Rouché's Theorem to show that g has precisely two zeros (possibly one double zero) in the unit disk $\mathbb{D}$. Indeed, in Theorem 22.9 we let $U := \mathbb{C}$, $\gamma(t) := e^{2\pi it}$ for $t \in [0, 1]$, and $f(z) := -9z^2$. Note that then $\Omega = \{z \in \mathbb{C} \setminus \gamma^* : \text{ind}_\gamma(z) = 1\} = \mathbb{D}$, and that for $z \in \gamma^* = \partial\mathbb{D}$ we have

$$\begin{aligned} |g(z) - f(z)| &= |z^9 - 2z^6 + z^4 + 3| \leq |z|^9 + 2|z|^6 + |z|^4 + 3 = 7 \\ &< 9 = |-9z^2| = |f(z)|. \end{aligned}$$

So N_g , the number of zeros of g in $\mathbb{D}$, is equal to $N_f = 2$, the number of zeros of f in $\mathbb{D}$.

(b) One can derive the Fundamental Theorem of Algebra from Rouché's Theorem. If

$$g(z) = z^n + a_{n-1}z^{n-1} + \cdots + a_0$$

is a polynomial of degree $n \geq 1$, we let $f(z) = z^n$, and $\gamma(t) = Re^{2\pi it}$ for $t \in [0, 1]$. For sufficiently large $R \geq 1$ and $M := |a_{n-1}| + \cdots + |a_0|$ we have for $z \in \gamma^*$,

$$|g(z) - f(z)| = |a_{n-1}z^{n-1} + \cdots + a_0| \leq MR^{n-1} < R^n = |z|^n = |f(z)|.$$

Since f has n zeros in $D(0, R)$, the polynomial g has also $n \geq 1$ zeros.

22.1 Exercises

Problem 22.1. Compute the following integrals by using the Residue Theorem:

- (a) $\int_{\gamma} \frac{z dz}{(z-1) \sin^2(\pi z)}$, where $\gamma(t) = \frac{5}{2}e^{it}$ for $t \in [0, 2\pi]$,
- (b) $\int_{-\infty}^{\infty} \frac{\cos x}{(x^2+1)^2} dx$,
- (c) $\int_1^{\infty} \frac{dx}{x\sqrt{x^2-1}}$. Hint: You need two branch cuts.

Problem 22.2. The value of the integral $I = \int_{-\infty}^{\infty} e^{-x^2} dx$ can easily be obtained from the computation

$$I^2 = \int_{\mathbb{R}^2} e^{-(x^2+y^2)} dx dy = 2\pi \int_0^{\infty} r e^{-r^2} dr = \pi \lim_{R \rightarrow \infty} [-e^{-r^2}]_0^R = \pi.$$

The purpose of this problem is to present an alternative method to compute I based on the Residue Theorem.

(a) Let $a := (1+i)\sqrt{\frac{\pi}{2}}$ (note that $a^2 = i\pi$). Find a meromorphic function f on $\mathbb{C}$ such that $f(z) - f(z+a) = e^{-z^2}$ for $z \in \mathbb{C}$. Hint: the function is of the form $f(z) = g(z)h(z)$, where the first factor $g(z)$ is the “obvious” one and the other factor $h(z)$ has an additional useful property (which motivated the choice of the constant a).

(b) Now determine I as follows: integrate f over the boundary of a conveniently chosen parallelogram, apply the Residue Theorem, and use suitable estimates.

Problem 22.3. (a) Fix $n \in \mathbb{N}$ and consider the square Q_n centered at 0 of sidelength $2n+1$ (so the vertices of Q_n are $\pm(n+1/2) \pm i(n+1/2)$). Show that

$$\max_{z \in \partial Q_n} |\cot(\pi z)| \leq 2.$$

(b) Let $f = P/Q$, where $P, Q \neq 0$ are polynomials with $\deg Q \geq \deg P + 2$. Assume that $Q^{-1}(0) \cap \mathbb{Z} = \emptyset$ and define $g(z) = f(z) \cot(\pi z)$ for $z \in \mathbb{C}$. Show that $\sum_{n=-\infty}^{\infty} f(n)$ converges and that $\sum_{n=-\infty}^{\infty} f(n) = -\pi \sum_{a \in Q^{-1}(0)} \text{Res}(g, a)$.

(c) Show that
$$\sum_{n=-\infty}^{\infty} \frac{1}{n^2 + 1} = \pi \frac{1 + e^{-2\pi}}{1 - e^{-2\pi}}.$$

Problem 22.4. (a) Let $z_0, w_0 \in \mathbb{C}$, and φ be a conformal map of a neighborhood of z_0 onto a neighborhood of w_0 . Show that if f is holomorphic in a punctured neighborhood of w_0 , then

$$\operatorname{Res}(f, w_0) = \operatorname{Res}((f \circ \varphi) \cdot \varphi', z_0).$$

(b) Let f be a function that is holomorphic in a punctured neighborhood of $\infty \in \widehat{\mathbb{C}}$, i.e., holomorphic outside a disk $\overline{B}(0, R)$, where $R > 0$ is sufficiently large. Let

$$f(z) = \sum_{n=-\infty}^{\infty} a_n z^n$$

be the unique Laurent series expansion of f on $\mathbb{C} \setminus \overline{B}(0, R)$. Then we define the *residue* $\operatorname{Res}(f, \infty)$ of f at ∞ as

$$\operatorname{Res}(f, \infty) := -a_{-1}.$$

Note that if $g(z) = f(1/z)$ for z near 0, then in general $\operatorname{Res}(f, \infty)$ does not agree with $\operatorname{Res}(g, 0)$, but the definition of $\operatorname{Res}(f, \infty)$ is consistent with the transformation rule for residues established in (a) (why?).

Show that if γ is a piecewise smooth loop in $\mathbb{C} \setminus \overline{B}(0, R)$, then

$$\int_{\gamma} f(z) dz = -2\pi i \cdot \operatorname{ind}_{\gamma}(0) \cdot \operatorname{Res}(f, \infty).$$

(c) Let R be a non-constant rational function. Show that

$$\sum_{z_0 \in \widehat{\mathbb{C}}} \operatorname{Res}(R, z_0) = 0.$$

Problem 22.5. Let $f: \overline{\mathbb{D}} \rightarrow \mathbb{C}$ be a non-constant continuous function such that $f|_{\mathbb{D}}$ is holomorphic. Define $\alpha(t) := f(e^{it})$ for $t \in [0, 2\pi]$.

(a) Prove the following version of the Argument Principle: Let $w \in \mathbb{C} \setminus \alpha^*$ and denote by N_w the number of times $f|_{\mathbb{D}}$ attains the value w counting multiplicities (so N_w is the number of zeros of $f - w$). Show that N_w is finite and $N_w = \operatorname{ind}_{\alpha}(w)$.

Assume in addition that $\text{ind}_\alpha(w) \in \{0, 1\}$ for all $w \in \mathbb{C} \setminus \alpha^*$, and define

$$\Omega_k = \{w \in \mathbb{C} \setminus \alpha^* : \text{ind}_\alpha(w) = k\}$$

for $k = 0, 1$. Suppose that $\alpha^* \subseteq \overline{\Omega}_0$.

(b) Show that f maps $\mathbb{D}$ conformally onto Ω_1 .

Problem 22.6. Let $\Gamma = \{n\omega_1 + k\omega_2 : k, n \in \mathbb{Z}\}$ be a rank-2 lattice where $\omega_1, \omega_2 \in \mathbb{C}$ are linearly independent over $\mathbb{R}$. Recall (see Problem 15.4) that then we obtain an equivalence relation $\sim$ on $\mathbb{C}$ by setting $z_0 \sim z_1$ for $z_0, z_1 \in \mathbb{C}$ iff $z_0 - z_1 \in \Gamma$. Again we denote by $[z_0] := \{z_1 \in \mathbb{C} : z_0 \sim z_1\}$ the equivalence class of a point $z_0 \in \mathbb{C}$, and by $\mathbb{C}/\Gamma := \{[z_0] : z_0 \in \mathbb{C}\}$ the corresponding quotient space.

(a) Show that if $u_0 \in \mathbb{C}$ and $F := \{u_0 + s\omega_1 + t\omega_2 : s, t \in [0, 1)\}$, then each equivalence class in $\mathbb{C}/\Gamma$ has a unique representative in F .

Now suppose that f is an *elliptic function with respect to* Γ , that is, f is a non-constant meromorphic function on $\mathbb{C}$ such that $f(z + \gamma) = f(z)$ for $z \in \mathbb{C}$ and $\gamma \in \Gamma$.

(b) Show that if $z_0, z_1 \in \mathbb{C}$ and $z_0 \sim z_1$, then f has a pole at z_0 if and only if f has a pole at z_1 . Moreover, in this case the multiplicities (=orders) of the poles at z_0 and z_1 are the same and we have $\text{Res}(f, z_0) = \text{Res}(f, z_1)$.

It follows from (b) that whether f has a pole at a point $z_0 \in \mathbb{C}$ and the multiplicity $m \in \mathbb{N}$ of the pole only depend on the equivalence class $[z_0]$. In this case, we say that f has a pole at $[z_0]$ with multiplicity m .

(c) Show that there are only finitely many equivalence classes in $\mathbb{C}/\Gamma$, where f has a pole.

We denote by N_∞ the (finite) sum of all multiplicities of poles in $\mathbb{C}/\Gamma$ and refer to it as the *number of poles of f in $\mathbb{C}/\Gamma$ counting multiplicities*.

(d) For a pole $[z_0] \in \mathbb{C}/\Gamma$ of f we set $\text{Res}(f, [z_0]) := \text{Res}(f, z_0)$. By (b) this is well-defined. Show that

$$\sum_{[z_0] \in \mathbb{C}/\Gamma} \text{Res}(f, [z_0]) = 0$$

(note that by (c) this sum has only finitely many non-zero terms). Hint: use (a) and the Residue Theorem.

(e) Let $w \in \mathbb{C}$. Then if $f(z_0) = w$ for some $z_0 \in \mathbb{C}$, we call the order of the zero of $f - w$ at z_0 the *multiplicity* at which the value w is taken at z_0 by f . Show that whether f takes the value w at a point $z_0 \in \mathbb{C}$ and the multiplicity $m \in \mathbb{N}$ of w at z_0 only depends on the equivalence class $[z_0]$. In this case, we say that f takes the value w at $[z_0]$ with multiplicity m .

(f) Let N_w be the sum of all multiplicities of $w \in \mathbb{C}$ over all $[z_0] \in \mathbb{C}/\Gamma$ at which w is attained. Show that $N_w = N_\infty$ and conclude that *the function f attains all values $w \in \hat{\mathbb{C}}$ in $\mathbb{C}/\Gamma$ the same (finite) number of times, counting multiplicities*.

(g) Show that $N_\infty \neq 1$ and so f attains each value $w \in \hat{\mathbb{C}}$ in $\mathbb{C}/\Gamma$ more than once, counting multiplicities.

Problem 22.7. Let $\Gamma = \{n\omega_1 + k\omega_2 : k, n \in \mathbb{Z}\}$ be a rank-2 lattice where $\omega_1, \omega_2 \in \mathbb{C}$ are linearly independent over $\mathbb{R}$, and $\wp$ be the associated Weierstrass $\wp$ -function as in Problem 16.1. We use the notation and terminology from Problem 22.6.

(a) Show that $\wp$ attains each value $w \in \hat{\mathbb{C}}$ in $\mathbb{C}/\Gamma$ twice, and $\wp'$ attains each value $w \in \hat{\mathbb{C}}$ in $\mathbb{C}/\Gamma$ three times, counting multiplicities.

(b) Show that if $z_0 \in \frac{1}{2}\Gamma$, then $\wp(z_0 + z) = \wp(z_0 - z)$ for all $z \in \mathbb{C}$.

(c) Show that $\wp'$ attains the value 0 in $\mathbb{C}/\Gamma$ precisely at $[\omega_1/2]$, $[\omega_2/2]$, and $[(\omega_1 + \omega_2)/2]$.

(d) Define $e_1 := \wp(\omega_1/2)$, $e_2 := \wp(\omega_2/2)$, $e_3 := \wp((\omega_1 + \omega_2)/2)$. Show that the values e_1, e_2, e_3 are distinct and satisfy

$$\wp'(z)^2 = 4(\wp(z) - e_1)(\wp(z) - e_2)(\wp(z) - e_3)$$

for $z \in \mathbb{C}$ and $e_1 + e_2 + e_3 = 0$. Hint: Problem 16.3.

23 Normal families

Definition 23.1 (Normal families). Let $U \subseteq \mathbb{C}$ be open, and $\mathcal{F}$ be a family of complex-valued continuous functions on U . We say that $\mathcal{F}$ is a *normal family* if every sequence $\{f_n\}$ of functions in $\mathcal{F}$ has a subsequence $\{f_{n_k}\}$ that converges to some function $f: U \rightarrow \mathbb{C}$ locally uniformly on U .

We do *not* require that the sublimit f belongs to $\mathcal{F}$.

Definition 23.2. Let $U \subseteq \mathbb{C}$ be open, and $\mathcal{F}$ be a family of complex-valued continuous functions on U . Then the family $\mathcal{F}$ is called:

- (a) *equicontinuous at a point* $z_0 \in U$ if for all $\epsilon > 0$ there exists $\delta > 0$ such that for all $f \in \mathcal{F}$ and all $z \in U$ we have the implication

$$|z - z_0| < \delta \Rightarrow |f(z) - f(z_0)| < \epsilon.$$

- (b) *uniformly bounded on* $A \subseteq U$ if there exists $M \geq 0$ such that for all $f \in \mathcal{F}$ and all $z \in A$ we have

$$|f(z)| \leq M.$$

The family is called *uniformly bounded at* $z_0 \in U$ if it is uniformly bounded on $A = \{z_0\}$. It is called *locally uniformly bounded* if every point $z_0 \in U$ has a neighborhood on which $\mathcal{F}$ is uniformly bounded.

The following theorem is a version of the Arzelà–Ascoli Theorem. It gives a complete characterization of normal families.

Theorem 23.3 (Arzelà–Ascoli). *Let $U \subseteq \mathbb{C}$ be open, and $\mathcal{F}$ be a family of complex-valued continuous functions on U . Then $\mathcal{F}$ is a normal family if and only if $\mathcal{F}$ is equicontinuous and uniformly bounded at each point $z_0 \in U$.*

Proof. $\Rightarrow$: Suppose $\mathcal{F}$ is a normal family. To show that it is equicontinuous and uniformly bounded at each point in U we argue by contradiction.

Suppose $\mathcal{F}$ is not uniformly bounded at some point $z_0 \in U$. Then for each $n \in \mathbb{N}$ there exists $f_n \in \mathcal{F}$ such that $|f_n(z_0)| \geq n$. Then the sequence $\{f_n\}$ cannot have a subsequence that converges locally uniformly on U , because no subsequence even has a pointwise limit at z_0 . This contradicts the normality of $\mathcal{F}$.

Suppose $\mathcal{F}$ is not equicontinuous at some point $z_0 \in U$. Then there exists $\epsilon_0 > 0$ (“bad ϵ ”) such that for each $n \in \mathbb{N}$ there exists $f_n \in \mathcal{F}$ and $z_n \in U$ with $|z_n - z_0| < 1/n$ such that

$$|f_n(z_n) - f_n(z_0)| \geq \epsilon_0 > 0. \quad (23.1)$$

Since $\mathcal{F}$ is normal, we may assume (by passing to a subsequence if necessary) that f_n converges to some continuous function $f: U \rightarrow \mathbb{C}$ locally uniformly on U . Then $f_n(z_0) \rightarrow f(z_0)$ as $n \rightarrow \infty$. Since $z_n \rightarrow z_0$, the convergence $f_n \rightarrow f$ is locally uniform, and f is continuous at z_0 , we also have $f_n(z_n) \rightarrow f(z_0)$ (exercise!). Hence $|f_n(z_n) - f_n(z_0)| \rightarrow 0$ as $n \rightarrow \infty$, contradicting (23.1).

$\Leftarrow$: Suppose $\mathcal{F}$ is equicontinuous and uniformly bounded at each point in U . Let $\{f_n\}$ be an arbitrary sequence in $\mathcal{F}$. In order to establish that $\mathcal{F}$ is a normal family, we have to show that $\{f_n\}$ has a subsequence that converges locally uniformly on U .

We choose a countable dense subset $D := \{p_n : n \in \mathbb{N}\}$ in U (for example the set of all points in U whose real and imaginary parts are rational numbers). Since $\mathcal{F}$ is uniformly bounded at each point in U , for each $k \in \mathbb{N}$ the sequence $\{f_n(p_k)\}$ is bounded. Hence we can find a subsequence $\{f_{1n}\}$ of $\{f_n\}$ such that $\{f_{1n}(p_1)\}$ converges. Using the boundedness of $\{f_{1n}(p_2)\}$, we can find a subsequence $\{f_{2n}\}$ of $\{f_{1n}\}$ such that $\{f_{2n}(p_2)\}$ converges. Repeated this argument, for each $k \in \mathbb{N}$ we can find a subsequence $\{f_{kn}\}$ of $\{f_n\}$ such that $\{f_{kn}(p_k)\}$ converges and such that $\{f_{k+1n}\}$ is a subsequence of $\{f_{kn}\}$. Consider the “diagonal” subsequence $\{g_n\}$ given by $g_n := f_{nn}$ for $n \in \mathbb{N}$. Then $\{g_n\}$ is a subsequence of $\{f_n\}$, and, if we disregard finitely many terms, $\{g_n(p_k)\}$ is a subsequence of $\{f_{kn}(p_k)\}$ for each $k \in \mathbb{N}$. Hence $\{g_n(p)\}$ converges for each $p \in D$.

Claim. For each compact set $K \subseteq U$ and each $\epsilon > 0$ there exists $N \in \mathbb{N}$ such that

$$|g_k(z) - g_n(z)| \leq \epsilon \quad \text{for all } z \in K \text{ and all } k, n \in \mathbb{N} \text{ with } k, n \geq N. \quad (23.2)$$

To see this, let $\epsilon > 0$ and $K \subseteq U$ be compact. Since $\mathcal{F}$ and hence also the subfamily $\{g_n\}$ is equicontinuous at each point in U , for each $w \in K$ there exists $\delta_w > 0$ such that

$$|g_n(z) - g_n(w)| \leq \epsilon/6 \quad \text{for all } n \in \mathbb{N} \text{ and all } z \in B(w, \delta_w) \subseteq U. \quad (23.3)$$

Since K is compact there exist finitely many points $w_1, \dots, w_m \in K$ such that the balls $B_j := B(w_j, \delta_{w_j})$, $j = 1, \dots, m$, cover K . Since D is dense in

U , for each $j \in \{1, \dots, m\}$ we can pick a point $p'_j \in D \cap B_j$. Then $\{g_n(p'_j)\}$ converges, and so is a Cauchy sequence for each $j = 1, \dots, m$. Hence there exists $N \in \mathbb{N}$ such that

$$|g_k(p'_j) - g_n(p'_j)| \leq \epsilon/3 \quad \text{whenever } j \in \{1, \dots, m\} \text{ and } k, n \geq N. \quad (23.4)$$

Now let $z \in K$ and $k, n \in \mathbb{N}$ with $k, n \geq N$ be arbitrary. Then there exists $j \in \{1, \dots, m\}$ with $z \in B_j$ and so inequality (23.3) shows that

$$|g_n(z) - g_n(p'_j)| \leq |g_n(z) - g_n(w_j)| + |g_n(w_j) - g_n(p'_j)| \leq \epsilon/6 + \epsilon/6 = \epsilon/3.$$

A similar inequality is also true if we replace n with k here.

It then follows from (23.4) that

$$\begin{aligned} |g_k(z) - g_n(z)| &\leq |g_k(z) - g_k(p'_j)| + |g_k(p'_j) - g_n(p'_j)| + |g_n(p'_j) - g_n(z)| \\ &\leq \epsilon/3 + \epsilon/3 + \epsilon/3 = \epsilon. \end{aligned}$$

This establishes the claim.

The claim implies that $\{g_n\}$ converges locally uniformly on U to some limit function. Indeed, first it shows that for each $z \in U$ the sequence $\{g_n(z)\}$ is a Cauchy sequence, and hence converges, say $g_n(z) \rightarrow g(z) \in \mathbb{C}$ as $n \rightarrow \infty$. Moreover, letting $k \rightarrow \infty$ in (23.2) we see that the convergence $g_n \rightarrow g$ is uniform on each compact set $K \subseteq U$. So the subsequence $\{g_n\}$ of our original sequence $\{f_n\}$ converges to some limit function locally uniformly on U . Hence $\mathcal{F}$ is a normal family. $\square$

It seems that this theorem completely solves all questions concerning normal families. In practice it is often not so easy to establish the relevant equicontinuity and uniform boundedness conditions in order to establish the normality of a family. For example, let $\mathcal{S}$ be the family of all holomorphic functions f on the unit disk $\mathbb{D}$ that are injective and satisfy $f(0) = f'(0) - 1 = 0$. Then one can show that $\mathcal{S}$ is a normal family (see Problem 23.1), but it is not immediately clear how to verify the conditions in the Arzelà–Ascoli theorem.

A condition that guarantees that a family is normal is called a *normality criterion*. The following theorem provides the basic normality criterion for holomorphic functions.

Theorem 23.4 (Montel’s “Little” Theorem). *Let $U \subseteq \mathbb{C}$ be open, and $\mathcal{F}$ be a family of holomorphic functions on U . Then $\mathcal{F}$ is a normal family if and only if $\mathcal{F}$ is locally uniformly bounded.*

Proof. $\Rightarrow$: This implication follows from the Arzelà–Ascoli Theorem (even if we replace the assumption that the functions in the family are holomorphic by the weaker assumption that they are only continuous). Indeed, if the family is normal, then it is equicontinuous and uniformly bounded at each point of U . This in turn implies that the family is locally uniformly bounded on U (exercise!).

$\Leftarrow$: By the Arzelà–Ascoli Theorem it suffices to show that $\mathcal{F}$ is equicontinuous at each point in U . So let $z_0 \in U$ be arbitrary. We can pick $r > 0$ such that $\overline{B}(z_0, r) \subseteq U$ and such that $\mathcal{F}$ is uniformly bounded on $\overline{B}(z_0, r)$. So there exists $M \geq 0$ such that

$$|f(z)| \leq M \quad \text{whenever } f \in \mathcal{F} \text{ and } z \in \overline{B}(z_0, r).$$

Define $\gamma(t) = z_0 + re^{it}$ for $t \in [0, 2\pi]$, and let $z \in B(z_0, r/2)$ and $f \in \mathcal{F}$ be arbitrary. Then

$$\begin{aligned} |f(z) - f(z_0)| &= \frac{1}{2\pi} \left| \int_{\gamma} f(w) \left(\frac{1}{w-z} - \frac{1}{w-z_0} \right) dw \right| \\ &\leq \frac{1}{2\pi} \cdot 2\pi r \cdot \max_{w \in \partial B(z_0, r)} \frac{|f(w)| \cdot |z - z_0|}{|w-z| \cdot |w-z_0|} \\ &= r \cdot \frac{M}{(r/2)r} |z - z_0| = \frac{2M}{r} |z - z_0|. \end{aligned}$$

Since M is independent of f and z , the equicontinuity of $\mathcal{F}$ at z_0 immediately follows. $\square$

Theorem 23.5 (Hurwitz’s Theorem). *Let $U \subseteq \mathbb{C}$ be a region, $f \in H(U)$, and $f_n \in H(U)$ for $n \in \mathbb{N}$. Suppose that $f_n \rightarrow f$ locally uniformly on U . If f_n is injective for each $n \in \mathbb{N}$, then f is constant or also injective.*

Proof. We argue by contradiction and assume that under the given hypotheses, f is neither constant nor injective. Then there exist $z_1, z_2 \in U$, $z_1 \neq z_2$, such that $f(z_1) = f(z_2)$. Without loss of generality, we may assume that $f(z_1) = f(z_2) = 0$.

We can choose a small $r > 0$ such that $B_1 := \overline{B}(z_1, r) \subseteq U$, $B_2 := \overline{B}(z_2, r) \subseteq U$, $B_1 \cap B_2 = \emptyset$, and so that f has no zeros on $\partial B_1 \cup \partial B_2$ (a choice satisfying the last condition is possible, because f is non-constant and so the zeros of f are isolated points). Then $m := \inf\{|f(z)| : z \in \partial B_1 \cup \partial B_2\} > 0$.

Since $f_n \rightarrow f$ uniformly on $\partial B_1 \cup \partial B_2$, we can find $N \in \mathbb{N}$ such that

$$|f_N(z) - f(z)| < m \leq |f(z)| \quad \text{for all } z \in \partial B_1 \cup \partial B_2.$$

By Rouché's Theorem f_N has the same number of zeros as f in $\text{int}(B_1)$ and in $\text{int}(B_2)$. So f_N has at least one zero, say w_1 , in $\text{int}(B_1)$, and one, say w_2 , in $\text{int}(B_2)$. Then $w_1 \neq w_2$, and $f_N(w_1) = 0 = f_N(w_2)$. This contradicts the fact that f_N is injective. $\square$

Remark 23.6. One can prove results similar to the one established for holomorphic functions also for meromorphic functions (considered as holomorphic maps into $\widehat{\mathbb{C}}$). In this case one uses the chordal metric in order to measure distances in the target.

One can show that a locally uniform limit of meromorphic functions on a region is again meromorphic (one has to allow the function that is identically equal to ∞ here).

The notion of a *normal family* of continuous maps into $\widehat{\mathbb{C}}$ is defined in the same way as in Definition 23.1. The corresponding version of the Arzelà–Ascoli Theorem then says that a family of such maps $\mathcal{F}$ is normal if and only if it is equicontinuous at each point. The proof of this statement is identical to the proof of Theorem 23.3. The uniform boundedness condition is not relevant here since every sequence in $\widehat{\mathbb{C}}$ has a convergent subsequence.

When one considers normal families one has to be careful about the metric used in the target. Every holomorphic function on a region U can also be considered as a map into $\widehat{\mathbb{C}} \supseteq \mathbb{C}$. If a family of holomorphic functions on a region U is normal as considered as a family of maps into $\mathbb{C}$, then it is also normal as considered as a family of maps into $\widehat{\mathbb{C}}$, but the converse is not true. One can show that a family $\mathcal{F} \subseteq H(U)$ is normal in the latter sense, if and only if every sequence $\{f_n\}$ in $\mathcal{F}$ has a subsequence that converges locally uniformly to the constant function ∞ or to a function holomorphic on U . Often this is taken as the definition of a normal family of holomorphic functions.

One can formulate sufficient conditions for the normality of a family of meromorphic functions that are similar to Montel's Theorem 23.4. A stronger statement is Montel's "Big" Theorem: if each function in a family of meromorphic functions on a region omits three distinct fixed values $a, b, c \in \widehat{\mathbb{C}}$, then it is normal. For example, a family of holomorphic functions omitting 0 and 1 (the third omitted value is ∞ here) is normal (as a family of maps into $\widehat{\mathbb{C}}$).

23.1 Exercises

Problem 23.1. Let $\mathcal{S}$ be the family of all conformal maps $f: \mathbb{D} \rightarrow f(\mathbb{D}) \subseteq \mathbb{C}$ satisfying $f(0) = f'(0) - 1 = 0$. The purpose of this problem is to show that $\mathcal{S}$ is a normal family.

(a) Show that if $f \in \mathcal{S}$ and $d_f := \inf\{|w| : w \in \mathbb{C} \setminus f(\mathbb{D})\}$, then the infimum is attained as a minimum and $0 < d_f \leq 1$.

(b) For $f \in \mathcal{S}$ consider the auxiliary function $g_f := f/w_f - 1$, where $w_f \in \mathbb{C} \setminus f(\mathbb{D})$ satisfies $|w_f| = d_f$. Then g_f has a holomorphic square root denoted by h_f , and by a suitable choice of the branch we may assume $h_f(0) = i$ (justify these statements!). Show that there exists $r > 0$ independent of f such that $B(-i, r) \cap h_f(\mathbb{D}) = \emptyset$.

(c) Show that $\mathcal{S}$ is a normal family. Hint: It suffices to show that $\mathcal{S}$ is locally uniformly bounded. Argue by contradiction, and suppose $\{f_n\}$ is a sequence in $\mathcal{S}$ and $\{z_n\}$ is a sequence of points in some compact set $K \subseteq \mathbb{D}$ such that $|f_n(z_n)| \geq n$ for all $n \in \mathbb{N}$. Now consider the functions $k_n := 1/(h_n + i)$, where $h_n := h_{f_n}$ for $n \in \mathbb{N}$ is defined as in (b).

(d) The previous argument actually shows that the family $\{g_f : f \in \mathcal{S}\}$ of the auxiliary maps as defined in (b) is locally uniformly bounded (why?). Use this to show that there exists $\rho_0 > 0$ such that $B(0, \rho_0) \subseteq f(\mathbb{D})$ for all $f \in \mathcal{S}$.

(The largest possible value of ρ_0 in the last statement is known as the *Koebe radius*. One can show that $\rho_0 = 1/4$.)

Problem 23.2. Let $f: \overline{\mathbb{D}} \rightarrow \mathbb{C}$ be a continuous function such that $f(0) = 0$ and $f|_{\mathbb{D}}$ is holomorphic.

(a) Show that if $r > 0$ and $|f(e^{it})| \geq r$ for all $t \in [0, 2\pi]$, then $B(0, r) \subseteq f(\mathbb{D})$.

(b) Let $\varphi: [0, 1) \rightarrow (0, \infty)$ be a continuous function and suppose that $\operatorname{Re}(f'(z)) \geq \varphi(|z|)$ for all $z \in \mathbb{D}$. Show that then $D(0, r) \subseteq f(\mathbb{D})$, where $r := \int_0^1 \varphi(x) dx$.

Problem 23.3 (Landau's constant). If f is a non-constant holomorphic function on $\mathbb{D}$, we denote by $L_f \in (0, \infty]$ the radius of the “largest” disk contained in the image $f(\mathbb{D})$; more precisely,

$$L_f := \sup\{r > 0 : \text{there exists } z_0 \in \mathbb{C} \text{ with } B(z_0, r) \subseteq f(\mathbb{D})\}.$$

The number

$$L := \inf\{L_f : f \in H(\mathbb{D}) \text{ and } f'(0) = 1\}$$

is known as *Landau's constant*. Its precise numerical value is not known. The purpose of this problem is to show that $L > 0$ by establishing an explicit positive lower bound for L .

(a) Let $\mathcal{B}$ be the family of all functions $f \in H(\mathbb{D})$ satisfying $f(0) = 0$, $f'(0) = 1$, and

$$|f'(z)| \leq \frac{1}{1 - |z|^2} \quad \text{for } z \in \mathbb{D}.$$

Show that $L = \inf\{L_f : f \in \mathcal{B}\}$.

Hint: First show that in the definition of L we may assume that the functions f are holomorphic in an open set containing $\overline{\mathbb{D}}$. For such a function consider a point $z_0 \in \mathbb{D}$, where $|f'(z)|(1 - |z|^2)$ attains a maximum on $\mathbb{D}$, and precompose f by a suitable map that sends 0 to z_0 .

(b) Let $g: \mathbb{D} \rightarrow \mathbb{C}$ be a holomorphic function with $g(0) = w_0 \geq 0$ and $|g(z)| \leq 1$ for $z \in \mathbb{D}$. Show that

$$\operatorname{Re}(g(z)) \geq \frac{w_0 - |z|}{1 - w_0|z|}$$

for all $z \in \mathbb{D}$ with $|z| \leq w_0$.

(c) Find an explicit number $c > 0$ such that $B(0, c) \subseteq f(\mathbb{D})$ for all $f \in \mathcal{B}$.

(d) Find an explicit positive lower bound for L .

Problem 23.4 (Picard's Theorem). (a) Show by using Problem 23.3 that every non-constant entire function f contains arbitrarily large disks in its image, i.e.,

$$L_f = \sup\{r > 0 : \text{there exists } z_0 \in \mathbb{C} \text{ with } B(z_0, r) \subseteq f(\mathbb{C})\} = \infty.$$

(b) Let f be an entire function omitting the values -1 and 1 . Show that then there exists an entire function g such that $f = \cos(g)$. Which values will g omit? Hint: You can construct g explicitly by taking suitable roots and logarithms.

(c) Prove *Picard's Theorem* stating that every entire function f omitting two distinct values in $\mathbb{C}$ is constant.

24 The Riemann Mapping Theorem

Lemma 24.1. *Let $\Omega \subseteq \mathbb{C}$ be a simply connected region, $\Omega \neq \mathbb{C}$. Then there exists a conformal map $g: \Omega \rightarrow \tilde{\Omega}$, where $\tilde{\Omega}$ is a Koebe region, i.e., $\tilde{\Omega}$ is a simply connected region with $0 \in \tilde{\Omega} \subseteq \mathbb{D}$.*

Proof. Pick $a \in \mathbb{C} \setminus \Omega$. Then $z \mapsto z - a$ is a zero-free holomorphic function on Ω . So by Corollary 20.9 this function has a holomorphic square root S , i.e., $S \in H(\Omega)$ and

$$S(z)^2 = z - a \quad \text{for all } z \in \Omega.$$

The function S is injective on Ω : if $z_1, z_2 \in \Omega$ are arbitrary, then we have the implications

$$\begin{aligned} S(z_1) = S(z_2) &\Rightarrow S(z_1)^2 = S(z_2)^2 \\ &\Rightarrow z_1 - a = z_2 - a \Rightarrow z_1 = z_2. \end{aligned}$$

So S is a holomorphic bijection $S: \Omega \rightarrow U := S(\Omega)$ onto its image U , and in particular a conformal map (Corollary 13.6). The set U is a region (see Corollary 13.5). It has the following property: if $w \in U$, then $-w \notin U$; indeed, suppose $w, -w \in U$. Then there exist $z_1, z_2 \in \Omega$ such that $w = S(z_1)$ and $-w = S(z_2)$. Hence

$$z_1 - a = S(z_1)^2 = w^2 = S(z_2)^2 = z_2 - a,$$

and so $z_1 = z_2$. This implies

$$w = S(z_1) = S(z_2) = -w,$$

and gives $w = 0$. This is impossible, because the map $z \mapsto z - a = S(z)^2$, and hence also S , is zero-free on Ω .

Since U is open, there exists $w_0 \in U$, and $r > 0$ such that $B(w_0, r) \subseteq U$. Then $B(w_1, r) \cap U = \emptyset$, where $w_1 := -w_0$.

Consider the Möbius transformation $T: U \rightarrow \mathbb{C}$,

$$T(w) := \frac{r}{w - w_1}.$$

Then $T: U \rightarrow V := T(U)$ is a conformal map of U onto its image $V = T(U) \subseteq \mathbb{D}$. Pick $u_0 \in V$, and let

$$R(u) := \frac{u - u_0}{1 - \overline{u_0}u}, \quad u \in \mathbb{D}.$$

Then $R \in \text{Aut}(\mathbb{D})$, and $0 \in \tilde{\Omega} := R(T(U)) \subseteq \mathbb{D}$. So we have the sequence of holomorphic bijections

$$\Omega \xrightarrow{S} U \xrightarrow{T} V \xrightarrow{R} \tilde{\Omega} \subseteq \mathbb{D}.$$

Then $g := R \circ T \circ S$ is a holomorphic bijection of Ω onto $\tilde{\Omega}$, and hence a conformal map from Ω onto $\tilde{\Omega}$. The set $\tilde{\Omega}$ is a region with $0 \in \tilde{\Omega}$. Moreover, $\tilde{\Omega}$ is simply connected, because this property of a region is invariant under conformal maps. So $\tilde{\Omega}$ is a Koebe region, and g is a conformal map from Ω onto $\tilde{\Omega}$. $\square$

If one replaces the condition that the regions in the previous statement are simply connected by the condition that every zero-free holomorphic function on the region has a holomorphic square root, then the previous proof goes through with the conclusion that $\tilde{\Omega}$ satisfies the latter property. Note this property of a region is invariant under conformal maps of the region.

Lemma 24.2. *Let $\Omega \subseteq \mathbb{D}$ be a Koebe region, and*

$$\mathcal{S}(\Omega) := \{f: \Omega \rightarrow \mathbb{D} \text{ conformal}, f(\Omega) \subseteq \mathbb{D}, f(0) = 0\}.$$

Then

$$\alpha := \sup\{|f'(0)| : f \in \mathcal{S}(\Omega)\} \quad (24.1)$$

is attained as a maximum, i.e., there exists $f_0 \in \mathcal{S}(\Omega)$ such that $|f'(0)| \leq |f'_0(0)| = \alpha$ for all $f \in \mathcal{S}(\Omega)$.

Proof. Note that $\alpha \geq 1$, because $\text{id}_\Omega \in \mathcal{S}(\Omega)$. Let $\{f_n\}$ be a sequence in $\mathcal{S}(\Omega)$ with $|f'_n(0)| \rightarrow \alpha$. Montel's Theorem implies that by passing to a subsequence if necessary we may assume that $f_n \rightarrow f_0 \in H(\Omega)$ locally uniformly on Ω . Then $f_0(0) = \lim_{n \rightarrow \infty} f_n(0) = 0$, and $|f'_0(0)| = \lim_{n \rightarrow \infty} |f'_n(0)| = \alpha > 0$. In particular, f_0 is not a constant function. By Hurwitz's Theorem f_0 is injective, and so f_0 is a conformal map of Ω onto $f_0(\Omega) \subseteq \mathbb{D}$. Since $f_0(\Omega)$ is open, we must have $f_0(\Omega) \subseteq \mathbb{D}$. Then $f_0 \in \mathcal{S}(\Omega)$ and $|f'_0(0)| = \alpha$ as desired. $\square$

Lemma 24.3. *Let $\Omega \subseteq \mathbb{D}$ be a Koebe region, and $f_0 \in \mathcal{S}(\Omega)$ be a map for which the supremum in (24.1) is attained. Then $f_0(\Omega) = \mathbb{D}$, i.e., f is a conformal map from Ω onto $\mathbb{D}$.*

Proof. We argue by contradiction; so suppose $U_0 := f_0(\Omega) \neq \mathbb{D}$. Since $0 \in U_0 \subseteq \mathbb{D}$, there exists $a \neq 0$ such that $a \in \mathbb{D} \setminus U_0$. Let R be the Möbius transformation given by

$$R(u) := \frac{u - a}{1 - \bar{a}u}.$$

Then $R \in \text{Aut}(\mathbb{D})$, and so R is a conformal map from U_0 onto $U_1 := R(U_0) \subseteq \mathbb{D}$. Then U_1 is a simply connected region. Since $R(a) = 0$, we have $0 \notin U_1$. So by Corollary 20.9 there exists a holomorphic branch of the square root function on U_1 ; i.e., a holomorphic map $S: U_1 \rightarrow \mathbb{C}$ such that

$$S(v)^2 = v \quad \text{for } v \in U_1.$$

Then for $v \in U_1$ we have $|S(v)|^2 = |v| < 1$, and so $U_2 := S(U_1) \subseteq \mathbb{D}$. Moreover, S is injective on U_1 ; indeed, if for $v_1, v_2 \in U_1$, we have $S(v_1) = S(v_2)$, then $v_1 = S(v_1)^2 = S(v_2)^2 = v_2$. It follows that S is a conformal map of U_1 onto the simply connected region $U_2 := S(U_1) \subseteq \mathbb{D}$.

Let $b := S(R(0)) \in S(U_1) \subseteq U_2 \subseteq \mathbb{D}$, and T be the Möbius transformation given by

$$T(w) := \frac{w - b}{1 - \bar{b}w}.$$

Then $T \in \text{Aut}(\mathbb{D})$, and so T is a conformal map from U_2 onto the simply connected region $U_3 := T(U_2) \subseteq \mathbb{D}$. Note that $0 = T(b) \in T(U_2) = U_3$, and so U_3 is a Koebe region.

We have constructed conformal maps

$$U_0 \xrightarrow{R} U_1 \xrightarrow{S} U_2 \xrightarrow{T} U_3.$$

Consider the composition $g := T \circ S \circ R$. Then g is a conformal map from U_0 onto U_3 with

$$g(0) = (T \circ S \circ R)(0) = T(b) = 0.$$

Moreover,

$$R'(u) = \frac{1 - |a|^2}{(1 - \bar{a}u)^2}, \quad |S'(v)| = \frac{1}{2\sqrt{|v|}}, \quad T'(w) = \frac{1 - |b|^2}{(1 - \bar{b}w)^2},$$

and so

$$|R'(0)| = 1 - |a|^2, \quad |S'(R(0))| = |S'(-a)| = \frac{1}{2\sqrt{|a|}}, \quad |T'(b)| = \frac{1}{1 - |b|^2}.$$

Note also that

$$|b|^2 = |S(R(0))^2| = |S(-a)^2| = |-a| = |a| < 1.$$

Putting this all together, we see that

$$\begin{aligned} |g'(0)| &= |T'(S(R(0)))| \cdot |S'(R(0))| \cdot |R'(0)| = |T'(b)| \cdot |S'(-a)| \cdot |R'(0)| \\ &= \frac{1}{1 - |b|^2} \cdot \frac{1}{2\sqrt{|a|}} \cdot (1 - |a|^2) = \frac{1 - |a|^2}{2\sqrt{|a|}(1 - |a|)} \\ &= \frac{1 + |a|}{2\sqrt{|a|}} = 1 + \frac{(1 - \sqrt{|a|})^2}{2\sqrt{|a|}} > 1. \end{aligned}$$

Then $f := g \circ f_0$ is a conformal map of Ω onto the Koebe region $g(f_0(\Omega)) = g(U_0) = U_3$, and $f(0) = g(f_0(0)) = g(0) = 0$; so $f \in \mathcal{S}(\Omega)$, but

$$|f'(0)| = |g'(0)| \cdot |f'_0(0)| > |f'_0(0)|.$$

This contradicts the maximality property of f_0 . $\square$

Remark 24.4. (a) Again the relevant property of Ω in the previous proof was that every zero-free holomorphic function on Ω has a holomorphic square root (which transfers to $U_0 = f_0(\Omega)$ and all the other regions considered).

(b) The computation in the previous proof make it seem to be a fortunate coincidence that we get the crucial inequality $|g'(0)| > 1$. One can actually see that this inequality must be true almost without any computation if one considers how the hyperbolic length element $ds_h = \frac{2|dz|}{1 - |z|^2}$ is distorted by the map. If f is a holomorphic map defined near a point $z \in \mathbb{D}$ and has values in $\mathbb{D}$, then this distortion at z is measured by the *derivative of f with respect to the hyperbolic metric* given by

$$D_h(f)(z) := \frac{f^*(ds_h)}{ds_h} = \frac{(1 - |z|^2)|f'(z)|}{1 - |f(z)|^2}.$$

This derivative has the expected properties (chain rule, behavior for inverse maps, etc.). Note also that if $f(0) = 0$ then $D_h(f)(0) = |f'(0)|$.

We know that $D_h(\varphi)(z) \equiv 1$ if $\varphi \in \text{Aut}(\mathbb{D})$. Moreover, if $f: \mathbb{D} \rightarrow \mathbb{D}$ is holomorphic, but $f \notin \text{Aut}(\mathbb{D})$, then $D_h(f)(z) < 1$ for all $z \in \mathbb{D}$. So if g is a branch of the inverse f^{-1} defined near a point $z \in \mathbb{D}$, then $D_h(g)(z) =$

$D_h(f)(g(z))^{-1} > 1$. In particular, this is true for the branch S of the square root function (an inverse branch of $z \mapsto z^2$ mapping $\mathbb{D}$ into $\mathbb{D}$) used in the previous proof. This can also be verified by direct computation:

$$|S'(z)| = \frac{1}{2\sqrt{|z|}} \quad \text{and} \quad |S(z)|^2 = |z|,$$

which implies

$$D_h(S)(z) = \frac{(1 - |z|^2)|S'(z)|}{1 - |S(z)|^2} = \frac{1 - |z|^2}{2\sqrt{|z|}(1 - |z|)} = \frac{1 + |z|}{2\sqrt{|z|}} > 1,$$

whenever $z \in \mathbb{D}$.

In the previous proof, the function g was given as $g := R \circ S \circ T$, where $R, T \in \text{Aut}(\mathbb{D})$. Moreover, $g(0) = 0$, and so

$$\begin{aligned} |g'(0)| &= D_h(g)(0) = D_h(R)(S(T(0))) \cdot D_h(S)(T(0)) \cdot D_h(T)(0) \\ &= D_h(S)(T(0)) > 1. \end{aligned}$$

Theorem 24.5 (Riemann Mapping Theorem). *Let $\Omega \subseteq \mathbb{C}$ be a simply connected region, $\Omega \neq \mathbb{C}$. Then there exists a conformal map $f: \Omega \rightarrow \mathbb{D}$ of Ω onto $\mathbb{D}$.*

If $z_0 \in \Omega$ is arbitrary, then f can be chosen so that

$$f(z_0) = 0 \quad \text{and} \quad f'(z_0) > 0. \quad (24.2)$$

Moreover, with this normalization the map f is uniquely determined.

Proof. By Lemma 24.1 there exists a conformal map $h_1: \Omega \rightarrow \tilde{\Omega}$ onto a Koebe region $\tilde{\Omega}$. By Lemma 24.3 there exists a conformal map $h_2: \tilde{\Omega} \rightarrow \mathbb{D}$. Then $f := h_2 \circ h_1$ is a conformal map from Ω onto $\mathbb{D}$.

If $z_0 \in \Omega$ is arbitrary, then $w_0 := f(z_0) \in \mathbb{D}$. For fixed $\theta \in [0, 2\pi]$ to be determined momentarily, consider the Möbius transformation

$$\varphi(w) = e^{i\theta} \frac{w - w_0}{1 - \overline{w_0}w}.$$

Then $\varphi \in \text{Aut}(\mathbb{D})$, and $\varphi(w_0) = 0$. So $\tilde{f} := \varphi \circ f$ is a conformal map from Ω onto $\mathbb{D}$ with $\tilde{f}(0) = \varphi(w_0) = 0$. Moreover,

$$\tilde{f}'(z_0) = \varphi'(w_0) \cdot f'(z_0) = e^{i\theta} \frac{f'(z_0)}{1 - |w_0|^2}.$$

By choosing $\theta \in [0, 2\pi]$ appropriately, we get $\tilde{f}'(z_0) > 0$. Then $\tilde{f}$ is a conformal map of Ω onto $\mathbb{D}$ with the desired normalization.

Suppose f_1 and f_2 are two conformal maps from Ω onto $\mathbb{D}$ satisfying the normalization (24.2). Then $\psi := f_2 \circ f_1^{-1} \in \text{Aut}(\mathbb{D})$, $\psi(0) = 0$, and

$$\psi'(0) = f_2'(f_1^{-1}(0)) \cdot (f_1^{-1})'(0) = \frac{f_2'(z_0)}{f_1'(z_0)} > 0.$$

So $\psi(z) = e^{i\alpha}z$, where $\alpha \in [0, 2\pi)$ and $\psi'(0) = e^{i\alpha} > 0$. This implies $\alpha = 0$, $\psi = \text{id}_{\mathbb{D}}$, and $f_1 = f_2$. $\square$

Based on the remarks after Lemmas 24.1 and 24.3 the argument in the previous proof actually shows that if $\Omega \subseteq \mathbb{C}$, $\Omega \neq \mathbb{C}$, is a region such that every zero-free holomorphic function has a holomorphic square root, then Ω is conformally equivalent to $\mathbb{D}$, i.e., there exists a conformal map of Ω onto $\mathbb{D}$.

Example 24.6. In some simple cases the *Riemann map*, i.e., the conformal map of the simply connected region of Ω onto $\mathbb{D}$ can be found explicitly. Here are two examples.

(a) Let $\Omega := \{re^{i\alpha} : 0 < r < 1, 0 < \alpha < \pi/4\}$. The region Ω is convex, and hence simply connected. The Riemann map can be found as a composition of conformal maps $U_0 = \Omega \rightarrow U_1 \rightarrow U_2 \rightarrow U_3 \rightarrow U_4 = \mathbb{D}$, where

$$\begin{aligned} z \in U_0 &\mapsto s := z^4 \in U_1 := \{s \in \mathbb{C} : |s| < 1, \text{Im } s > 0\}, \\ s \in U_1 &\mapsto t := \frac{1+s}{1-s} \in U_2 := \{t \in \mathbb{C} : \text{Re } t > 0, \text{Im } t > 0\}, \\ t \in U_2 &\mapsto u := t^2 \in U_3 := \{u \in \mathbb{C} : \text{Im } u > 0\}, \\ u \in U_3 &\mapsto w := \frac{1+iu}{1-iu} \in U_4 := \mathbb{D}. \end{aligned}$$

So $w = f(z) = \frac{1 + i\left(\frac{1+z^4}{1-z^4}\right)^2}{1 - i\left(\frac{1+z^4}{1-z^4}\right)^2}$ gives a conformal map of Ω onto $\mathbb{D}$.

(b) Let $\Omega := \{z \in \mathbb{C} : -\pi < \text{Im } z < \pi\} \setminus (-\infty, 0]$. The region Ω is *starlike with respect to* $p = 1$. This means that $[p, z] \subseteq \Omega$ for all $z \in \Omega$ and implies that Ω is simply connected. Again the Riemann map can be found

as a composition of conformal maps $U_0 = \Omega \rightarrow U_1 \rightarrow U_2 \rightarrow U_3 = \mathbb{D}$. Here

$$\begin{aligned} z \in U_0 &\mapsto u := e^z \in U_1 := \mathbb{C} \setminus (-\infty, 1], \\ u \in U_1 &\mapsto v := \sqrt{u-1} \in U_2 := \{v \in \mathbb{C} : \operatorname{Re} v > 0\} \quad (\text{principal branch}), \\ v \in U_2 &\mapsto w := \frac{1-v}{1+v} \in U_3 := \mathbb{D}. \end{aligned}$$

So $w = f(z) = \frac{1 - \sqrt{e^z - 1}}{1 + \sqrt{e^z - 1}}$ gives a conformal map of Ω onto $\mathbb{D}$.

Remark 24.7. Since the upper half-plane $\mathbb{H} := \{z \in \mathbb{C} : \operatorname{Im} z > 0\}$ is conformally equivalent to $\mathbb{D}$, for every simply connected region $\Omega \subseteq \mathbb{C}$, $\Omega \neq \mathbb{C}$, there exists a conformal map $f: \mathbb{H} \rightarrow \Omega$. A general class of simply connected regions Ω for which such a map f can be obtained quite explicitly are those bounded by polygonal paths (consisting of straight line segments).

Suppose that $\partial\Omega$ is a finite union of line segments $\partial\Omega = \bigcup_{k=1}^n [p_k, p_{k+1}]$ with pairwise disjoint interiors, where $p_{n+1} = p_1$, and let $\alpha_k\pi$ be the interior angle at the corner p_k with $\alpha_k \in (0, 2)$. Then one can show that there exist points $a_1 < a_2 < \cdots < a_n$ on the real axis and constants $A \in \mathbb{C}$ and $B \in \mathbb{C} \setminus \{0\}$ such that f is given by the *Schwarz–Christoffel map*

$$f(z) = A + B \int_{[0, z]} \prod_{k=1}^n (\zeta - a_k)^{\alpha_k - 1} d\zeta.$$

Here one has to choose suitable branches of the power functions, for example those that attain positive values for large positive values of ζ . Moreover, this conformal map of $\mathbb{H}$ onto Ω has a continuous extension to $\overline{\mathbb{H}} = \{z \in \mathbb{C} : \operatorname{Im} z \geq 0\} \cup \{\infty\}$ such that $f(a_k) = p_k$ for $k = 1, \dots, n$.

For example, let $k \in (0, 1)$, and define

$$f(z) = \int_{[0, z]} \frac{d\zeta}{\sqrt{(1 - \zeta^2)(1 - k^2\zeta^2)}}$$

for $z \in \overline{\mathbb{H}}$. Here the branch of the root is such that for $\zeta = 0$ we have $\sqrt{1} = 1$. One can show that then f is homeomorphism of $\overline{\mathbb{H}}$ onto the rectangle

$$R := \{w \in \mathbb{C} : -K \leq \operatorname{Re} w \leq K, 0 \leq \operatorname{Im} w \leq K'\}$$

and a conformal map of $\mathbb{H}$ onto $\text{int}(R)$. The quantities K and K' are given by *complete elliptic integrals of the first kind*:

$$K = \int_0^{\pi/2} \frac{dt}{\sqrt{1 - k^2 \sin^2 t}}, \quad \text{and} \quad K' = \int_0^{\pi/2} \frac{dt}{\sqrt{1 - k'^2 \sin^2 t}},$$

where $k' := \sqrt{1 - k^2} \in (0, 1)$. The map f sends the points ± 1 and $\pm 1/k$ to the corners of R ; more precisely, we have $f(\pm 1) = \pm K$ and $f(\pm 1/k) = \pm K + iK'$.

Theorem 24.8 (Schwarz Reflection Principle). *Let $U^+ \subseteq H^+ := \{z \in \mathbb{C} : \text{Im } z > 0\}$ be open, $U^- := \{\bar{z} : z \in U^+\}$, and $\alpha \subseteq \mathbb{R}$ be open (as a subset of $\mathbb{R}$). Define $\Omega = U^+ \cup \alpha \cup U^-$, and assume that $\alpha \subseteq \text{int}(\Omega)$ (which implies that Ω is open).*

Let $f: U^+ \cup \alpha \rightarrow \mathbb{C}$ be continuous and holomorphic on U^+ . If $f(\alpha) \subseteq \mathbb{R}$, then the extension $F: \Omega \rightarrow \mathbb{C}$ of f given by

$$F(z) = \begin{cases} f(z) & \text{for } z \in U^+ \cup \alpha, \\ \overline{f(\bar{z})} & \text{for } z \in U^-, \end{cases}$$

is holomorphic on Ω .

If in addition f is injective on $U^+ \cup \alpha$ and $f(U^+) \subseteq H^+$, then F is injective on Ω , and so F is a conformal map of Ω onto its image $F(\Omega)$.

Proof. Obviously, F is continuous on Ω . Moreover, $F|_{U^-}$ is holomorphic on U^- . Indeed, if $a \in U^-$ is arbitrary, then $\bar{a} \in U^+$, and so f has a power series representation

$$f(z) = \sum_{n=0}^{\infty} a_n (z - \bar{a})^n \quad \text{for } z \text{ near } \bar{a}.$$

Then

$$F(z) = \overline{f(\bar{z})} = \sum_{n=0}^{\infty} \bar{a}_n (z - a)^n \quad \text{for } z \text{ near } a,$$

and so the holomorphicity of F near a follows.

The holomorphicity of F on the whole set Ω easily follows from its continuity on Ω , its holomorphicity of F on $U^+ \cup U^-$, and Morera's Theorem.

Finally, if f is injective on $U^+ \cup \alpha$ and $f(U^+) \subseteq H^+$, then F is injective on U^- and $F(U^-) \subseteq H^- := \{z \in \mathbb{C} : \text{Im } z < 0\}$. This implies that F is injective on Ω . $\square$

Remark 24.9. Using auxiliary Möbius transformations one can easily formulate and prove a more general version of the Schwarz Reflection Principle for meromorphic functions, where the real line in source and target is replaced by two circles C and C' . One then assumes that $\alpha \subseteq C$ and $f(\alpha) \subseteq C'$. If R and S are the reflections in the circles C and C' , respectively, then the extension F will be defined as $F(z) = S(f(R(z)))$ for $z \in U^- := R(U^+)$.

Example 24.10. Let $k \in (0, 1)$, and as in Remark 24.7 define

$$f(z) = \int_{[0,z]} \frac{d\zeta}{\sqrt{(1-\zeta^2)(1-k^2\zeta^2)}}$$

for $z \in \overline{\mathbb{H}} = \mathbb{H} \cup \widehat{\mathbb{R}} = \{z \in \mathbb{C} : \operatorname{Im} z \geq 0\} \cup \{\infty\}$. Then f maps $\overline{\mathbb{H}}$ homeomorphically onto

$$R = \{w \in \mathbb{C} : -K \leq \operatorname{Re} w \leq K, 0 \leq \operatorname{Im} w \leq K'\},$$

and $\mathbb{H}$ conformally onto $\operatorname{int}(R)$, where K and K' are as in Remark 24.7. Moreover, $f(\pm 1) = \pm K$, and $f(\pm 1/k) = \pm K + iK'$. The inverse map $u \in R \mapsto z \in \overline{\mathbb{H}}$ is denoted by $\operatorname{sn}(u; k)$ or simply by $\operatorname{sn} u$ if the parameter k is understood. This function is called the *sinus amplitudinis* (Jacobi's original Latin phrase), the *sine amplitude*, or the *modular sine* (for parameter $k \in (0, 1)$).

Under the this map reflection in the sides of R , and the sides of rectangles obtained by successive reflections, corresponds to reflection in $\widehat{\mathbb{R}}$ on the target side. Using this and applying the Schwarz Reflection Principle repeatedly, one can show that sn can be extended to a meromorphic function on $\mathbb{C}$. From this reflection process it also follows that

$$\operatorname{sn}(u + 4Km + 2iK'n) = \operatorname{sn}(u) \quad \text{for } u \in \mathbb{C}, m, n \in \mathbb{Z}.$$

So this function has the two periods $4K$ and $2iK$ that are linearly independent over $\mathbb{R}$. Hence it is *doubly periodic*. Functions that are meromorphic on $\mathbb{C}$ and doubly periodic are called *elliptic functions*. So sn is an example of an elliptic function.

24.1 Exercises

Problem 24.1. The purpose of this and the next problem is to find a conformal map from the open upper half-plane $\mathbb{H} = \{z \in \mathbb{C} : \operatorname{Im} z > 0\}$ onto a given rectangle.

Let $k \in (0, 1)$ and $U = \mathbb{C} \setminus \{x \in \mathbb{R} : |x| \geq 1\}$.

(a) Show that there exists a unique holomorphic function $f: U \rightarrow \mathbb{C}$ such that $f(0) = 0$ and

$$f'(z) = \frac{1}{\sqrt{(1-z^2)(1-k^2z^2)}}.$$

Here $g(z) = \sqrt{(1-z^2)(1-k^2z^2)}$ is the unique branch that is holomorphic on U and satisfies $g(0) = 1$ (justify existence and uniqueness of g !).

(b) Show the following equality for improper integrals:

$$A := \int_0^1 \frac{dx}{\sqrt{(1-x^2)(1-k^2x^2)}} = \int_{1/k}^\infty \frac{dx}{\sqrt{(x^2-1)(k^2x^2-1)}}.$$

(c) Show that $f|_{\mathbb{H}}$ has a continuous extension F to $\overline{\mathbb{H}} = \mathbb{H} \cup \mathbb{R} \cup \{\infty\}$. Setting

$$B := \int_1^{1/k} \frac{dx}{\sqrt{(x^2-1)(1-k^2x^2)}}$$

show that this continuous extension is given by

$$F(u) = A + iB - \int_{1/k}^u \frac{dx}{\sqrt{(x^2-1)(k^2x^2-1)}} \quad \text{for } u \in [1/k, \infty].$$

Find similar expressions for other values of $u \in \widehat{\mathbb{R}} = \mathbb{R} \cup \{\infty\}$.

(d) Show that F is a homeomorphism of $\widehat{\mathbb{R}}$ onto ∂R , where R is the rectangle

$$R := \{w \in \mathbb{C} : -A \leq \operatorname{Re} w \leq A, 0 \leq \operatorname{Im} w \leq B\}$$

with sidelengths $2A$ and B .

(e) Show that $F|_{\mathbb{H}} = f|_{\mathbb{H}}$ is a conformal map of $\mathbb{H}$ onto $\operatorname{int}(R)$.

Problem 24.2. (a) Show that A and B as defined in Problem 24.1 are given by the following *complete elliptic integrals of the first kind*

$$A = A(k) = \int_0^{\pi/2} \frac{dt}{\sqrt{1-k^2 \sin^2 t}}, \quad B = B(k) = \int_0^{\pi/2} \frac{dt}{\sqrt{1-k'^2 \sin^2 t}},$$

where $k' = \sqrt{1 - k^2}$.

(b) Show that the function $k \in (0, 1) \mapsto A(k)/B(k)$ is strictly increasing and attains all values in $(0, \infty)$.

(c) Based on the previous results describe how to find a conformal map of the open upper half-plane $\mathbb{H}$ onto (the interior of) a given rectangle R .

(d) Find the unique value $k \in (0, 1)$ for which R is a square, i.e., for which $A(k)/B(k) = 1/2$.

Problem 24.3. The purpose of this problem is to develop the necessary methods to prove that the modular sine function has an extension to a meromorphic function on $\mathbb{C}$.

Let $a, b > 0$ and $T_0 := \{u \in \mathbb{C} : 0 \leq \operatorname{Re} u \leq a, 0 \leq \operatorname{Im} u \leq b\}$. Consider the collection $\mathcal{R}$ of rectangles obtained from the natural tiling of $\mathbb{C}$ by translates of T_0 , i.e.,

$$\mathcal{R} = \{T_0 + ka + ilb : k, l \in \mathbb{Z}\}.$$

We call a map f defined on a rectangle $R \in \mathcal{R}$ *admissible* if it is a conformal map of $\operatorname{int}(R)$ onto the open upper or lower half-plane $H_{\pm}$, and extends to a homeomorphism of R onto $\overline{H}_{\pm} \subseteq \widehat{\mathbb{C}}$ (so in particular f is a homeomorphism of ∂R onto $\widehat{\mathbb{R}} = \mathbb{R} \cup \{\infty\}$).

A *gallery* $\mathcal{G}$ is a finite sequence of rectangles $R_1, \dots, R_n \in \mathcal{R}$ such that successive rectangles are distinct and have a side in common.

(a) Let G be the group generated by all the reflections in the sides of the rectangles $R \in \mathcal{R}$. Describe explicitly which maps belong to G and use this to show that if $\varphi \in G$, $R \in \mathcal{R}$, and $\varphi(R) = R$, then $\varphi = \operatorname{id}_{\mathbb{C}}$.

(b) If $R_1, \dots, R_n \in \mathcal{R}$ form a gallery $\mathcal{G}$ and f_1 is an admissible map on R_1 , then we define an admissible maps f_k on R_k for $k = 1, \dots, n$ successively by Schwarz Reflection as follows: if f_k is already defined on R_k and $k < n$, then we set

$$f_{k+1}(u) := \psi(f_k(\varphi(u))) \quad \text{for } u \in R_{k+1}.$$

Here φ is the reflection in the common side of R_k and R_{k+1} , and ψ is reflection in $\widehat{\mathbb{R}}$, i.e., complex conjugation.

Show that in this way we get an admissible map f_k on R_k for each $k = 1, \dots, n$. We say that the map f_n on the last rectangle R_n is obtained by *analytic continuation of f_1 along the gallery $\mathcal{G}$* .

(c) Suppose $\mathcal{G}$ is a gallery whose first rectangle R coincides with its last rectangle. Show that if f is an admissible map on R , and g is the map on R obtained by analytic continuation of f along $\mathcal{G}$, then $f = g$.

(d) Let f_0 be an admissible map on T_0 . We define an admissible map f_R on each $R \in \mathcal{R}$ as follows: pick a gallery $\mathcal{G}$ whose first rectangle is T_0 and whose last rectangle is R (why does such a gallery exist?). Let f_R be the analytic continuation of f_0 along $\mathcal{G}$. Show that f_R is well-defined, and that if $R, S \in \mathcal{R}$, then

$$f_R(u) = f_S(u) \quad \text{for all } u \in S \cap R.$$

Hint: For the last identity first consider the case where R and S share a side.

(e) Show that if f_0 be an admissible map on T_0 , then there exists a unique meromorphic function F on $\mathbb{C}$ such that $F|_{T_0} = f_0$.

(f) Explain how to use the last fact to get a meromorphic extension of the modular sine sn for a given module $k \in (0, 1)$ (originally defined on the rectangle $R = \{u \in \mathbb{C} : -K \leq \text{Re } u \leq K, 0 \leq \text{Im } u \leq K'\}$) to the whole complex plane.

Problem 24.4 (The absolute invariant J). Let $\Gamma \subseteq \mathbb{C}$ be a rank-2 lattice and $g_2 = g_2(\Gamma)$ and $g_3 = g_3(\Gamma)$ be defined as in Problem 16.3.

(a) Show that if we define $\Delta = \Delta(\Gamma) := g_2^3 - 27g_3^2$, then $\Delta \neq 0$. Hint: Express Δ in terms of the quantities e_1, e_2, e_3 defined as in Problem 22.7.

(b) The *absolute invariant* $J = J(\Gamma)$ of the lattice Γ is defined as

$$J := g_2^3/\Delta. \tag{24.3}$$

Show that if $\tilde{\Gamma} \subseteq \mathbb{C}$ is another rank-2 lattice, then $J(\Gamma) = J(\tilde{\Gamma})$ if and only if there exists $\alpha \in \mathbb{C} \setminus \{0\}$ such that $\tilde{\Gamma} = \alpha\Gamma$.

(c) Let $\tau \in \mathbb{H} = \{z \in \mathbb{C} : \text{Im } z > 0\}$ and consider the rank-2 lattice $\Gamma_\tau := \{n + k\tau : n, k \in \mathbb{Z}\}$. Abusing notation, we set $J(\tau) := J(\Gamma_\tau)$. Show that

$$J(\tau) = \frac{E_4^3(\tau)}{E_4^3(\tau) - E_6^2(\tau)},$$

where E_4 and E_6 are the normalized Eisenstein series as in Problem 21.8.

(d) Show that $J(\tau)$ is a holomorphic function of $\tau \in \mathbb{H}$ with the following property: if $\tau, \tau' \in \mathbb{H}$, then $J(\tau) = J(\tau')$ if and only if τ and τ' are equivalent

under the modular group, that is, if and only if there exist $a, b, c, d \in \mathbb{Z}$ with $ad - bc = 1$ such that

$$\tau' = \frac{a\tau + b}{c\tau + d}.$$

Problem 24.5. Let $J: \mathbb{H} \rightarrow \mathbb{C}$ be the function defined in Problem 24.4 and

$$T := \{\tau \in \mathbb{H} : 0 \leq \operatorname{Re} \tau \leq 1/2, |\tau| \geq 1\}.$$

- (a) Show that J is a homeomorphism on ∂T onto $\mathbb{R}$.
- (b) Show that J is a homeomorphism of the interior T onto the closed lower half-plane $\{w \in \mathbb{C} : \operatorname{Im} w \leq 0\}$ and a conformal map of the interior $\operatorname{int}(T)$ onto the open lower half-plane $\{w \in \mathbb{C} : \operatorname{Im} w < 0\}$.
- (c) Let $g_2, g_3 \in \mathbb{C}$ and $g_2^3 - 27g_3^2 \neq 0$. Show that then there exists a unique rank-2 lattice $\Gamma \subseteq \mathbb{C}$ such that $g_2(\Gamma) = g_2$ and $g_3(\Gamma) = g_3$.

Problem 24.6 (Simple gravity pendulum). The *simple gravity pendulum* is an idealized mathematical model for the motion of a pendulum given by a pointlike body of mass m suspended from a massless chord or rod of length L from a pivot while gravity pulls the body downwards. The point mass can make an oscillatory motion in a vertical plane containing the pivot to which the rod is attached. We assume that there is no friction or air drag.

We can describe the motion of the mass point by considering the amplitude φ that the rod makes with the vertical direction oriented downward. We denote by $\varphi_0 \in (0, \pi)$ the maximal amplitude and consider $\varphi = \varphi(t)$ as a (smooth) function of time, where the time origin is chosen so that $\varphi(0) = 0$.

The speed of the point mass is $v = L\dot{\varphi}(t)$ and so it has kinetic energy

$$\frac{1}{2}mv^2 = \frac{1}{2}mL^2\dot{\varphi}(t)^2.$$

On the other hand, the potential energy is given

$$-mgL \cos \varphi(t)$$

when we consider the horizontal position of the pivot corresponding to 0 potential energy and

$$g = 9.81 \frac{\text{kg}}{\text{sec}^2}$$

is *acceleration of gravity*. Since the total energy E of the moving body is preserved (due to the fact that there is no friction or air drag), we have

$$\frac{1}{2}mL^2\dot{\varphi}(t)^2 - mgL \cos \varphi(t) = E. \quad (24.4)$$

On the other hand, at the time when the maximal amplitude φ_0 is reached the velocity of the point mass is equal to 0 and so

$$E = -mgL \cos \varphi_0.$$

We will make the assumption here that $\varphi_0 \in (0, \pi)$ which corresponds to $E < mgL$. If $E > mgL$, then the pendulum will have enough energy to swing over the top and will make a circular motion (we will not discuss this case).

So we arrive at the differential equation

$$\frac{1}{2}mL^2\dot{\varphi}(t)^2 - mgL \cos \varphi(t) = -mgL \cos \varphi_0.$$

(a) Show that

$$\ddot{\varphi}(t) + \frac{g}{L} \sin \varphi(t) = 0. \quad (24.5)$$

(b) Assume that the maximal amplitude φ_0 is small. Since $\varphi(t) \in [-\varphi_0, \varphi_0]$, we can use the approximation $\sin \varphi(t) \approx \varphi(t)$. Then (24.5) becomes

$$\ddot{\varphi}(t) + \frac{g}{L} \varphi(t) = 0.$$

Find the general solution of this differential equation. Show that under our assumptions

$$\varphi(t) = \varphi_0 \sin(\sqrt{g/L} t)$$

and that the motion of the pendulum is periodic with period

$$T = 2\pi \sqrt{\frac{L}{g}}. \quad (24.6)$$

Remark: For the famous *Foucault pendulum* in Paris we have $m = 28\text{kg}$, $L = 67\text{m}$, $\varphi_0 \approx 4.3^\circ$, and $T \approx 16\text{sec}$. This pendulum is *not* restricted to a movement in a fixed plane as in our model, but this can be ignored for a few oscillations. Over longer time periods the plane in which the pendulum moves slowly revolves around a vertical axis due to the rotation of the earth.

Problem 24.7. The purpose of this problem is to give a precise mathematical model of the simple gravity pendulum using Weierstrass $\wp$ -functions. For this we introduce an x - y coordinate system with the pivot as the center and

the y -axis parallel to the vertical direction. Then the location of the point mass at time t is described by its coordinates $x(t)$ and $y(t)$, where

$$x(t)^2 + y(t)^2 = L^2.$$

We assume that $x(t)$ and $y(t)$ are smooth functions of t .

(a) Show that

$$\dot{x}(t) = -\dot{y}(t)y(t)/x(t)$$

and that the kinetic energy of the system is given by

$$\frac{1}{2}m(\dot{x}(t)^2 + \dot{y}(t)^2) = \frac{1}{2}m\dot{y}(t)^2 \frac{L^2}{L^2 - y(t)^2}.$$

(b) If E denotes the total energy of the system, show that

$$\frac{1}{2}m\dot{y}(t)^2 \frac{L^2}{L^2 - y(t)^2} + mgy(t) = E.$$

(c) We now introduce the dimensionless parameter

$$\lambda := \frac{E}{mgL} \in (-1, \infty).$$

Show that then

$$\frac{\dot{y}(t)^2}{(y(t) - L)(y(t) + L)(y(t) - \lambda L)} = 2g/L^2. \quad (24.7)$$

In the following we assume that $\lambda < 1$. This corresponds to the inequality $E < mgL$ and means that the point mass does not reach the top position. Moreover, we then have

$$-L \leq y(t) \leq \lambda L < L$$

for all times t . It follows that the denominator of the left hand side in (24.7) only takes positive values and we can take the square root in (24.7). We now introduce a new function $z = z(t)$ by the equation

$$y = Lz + L\lambda/3.$$

Show that then

$$\frac{\dot{z}(t)}{\sqrt{4(z(t) - e_1)(z(t) - e_2)(z(t) - e_3)}} = \sqrt{g/(2L)},$$

where

$$e_1 := 1 - \lambda/3, \quad e_2 := 2\lambda/3, \quad e_3 := -1 - \lambda/3.$$

(d) Show that $e_1 + e_2 + e_3 = 0$, $e_1 > e_2 > e_3$, and that $e_3 \leq z(t) \leq e_2$ for all times t .

We put the time origin so that $y(0) = -L$ which is equivalent to $z(0) = e_3$. Then after integration we have

$$\int_{e_3}^{z(t)} \frac{dz}{\sqrt{4(z - e_1)(z - e_2)(z - e_3)}} = \sqrt{g/(2L)}t.$$

(e) Show that there exists a unique rank-2 lattice $\Gamma \subseteq \mathbb{C}$ such that the corresponding Weierstrass $\wp$ -function satisfies the differential equation

$$\wp'(u)^2 = 4(\wp(u) - e_1)(\wp(u) - e_2)(\wp(u) - e_3)$$

for the given values of e_1, e_2, e_3 .

One can show (you can use this without proof!) that this lattice is generated by numbers ω and ω' such that ω is positive real and ω' is positive imaginary and

$$\wp(\omega/2) = e_1, \quad \wp((\omega + \omega')/2) = e_2, \quad \wp(\omega'/2) = e_3.$$

(f) Show that $\wp(\omega'/2 + s)$ takes real values between e_3 and e_2 for $s \in \mathbb{R}$.

(g) Show that $z(t) = \wp(u(t)) = \wp(\omega'/2 + \sqrt{g/(2L)}t)$ and

$$y(t) = L\wp(\omega'/2 + \sqrt{g/(2L)}t) + L\lambda/3.$$

(h) Show that the quarter-period $T/4$ of the motion is given by the time for z to move from e_3 to e_2 and

$$T = 2\sqrt{\frac{2L}{g}}\omega,$$

where

$$\omega = 2 \int_{e_3}^{e_2} \frac{dz}{\sqrt{4(z - e_1)(z - e_2)(z - e_3)}}.$$

(i) Show that

$$\omega = \frac{2}{\sqrt{e_1 - e_3}} \int_0^{\pi/2} \frac{d\alpha}{\sqrt{1 - \ell^2 \sin^2 \alpha}}$$

with

$$\ell = \sqrt{\frac{e_2 - e_3}{e_1 - e_3}} = \sqrt{(1 + \lambda)/2}.$$

(j) Show that

$$T = 4\sqrt{\frac{L}{g}} K(\ell),$$

where $K(\ell)$ is the complete elliptic integral of the first kind as in Problem 24.2. Why does this formula agree with (24.6) for small maximal amplitudes?

25 The Cauchy transform

In this section we use some basic results of measure theory such as Lebesgue's theorem on dominated convergence or Fubini's Theorem. We denote by A Lebesgue measure on $\mathbb{R}^2 \cong \mathbb{C}$, and by dA integration with respect to this measure.

We first fix some notation for function spaces. All functions are complex-valued. Let $U \subseteq \mathbb{C}$ be open. Then $L^1(U)$ and $L^1_{\text{loc}}(U)$ denote the spaces of integrable and locally integrable functions on U , respectively. The space of continuous functions on U is denoted by $C(U)$; we also use this notation if U is an arbitrary metric space. For $k \in \mathbb{N}$ we let $C^k(U)$ be the space of functions on U that have continuous partial derivatives up to order k . As usual $C^\infty(U)$ stands for the space of functions on U that have continuous partial derivatives of all orders. A subscript “ c ” indicates compact support; so for example, $C_c^\infty(U)$ is the space of all functions $f \in C^\infty(U)$ whose support

$$\text{supp}(f) := \overline{\{z \in U : f(z) \neq 0\}}$$

is a compact subset of U .

Definition 25.1 (Cauchy transform). Let $u \in C_c(\mathbb{C})$. Then the *Cauchy transform* $Tu: \mathbb{C} \rightarrow \mathbb{C}$ of u is defined as

$$(Tu)(w) = -\frac{1}{\pi} \int_{\mathbb{C}} \frac{u(z)}{z - w} dA(z) \quad \text{for } w \in \mathbb{C}.$$

Note that the *Cauchy kernel*

$$K(z) := \frac{1}{\pi z}$$

is locally integrable. This implies that the Cauchy transform exists for each $w \in \mathbb{C}$. Actually, $Tu = K * u$, where the *convolution* $K * u$ is given by

$$(K * u)(w) := \int_{\mathbb{C}} K(w - z)u(z) dA(z) \quad \text{for } w \in \mathbb{C}.$$

In the following we use z to denote a typical point in $\mathbb{C}$, and use the notation $\frac{\partial f}{\partial z}$ and $\frac{\partial f}{\partial \bar{z}}$, or simply f_z and $f_{\bar{z}}$, for the z - and $\bar{z}$ -derivatives of a function f as defined in Remark 3.13.

The Cauchy transform is in some sense the inverse operator of the differential operator $\frac{\partial}{\partial \bar{z}}$ as the following two lemmas show.

Lemma 25.2. *If $f \in C_c^1(\mathbb{C})$, then $f = T(f_{\bar{z}})$.*

Proof. Note that $f_{\bar{z}} \in C_c(\mathbb{C})$; so $T(f_{\bar{z}})$ is defined.

In order to compute $(Tf_{\bar{z}})(w)$, we may assume that $w = 0$. We introduce polar coordinates in the z -plane: $z = re^{i\alpha}$ and $\bar{z} = re^{-i\alpha}$, where $r \geq 0$ and $\alpha \in [0, 2\pi]$. Then $r = (z \cdot \bar{z})^{1/2}$ and $\alpha = \frac{1}{2i} \log(z/\bar{z})$ (locally for some branch of the logarithm). Then by the chain rule,

$$\frac{\partial}{\partial \bar{z}} = \frac{\partial}{\partial r} \cdot \frac{\partial r}{\partial \bar{z}} + \frac{\partial}{\partial \alpha} \cdot \frac{\partial \alpha}{\partial \bar{z}}.$$

Here

$$\frac{\partial r}{\partial \bar{z}} = \frac{\partial((z \cdot \bar{z})^{1/2})}{\partial \bar{z}} = \frac{1}{2} \cdot \frac{z^{1/2}}{\bar{z}^{1/2}} = \frac{1}{2} \cdot \frac{(z\bar{z})^{1/2}}{\bar{z}} = \frac{1}{2} e^{i\alpha}$$

(if we choose local branches of the square root appropriately), and

$$\frac{\partial \alpha}{\partial \bar{z}} = \frac{1}{2i} \frac{\partial(\log(z/\bar{z}))}{\partial \bar{z}} = \frac{1}{2i} \cdot \frac{-1}{\bar{z}} = \frac{i}{2r} e^{i\alpha}.$$

So

$$\frac{\partial}{\partial \bar{z}} = \frac{1}{2} e^{i\alpha} \cdot \frac{\partial}{\partial r} + \frac{i}{2r} e^{i\alpha} \cdot \frac{\partial}{\partial \alpha}.$$

Then by integrating in polar coordinates we obtain

$$\begin{aligned} T(f_{\bar{z}})(0) &= -\frac{1}{\pi} \lim_{\epsilon \rightarrow 0} \int_{\mathbb{C} \setminus B(0, \epsilon)} \frac{f_{\bar{z}}(z)}{z} dA(z) \quad (\text{by Lebesgue dominated conv.}) \\ &= -\frac{1}{\pi} \lim_{\epsilon \rightarrow 0} \int_{\epsilon}^{\infty} \int_0^{2\pi} \left(\frac{1}{2} e^{i\alpha} \frac{\partial f(re^{i\alpha})}{\partial r} + \frac{i}{2r} e^{i\alpha} \frac{\partial f(re^{i\alpha})}{\partial \alpha} \right) e^{-i\alpha} dr d\alpha \\ &= A + B, \end{aligned}$$

where

$$A = -\frac{1}{2\pi} \lim_{\epsilon \rightarrow 0} \int_0^{2\pi} \underbrace{\left(\int_{\epsilon}^{\infty} \frac{\partial f(re^{i\alpha})}{\partial r} dr \right)}_{= -f(\epsilon e^{i\alpha})} d\alpha = \lim_{\epsilon \rightarrow 0} \frac{1}{2\pi} \int_0^{2\pi} f(\epsilon e^{i\alpha}) d\alpha = f(0)$$

by continuity of f at 0, and

$$B = -\frac{i}{2\pi} \lim_{\epsilon \rightarrow 0} \int_{\epsilon}^{\infty} \underbrace{\left(\int_0^{2\pi} \frac{1}{r} \frac{\partial f(re^{i\alpha})}{\partial \alpha} d\alpha \right)}_{= 0} dr = 0.$$

Here we introduced the limit as $\epsilon \rightarrow 0$ in order to avoid possible problems with the factor $1/r$ in the 1-dimensional integrals.

We get $T(f_{\bar{z}})(0) = A + B = f(0)$ as desired. $\square$

Lemma 25.3. *If $u \in C_c^1(\mathbb{C})$, then $Tu \in C^1(\mathbb{C})$, and $(Tu)_{\bar{z}} = u$.*

Proof. The regularity statement is based on some standard fact, and we only give an outline. Namely, if K is a locally integrable kernel on $\mathbb{R}^n$ and u is C^1 -smooth on $\mathbb{R}^n$ with compact support, then $K * u$ is C^1 -smooth; indeed, if ∂_k denotes the partial derivative with respect to the k -th coordinate on $\mathbb{R}^n$, then we have $\partial_k(K * u) = K * (\partial_k u)$ for $k = 1, \dots, n$. This can easily be proved by considering the relevant difference quotients, and passing to the limit under the integral based on Lebesgue's theorem on dominated convergence.

If K denotes the Cauchy kernel, then in our situation we have $Tu = K * u \in C^1(\mathbb{C})$. Moreover, since the operator $\partial_{\bar{z}}$ is a linear combination of the usual partial differential operators on $\mathbb{R}^2 \cong \mathbb{C}$, we conclude

$$(Tu)_{\bar{z}} = (K * u)_{\bar{z}} = K * u_{\bar{z}} = T(u_{\bar{z}}) = u,$$

where we used Lemma 25.2 in the last step. $\square$

Lemma 25.4. (a) *Let $\gamma: [0, 1] \rightarrow \mathbb{C}$ be a C^1 -smooth path, and $u \in C_c(\mathbb{C})$. Define $F: \mathbb{C} \times [0, 1] \rightarrow \mathbb{C}$ by*

$$F(z, t) = \begin{cases} \frac{u(z) \cdot \gamma'(t)}{z - \gamma(t)} & \text{for } z \neq \gamma(t), \\ 0 & \text{for } z = \gamma(t). \end{cases}$$

Then F is integrable on $\mathbb{C} \times [0, 1]$ with respect to the product of Lebesgue measures on $\mathbb{C}$ and on $[0, 1]$.

(b) *Let Γ be a piecewise smooth cycle in $\mathbb{C}$. Then the function $z \in \mathbb{C} \mapsto \text{ind}_{\Gamma}(z)$ is almost everywhere defined and integrable.*

Proof. (a) The exceptional set $E := \{(z, t) \in \mathbb{C} \times [0, 1] : z = \gamma(t)\}$ is closed, and hence a Borel set. Moreover, F is continuous on the complement of E . From this it follows that F is a Borel function, and hence measurable.

We can pick constants $R > 0$ and $M \geq 0$ such that the sets $\text{supp}(u)$ and γ^* both lie in $\overline{B}(0, R)$, and $|u(z)| \leq M$ for $z \in \overline{B}(0, R)$.

Then we have

$$\begin{aligned}
 \int_0^1 \left(\int_{\mathbb{C}} |F(z, t)| dA(z) \right) dt &\leq \int_0^1 \left(\int_{\mathbb{C}} \frac{|u(z)| \cdot |\gamma'(t)|}{|z - \gamma(t)|} dA(z) \right) dt \\
 &\leq M \int_0^1 \left(\int_{\overline{B}(0, R)} \frac{|\gamma'(t)|}{|z - \gamma(t)|} dA(z) \right) dt \\
 &\leq M \int_0^1 \left(\int_{\overline{B}(0, 2R)} \frac{|\gamma'(t)|}{|u|} dA(u) \right) dt \\
 &= M \left(\int_{\overline{B}(0, 2R)} \frac{1}{|u|} dA(u) \right) \left(\int_0^1 |\gamma'(t)| dt \right) < \infty.
 \end{aligned}$$

This implies that F is integrable on $\mathbb{C} \times [0, 1]$.

(b) If γ is a smooth path, then a simple covering argument shows that $A(\gamma^*) = 0$. This implies that $A(\Gamma^*) = 0$. The function $z \mapsto \text{ind}_{\Gamma}(z)$ is defined on $\mathbb{C} \setminus \Gamma^*$, and hence almost everywhere. Since this function is locally constant on $\mathbb{C} \setminus \Gamma^*$ as follows from Theorem 19.14, it is measurable.

To show that it is also integrable, we may assume that Γ consists of one loop α , and represent the winding number as an integral as in Theorem 19.8. Pick $R > 0$ large enough so that $\alpha^* \subseteq B(0, R)$. Then $\mathbb{C} \setminus B(0, R)$ lies in the unbounded component of $\mathbb{C} \setminus \alpha^*$, and so $\text{ind}_{\alpha}(z) = 0$ for $z \in \mathbb{C} \setminus B(0, R)$ (Theorem 19.14). By breaking up the loop α into finitely many C^1 -smooth paths, we are reduced to proving that if $\gamma: [0, 1] \rightarrow B(0, R)$ is a C^1 -smooth path and if we define

$$h(z) = \int_{\gamma} \frac{d\zeta}{\zeta - z} = \int_0^1 \frac{\gamma'(t)}{\gamma(t) - z} dt$$

for $z \in B(0, R)$, then $h \in L^1(B(0, R))$. Note that this function is defined on $B(0, R) \setminus \gamma^*$, and hence almost everywhere on $B(0, R)$. It is easy to see that it is continuous on $B(0, R) \setminus \gamma^*$, and so measurable.

The integrability of h follows from part (a); indeed, we can pick a function $u \in C_c(\mathbb{C})$ such that $u(z) = 1$ for $z \in B(0, R)$. If F is defined as in (a), then by Fubini's Theorem the function $z \mapsto \int_0^1 |F(z, t)| dt$ is almost everywhere

defined and integrable on $\mathbb{C}$. Hence

$$\begin{aligned}
 \int_{B(0,R)} |h(z)| dA(z) &= \int_{B(0,R) \setminus \gamma^*} |h(z)| dA(z) \\
 &= \int_{B(0,R) \setminus \gamma^*} \left| \int_0^1 \frac{\gamma'(t)}{\gamma(t) - z} dt \right| dA(z) \\
 &= \int_{B(0,R) \setminus \gamma^*} \left(\int_0^1 \frac{|u(z)| \cdot |\gamma'(t)|}{|\gamma(t) - z|} dt \right) dA(z) \\
 &\leq \int_{\mathbb{C}} \left(\int_0^1 |F(z, t)| dt \right) dA(z) < \infty. \quad \square
 \end{aligned}$$

Proposition 25.5 (Gauss–Green Formula; preliminary version). Let $u \in C_c^1(\mathbb{C})$, and Γ be a piecewise smooth cycle in $\mathbb{C}$. Then

$$\frac{1}{2i} \int_{\Gamma} u(z) dz = \int_{\mathbb{C}} \text{ind}_{\Gamma}(z) \cdot u_{\bar{z}}(z) dA(z).$$

Note that $u_{\bar{z}}$ is a continuous function with compact support. This and Lemma 25.4 (b) imply that the integrand on the right hand side is an almost everywhere defined and integrable function.

Proof. By Lemma 25.2 we know that

$$u(w) = -\frac{1}{\pi} \int_{\mathbb{C}} \frac{u_{\bar{z}}(z)}{z - w} dA(z)$$

for all $w \in \mathbb{C}$. Integrating over Γ and interchanging the order of integration (which is justified by Fubini's Theorem and Lemma 25.4 (a) applied to paths obtained from splitting up the loops in Γ into C^1 -smooth pieces), we obtain

$$\begin{aligned}
 \frac{1}{2i} \int_{\Gamma} u(w) dw &= -\frac{1}{2\pi i} \int_{\Gamma} \left(\int_{\mathbb{C}} \frac{u_{\bar{z}}(z)}{z - w} dA(z) \right) dw \\
 &= -\frac{1}{2\pi i} \int_{\mathbb{C}} \left(\int_{\Gamma} \frac{u_{\bar{z}}(z)}{z - w} dw \right) dA(z) \\
 &= \int_{\mathbb{C}} \text{ind}_{\Gamma}(z) \cdot u_{\bar{z}}(z) dA(z). \quad \square
 \end{aligned}$$

25.6. Partitions of unity. Let $V \subseteq \mathbb{C}$ be open, and $\mathcal{U} = \{U_i \in I\}$ be a cover of V by open sets $U_i \subseteq V$ indexed by a set I . We need the following two statements about partitions of unity.

Version I (“with compact supports”). There exists a countable family $\{\varphi_k : k \in \mathbb{N}\}$ of functions such that

- (i) $\varphi_k \in C^\infty(V)$, $0 \leq \varphi_k \leq 1$, and $\text{supp}(\varphi_k) \subseteq V$ is compact for all $k \in \mathbb{N}$,
- (ii) the supports $\text{supp}(\varphi_k)$, $k \in \mathbb{N}$, form a *locally finite* family, i.e., every point $p \in V$ has a neighborhood N such that $\text{supp}(\varphi_k) \cap N \neq \emptyset$ for only finitely many $k \in \mathbb{N}$,
- (iii) the supports $\text{supp}(\varphi_k)$, $k \in \mathbb{N}$, form a family that is *subordinate* to $\mathcal{U}$, i.e., for every $k \in \mathbb{N}$ there exists $i(k) \in I$ with $\text{supp}(\varphi_k) \subseteq U_{i(k)}$,
- (iv) $\sum_{k \in \mathbb{N}} \varphi_k(z) = 1$ for all $z \in V$.

Note that in the last sum only finitely many terms are non-zero near each point by property (ii). So locally we can treat this as a finite sum and take partial derivatives term-by-term, for example.

The phrase “partition of unity” is of course explained by the identity (iv). It often allows one to break up a given function into “bumps” with well-localized support.

Version II (“with same index set”). There exists a family $\{\varphi_i : i \in I\}$ of functions having the above properties (i)–(iv) with the following modifications: the functions φ_i do not necessarily have compact support, and (iii) is replaced by $\text{supp}(\varphi_i) \cap V \subseteq U_i$ for all $i \in I$.

So in Version I the functions of the partition of unity have compact support, while in Version II we can index the family by the same index set as the cover $\mathcal{U}$. This is not always possible if one insists on compact support of the functions.

Lemma 25.7. *Let $U \subseteq \mathbb{C}$ be open, and $K \subseteq U$ be compact. Then there exists $\varphi \in C_c^\infty(U)$ such that $0 \leq \varphi \leq 1$, $\varphi(z) = 1$, and $\nabla\varphi(z) = 0$ for all $z \in K$.*

Proof. We can choose an open set $V \subseteq \mathbb{C}$ such that $\overline{V}$ is compact and $K \subseteq V \subseteq \overline{V} \subseteq U$. Then $\mathcal{U} = \{U \setminus K, V\}$ is an open cover of U . Hence there exists a partition of unity on V subordinate to $\mathcal{U}$; more precisely, there exist two functions $\varphi, \psi \in C^\infty(U)$ with $0 \leq \varphi, \psi \leq 1$, $\text{supp}(\varphi) \cap U \subseteq V$,

$\text{supp}(\psi) \cap U \subseteq U \setminus K$, and $\varphi + \psi = 1$ on U . Then φ vanishes on $U \setminus V$, and so $\text{supp}(\varphi) \subseteq \overline{V}$. This implies $\varphi \in C_c^\infty(U)$.

Moreover, $\text{supp}(\psi) \cap K = \emptyset$, and so $\psi(z) = 0$ for all points z in an open neighborhood of K . So for $z \in K$ we have $\varphi(z) = 1 - \psi(z) = 1$, and $\nabla\varphi(z) = -\nabla\psi(z) = 0$. $\square$

Theorem 25.8. *Let $\Omega \subseteq \mathbb{C}$ be open, $f \in C^1(\Omega)$, and Γ be a piecewise smooth cycle in Ω that is null-homologous in Ω . Then*

$$\frac{1}{2i} \int_{\Gamma} f(z) dz = \int_{\Omega} \text{ind}_{\Gamma}(z) \cdot f_{\bar{z}}(z) dA(z) \quad (\text{Gauss-Green Formula}),$$

and for all $z_0 \in \Omega \setminus \Gamma^*$,

$$\text{ind}_{\Gamma}(z_0) \cdot f(z_0) = \frac{1}{2\pi i} \int_{\Gamma} \frac{f(z)}{z - z_0} dz - \frac{1}{\pi} \int_{\Omega} \text{ind}_{\Gamma}(z) \cdot \frac{f_{\bar{z}}(z)}{z - z_0} dA(z)$$

(Cauchy-Green Formula or Pompeiu's Formula).

Proof. We have $\text{ind}_{\Gamma}(z) = 0$ for all $z \in \mathbb{C}$ near $\partial\Omega \subseteq \mathbb{C} \setminus \Omega$ (since Γ is null-homologous in Ω), and also $\text{ind}_{\Gamma}(z) = 0$ for all $z \in \mathbb{C}$ with $|z|$ large. Hence we can pick a compact set $K \subseteq \Omega$ such that $\Gamma^* \subseteq K$ and $\text{ind}_{\Gamma}(z) = 0$ for all $z \in \mathbb{C} \setminus K$. Actually, one can take K of the form

$$K = \{z \in \Omega : \text{dist}(z, \partial\Omega) \geq \delta \text{ and } |z| \leq R\},$$

where $\delta > 0$ is small enough and $R > 0$ is large enough.

By Lemma 25.7 there exists $\varphi \in C_c^\infty(\Omega)$ with $0 \leq \varphi \leq 1$ such that $\varphi(z) - 1 = \varphi_{\bar{z}}(z) = 0$ for $z \in K$. We now apply Proposition 25.5 to $u := \varphi f$ (considered as a function on $\mathbb{C}$ by setting it $\equiv 0$ on $\mathbb{C} \setminus \Omega$). Then $u \in C_c^1(\mathbb{C})$, $u(z) = f(z)$ for $z \in \Gamma^* \subseteq K$, and

$$u_{\bar{z}} = \varphi_{\bar{z}} f + \varphi f_{\bar{z}} = f_{\bar{z}} \quad \text{on } K.$$

Hence

$$\begin{aligned} \frac{1}{2i} \int_{\Gamma} f(z) dz &= \frac{1}{2i} \int_{\Gamma} u(z) dz = \int_{\mathbb{C}} \text{ind}_{\Gamma}(z) \cdot u_{\bar{z}}(z) dA(z) \\ &= \int_K \text{ind}_{\Gamma}(z) \cdot u_{\bar{z}}(z) dA(z) = \int_K \text{ind}_{\Gamma}(z) \cdot f_{\bar{z}}(z) dA(z) \\ &= \int_{\Omega} \text{ind}_{\Gamma}(z) \cdot f_{\bar{z}}(z) dA(z). \end{aligned}$$

For the proof of the Cauchy–Green Formula we let $\tilde{\Gamma} := \Gamma - \text{ind}_{\Gamma}(z_0)\gamma$, where $\gamma(t) := z_0 + re^{it}$ for $t \in [0, 2\pi]$ and $r > 0$ is small. Here γ depends on r , but we suppress this in our notation.

Define

$$g(z) = \frac{1}{\pi} \frac{f(z)}{z - z_0}$$

for $z \in \tilde{\Omega} := \Omega \setminus \{z_0\}$. Then $g \in C^1(\tilde{\Omega})$ and $\tilde{\Gamma}$ is null-homologous in $\tilde{\Omega}$. Note that

$$g_{\bar{z}}(z) = \frac{1}{\pi} \cdot \frac{\partial}{\partial \bar{z}} \left(\frac{f(z)}{z - z_0} \right) = \frac{1}{\pi} \cdot \frac{f_{\bar{z}}(z)}{z - z_0} \quad \text{for } z \in \tilde{\Omega}.$$

By the first part we have,

$$\begin{aligned} \frac{1}{2i} \int_{\tilde{\Gamma}} g(z) dz &= \frac{1}{2\pi i} \int_{\Gamma} \frac{f(z)}{z - z_0} dz - \text{ind}_{\Gamma}(z_0) \cdot \frac{1}{2\pi i} \int_{\gamma} \frac{f(z)}{z - z_0} dz \quad (25.1) \\ &= \int_{\tilde{\Omega}} \text{ind}_{\tilde{\Gamma}}(z) \cdot g_{\bar{z}}(z) dA(z) \\ &= \frac{1}{\pi} \int_{\Omega} \text{ind}_{\Gamma}(z) \cdot \frac{f_{\bar{z}}(z)}{z - z_0} dA(z) - \text{ind}_{\Gamma}(z_0) \cdot \frac{1}{\pi} \int_{\Omega} \text{ind}_{\gamma}(z) \cdot \frac{f_{\bar{z}}(z)}{z - z_0} dA(z) \\ &= \frac{1}{\pi} \int_{\Omega} \text{ind}_{\Gamma}(z) \cdot \frac{f_{\bar{z}}(z)}{z - z_0} dA(z) - \text{ind}_{\Gamma}(z_0) \cdot \frac{1}{\pi} \int_{B(z_0, r)} \frac{f_{\bar{z}}(z)}{z - z_0} dA(z), \end{aligned}$$

where in the last step we used that

$$\text{ind}_{\gamma}(z) = \begin{cases} 0 & \text{for } z \in \mathbb{C} \setminus \overline{B}(z_0, r), \\ 1 & \text{for } z \in B(z_0, r). \end{cases}$$

Note that

$$\lim_{r \rightarrow 0} \frac{1}{2\pi i} \int_{\gamma} \frac{f(z)}{z - z_0} dz = \lim_{r \rightarrow 0} \frac{1}{2\pi} \int_0^{2\pi} f(z_0 + re^{it}) dt = f(z_0)$$

by continuity of f at $z_0 \in \Omega$, and

$$\lim_{r \rightarrow 0} \int_{B(z_0, r)} \frac{f_{\bar{z}}(z)}{z - z_0} dA(z) = 0$$

as easily follows from Lebesgue's theorem on dominated convergence.

If we let $r \rightarrow 0$ in (25.1) and use these limit identities, then the Cauchy–Green formula follows. $\square$

Corollary 25.9 (Area formula). Let γ be a piecewise smooth loop in $\mathbb{C}$ and assume that $\text{ind}_\gamma(z) \in \{0, 1\}$ for all $z \in \mathbb{C} \setminus \gamma^*$. Define

$$\Omega = \{z \in \mathbb{C} \setminus \gamma^* : \text{ind}_\gamma(z) = 1\}.$$

Then

$$A(\Omega) = \frac{1}{2i} \int_\gamma \bar{z} dz.$$

Proof. Note that Ω is an open set, and hence measurable. We apply the Gauss–Green Formula on $\mathbb{C}$, where $\Gamma = \gamma$ and $f(z) = \bar{z}$ for $z \in \mathbb{C}$. Then

$$\frac{1}{2i} \int_\gamma \bar{z} dz = \int_{\mathbb{C}} \text{ind}_\gamma(z) dA(z) = \int_\Omega dA(z) = A(\Omega). \quad \square$$

A consequence of the previous corollary is that under the given assumptions the integral $\int_\gamma \bar{z} dz$ is purely imaginary. This can also be seen directly as follows. Suppose $\gamma: [a, b] \rightarrow \mathbb{C}$ is a parametrization of the piecewise smooth loop γ . Then

$$\begin{aligned} \text{Re} \left(\int_\gamma \bar{z} dz \right) &= \text{Re} \left(\int_a^b \overline{\gamma(t)} \gamma'(t) dt \right) \\ &= \frac{1}{2} \int_a^b (\overline{\gamma(t)} \gamma'(t) + \gamma(t) \overline{\gamma'(t)}) dt \\ &= \frac{1}{2} \int_a^b \frac{d}{dt} (\overline{\gamma(t)} \gamma(t)) dt = [|\gamma(t)|^2]_a^b = 0. \end{aligned}$$

26 Runge's Approximation Theorem

26.1. Background from Approximation Theory. Let $\mathcal{F}$ be a class of functions on a set, say on a compact set $K \subseteq \mathbb{R}^n$ or an open set $U \subseteq \mathbb{R}^n$, and let $\mathcal{S} \subseteq \mathcal{F}$ be a subclass of particularly “nice” or “simple” functions. The basic problem of approximation theory is under what conditions we can approximate functions in $\mathcal{F}$ by functions in $\mathcal{S}$.

An instance for a result in this direction is the *Weierstrass Approximation Theorem*. In this case $\mathcal{F}$ is the set $C(K)$ of all (complex-valued) continuous functions on a compact set $K \subseteq \mathbb{R}^n$, and $\mathcal{S}$ is the set of all polynomials (with complex coefficients) in the standard coordinate functions $x_1, \dots, x_n$ on $\mathbb{R}^n$. Then according to this theorem for all $f \in \mathcal{F} = C(K)$ and all $\epsilon > 0$ there exists a polynomial $P \in \mathcal{S}$ such that $|f(u) - P(u)| < \epsilon$ for all $u \in K$.

For $n = 1$ and $K = [a, b] \subseteq (0, 1) \subseteq \mathbb{R}$ the proof can be outlined as follows. If $f \in C([a, b])$, then we can extend f to a continuous function on $\mathbb{R}$ with support contained in $[0, 1]$. For $l \in \mathbb{N}$ we define

$$P_l(u) = \frac{\int_0^1 f(x)(1 - (u - x)^2)^l dx}{\int_{-1}^1 (1 - x^2)^l dx}, \quad u \in [0, 1].$$

Then P_l is a polynomial in u and it not difficult to see that $P_l \rightarrow f$ uniformly on K as $l \rightarrow \infty$.

In higher dimensions one can implement a proof along very similar lines if one uses the fact that every continuous function on a closed set A in $\mathbb{R}^n$ can be extended to a continuous function on $\mathbb{R}^n$ (one either uses *Tietze's Extension Theorem* to justify this fact or gives a direct simple argument based on $A \subseteq \mathbb{R}^n$).

From the Weierstrass Approximation Theorem one can derive the following consequence. Let $U \subseteq \mathbb{R}^n$ be open, and K_l , $l \in \mathbb{N}$, be a compact exhaustion of U (for this terminology see the proof of Theorem 11.3 and Lemma 26.5 below). Then there exists a polynomial P_l in the standard coordinate functions $x_1, \dots, x_n$ on $\mathbb{R}^n$ such that $|f(u) - P_l(u)| < 1/l$ for $l \in \mathbb{N}$ and $u \in K_l$. In particular, we can find a sequence of polynomials P_l such that $P_l \rightarrow f$ locally uniformly on U as $l \rightarrow \infty$.

In $\mathbb{R}^2 \cong \mathbb{C}$ we can replace the standard coordinate functions x and y on $\mathbb{R}^2$ by $z = x + iy$ and $\bar{z} = x - iy$. So every continuous function f on an open

set $U \subseteq \mathbb{C}$ is the locally uniform limit of a sequence of polynomials in z and $\bar{z}$. An obvious question here is whether one can strengthen this statement and show that every *holomorphic* function f on U is the locally uniform limit of polynomials in z *alone*.

As the following example shows this is not true in general. Let $U = \mathbb{C} \setminus \{0\}$, and $f(z) = 1/z$. Then $f \in H(U)$, but there is no sequence $\{P_n\}$ of polynomials in z such that $P_n \rightarrow f$ locally uniformly on U . Indeed, suppose there was such a sequence, and let $\gamma(t) = e^{it}$ for $t \in [0, 2\pi]$. Then

$$2\pi i = \int_{\gamma} \frac{dz}{z} = \int_{\gamma} f(z) dz = \lim_{n \rightarrow \infty} \underbrace{\int_{\gamma} P_n(z) dz}_{=0} = 0,$$

which is a contradiction.

As we will prove in this section (see Theorem 26.6), one has to impose some restrictions on U to get a statement in this direction, or one has to allow rational functions instead of polynomials.

Lemma 26.2 (Basic approximation lemma). *Let $U \subseteq \mathbb{C}$ be open, $K \subseteq U$ be compact, and $f \in H(U)$. Then for all $\epsilon > 0$ there exists a rational function R of the form*

$$R(w) = \sum_{k=1}^N \frac{a_k}{w - z_k},$$

where $N \in \mathbb{N}$, $a_1, \dots, a_N \in \mathbb{C}$, and $z_1, \dots, z_N \in \mathbb{C} \setminus K$, such that

$$|f(w) - R(w)| < \epsilon \quad \text{for } w \in K.$$

Proof. The basic idea for the proof is very simple. We want to represent f on K as a convolution integral of the Cauchy kernel and a function supported away from K , and then approximate the integral by a Riemann sum.

To implement this idea, we pick a compact set K' such that

$$K \subseteq \text{int}(K') \subseteq K' \subseteq U.$$

Then we can find $\varphi \in C_c^\infty(U)$ such that $\varphi(z) - 1 = \varphi_{\bar{z}}(z) = 0$ for $z \in K'$ (Lemma 25.7). Then $u := \varphi f \in C_c^1(\mathbb{C})$ (where it is understood that this function is extended to $\mathbb{C}$ by $u \equiv 0$ outside U). We have $u(w) = f(w)$ for $w \in K$, and $u_{\bar{z}} = \varphi_{\bar{z}} f$ by holomorphicity of f , and so

$$S := \text{supp}(u_{\bar{z}}) \subseteq U \setminus \text{int}(K') \subseteq U \setminus K.$$

Hence $D := \text{dist}(S, K) > 0$.

By Lemma 25.2

$$f(w) = u(w) = -\frac{1}{\pi} \int_{\mathbb{C}} \frac{u_{\bar{z}}(z)}{z - w} dA(z) = \frac{1}{\pi} \int_S \frac{u_{\bar{z}}(z)}{w - z} dA(z)$$

for $w \in K$. Now let $\delta \in (0, D/10)$ be small. We can subdivide $\mathbb{C}$ into squares of side-length δ that are *non-overlapping*, i.e., have disjoint interior. Let $Q_1, \dots, Q_N$ be the collection of these squares that meet S , let z_k be the center of Q_k , and

$$a_k := \frac{1}{\pi} \int_{Q_k} u_{\bar{z}}(z) dA(z)$$

for $k = 1, \dots, N$. Then $S \subseteq Q_1 \cup \dots \cup Q_N$, and $\text{dist}(K, Q_k) \geq D/2$ for $k = 1, \dots, N$. So for $z \in Q_k$ and $w \in K$ we have

$$\left| \frac{1}{w - z} - \frac{1}{w - z_k} \right| = \frac{|z - z_k|}{|w - z| \cdot |w - z_k|} \leq \frac{\delta}{(D/2) \cdot (D/2)} = \frac{4\delta}{D^2}.$$

Hence for $w \in K$

$$\begin{aligned} \left| f(w) - \sum_{k=1}^N \frac{a_k}{w - z_k} \right| &= \left| \frac{1}{\pi} \sum_{k=1}^N \int_{Q_k} \left(\frac{u_{\bar{z}}(z)}{w - z} - \frac{u_{\bar{z}}(z)}{w - z_k} \right) dA(z) \right| \\ &\leq \frac{4\delta}{\pi D^2} \sum_{k=1}^N \int_{Q_k} |u_{\bar{z}}(z)| dA(z) \\ &= \frac{4\delta}{\pi D^2} \underbrace{\int_S |u_{\bar{z}}(z)| dA(z)}_{=: C} = \frac{4C}{\pi D^2} \delta < \epsilon, \end{aligned}$$

if δ is chosen small enough (note that C and D are independent of δ). The claim follows. $\square$

The previous lemma allows us to approximate f on K by a rational function whose poles are outside K . We want to show a stronger statement, where the poles are in a prescribed position depending on K , but independent of f . For this we need the following lemma.

Lemma 26.3 (Pole-Pushing Lemma). *Let $K \subseteq \mathbb{C}$ be compact, V be a component of $\mathbb{C} \setminus K$, and $a \in V$. Define*

$$F_u(w) = \frac{1}{w - u} \quad \text{for } w \in K \text{ and } u \in V.$$

Then for each $u \in V$ the function F_u is uniformly approximable on K by polynomials in F_a , i.e., for each $\epsilon > 0$ there exists $n \in \mathbb{N}$, and $a_0, \dots, a_n \in \mathbb{C}$ such that

$$\left| \frac{1}{w-u} - \sum_{k=0}^n \frac{a_k}{(w-a)^k} \right| < \epsilon \quad \text{for all } w \in K.$$

If V is the unbounded component of $\mathbb{C} \setminus K$, then for each $u \in V$ the function F_u is uniformly approximable on K by polynomials, i.e., for each $\epsilon > 0$ there exists $n \in \mathbb{N}$, and $a_0, \dots, a_n \in \mathbb{C}$ such that

$$\left| \frac{1}{w-u} - \sum_{k=0}^n a_k w^k \right| < \epsilon \quad \text{for all } w \in K.$$

Proof. The proof consists of several steps. In the following w will always be a variable point in K .

Step 1. For $a \in V$ let $P_a(K) \subseteq C(K)$ be the set of all functions on K that can be written as polynomials in F_a , and $A_a(K) \subseteq C(K)$ be the set of all functions that are uniformly approximable on K by functions in $P_a(K)$ (note that the functions in $A_a(K)$ are continuous as uniform limits of continuous functions). The space $C(K)$ carries a natural metric d induced by the supremum norm on $C(K)$, i.e.,

$$d(g, h) := \sup_{w \in K} |g(w) - h(w)|, \quad g, h \in C(K).$$

Then $f \in A_a(K)$ iff there exists a sequence $\{f_n\}$ in $P_a(K)$ such that $f_n \rightarrow f$ uniformly on K iff $d(f_n, f) \rightarrow 0$ as $n \rightarrow \infty$ iff $f \in \overline{P_a(K)}$. So $A_a(K) = \overline{P_a(K)}$ (where the bar refers to closure in the metric space $(C(K), d)$).

It is clear that $P_a(K)$ is an algebra, i.e., a complex vector space of functions with the additional property that if $f, g \in P_a(K)$, then $fg \in P_a(K)$. Since $A_a(K) = \overline{P_a(K)}$, the space $A_a(K)$ is also an algebra.

Another immediate observation is that a function $f: K \rightarrow \mathbb{C}$ is uniformly approximable by functions in $A_a(K)$ iff $f \in \overline{A_a(K)} = A_a(K) = \overline{P_a(K)}$ iff f is uniformly approximable by functions in $P_a(K)$.

Step 2. Now suppose that $u \in V$, $r > 0$, $B(u, r) \subseteq V$, $F_u \in A_a(K)$, and $v \in B(u, r)$. We claim that $F_v \in A_a(K)$. By the previous discussion it is enough to show that F_v is uniformly approximable on K by polynomials in F_u (because then F_v is uniformly approximable by functions in the algebra $A_a(K)$, and hence lies in $A_a(K)$ itself).

Now for $w \in K$ we have

$$\begin{aligned} F_v(w) &= \frac{1}{w-v} = \frac{1}{(w-u)-(v-u)} \\ &= \frac{1}{(w-u)} \cdot \frac{1}{1 - \frac{v-u}{w-u}} = \sum_{n=0}^{\infty} \frac{(v-u)^n}{(w-u)^{n+1}}. \end{aligned} \quad (26.1)$$

Note that $|w-u| \geq r$ for $w \in K$, and so

$$\left| \frac{v-u}{w-u} \right| \leq \frac{|v-u|}{r} < 1.$$

Hence the convergence of the series in (26.1) is uniform for $w \in K$ by the Weierstrass M -test. We conclude that F_v is uniformly approximable on K by partial sums of this series and hence by polynomials in F_u as desired.

Step 3. We claim that $F_u \in A_a(K)$ for all $u \in V$. To see this, let $u \in V$ be arbitrary. Since V is open and connected, there exists a polygonal path P in V joining a and u (Theorem 2.31). Then $r := \text{dist}(P, \mathbb{C} \setminus V) > 0$. On P we can find points $u_0 = a, u_1, \dots, u_n = u$ such that $|u_k - u_{k-1}| < r$, and so $u_k \in B(u_{k-1}, r) \subseteq V$ for all $k = 1, \dots, n$.

Note that $F_{u_0} = F_a \in P_a(K) \subseteq A_a(K)$, and so $F_{u_1} \in A_a(K)$ by what we have seen in Step 2. Repeating this argument, we conclude that

$$F_{u_0}, F_{u_1}, \dots, F_{u_n} = F_u \in A_a(K).$$

This establishes the first part of the lemma.

Step 4. Now suppose that V is the unbounded component of K . We denote by $P_{\infty}(K) \subseteq C(K)$ the set of all functions that can be written as polynomials (in w) on K , and by $A_{\infty}(K) \subseteq C(K)$ the set of all functions that are uniformly approximable on K by functions in $P_{\infty}(K)$. Again $P_{\infty}(K)$ and $A_{\infty}(K)$ are algebras, and $\overline{A_{\infty}(K)} = A_{\infty}(K) = \overline{P_{\infty}(K)}$.

We can pick $a \in V$ such that

$$|a| > R := \sup\{|w| : w \in K\}.$$

Then for $w \in K$ we have

$$F_a(w) = \frac{1}{w-a} = -\frac{1}{a} \cdot \frac{1}{1-w/a} = -\sum_{n=0}^{\infty} \frac{w^n}{a^{n+1}}. \quad (26.2)$$

Note that $|w/a| \leq R/|a| < 1$, and so the convergence of the series in (26.2) is uniform on K by the Weierstrass M -test. Hence F_a is uniformly approximable on K by partial sums of this series, and so by elements in $P_\infty(K)$. This shows that $F_a \in A_\infty(K)$. Since $A_\infty(K)$ is an algebra, we conclude that $P_a(K) \subseteq A_\infty(K)$, and so $A_a(K) = \overline{P_a(K)} \subseteq \overline{A_\infty(K)} = A_\infty(K)$.

Now if $u \in V$ is arbitrary, then $F_u \in A_a(K) \subseteq A_\infty(K)$ by what we have seen in Step 2. This shows that F_u is uniformly approximable on K by polynomials as claimed. $\square$

Lemma 26.4 (Improved approximation lemma). *Let $U \subseteq \mathbb{C}$ be open, $K \subseteq U$ be compact, and $f \in H(U)$. Suppose $A \subseteq \mathbb{C} \setminus K$ is a set that meets each bounded component of $\mathbb{C} \setminus K$.*

Then for all $\epsilon > 0$ there exists a rational function R that has no poles outside the set $A \cup \{\infty\}$ such that

$$|f(w) - R(w)| < \epsilon \quad \text{for } w \in K.$$

If $\mathbb{C} \setminus K$ has no bounded component, then for all $\epsilon > 0$ there exists a polynomial P such that

$$|f(w) - P(w)| < \epsilon \quad \text{for } w \in K.$$

The second part of this lemma can be considered as a special case of the first part if one chooses $A = \emptyset$. Here we consider rational functions as holomorphic maps on $\widehat{\mathbb{C}}$. The non-constant polynomials are precisely those rational functions that have a pole at ∞ and no other poles.

Proof. This immediately follows from Lemma 26.2 if one “pushes” the poles z_k of the approximating rational function R to a location in A . This can be done with arbitrarily small error by Lemma 26.3. If $\mathbb{C} \setminus K$ has no bounded component, then one can push each pole to ∞ , i.e., get a polynomial approximation, by the second part of Lemma 26.3. $\square$

Lemma 26.5 (Compact exhaustions with good properties). *Let $U \subseteq \mathbb{C}$ be open. Then there exist compact sets $K_n \subseteq U$ for $n \in \mathbb{N}$ such that*

$$(i) \quad K_n \subseteq \text{int}(K_{n+1}) \text{ for } n \in \mathbb{N},$$

$$(ii) \quad U = \bigcup_{n \in \mathbb{N}} K_n,$$

- (ii) every bounded component of $\mathbb{C} \setminus K_n$ for $n \in \mathbb{N}$ contains a bounded component of $\mathbb{C} \setminus U$.

The first two properties (i) and (ii) just say that the sequence K_n , $n \in \mathbb{N}$, is a compact exhaustion of K . The additional property (iii) intuitively says that each “hole” (=bounded complementary component) of a set K_n contains at least one “hole” of U . So the sets K_n may have fewer holes than U , but not additional unnecessary holes.

Proof. For $n \in \mathbb{N}$ we define

$$V_n = \bigcup_{a \in \mathbb{C} \setminus U} B(a, 1/n) \cup \{z \in \mathbb{C} : |z| > n\}, \quad (26.3)$$

and $K_n = \mathbb{C} \setminus V_n$. Then the sets K_n , $n \in \mathbb{N}$, are compact and have the properties (i) and (ii) (exercise!).

Suppose that $n \in \mathbb{N}$ and C is a bounded component of $\mathbb{C} \setminus K_n = V_n$. Since the sets in the union (26.3) are connected, it follows that C contains each of these sets that it meets. Since C is also bounded, it follows that

$$C = \bigcup_{a \in A} B(a, 1/n)$$

for some non-empty set $A \subseteq \mathbb{C} \setminus U$. Pick $a \in A$, and let C' be the component of $\mathbb{C} \setminus U$ that contains a . Then $C' \subseteq \mathbb{C} \setminus U \subseteq \mathbb{C} \setminus K_n$ is connected and meets C . Hence $C' \subseteq C$, and so C' is bounded. This shows that C contains a bounded component of $\mathbb{C} \setminus U$. $\square$

Theorem 26.6 (Runge's Approximation Theorem). *Let $U \subseteq \mathbb{C}$ be open, $f \in H(U)$, and $A \subseteq \mathbb{C} \setminus U$ be a set that meets each bounded component of $\mathbb{C} \setminus U$.*

Then there exists a sequence $\{R_n\}$ of rational functions that have no poles outside the set $A \cup \{\infty\}$ such that $R_n \rightarrow f$ locally uniformly on U .

If $\mathbb{C} \setminus U$ has no bounded component, then there exists a sequence $\{P_n\}$ of polynomials such that $P_n \rightarrow f$ locally uniformly on U .

Again we can consider the second part as a special case of the first part if we choose $A = \emptyset$, but is useful to state this case explicitly.

Proof. Let K_n , $n \in \mathbb{N}$, be a compact exhaustion of U as in Lemma 26.5. Then every bounded component of one of the sets $\mathbb{C} \setminus K_n$ contains a bounded component of $\mathbb{C} \setminus U$, and hence a point in A . By Lemma 26.4 there exists a rational function R_n with no poles outside $A \cup \{\infty\}$ such that

$$|f(w) - R_n(w)| < 1/n \quad \text{for } w \in K_n.$$

Since every compact set $K \subseteq U$ lies in each set K_n for n large enough, it follows that $R_n \rightarrow f$ compactly and hence locally uniformly on U .

If $\mathbb{C} \setminus U$ has no bounded component, then none of the sets $\mathbb{C} \setminus K_n$ has a bounded component either. The second part of Lemma 26.4 then shows that we can find sequence $\{P_n\}$ of polynomials such that

$$|f(w) - P_n(w)| < 1/n \quad \text{for } w \in K_n.$$

Again this implies $P_n \rightarrow f$ locally uniformly on U as desired. $\square$

Corollary 26.7. Let $U \subseteq \mathbb{C}$ be a region such that $\widehat{\mathbb{C}} \setminus U$ is connected, and $f \in H(U)$. Then there exists a sequence of polynomials $\{P_n\}$ such that $P_n \rightarrow f$ locally uniformly on U .

Proof. Let K_n , $n \in \mathbb{N}$, be a compact exhaustion of U as in Lemma 26.5. Then for all $n \in \mathbb{N}$ the set $\mathbb{C} \setminus K_n$ has no bounded component.

Indeed, suppose V is a bounded component of one of the sets $\mathbb{C} \setminus K_n$. Then there exists a bounded component C of $\mathbb{C} \setminus U$ such that $C \subseteq V$, and so $(\widehat{\mathbb{C}} \setminus U) \cap V \neq \emptyset$.

Let $V' \subseteq \widehat{\mathbb{C}}$ be the union of all other components of $\mathbb{C} \setminus K_n$ and $\{\infty\}$. Then V and V' are open subsets of $\widehat{\mathbb{C}}$ (Note that $\mathbb{C} \setminus K_n$ has precisely one unbounded component. Together with ∞ it provides an open neighborhood of ∞ that lies in V').

We have $V \cap V' = \emptyset$, and

$$V \cup V' \supseteq (\mathbb{C} \setminus K_n) \cup \{\infty\} = \widehat{\mathbb{C}} \setminus K_n \supseteq \widehat{\mathbb{C}} \setminus U.$$

Since $(\widehat{\mathbb{C}} \setminus U) \cap V \neq \emptyset$ and $\infty \in (\widehat{\mathbb{C}} \setminus U) \cap V' \neq \emptyset$, we get a contradiction to our assumption that $\widehat{\mathbb{C}} \setminus U$ is connected.

Since no set $\mathbb{C} \setminus K_n$ has a bounded component, the improved approximation lemma (Lemma 26.4) provides polynomials P_n such that

$$|f(w) - P_n(w)| < 1/n \quad \text{for } w \in K_n.$$

Then $P_n \rightarrow f$ locally uniformly on U . $\square$

27 Characterizations of simply connected regions

Lemma 27.1. *Let $\Omega \subseteq \mathbb{C}$ be a region. The the following statements are true:*

- (a) *If Ω is homeomorphic to $\mathbb{D}$, then $\partial\Omega$ (as a subset of $\widehat{\mathbb{C}}$) is connected.*
- (b) *If $\partial\Omega$ is connected, then $\widehat{\mathbb{C}} \setminus \Omega$ is connected.*

Proof. (a) Let $\varphi: \mathbb{D} \rightarrow \Omega$ be a homeomorphism.

Claim: for $w \in \partial\Omega$, there exists a sequence $z_n \in \mathbb{D}$ with $|z_n| \rightarrow 1$ and $\varphi(z_n) \rightarrow w$.

Indeed, if $w \in \partial\Omega$, then $w \in \overline{\Omega}$, so there is a sequence $z_n \in \mathbb{D}$ such that $\varphi(z_n) \rightarrow w$. We claim $z_n \rightarrow a \in \overline{\mathbb{D}}$ with $|a| = 1$. If not, then some subsequence converges to $a \in \mathbb{D}$, and continuity would give $\varphi(a) = w \in \Omega$, contradicting $w \in \partial\Omega$. Conversely, if there is a sequence $z_n \in \mathbb{D}$ with $|z_n| \rightarrow 1$ and $\varphi(z_n) \rightarrow w$, then $w \in \overline{\Omega}$. If $w \in \Omega$, write $a = \varphi^{-1}(w) \in \mathbb{D}$; then continuity of φ^{-1} would imply $z_n \rightarrow a$, contradicting $|z_n| \rightarrow 1$. Hence $w \in \partial\Omega$.

Now let

$$A_n := \{z \in \mathbb{D} : 1 - 1/n < |z| < 1\}.$$

The Claim implies that

$$\partial\Omega = \bigcap_{n \in \mathbb{N}} \overline{\varphi(A_n)}.$$

Each set $\overline{\varphi(A_n)}$ is compact and connected, and the family is decreasing. Therefore $\partial\Omega$ is connected.

(b) Let $\text{ext}(\Omega)$ denote the set of exterior points of Ω . Then we have the disjoint union

$$\widehat{\mathbb{C}} = \Omega \sqcup \partial\Omega \sqcup \text{ext}(\Omega).$$

Suppose $\widehat{\mathbb{C}} \setminus \Omega = U \cup V$, where $U, V \subseteq \widehat{\mathbb{C}}$ are open and $U \cap V = \emptyset$. Since $\partial\Omega \subseteq \widehat{\mathbb{C}} \setminus \Omega$ is connected, either $\partial\Omega \subseteq U$ or $\partial\Omega \subseteq V$; say $\partial\Omega \subseteq U$. Set

$$U' := \Omega \cup U, \quad V' := V \setminus \overline{\Omega}.$$

Then U' and V' are open, disjoint, and

$$U' \cup V' \supseteq \Omega \cup \partial\Omega \cup \text{ext}(\Omega) = \widehat{\mathbb{C}}.$$

Hence $V' = \emptyset$. In particular $V \subseteq \overline{\Omega}$ and $V \cap \text{ext}(\Omega) = \emptyset$, so $V = \emptyset$. Thus $\widehat{\mathbb{C}} \setminus \Omega$ is connected. $\square$

Theorem 27.2 (Main Theorem of Classical Complex Analysis). *Let $\Omega \subseteq \mathbb{C}$ be a region. Then the following conditions are equivalent:*

- (a) Ω is simply connected.
- (b) Every loop in Ω is null-homotopic.
- (c) Every loop in Ω is null-homologous.
- (d) $\int_{\gamma} f(z) dz = 0$ whenever $f \in H(\Omega)$ and γ is a piecewise smooth loop in Ω .
- (e) $\int_{\alpha} f = \int_{\beta} f$ whenever $f \in H(\Omega)$ and α, β are piecewise smooth paths in Ω with the same endpoints.
- (f) Every holomorphic function on Ω has a primitive.
- (g) For every harmonic function u on Ω there exists $f \in H(\Omega)$ such that $u = \operatorname{Re} f$.
- (h) Every harmonic function on Ω has a harmonic conjugate.
- (i) Every zero-free holomorphic function on Ω has a holomorphic logarithm.
- (j) Every zero-free holomorphic function on Ω has a holomorphic square root.
- (k) Either $\Omega = \mathbb{C}$, or Ω is conformally equivalent to $\mathbb{D}$.
- (l) Ω is homeomorphic to $\mathbb{D}$.
- (m) $\partial\Omega$ is connected (as a subset of $\widehat{\mathbb{C}}$).
- (n) $\widehat{\mathbb{C}} \setminus \Omega$ is connected.
- (o) For every $f \in H(\Omega)$ there exists a sequence of polynomials (p_n) such that $p_n \rightarrow f$ locally uniformly on Ω .

Proof. We will prove

$$(a) \Leftrightarrow (b) \Rightarrow (c) \Rightarrow (d) \Rightarrow (e) \Rightarrow (f) \Rightarrow (g) \Rightarrow (h) \Rightarrow (i) \Rightarrow (j) \Rightarrow (k) \Rightarrow (l) \Rightarrow (b).$$

and $(l) \Rightarrow (m) \Rightarrow (n) \Rightarrow (o) \Rightarrow (d)$.

(a) $\Leftrightarrow$ (b): by definition.

(b) $\Rightarrow$ (c): follows from Corollary 20.8.

(c) $\Rightarrow$ (d): Cauchy's integral theorem.

(d) $\Rightarrow$ (e): if α and β have the same endpoints, then

$$\int_{\alpha} f - \int_{\beta} f = \int_{\alpha-\beta} f = 0.$$

(e) $\Rightarrow$ (f): Fix a base point $a \in \Omega$ and define

$$F(z) := \int_{\alpha} f(w) dw,$$

where α is any piecewise smooth path from a to z . By (e), F is well defined, and standard arguments (as in Corollary 20.9 (a)) show that $F \in H(\Omega)$ and $F' = f$.

(f) $\Rightarrow$ (g): Let u be harmonic on Ω ; so $u \in C^2(\Omega)$ and $\Delta u = u_{xx} + u_{yy} = 0$. Define

$$g := u_x - iu_y.$$

Then $g \in C^1(\Omega)$ and, writing $a := \operatorname{Re} g = u_x$, $b := \operatorname{Im} g = -u_y$, we obtain

$$a_x = u_{xx} = -u_{yy} = b_y, \quad a_y = u_{xy} = u_{yx} = -b_x.$$

Thus g satisfies the Cauchy–Riemann equations, hence $g \in H(\Omega)$. By (f), there exists $h \in H(\Omega)$ with $h' = g$. Therefore

$$h' = (\operatorname{Re} h)_x + i(\operatorname{Im} h)_x = (\operatorname{Re} h)_x - i(\operatorname{Re} h)_y = g = u_x - iu_y.$$

So $\nabla u = \nabla(\operatorname{Re} h)$, and hence $u = \operatorname{Re} h + c = \operatorname{Re}(h + c)$ for some constant $c \in \mathbb{R}$.

(g) $\Rightarrow$ (h): if $u = \operatorname{Re} f$, where $f \in H(\Omega)$, then $v := \operatorname{Im} f$ is a harmonic conjugate of u .

(h) $\Rightarrow$ (i): let $f \in H(\Omega)$ be zero-free. Set

$$u := \log |f| = \frac{1}{2} \log |f|^2.$$

Locally one may choose a branch of $\log f$, which implies

$$\Delta u = 4u_{z\bar{z}} = 2(\log f + \log \bar{f})_{z\bar{z}} = 0.$$

Hence u is harmonic. By (h), u has a harmonic conjugate v , so

$$g := u + iv \in H(\Omega).$$

Then e^g/f is holomorphic and has modulus 1 everywhere on U :

$$\left| \frac{e^g}{f} \right| = \frac{e^{\operatorname{Re} g}}{|f|} = \frac{e^u}{|f|} = 1.$$

By the maximum principle e^g/f is constant, say $= e^{i\theta}$ with $\theta \in \mathbb{R}$. Therefore,

$$f = e^{g-i\theta},$$

and $g - i\theta$ is a holomorphic logarithm of f .

(i) $\Rightarrow$ (j): if $f = e^g$, then $s := e^{g/2}$ is a holomorphic square root of f .

(j) $\Rightarrow$ (k): we have seen in the proof of the Riemann mapping theorem that if $\Omega \neq \mathbb{C}$ and every zero-free holomorphic function on Ω has a holomorphic square root, then Ω is conformally equivalent to $\mathbb{D}$.

(k) $\Rightarrow$ (l): if Ω conformally equivalent to $\mathbb{D}$, then it is homeomorphic to $\mathbb{D}$; and if $\Omega = \mathbb{C}$, then Ω is also homeomorphic to $\mathbb{D}$ (exercise).

(l) $\Rightarrow$ (m): Lemma 27.1 (a).

(m) $\Rightarrow$ (n): Lemma 27.1 (b).

(n) $\Rightarrow$ (o): This follows from Runge approximation by polynomials as in Corollary 26.7, since $\widehat{\mathbb{C}} \setminus \Omega$ is connected.

(o) $\Rightarrow$ (d): let $f \in H(\Omega)$ and let p_n be polynomials converging locally uniformly to f on Ω . If γ is a piecewise smooth loop in Ω , then

$$\int_{\gamma} f = \lim_{n \rightarrow \infty} \int_{\gamma} p_n = 0$$

by uniform convergence on γ^* and Cauchy's integral theorem for polynomials.

□

28 The Gamma Function

Definition 28.1. For $\operatorname{Re}(s) > 0$, the Gamma function is defined by

$$\Gamma(s) := \int_0^\infty t^{s-1} e^{-t} dt.$$

Definition 28.2. The Euler–Mascheroni constant is defined by

$$\gamma := \lim_{n \rightarrow \infty} \left(\sum_{k=1}^n \frac{1}{k} - \log n \right).$$

Proposition 28.3. The limit defining γ exists.

Proof. Let

$$a_n = \sum_{k=1}^n \frac{1}{k} - \log n.$$

Then

$$a_{n+1} - a_n = \frac{1}{n+1} - \log \left(1 + \frac{1}{n} \right).$$

Using $\log(1+x) = x - \frac{x^2}{2} + O(x^3)$, we get

$$a_{n+1} - a_n = -\frac{1}{2n^2} + O(n^{-3}) < 0$$

for large n , so (a_n) is eventually decreasing. It is bounded below, hence convergent. $\square$

Theorem 28.4 (Stirling's formula). As $|z| \rightarrow \infty$ in $|\arg z| \leq \pi - \varepsilon$,

$$\Gamma(z) = \sqrt{2\pi} z^{z-\frac{1}{2}} e^{-z} \left(1 + O\left(\frac{1}{z}\right) \right).$$

Proof. We sketch a standard proof using Laplace's method.

Write

$$\Gamma(z) = \int_0^\infty e^{z \log t - t} dt.$$

Set $t = zu$, then

$$\Gamma(z) = z^z \int_0^\infty e^{z(\log u - u)} du.$$

The function $\phi(u) = \log u - u$ has a maximum at $u = 1$.

Expand near $u = 1$:

$$\phi(u) = -1 - \frac{(u-1)^2}{2} + O((u-1)^3).$$

Thus

$$\Gamma(z) \sim z^z e^{-z} \int_{-\infty}^{\infty} e^{-z(u-1)^2/2} du.$$

This Gaussian integral gives

$$\Gamma(z) \sim z^z e^{-z} \sqrt{\frac{2\pi}{z}}.$$

□

Theorem 28.5 (Weierstrass product).

$$\frac{1}{\Gamma(z)} = ze^{\gamma z} \prod_{n=1}^{\infty} \left(1 + \frac{z}{n}\right) e^{-z/n}.$$

Proof. Define

$$G(z) = ze^{\gamma z} \prod_{n=1}^{\infty} \left(1 + \frac{z}{n}\right) e^{-z/n}.$$

Using estimates

$$\log\left(1 + \frac{z}{n}\right) - \frac{z}{n} = O(n^{-2}),$$

the product converges uniformly on compact sets.

Compute logarithmic derivative:

$$\frac{G'}{G} = \frac{1}{z} + \gamma + \sum_{n=1}^{\infty} \left(\frac{1}{n+z} - \frac{1}{n}\right).$$

One shows this equals $-\Gamma'/\Gamma$, hence $G = 1/\Gamma$.

□

Theorem 28.6 (Reflection formula).

$$\Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin(\pi z)}.$$

Proof. Define $F(z) = \Gamma(z)\Gamma(1-z)\sin(\pi z)$.

Using the functional equation and $\sin(\pi(z+1)) = -\sin(\pi z)$, we get $F(z+1) = F(z)$.

Using Stirling's formula, F is bounded in vertical strips, hence bounded entire and thus constant.

Evaluate at $z = 1/2$ to obtain the constant π . $\square$

Theorem 28.7 (Duplication formula).

$$\Gamma(z)\Gamma\left(z + \frac{1}{2}\right) = 2^{1-2z}\sqrt{\pi}\Gamma(2z).$$

Proof. Define

$$H(z) = \frac{\Gamma(z)\Gamma\left(z + \frac{1}{2}\right)}{\Gamma(2z)}.$$

Using the functional equation, $H(z+1) = 2^{-2}H(z)$, hence $2^{2z}H(z)$ is periodic.

Using Stirling's formula, it is bounded, hence constant.

Evaluate at $z = 1/2$. $\square$

Theorem 28.8 (Characterization (Problem 6)). *Let f be holomorphic on $\operatorname{Re}(s) > 0$ satisfying:*

1. $f(1) = 1$,
2. $f(s+1) = sf(s)$,
3. f has at most exponential growth in vertical strips.

Then $f(s) = \Gamma(s)$.

Proof. Define $g(s) = f(s)/\Gamma(s)$.

Then $g(s+1) = g(s)$, so g is periodic.

Using Stirling's formula, g is bounded in vertical strips, hence bounded entire, so constant.

Since $g(1) = 1$, we get $g \equiv 1$. $\square$

29 Solutions and hints for selected exercises

Problem 1.6: Let $z_0 \in \mathbb{C}$ be a root of the polynomial P . In order to show that $\operatorname{Re}(z_0) < 0$ or $|z_0| < (1 + \sqrt{5})/2$, we may assume that $\operatorname{Re}(z_0) \geq 0$ and $|z_0| > 1$ (otherwise, there is nothing to prove). Then $z_0 \neq 0$ and

$$\operatorname{Re}(a_{n-1}/z_0) = a_{n-1} \operatorname{Re}(1/z_0) = \frac{a_{n-1}}{|z_0|^2} \operatorname{Re}(\overline{z_0}) = \frac{a_{n-1}}{|z_0|^2} \operatorname{Re}(z_0) \geq 0.$$

Here we used that $a_{n-1} \in [0, 1]$ and $\operatorname{Re}(z_0) \geq 0$.

It follows that

$$\begin{aligned} 1 &\leq 1 + \operatorname{Re}(a_{n-1}/z_0) = \operatorname{Re}(1 + a_{n-1}/z_0) \\ &\leq |1 + a_{n-1}/z_0| = |1 + a_{n-1}/z_0 - P(z_0)/z_0^n| \\ &= |a_{n-2}/z_0^2 + \cdots + a_0/z_0^n| \\ &\leq \frac{1}{|z_0|^2} + \cdots + \frac{1}{|z_0|^n} < \frac{1}{|z_0|^2} \sum_{k=0}^{\infty} \frac{1}{|z_0|^k} \\ &= \frac{1}{|z_0|^2(1 - 1/|z_0|)} = \frac{1}{|z_0|^2 - |z_0|}. \end{aligned}$$

Hence $|z_0|^2 - |z_0| < 1$, or equivalently,

$$0 > |z_0|^2 - |z_0| - 1 = \left(|z_0| - (1 + \sqrt{5})/2\right) \left(|z_0| - (1 - \sqrt{5})/2\right).$$

Here the second factor is positive, and so the first factor must be negative, that is, $|z_0| < (1 + \sqrt{5})/2$, as desired.

Problem 1.7: (a) We have

$$P(z) = z^n + a_{n-1}z^{n-1} + a_{n-2}z^{n-2} + \cdots + a_0 = \prod_{j=1}^n (z - z_j).$$

By multiplying out the last product and comparing coefficients of corresponding powers of z , we see that

$$\sum_j z_j = -a_{n-1}, \text{ and } \sum_{j < k} z_j z_k = a_{n-2}.$$

It follows that

$$\sum_j z_j^2 = \left(\sum_j z_j \right)^2 - 2 \sum_{j < k} z_j z_k = a_{n-1}^2 - 2a_{n-2}.$$

(b) The basic idea is to find a monic polynomial P (that is, one with leading coefficient 1) that has exactly the roots $u_k := 1/(\zeta_k - 1)$, $k = 1, \dots, n-1$ (for fixed $n \in \mathbb{N}$) and then apply (a) to find $\sum_k u_k^2$.

Note first that the numbers $\zeta_k = \exp\left(\frac{2\pi i k}{n}\right)$, $k = 1, \dots, n-1$, are exactly the n -th roots of unity except 1. So a polynomial that has exactly these roots is given by

$$Q(z) = \frac{z^n - 1}{z - 1} = z^{n-1} + z^{n-2} + \dots + 1.$$

Since $\zeta_k = 1 + 1/u_k$ the expression $Q(1 + 1/z)$ has exactly the zeros u_k , $k = 1, \dots, n-1$. This is not a polynomial in z , but a polynomial of $1/z$ of degree $n-1$. To make it a polynomial in z we multiply the expression by z^{n-1} . It follows that the polynomial

$$\begin{aligned} \tilde{P}(z) &= z^{n-1}Q(1 + 1/z) = z^{n-1} \frac{(1 + 1/z)^n - 1}{(1 + 1/z) - 1} \\ &= z^n((1 + 1/z)^n - 1) = (z + 1)^n - z^n \\ &= \binom{n}{1} z^{n-1} + \binom{n}{2} z^{n-2} + \binom{n}{3} z^{n-3} + \dots + 1 \end{aligned}$$

has exactly the roots $u_k := 1/(\zeta_k - 1)$, $k = 1, \dots, n-1$. In order to make this polynomial monic, we divide it by $\binom{n}{1} = n$. It follows that

$$P(z) = \frac{1}{n} \tilde{P}(z) = z^{n-1} + \frac{n-1}{2} z^{n-2} + \frac{(n-1)(n-2)}{6} z^{n-3} + \dots$$

is the polynomial we are looking for. It now follows from (a) (applied to $n-1$ instead of n) that

$$\begin{aligned} \sum_{k=1}^{n-1} \frac{1}{(\zeta_k - 1)^2} &= \sum_{k=1}^{n-1} u_k^2 = \left(\frac{n-1}{2} \right)^2 - \frac{(n-1)(n-2)}{3} \\ &= (n-1)(5-n)/12. \end{aligned}$$

Let $U \subseteq \mathbb{C}$ be an open set with $0 \in U$. Show that then it is impossible to define a continuous square root function on U ; more precisely, show that

there exists no continuous function $S: U \rightarrow \mathbb{C}$ such that $S(z)^2 = z$ for each $z \in U$. Hint: Consider a suitable parametrization of a small circle centered at 0.

Problem 2.6: We argue by contradiction and assume that there is a continuous function $S: U \rightarrow \mathbb{C}$ such that $S(z)^2 = z$ for all $z \in U$, where U is an open subset of $\mathbb{C}$ containing 0. Then there exists $r > 0$ such that the circle $C := \{z \in \mathbb{C} : |z| = r\}$ lies in U and so we can define a function $g: \mathbb{R} \rightarrow \mathbb{C}$ as

$$g(t) = \frac{1}{\sqrt{r}} e^{-it/2} S(re^{it})$$

for $t \in \mathbb{R}$. Since S and the exponential function are continuous, g is also continuous. Moreover, for $t \in \mathbb{R}$ we have

$$g(t)^2 = \frac{1}{r} e^{-it} S(re^{it})^2 = 1$$

and so $g(t) \in \{-1, 1\}$. Since the latter set is disconnected, but $g(\mathbb{R})$ is connected as the continuous image of the connected set $\mathbb{R}$, we must have $g(\mathbb{R}) = \{1\}$ or $g(\mathbb{R}) = \{-1\}$. In other words, $g(t) = 1$ for all $t \in \mathbb{R}$ or $g(t) = -1$ for all $t \in \mathbb{R}$. Now both cases lead to a contradiction.

Indeed, in the first case we have $S(re^{it}) = \sqrt{r}e^{it/2}$ for all $t \in \mathbb{R}$. But this is impossible as one immediately sees by evaluating both sides of this identity at $t = 0$ and at $t = 2\pi$: we obtain $S(r) = \sqrt{r}$ and $S(r) = -\sqrt{r}$ which is absurd.

Similarly, in the second case we have $S(re^{it}) = -\sqrt{r}e^{it/2}$ for all $t \in \mathbb{R}$ which again leads to a contradiction if we evaluate at $t = 0$ and at $t = 2\pi$.

Problem 3.7: As usual, let $z = x + iy$, where $x, y \in \mathbb{R}$. In the following, we denote the application of a partial differential operator such as $\frac{\partial}{\partial z}$, $\frac{\partial}{\partial x}$, etc. to a function by a corresponding subscript on the function.

If k is a C^2 -smooth function of z , then we have

$$\begin{aligned} k_{z\bar{z}} &= \frac{1}{2}(k_x - ik_y)_{\bar{z}} \\ &= \frac{1}{4}((k_x - ik_y)_x + i(k_x - ik_y)_y) \\ &= \frac{1}{4}(k_{xx} + k_{yy}) = \frac{1}{4}\Delta k. \end{aligned}$$

So symbolically we have

$$\Delta = 4 \frac{\partial^2}{\partial z \partial \bar{z}}.$$

Now suppose that h is a C^2 -smooth function and f a holomorphic function such that $g = h \circ f$ is defined. Then $f_{\bar{z}} = 0$ and so

$$(\bar{f})_z = \overline{(f_{\bar{z}})} = 0.$$

We consider h as a function of a complex variable w . Then g is obtained by setting $w = f(z)$. The chain rule then implies that

$$\begin{aligned} g_z &= (h \circ f)_z \\ &= h_w \cdot w_z + h_{\bar{w}} \cdot (\bar{w})_z \\ &= h_w \cdot f_z + h_{\bar{w}} \cdot (\bar{f})_z \\ &= h_w \cdot f_z = h_w \cdot f'. \end{aligned}$$

Now f' is also holomorphic and so $(f')_{\bar{z}} = 0$. It follows that

$$\begin{aligned} \frac{1}{4}\Delta(h \circ f) &= \frac{1}{4}\Delta g = (g_z)_{\bar{z}} = (h_w \cdot f')_{\bar{z}} \\ &= (h_w)_{\bar{z}} \cdot f' + h_w \cdot (f')_{\bar{z}} \\ &= (h_w)_{\bar{z}} \cdot f' \\ &= f' \cdot (h_{ww} \cdot w_{\bar{z}} + h_{w\bar{w}} \cdot (\bar{w})_{\bar{z}}) \\ &= f' \cdot (h_{ww} \cdot f_{\bar{z}} + h_{w\bar{w}} \cdot (\bar{f})_{\bar{z}}) \\ &= f' \cdot h_{w\bar{w}} \cdot \bar{f}_z = f' \cdot \bar{f}' \cdot h_{w\bar{w}} = |f'|^2 h_{w\bar{w}} \\ &= \frac{1}{4}|f'|^2((\Delta h) \circ f). \end{aligned}$$

We obtain the following chain rule for Laplacians under precomposition with a holomorphic map f :

$$\Delta(h \circ f) = |f'|^2((\Delta h) \circ f).$$

Now if h is harmonic, then $\Delta h = 0$. It follows that for $g = h \circ f$ we also have $\Delta g = 0$, and so g is harmonic.

Problem 3.8: We use the chain rule for Laplacians proved in Problem 3.7.

For this note that $s(z) = h(f(z))$, where f is holomorphic and

$$h(w) = \log(1 + |w|^2) = \log(1 + w\bar{w}).$$

Now we have

$$\begin{aligned} \Delta h &= 4h_{w\bar{w}} = 4(h_w)_{\bar{w}} = 4\left(\frac{\bar{w}}{1 + w\bar{w}}\right)_{\bar{w}} \\ &= 4\frac{(1 + w\bar{w}) - \bar{w}w}{(1 + w\bar{w})^2} = \frac{4}{(1 + |w|^2)^2}, \end{aligned}$$

and so

$$\begin{aligned}\Delta s(z) &= \Delta(h \circ f)(z) = |f'(z)|^2(\Delta h)(f(z)) \\ &= \frac{4|f'(z)|^2}{(1 + |f(z)|^2)^2} \geq 0.\end{aligned}$$

It follows that s is indeed a subharmonic function.

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